

voloshin-hypergraph: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

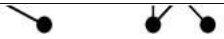
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00001	9	■ 1.000	$K_{\{5\}}$	K_5	
voloshin-hypergraph_F00002	9	■ 1.000	$K_{\{3,3\}}$	$K_{3,3}$	
voloshin-hypergraph_F00003	15	■ 0.994	$\rightarrow$	$\Rightarrow$	• bold type is used for definitions and paragraph headings, <i>italic type</i> is used to call a special attention and symbol $\square$ is used to indicate the end of proofs
voloshin-hypergraph_F00004	21	■ 0.999	$A, B, \ldots$	$A, B, \ldots$	
voloshin-hypergraph_F00005	21	■ 1.000	X, Y, Z	X, Y, Z	
voloshin-hypergraph_F00006	21	■ 1.000	$a, b, \ldots, e, \ldots, x, y, z$	$a, b, \ldots, e, \ldots, x, y, z$	
voloshin-hypergraph_F00007	22	■ 1.000	$G=(X, E)$	$G=(X, E)$	f elements in the list does not matter
voloshin-hypergraph_F00008	22	■ 1.000	A	A	In a graph, the set of vertices is denoted by X and is written as $X=\{x_1, x_2, \ldots, x_n\}$ where x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00009	22	■ 1.000	$ A $	$ A $	In a graph, the set of vertices is denoted by X and is written as $X=\{x_1, x_2, \ldots, x_n\}$ where x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00010	22	■ 0.883	$\boldsymbol{\emptyset}$	$\emptyset$	x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is written as $E=\{e_1, e_2, \ldots, e_m\}$ where e_i is the i -th edge and m is the number of edges. Each
voloshin-hypergraph_F00011	22	■ 0.967	X	X	to “invent” the last letter to denote the entire graph: G . Now we are ready to present the
voloshin-hypergraph_F00012	22	■ 0.967	$X=\left\{x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right\}$	$X=\left\{x_1, x_2, \ldots, x_n\right\}$	to “invent” the last letter to denote the entire graph: G . Now we are ready to present the
voloshin-hypergraph_F00013	22	■ 0.971	$x_{\{i\}}$	x_i	formal definition of a graph:
voloshin-hypergraph_F00014	22	■ 0.971	i	i	formal definition of a graph:
voloshin-hypergraph_F00015	22	■ 0.971	n	n	formal definition of a graph:
voloshin-hypergraph_F00016	22	■ 0.971	E	E	formal definition of a graph:
voloshin-hypergraph_F00017	22	■ 0.996	$E=\left\{e_{\{1\}}, e_{\{2\}}, \ldots, e_{\{m\}}\right\}$	$E=\left\{e_1, e_2, \ldots, e_m\right\}$	Definition 1.1.1 A graph G is a set X of vertices together with a set E of edges. It is
voloshin-hypergraph_F00018	22	■ 0.996	$e_{\{i\}}$	e_i	Definition 1.1.1 A graph G is a set X of vertices together with a set E of edges. It is
voloshin-hypergraph_F00019	22	■ 0.996	m	m	Definition 1.1.1 A graph G is a set X of vertices together with a set E of edges. It is
voloshin-hypergraph_F00020	22	■ 1.000	G	G	$G=(X, E)$.
voloshin-hypergraph_F00021	22	■ 1.000	$n=5$	$n=5$	are called disjoint . For example, vertices x_1 and x_2 are adjacent, but vertices x_1 and x_3 are disjoint. For a given vertex x , the number of all vertices adjacent to it is called degree of the
voloshin-hypergraph_F00022	22	■ 1.000	$m=7$	$m=7$	are called disjoint . For example, vertices x_1 and x_2 are adjacent, but vertices x_1 and x_3 are disjoint. For a given vertex x , the number of all vertices adjacent to it is called degree of the
voloshin-hypergraph_F00023	22	■ 1.000	$X=\left\{x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}, x_{\{5\}}\right\}$	$X=\left\{x_1, x_2, x_3, x_4, x_5\right\}$	disjoint. For a given vertex x , the number of all vertices adjacent to it is called degree of the vertex x , denoted by $d(x)$. In our example, $d(x_1)=2$, $d(x_3)=4$, and so on. The maximum

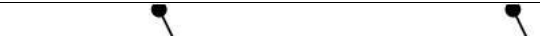
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00024	22	■ 1.000	$E=\left\{e_{\{1\}}, e_{\{2\}}, e_{\{3\}}, e_{\{4\}}, e_{\{5\}}, e_{\{6\}}, e_{\{7\}}\right\}$	$E=\left\{e_1, e_2, e_3, e_4, e_5, e_6, e_7\right\}$	disjoint. For a given vertex x , the number of all vertices adjacent to it is called degree of the vertex x , denoted by $d(x)$. In our example, $d(x_1)=2$, $d(x_2)=4$, and so on. The maximum
voloshin-hypergraph_F00025	22	■ 0.998	$e_{\{1\}}=\left\{x_{\{1\}}, x_{\{2\}}\right\}, e_{\{2\}}=\left\{x_{\{2\}}, x_{\{3\}}\right\}, e_{\{3\}}=\left\{x_{\{3\}}, x_{\{4\}}\right\}, e_{\{4\}}=\left\{x_{\{4\}}, x_{\{5\}}\right\}, e_{\{5\}}=\left\{x_{\{5\}}, x_{\{1\}}\right\}$	$e_1=\left\{x_1, x_2\right\}, e_2=\left\{x_2, x_3\right\}, e_3=\left\{x_3, x_4\right\}, e_4=\left\{x_4, x_5\right\}, e_5=\left\{x_5, x_1\right\}$	degree over all vertices is called the maximum degree of G , denoted by $\Delta(G)$. For graph
voloshin-hypergraph_F00026	22	■ 0.999	$e_{\{6\}}=\left\{x_{\{2\}}, x_{\{5\}}\right\}, e_{\{7\}}=\left\{x_{\{2\}}, x_{\{4\}}\right\}$	$e_6=\left\{x_2, x_5\right\}, e_7=\left\{x_2, x_4\right\}$	G , see Figure 1.2, $\Delta(G)=d(x_2)=4$.
voloshin-hypergraph_F00027	22	■ 0.999	$E=\left\{\left\{x_{\{1\}}, x_{\{2\}}\right\},\left\{x_{\{2\}}, x_{\{3\}}\right\},\left\{x_{\{3\}}, x_{\{4\}}\right\},\left\{x_{\{4\}}, x_{\{5\}}\right\},\left\{x_{\{5\}}, x_{\{1\}}\right\}\right\}$	$E=\left\{\left\{x_1, x_2\right\},\left\{x_2, x_3\right\},\left\{x_3, x_4\right\},\left\{x_4, x_5\right\},\left\{x_5, x_1\right\}\right\}$	G , see Figure 1.2, $\Delta(G)=d(x_2)=4$.
voloshin-hypergraph_F00028	22	■ 1.000	$\left\{\left\{x_{\{2\}}, x_{\{5\}}\right\},\left\{x_{\{2\}}, x_{\{4\}}\right\}\right\}$	$\left\{x_2, x_5\right\},\left\{x_2, x_4\right\}$	
voloshin-hypergraph_F00029	22	■ 1.000	$x_{\{1\}}$	x_1	
voloshin-hypergraph_F00030	22	■ 1.000	$x_{\{2\}}$	x_2	
voloshin-hypergraph_F00031	22	■ 1.000	$x_{\{3\}}$	x_3	
voloshin-hypergraph_F00032	22	■ 1.000	x	x	
voloshin-hypergraph_F00033	22	■ 1.000	$d(x)$	$d(x)$	
voloshin-hypergraph_F00034	22	■ 1.000	$d\left(x_{\{1\}}\right)=2, d\left(x_{\{2\}}\right)=4$	$d\left(x_1\right)=2, d\left(x_2\right)=4$	
voloshin-hypergraph_F00035	22	■ 1.000	$\Delta(G)$	$\Delta(G)$	
voloshin-hypergraph_F00036	22	■ 1.000	$\Delta(G)=d\left(x_{\{2\}}\right)=4$	$\Delta(G)=d\left(x_2\right)=4$	—
voloshin-hypergraph_F00037	23	■ 1.000	$N(x)$	$N(x)$	noted by $N(x)$. In our graph, for example, $N(x_1)=\left\{x_2, x_5\right\}$. Evidently, the degree of a
voloshin-hypergraph_F00038	23	■ 1.000	$N\left(x_{\{1\}}\right)=\left\{x_{\{2\}}, x_{\{5\}}\right\}$	$N\left(x_1\right)=\left\{x_2, x_5\right\}$	noted by $N(x)$. In our graph, for example, $N(x_1)=\left\{x_2, x_5\right\}$. Evidently, the degree of a
voloshin-hypergraph_F00039	23	■ 1.000	$d\left(x_{\{1\}}\right)=\left N\left(x_{\{1\}}\right)\right =2$	$d\left(x_1\right)=\left N\left(x_1\right)\right =2$	vertex is the cardinality (the number of elements) of its neighborhood: $d(x_1)=\left N(x_1)\right =2$,
voloshin-hypergraph_F00040	23	■ 1.000	$d\left(x_{\{2\}}\right)=\left N\left(x_{\{2\}}\right)\right =4$	$d\left(x_2\right)=\left N\left(x_2\right)\right =4$	$d(x_2)=\left N(x_2)\right =4$, and so on.
voloshin-hypergraph_F00041	23	■ 1.000	$e_{\{1\}}$	e_1	x belongs to an edge e , then we say that they are incident to each other. In the example
voloshin-hypergraph_F00042	23	■ 1.000	$e_{\{2\}}$	e_2	x belongs to an edge e , then we say that they are incident to each other. In the example
voloshin-hypergraph_F00043	23	■ 1.000	$e_{\{3\}}$	e_3	x belongs to an edge e , then we say that they are incident to each other. In the example
voloshin-hypergraph_F00044	23	■ 1.000	e	e	above, edge e_1 is incident to vertex x_1 and is not incident to vertex x_5 and so on. As one can see, adjacency is referred to the elements of the same type and incidence is referred to the
voloshin-hypergraph_F00045	23	■ 1.000	$x_{\{5\}}$	x_5	elements of different types.
voloshin-hypergraph_F00046	23	■ 1.000	$V(G)$	$V(G)$	by $n=n(G)$, and the number of edges by $m=m(G)$. The set of edges containing a
voloshin-hypergraph_F00047	23	■ 1.000	$E(G)$	$E(G)$	vertex is denoted by $E(x)$. In our example, $V(G)=X=\left\{x_1, x_2, x_3, x_4, x_5\right\}, X =n=5$,
voloshin-hypergraph_F00048	23	■ 1.000	$G=(V(G), E(G))$	$G=(V(G), E(G))$	by $n=n(G)$, and the number of edges by $m=m(G)$. The set of edges containing a
voloshin-hypergraph_F00049	23	■ 1.000	$n=n(G)$	$n=n(G)$	vertex is denoted by $E(x)$. In our example, $V(G)=X=\left\{x_1, x_2, x_3, x_4, x_5\right\}, X =n=5$, $E(G)=E=\left\{e_1, e_2, e_3, e_4, e_5, e_6, e_7\right\}, E =m=7$ $E(x_1)=\left\{e_1, e_5\right\}$, and so on.

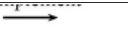
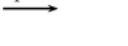
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00050	23	■ 1.000	$m=m(G)$	$m = m(G)$	$E(x_1) = \{e_1, e_5\}$, and so on.
voloshin-hypergraph_F00051	23	■ 1.000	$E(x)$	$E(x)$	
voloshin-hypergraph_F00052	23	■ 1.000	$V(G)=X=\left\{x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}, x_{\{5\}}\right\}\backslash\right\}, X =n=5$	$V(G) = X = \{x_1, x_2, x_3, x_4, x_5\}, X = n = 5$	
voloshin-hypergraph_F00053	23	■ 0.803	$E(G)=E=\left\{\left\{x_{\{1\}}, x_{\{2\}}\right\}\backslash\right\},\left\{x_{\{2\}}, x_{\{3\}}\right\}\backslash\right\},\left\{x_{\{3\}}, x_{\{4\}}\right\}\backslash\right\},\left\{x_{\{4\}}, x_{\{5\}}\right\}\backslash\right\},\left\{x_{\{5\}}, x_{\{1\}}\right\}\backslash\right\},\left\{x_{\{2\}}, x_{\{5\}}\right\}\backslash\right\},\left\{x_{\{2\}}, x_{\{4\}}\right\}\backslash\right\}\backslash\right\}, E =m=7$	$E(G) = E = \{\{x_1, x_2\}, \{x_2, x_3\}, \{x_3, x_4\}, \{x_4, x_5\}, \{x_5, x_1\}, \{x_2, x_5\}, \{x_2, x_4\}\}, E = m = 7$	Proposition 1.1.1 (Degree equality) <i>For any graph $G = (X, E)$, the following equality</i>
voloshin-hypergraph_F00054	23	■ 1.000	$E\left(x_{\{1\}}\right)=\left\{e_{\{1\}}, e_{\{5\}}\right\}\backslash\right\}$	$E\left(x_1\right)=\left\{e_1, e_5\right\}$	<i>notas:</i>
voloshin-hypergraph_F00055	23	—	$\backslash square$	$\square$	—
voloshin-hypergraph_F00056	23	■ 1.000	$n-1$	$n-1$	
voloshin-hypergraph_F00057	23	■ 1.000	$0 \leq d(x) \leq n-1$	$0 \leq d(x) \leq n-1$	
voloshin-hypergraph_F00058	23	■ 0.998	$n-1.0$	$n-1.0$	
voloshin-hypergraph_F00059	23	■ 1.000	$d\left(x_{\{1\}}\right)=d\left(x_{\{3\}}\right)=2$	$d\left(x_1\right)=d\left(x_3\right)=2$	
voloshin-hypergraph_F00060	23	■ 1.000	$d\left(x_{\{4\}}\right)=d\left(x_{\{5\}}\right)=3$	$d\left(x_4\right)=d\left(x_5\right)=3$	
voloshin-hypergraph_F00061	24	■ 1.000	$G_{\{1\}}, G_{\{2\}}$	G_1, G_2	
voloshin-hypergraph_F00062	24	■ 1.000	$G_{\{3\}}$	G_3	
voloshin-hypergraph_F00063	24	■ 0.999	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}$	x_1, x_2, x_3, x_4	
voloshin-hypergraph_F00064	25	■ 1.000	$x_{\{4\}}$	x_4	Let $X = \{x_1, x_2, x_3, x_4, x_5\}$ have the smallest number of parts, let the subsets, in such a way that no subset contains adjacent vertices? Each such subset will be considered as the subset
voloshin-hypergraph_F00065	25	■ 1.000	$\left\{x_{\{1\}}\right\}\backslash\right\},\left\{x_{\{2\}}\right\}\backslash\right\},\left\{x_{\{3\}}\right\}\backslash\right\}$	$\left\{x_1\right\},\left\{x_2\right\},\left\{x_3\right\}$	one can see that any partition of X into two subsets leaves edges in one of the subsets and
voloshin-hypergraph_F00066	25	■ 1.000	$\left\{x_{\{4\}}\right\}\backslash\right\},\left\{x_{\{5\}}\right\}\backslash\right\}$	$\left\{x_4\right\},\left\{x_5\right\}$	therefore is not good. Strict mathematical proof would be the following: consider triangle
voloshin-hypergraph_F00067	25	■ 1.000	$\left\{x_{\{1\}}, x_{\{3\}}\right\}\backslash\right\},\left\{x_{\{2\}}\right\}\backslash\right\},\left\{x_{\{4\}}\right\}\backslash\right\},\left\{x_{\{5\}}\right\}\backslash\right\}$	$\left\{x_1, x_3\right\},\left\{x_2\right\},\left\{x_4\right\},\left\{x_5\right\}$	formed by vertices x_1, x_2 , and x_5 ; any partition of it into two parts (even if one part is empty)
voloshin-hypergraph_F00068	25	■ 1.000	$\left\{x_{\{1\}}, x_{\{4\}}\right\}\backslash\right\},\left\{x_{\{2\}}\right\}\backslash\right\},\left\{x_{\{3\}}, x_{\{5\}}\right\}\backslash\right\}$	$\left\{x_1, x_4\right\},\left\{x_2\right\},\left\{x_3, x_5\right\}$	leaves an edge in one of the parts. So, two storage areas is not a possible solution, and three storage areas are sufficient and represent the optimal solution of the problem. Looking at
voloshin-hypergraph_F00069	25	■ 1.000	$x_{\{1\}}, x_{\{2\}}$	x_1, x_2	ask not only for the optimal but for all possible solutions, can efficiently be solved. In all applications, this is the example where identification is the key feature. Using notation
voloshin-hypergraph_F00070	25	■ 1.000	$\left\{x_{\{1\}}, x_{\{3\}}\right\}\backslash\right\}$	$\left\{x_1, x_3\right\}$	translated into the language of Graph Theory and solved mathematically. Mathematicians
voloshin-hypergraph_F00071	25	■ 0.993	$\left\{x_{\{2\}}, x_{\{4\}}\right\}\backslash\right\}$	$\left\{x_2, x_4\right\}$	then provide algorithms for finding optimal solutions which are implemer
voloshin-hypergraph_F00072	25	■ 0.993	$\left\{x_{\{5\}}\right\}\backslash\right\}$	$\left\{x_5\right\}$	then provide algorithms for finding optimal solutions which are implemer
voloshin-hypergraph_F00073	28	■ 0.982	$x_{\{1\}}: x_{\{2\}}, x_{\{5\}}$	$x_1: x_2, x_5$	• For $x_5: x$
voloshin-hypergraph_F00074	28	■ 0.990	$x_{\{2\}}: x_{\{1\}}, x_{\{3\}}, x_{\{4\}}, x_{\{5\}}$	$x_2: x_1, x_3, x_4, x_5$	
voloshin-hypergraph_F00075	28	■ 0.919	$x_{\{3\}}: x_{\{2\}}, x_{\{4\}}$	$x_3: x_2, x_4$	

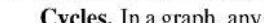
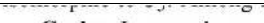
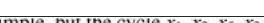
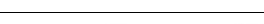
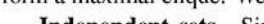
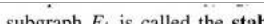
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voloshin-hypergraph_F00076	28	■ 0.999	$x_{\{4\}}: x_{\{2\}}, x_{\{3\}}, x_{\{5\}}$	$x_4 : x_2, x_3, x_5$	
voloshin-hypergraph_F00077	28	■ 0.996	$x_{\{5\}}: x_{\{1\}}, x_{\{2\}}, x_{\{4\}}$	$x_5 : x_1, x_2, x_4$	
voloshin-hypergraph_F00078	29	■ 1.000	$L(G)$	$L(G)$	$\{x_2, x_3, x_5\}$
voloshin-hypergraph_F00079	29	■ 1.000	$x_{\{j\}}$	x_j	$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \end{pmatrix}$
voloshin-hypergraph_F00080	29	■ 1.000	(i, j)	(i, j)	$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \end{pmatrix}$
voloshin-hypergraph_F00081	29	■ 1.000	$A(G)$	$A(G)$	$\begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}$
voloshin-hypergraph_F00082	29	■ 1.000	$A^{\backslash\prime}$	A'	
voloshin-hypergraph_F00083	29	■ 1.000	$A=A^{\backslash\prime}$	$A = A'$	
voloshin-hypergraph_F00084	30	■ 1.000	$e_{\{j\}}$	e_j	Incidence matrix. It is a matrix which has one row for each vertex and one column for each edge of a graph. If vertex x_i is incident to edge e_j , then the (i, j) -entry in the matrix
voloshin-hypergraph_F00085	30	■ 0.789	$I(G)$	$I(G)$	is 1, otherwise it is 0. For our graph G , the incidence matrix denoted by $I(G)$ has 5 rows
voloshin-hypergraph_F00086	30	■ 0.910	I	I	Edge lists. One can describe graph by giving just the list of all of its edges. For graph
voloshin-hypergraph_F00087	30	■ 1.000	$J(G)$	$J(G)$	$\mathcal{E}(G) = \{(x_1, x_2), (x_2, x_3), (x_3, x_5), (x_5, x_1)\}$
voloshin-hypergraph_F00088	31	■ 1.000	$\backslash{a, b}\backslash=\backslash{b, a}\backslash$	$\{a, b\} = \{b, a\}$	
voloshin-hypergraph_F00089	32	■ 0.677	$\backslash{\{$	$\{$	x_j is called the terminal vertex . A graph in which all edges are ordered pairs is called a directed graph , or digraph . Even loops may be ordered, see Figure 1.7.
voloshin-hypergraph_F00090	32	■ 0.677	$\backslash{\} , one use parentheses (,). So now (a, b) \neq(b, a)$	$\}, one use parentheses (,). So now (a, b) \neq (b, a)$	x_j is called the terminal vertex . A graph in which all edges are ordered pairs is called a directed graph , or digraph . Even loops may be ordered, see Figure 1.7.
voloshin-hypergraph_F00091	32	■ 0.984	(x, y)	(x, y)	Comparing edges and arcs one can accept the following point of view: an edge of a
voloshin-hypergraph_F00092	32	■ 1.000	y	y	graph is equivalent to two arcs going in opposite directions, see Figure 1.8.
voloshin-hypergraph_F00093	32	■ 1.000	L	L	An example of a directed multigraph G with a loop is shown in Figure 1.9. The adjacency list of G is:
voloshin-hypergraph_F00094	32	■ 1.000	J	J	$L(G) = \{\emptyset, \{2\}, \{2, 4\}, \{1, 1, 3\}\};$
voloshin-hypergraph_F00095	33	■ 0.546	a, b, c, d, e, f	a, b, c, d, e, f	the vertices are street crossings in the city map; crossings x_i and x_j form an arc (x_i, x_j) if the traffic is allowed in the direction from x_i to x_j;
voloshin-hypergraph_F00096	33	■ 0.889	d	d	
voloshin-hypergraph_F00097	33	■ 0.512	$\backslashleft(x_{\{i\}}, x_{\{j\}}\backslashright)$	(x_i, x_j)	the vertices are all web pages in the internet; the web pages x_i and x_j form an arc (x_i, x_j) if there is a hypertext link from page x_i to page x_j.
voloshin-hypergraph_F00098	34	■ 1.000	$E_{\{n\}}$	E_n	empty, then it has at least one edge. Notice that an empty graph is not the same as the empty set which is always denoted by $\emptyset$.
voloshin-hypergraph_F00099	34	■ 1.000	$E_{\{4\}}$	E_4	empty set which is always denoted by $\emptyset$.
voloshin-hypergraph_F00100	34	■ 0.977	$\backslashemptyset\backslash$	$\emptyset$	
voloshin-hypergraph_F00101	34	■ 1.000	$P_{\{5\}}$	P_5	edges, is called a path . The number of edges in a path is its length . A path
voloshin-hypergraph_F00102	34	■ 0.994	$x_{\{1\}}, x_{\{2\}}, \backslashldots, x_{\{n\}}$	$x_1, x_2, \dots, x_n$	and x_n . Generally, any path connecting vertices x and y is called (x,y)-path . In Figure 1.10,

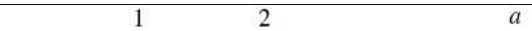
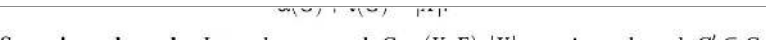
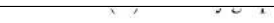
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00103	34	■ 1.000	$P_{\{n\}}$	P_n	
voloshin-hypergraph_F00104	34	■ 1.000	$m=n-1$	$m = n - 1$	
voloshin-hypergraph_F00105	34	■ 1.000	$P_{\{2\}}$	P_2	
voloshin-hypergraph_F00106	34	■ 1.000	$x_{\{n\}}$	x_n	
voloshin-hypergraph_F00107	34	■ 0.736	$\left(x_{\{1\}}, x_{\{5\}}\right)$	(x_1, x_5)	
voloshin-hypergraph_F00108	34	■ 1.000	$G_{\{1\}}$	G_1	
voloshin-hypergraph_F00109	34	■ 1.000	$G_{\{2\}}$	G_2	
voloshin-hypergraph_F00110	35	■ 1.000	$G^{\{\prime\}}$	G'	same graph, say G, and if G has k connected components, then we write: $G = kG'$. Cvcles. A connected graph in which every vertex has degree 2 is called a cvcle. (some-
voloshin-hypergraph_F00111	35	■ 1.000	k	k	same graph, say G, and if G has k connected components, then we write: $G = kG'$. Cvcles. A connected graph in which every vertex has degree 2 is called a cvcle. (some-
voloshin-hypergraph_F00112	35	■ 1.000	$G=k\ G^{\{\prime\}}$	$G = kG'$	same graph, say G, and if G has k connected components, then we write: $G = kG'$. Cvcles. A connected graph in which every vertex has degree 2 is called a cvcle. (some-
voloshin-hypergraph_F00113	35	■ 1.000	$C_{\{n\}}$	C_n	
voloshin-hypergraph_F00114	35	■ 1.000	$C_{\{1\}}, C_{\{2\}}, C_{\{3\}}, C_{\{4\}}, C_{\{5\}}$	C_1, C_2, C_3, C_4, C_5	figure 1.12. C_1 represents an edge connecting the vertex
voloshin-hypergraph_F00115	35	■ 1.000	$C_{\{6\}}$	C_6	figure 1.12. C_1 represents an edge connecting the vertex
voloshin-hypergraph_F00116	35	■ 1.000	$C_{\{1\}}$	C_1	
voloshin-hypergraph_F00117	35	■ 1.000	$C_{\{2\}}$	C_2	
voloshin-hypergraph_F00118	35	■ 1.000	$C_{\{3\}}$	C_3	
voloshin-hypergraph_F00119	35	■ 1.000	$W_{\{6\}}$	W_6	
voloshin-hypergraph_F00120	35	■ 1.000	$C_{\{k\}}, k \geq 3$	$C_k, k \geq 3$	—
voloshin-hypergraph_F00121	36	■ 1.000	$C_{\{k\}}$	C_k	vertices of C_k , then the graph obtained is called a wheel, denoted by W_k . The wheel W_k is
voloshin-hypergraph_F00122	36	■ 1.000	$W_{\{k+1\}}$	W_{k+1}	vertices of C_k , then the graph obtained is called a wheel, denoted by W_k . The wheel W_k is
voloshin-hypergraph_F00123	36	■ 1.000	$K_{\{n\}}$	K_n	plete , denoted by K_n where as usually, n is the number of vertices. It is called complete
voloshin-hypergraph_F00124	36	■ 1.000	$n \geq 1$	$n \geq 1$	because we cannot add any new edge to it and obtain a simple graph. For every $n \geq 1$, the
voloshin-hypergraph_F00125	36	■ 1.000	$K_{\{1\}}, K_{\{2\}}, K_{\{3\}}, K_{\{4\}}, K_{\{5\}}$	K_1, K_2, K_3, K_4, K_5	
voloshin-hypergraph_F00126	36	■ 1.000	$K_{\{6\}}$	K_6	
voloshin-hypergraph_F00127	36	■ 1.000	$n(n-1)=2\ m$	$n(n-1) = 2m$	vertices is denoted by T_n . In contrast to P_n, K_n, C_n and W_n there are many distinct trees on n
voloshin-hypergraph_F00128	36	■ 1.000	$m=6(6-1) / 2=15$	$m = 6(6-1)/2 = 15$	vertices is denoted by T_n . In contrast to P_n, K_n, C_n and W_n there are many distinct trees on n
voloshin-hypergraph_F00129	36	■ 1.000	$K_{\{2\}}=$	$K_2 =$	to distinguish graphs.

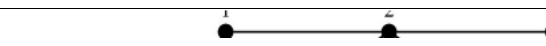
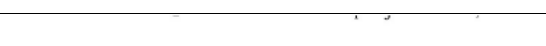
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00130	36	■ 1.000	$P_{\{2\}}, K_{\{3\}}=C_{\{3\}}$	$P_2, K_3 = C_3$	
voloshin-hypergraph_F00131	36	■ 1.000	$K_{\{4\}}=W_{\{4\}}$	$K_4 = W_4$	
voloshin-hypergraph_F00132	36	■ 1.000	$T_{\{n\}}$	T_n	
voloshin-hypergraph_F00133	36	■ 1.000	$P_{\{n\}}, K_{\{n\}}, C_{\{n\}}$	P_n, K_n, C_n	
voloshin-hypergraph_F00134	36	■ 1.000	$W_{\{n\}}$	W_n	
voloshin-hypergraph_F00135	36	■ 1.000	$P_{\{4\}}$	P_4	—
voloshin-hypergraph_F00136	37	■ 1.000	$X_{\{1\}}$	X_1	drawing, it is possible to color the vertices using just two colors (say blue and red) in such a way that adjacent vertices have different colors. If a graph is not bipartite, then in any
voloshin-hypergraph_F00137	37	■ 1.000	$X_{\{2\}}$	X_2	drawing, it is possible to color the vertices using just two colors (say blue and red) in such a way that adjacent vertices have different colors. If a graph is not bipartite, then in any
voloshin-hypergraph_F00138	37	■ 1.000	$G=\left(X_{\{1\}}, X_{\{2\}} ; E\right)$	$G=\left(X_1, X_2 ; E\right)$	and $ X_2 =s$, then the graph itself is denoted by $K_{r,s}$. The number of edges in $K_{r,s}$ clearly
voloshin-hypergraph_F00139	37	■ 1.000	$\left X_{\{1\}}\right =r$	$\left X_1\right =r$	be partitioned” (as in definition) and “is partitioned”.
voloshin-hypergraph_F00140	37	■ 0.987	$\left X_{\{2\}}\right =s$	$\left X_2\right =s$	Regular graphs. A graph in which every vertex has the same degree k is called regular of degree k or k -regular. Empty graph E is 0-regular, cycle C is 2-regular, and K is
voloshin-hypergraph_F00141	37	■ 0.987	$K_{\{r, s\}}$	$K_{r, s}$	Regular graphs. A graph in which every vertex has the same degree k is called regular of degree k or k -regular. Empty graph E is 0-regular, cycle C is 2-regular, and K is
voloshin-hypergraph_F00142	37	■ 1.000	$r \cdot s$	$r s$	$\frac{rs}{n}$ (n
voloshin-hypergraph_F00143	37	■ 0.999	$K_{\{1,3\}}$	$K_{1,3}$	same length, draw one inside another and connect the respective vertices by edges called
voloshin-hypergraph_F00144	37	■ 0.999	$C_{\{4\}}$	C_4	same length, draw one inside another and connect the respective vertices by edges called
voloshin-hypergraph_F00145	37	■ 0.999	$X_{\{1\}}, X_{\{2\}}$	X_1, X_2	same length, draw one inside another and connect the respective vertices by edges called
voloshin-hypergraph_F00146	37	■ 0.998	$(n-1)$	$(n-1)$	
voloshin-hypergraph_F00147	38	■ 0.999	$C_{\{5\}}$	C_5	
voloshin-hypergraph_F00148	39	■ 1.000	$G_{\{1\}}=\left(X_{\{1\}}, E_{\{1\}}\right)$	$G_1=\left(X_1, E_1\right)$	main concept of isomorphism (Greek: iso = equal, and morphic = shape). Two simple graphs $G_1=\left(X_1, E_1\right)$ and $G_2=\left(X_2, E_2\right)$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F00149	39	■ 1.000	$G_{\{2\}}=\left(X_{\{2\}}, E_{\{2\}}\right)$	$G_2=\left(X_2, E_2\right)$	main concept of isomorphism (Greek: iso = equal, and morphic = shape). Two simple graphs $G_1=\left(X_1, E_1\right)$ and $G_2=\left(X_2, E_2\right)$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F00150	39	■ 1.000	$G_{\{1\}} \cong G_{\{2\}}$	$G_1 \cong G_2$	G_1 is isomorphic to G_2 , and G_2 is isomorphic to G_1 , and we write $G_1=G_2$. Suppose $ V_1 = V_2 =n$. Generally, how many one-to-one correspondences
voloshin-hypergraph_F00151	39	■ 0.999	$\left X_{\{1\}}\right =\left X_{\{2\}}\right =n$	$\left X_1\right =\left X_2\right =n$	Suppose $ X_1 = X_2 =n$. Generally, how many one-to-one correspondences do exist. The first vertex of X_1 can be mapped into any of n vertices of X_2 , so we have n possibilities
voloshin-hypergraph_F00152	39	■ 1.000	$n-2$	$n-2$	$2) \cdots 3 \cdot 2 \cdot 1=n!$. If in at least one of these $n!$ cases there is a complete “coincidence”
voloshin-hypergraph_F00153	39	■ 1.000	$n(n-1)(n-$	$n(n-1)(n-$	$2) \cdots 3 \cdot 2 \cdot 1=n!$. If in at least one of these $n!$ cases there is a complete “coincidence”
voloshin-hypergraph_F00154	39	■ 0.979	$2) \cdot \cdots \cdot 3 \cdot 2 \cdot 1=n!$	$2) \cdot \cdots \cdot 3 \cdot 2 \cdot 1=n!$	of G_1 and G_2 , then they are isomorphic. Otherwise they are not. So, two graphs are not isomorphic, if no matter how we map the vertices of one into the vertices of another, there
voloshin-hypergraph_F00155	39	■ 0.979	$n!$	$n!$	of G_1 and G_2 , then they are isomorphic. Otherwise they are not. So, two graphs are not isomorphic, if no matter how we map the vertices of one into the vertices of another, there
voloshin-hypergraph_F00156	39	■ 0.990	$1,2, \ldots n!$	$1,2, \ldots n!$	Suppose $ V_1 = V_2 =n$. Generally, how many one-to-one correspondences do exist. there are two isomorphisms between any two paths P_n . On the other hand, the second tree

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00157	39	0.840	$2n$	$2n$	
voloshin-hypergraph_F00158	39	1.000	T_1	T_1	one-to-one correspondence from the definition.
voloshin-hypergraph_F00159	39	1.000	T_2	T_2	one-to-one correspondence from the definition.
voloshin-hypergraph_F00160	39	1.000	$8!=40320$	$8! = 40320$	
voloshin-hypergraph_F00161	40	1.000	$\boldsymbol{\sigma}$	$\boldsymbol{\sigma}$	
voloshin-hypergraph_F00162	40	1.000	a	a	one-to-one correspondences.
voloshin-hypergraph_F00163	40	1.000	b	b	one-to-one correspondences.
voloshin-hypergraph_F00164	40	0.665	$\binom{6}{2}=15$	$\binom{6}{2} = 15$	graph property, then investigate all graphs having that property. There are hundreds of graph classes that have been investigated. In the simplest classes such as E_n, K_n, C_n, W_n .
voloshin-hypergraph_F00165	40	0.665	$6!=720$	$6! = 720$	graph property, then investigate all graphs having that property. There are hundreds of graph classes that have been investigated. In the simplest classes such as E_n, K_n, C_n, W_n .
voloshin-hypergraph_F00166	40	1.000	E_n, K_n, C_n, W_n	E_n, K_n, C_n, W_n	
voloshin-hypergraph_F00167	40	0.987	r	r	matrix.
voloshin-hypergraph_F00168	40	0.987	$s, 1 \leq r, s \leq 5$	$s, 1 \leq r, s \leq 5$	matrix.
voloshin-hypergraph_F00169	40	1.000	$K_{s, r}$	$K_{s,r}$	
voloshin-hypergraph_F00170	41	1.000	$L(G), A(G), I(G), J(G)$	$L(G), A(G), I(G), J(G)$	4. Given any of $L(G), A(G), I(G), J(G)$, recognize if G is a complete
voloshin-hypergraph_F00171	41	1.000	$L(G_1), A(G_1), I(G_1)$	$L(G_1), A(G_1), I(G_1)$	In many proofs and algorithms, one often apply graph operations that allow to obtain one
voloshin-hypergraph_F00172	41	1.000	$J(G_1)$	$J(G_1)$	In many proofs and algorithms, one often apply graph operations that allow to obtain one
voloshin-hypergraph_F00173	41	1.000	$L(G_2), A(G_2), I(G_2)$	$L(G_2), A(G_2), I(G_2)$	In many proofs and algorithms, one often apply graph operations that allow to obtain one
voloshin-hypergraph_F00174	41	1.000	$J(G_2)$	$J(G_2)$	In many proofs and algorithms, one often apply graph operations that allow to obtain one
voloshin-hypergraph_F00175	41	1.000	σ	σ	graph from another. We now consider some of them.
voloshin-hypergraph_F00176	41	1.000	$x \in X$	$x \in X$	
voloshin-hypergraph_F00177	41	0.977	$X_1=X-\{x\}$	$X_1 = X - \{x\}$	
voloshin-hypergraph_F00178	41	1.000	$E_1=E-E(x)$	$E_1 = E - E(x)$	—
voloshin-hypergraph_F00179	42	1.000	$G_1=G-x$	$G_1 = G - x$	another vertex to obtain a graph G_2 and so on, sequential deletion of vertices leads to a sequence of graphs
voloshin-hypergraph_F00180	42	1.000	$E(x)=\{e_1, e_2, e_3\}$	$E(x) = \{e_1, e_2, e_3\}$	 delete x weakly
voloshin-hypergraph_F00181	42	1.000	$G_1=G-e_3$	$G_1 = G - e_3$	
voloshin-hypergraph_F00182	43	1.000	$e=\{x, y\} \in E$	$e = \{x, y\} \in E$	2. All edges of G having one end at x or y will have this end at xy with other end
voloshin-hypergraph_F00183	43	1.000	xy	xy	
voloshin-hypergraph_F00184	44	0.926	$N(x) \cap N(y) \neq \emptyset$	$N(x) \cap N(y) \neq \emptyset$	If G is a simple graph and $N(x) \cap N(y) \neq \emptyset$ then each vertex from $N(x) \cap N(y)$ will be

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00185	44	■ 0.926	$N(x) \cap N(y)$	$N(x) \cap N(y)$	If G is a simple graph and $M(x) \cap M(y) \neq \emptyset$, then each vertex from $M(x) \cap M(y)$ will be
voloshin-hypergraph_F00186	44	■ 1.000	$e^{\prime}$	e'	At last, if G is not simple and had another edge e' connecting x and y , then e' becomes
voloshin-hypergraph_F00187	44	■ 1.000	$G_{1}=G \cdot e$	$G_1 = G \cdot e$	to a graph G_2 if G_2 may be o Example: Let $G = K_4$ is a
voloshin-hypergraph_F00188	44	■ 1.000	$C_{\{k\}}, k \geq 4$	$C_k, k \geq 4$	The maximum value of n such that a graph G is contractible to K_n is called the Hadwiger
voloshin-hypergraph_F00189	44	■ 1.000	$C_{\{3\}}=K_{\{3\}}$	$C_3 = K_3$	The maximum value of n such that a graph G is contractible to K_n is called the Hadwiger
voloshin-hypergraph_F00190	44	■ 1.000	$K_{\{2\}}$	K_2	number of the graph G , denoted by $\eta(G)$.
voloshin-hypergraph_F00191	44	■ 1.000	$K_{\{m\}}$	K_m	number of the graph G , denoted by $\eta(G)$.
voloshin-hypergraph_F00192	44	■ 1.000	$n \geq m$	$n \geq m$	number of the graph G , denoted by $\eta(G)$.
voloshin-hypergraph_F00193	44	■ 1.000	$K_{\{4\}}$	K_4	taking the complement. Consider a simple graph $G = (X, E)$ with $ X = n$. The com-
voloshin-hypergraph_F00194	44	■ 1.000	$K_{\{3\}}$	K_3	taking the complement. Consider a simple graph $G = (X, E)$ with $ X = n$. The com-
voloshin-hypergraph_F00195	44	■ 1.000	$\eta(G)$	$\eta(G)$	where E' is such an edge set that $E \cup E'$ for
voloshin-hypergraph_F00196	44	■ 1.000	$ X =n$	$ X = n$	complement of G . Evidently, $\overline{\overline{G}} = G$, $\overline{E_n} = K_n$ and $\overline{K_n} = E_n$. Notice that P_4 is isomorphic
voloshin-hypergraph_F00197	44	■ 1.000	$\bar{G}$	$\bar{G}$	to $\overline{P_4}$, C_5 is isomorphic to $\overline{C_5}$, and $\overline{K_{r,s}} = \{K_r, K_s\}$. It may happen that G is connected graph
voloshin-hypergraph_F00198	44	■ 1.000	$\bar{G}=\left(X, E^{\prime}\right)$	$\bar{G} = (X, E')$	but $\bar{G}$ is not, see Figure 1.27.
voloshin-hypergraph_F00199	44	■ 1.000	$E^{\prime}$	E'	
voloshin-hypergraph_F00200	44	■ 1.000	$E \cup E^{\prime}$	$E \cup E'$	
voloshin-hypergraph_F00201	44	■ 1.000	$\overline{\bar{G}}=G, \bar{E}_{\{n\}}=K_{\{n\}}$	$\bar{\bar{G}} = G, \bar{E}_n = K_n$	 complement 
voloshin-hypergraph_F00202	44	■ 1.000	$\bar{K}_{\{n\}}=E_{\{n\}}$	$\bar{K}_n = E_n$	 complement 
voloshin-hypergraph_F00203	44	■ 1.000	$\overline{P_{\{4\}}}, C_{\{5\}}$	$\overline{P_4}, C_5$	 
voloshin-hypergraph_F00204	44	■ 1.000	$\overline{C_{\{5\}}}$	$\overline{C_5}$	 
voloshin-hypergraph_F00205	44	■ 1.000	$\overline{K_{\{r, s\}}}=\left\{K_{\{r\}}, K_{\{s\}}\right\}$	$\overline{K_{r,s}} = \{K_r, K_s\}$	 
voloshin-hypergraph_F00206	45	■ 1.000	$C_{\{4\}}, W_{\{5\}}, K_{\{5\}}, K_{\{3,3\}}$	$C_4, W_5, K_5, K_{3,3}$	1. Implement a weak sequential (in any order) deletion of vertices of C_4 , W_5 , K_5 , $K_{3,3}$ and
voloshin-hypergraph_F00207	45	■ 1.000	$C_{\{5\}}, W_{\{5\}}, K_{\{2,3\}}$	$C_5, W_5, K_{2,3}$	vertices, $E_5, 2C_3$.
voloshin-hypergraph_F00208	45	■ 1.000	$C_{\{3\}}, C_{\{4\}}, C_{\{5\}}, C_{\{6\}}, K_{\{5\}}, K_{\{3,5\}}$	$C_3, C_4, C_5, C_6, K_5, K_{3,5}$	8. Show that $\overline{C_5} \cong C_5$.
voloshin-hypergraph_F00209	45	■ 1.000	$P_{\{7\}}$	P_7	8. Show that $\overline{C_5} \cong C_5$.
voloshin-hypergraph_F00210	45	■ 1.000	$E_{\{5\}}, 2 C_{\{3\}}$	$E_5, 2C_3$	9. Show that $\bar{C}$
voloshin-hypergraph_F00211	45	■ 0.999	$\bar{C}_{\{5\}} \cong C_{\{5\}}$	$\bar{C}_5 \cong C_5$	10. How are the de

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00212	45	■ 1.000	$\bar{C}_6$	$\bar{C}_6$	
voloshin-hypergraph_F00213	45	■ 1.000	$G^{\prime}=\left(X^{\prime}, E^{\prime}\right)$	$G' = (X', E')$	
voloshin-hypergraph_F00214	45	■ 1.000	$X^{\prime} \subseteq X$	$X' \subseteq X$	
voloshin-hypergraph_F00215	45	■ 1.000	$E^{\prime} \subseteq E$	$E' \subseteq E$	
voloshin-hypergraph_F00216	45	■ 1.000	$G^{\prime} \subseteq G$	$G' \subseteq G$	
voloshin-hypergraph_F00217	45	■ 1.000	$X^{\prime}$	X'	
voloshin-hypergraph_F00218	45	■ 1.000	$X-X^{\prime}$	$X-X'$	
voloshin-hypergraph_F00219	45	■ 1.000	$E-E^{\prime}$	$E-E'$	
voloshin-hypergraph_F00220	45	■ 1.000	$\left\{x_3, x_5\right\}$	$\{x_3, x_5\}$	
voloshin-hypergraph_F00221	46	■ 1.000	$Y \subseteq X$	$Y \subseteq X$	
voloshin-hypergraph_F00222	46	■ 1.000	G_Y	G_Y	
voloshin-hypergraph_F00223	46	■ 1.000	$k \geq 3$	$k \geq 3$	
voloshin-hypergraph_F00224	46	■ 0.996	x_2, x_3, x_4, x_5, x_2	x_2, x_3, x_4, x_5, x_2	
voloshin-hypergraph_F00225	46	■ 1.000	$x_1, x_2, x_3, x_4, x_5, x_1$	$x_1, x_2, x_3, x_4, x_5, x_1$	
voloshin-hypergraph_F00226	46	■ 1.000	$x_1, x_2, x_5, x_3, x_4, x_5, x_1$	$x_1, x_2, x_5, x_3, x_4, x_5, x_1$	
voloshin-hypergraph_F00227	46	■ 1.000	$K_1, K_2, K_3, \ldots, K_n$	$K_1, K_2, K_3, \dots, K_n$	
voloshin-hypergraph_F00228	46	■ 1.000	K_r	K_r	
voloshin-hypergraph_F00229	46	■ 1.000	K_{r+1}	K_{r+1}	
voloshin-hypergraph_F00230	47	■ 0.999	$\omega(G)$	$\omega(G)$	
voloshin-hypergraph_F00231	47	■ 0.998	$G, 1 \leq \omega(G) \leq n$	$G, 1 \leq \omega(G) \leq n$	
voloshin-hypergraph_F00232	47	■ 1.000	x_2, x_3, x_5	x_2, x_3, x_5	
voloshin-hypergraph_F00233	47	■ 0.958	$\omega(G)=3$	$\omega(G) = 3$	
voloshin-hypergraph_F00234	47	■ 0.958	$\omega\left(G_1\right)=\omega\left(G_2\right)=2$	$\omega\left(G_1\right)=\omega\left(G_2\right)=2$	
voloshin-hypergraph_F00235	47	■ 1.000	$E_1, E_2, \ldots, E_n$	$E_1, E_2, \dots, E_n$	
voloshin-hypergraph_F00236	47	■ 1.000	E_k	E_k	
voloshin-hypergraph_F00237	47	■ 0.801	$\alpha(G)$	$\alpha(G)$	
voloshin-hypergraph_F00238	47	■ 1.000	$G, 1 \leq \alpha(G) \leq n$	$G, 1 \leq \alpha(G) \leq n$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00239	47	■ 1.000	$\omega(G)=\alpha(\bar{G})$	$\omega(G) = \alpha(\bar{G})$	For graph G , see Figure 1.28, vertices x_1 and x_3 form a stable set, vertices x_2 and x_4
voloshin-hypergraph_F00240	47	■ 1.000	$\alpha(G)=\omega(\bar{G})$	$\alpha(G) = \omega(\bar{G})$	For graph G , see Figure 1.28, vertices x_1 and x_3 form a stable set, vertices x_2 and x_4
voloshin-hypergraph_F00241	47	■ 1.000	$\alpha(G)=2$	$\alpha(G) = 2$	
voloshin-hypergraph_F00242	47	■ 1.000	$\omega\left(G_{1}\right)=3$	$\omega\left(G_{1}\right)=3$	(minimum) in the sense of largest (smallest) over all maximal (minimal) . The difference
voloshin-hypergraph_F00243	47	■ 1.000	a, b	a, b	(minimum) in the sense of largest (smallest) over all maximal (minimal) . The difference
voloshin-hypergraph_F00244	47	■ 1.000	a, b, e	a, b, e	is similar to the difference between the local and global maximum (minimum) of a function. Transversals (vertex covers). In a graph $G=(V, E)$ a subset of vertices $T \subseteq V$ is
voloshin-hypergraph_F00245	47	■ 1.000	f	f	is similar to the difference between the local and global maximum (minimum) of a function. Transversals (vertex covers). In a graph $G=(V, E)$ a subset of vertices $T \subseteq V$ is
voloshin-hypergraph_F00246	47	■ 0.982	$\alpha\left(G_{2}\right)=4$	$\alpha\left(G_{2}\right)=4$	Transversals (vertex covers). In a graph $G=(V, E)$ a subset of vertices $T \subseteq V$ is called a transversal (vertex cover) if its complemen
voloshin-hypergraph_F00247	47	■ 1.000	$T \subseteq X$	$T \subseteq X$	
voloshin-hypergraph_F00248	47	■ 1.000	$X \backslash T$	$X \backslash T$	Spanning subgraphs. Let us have a graph $G=(V, E)$, $ V =n$. Any subgraph $G' \subseteq G$ such that $G'=(V, E')$ is called a spanning subgraph . Thus spanning subgraphs have the
voloshin-hypergraph_F00249	47	■ 1.000	T	T	same vertex set as the graph itself. Spanning subgraph which is a tree is called spanning
voloshin-hypergraph_F00250	47	■ 1.000	$\tau(G)$	$\tau(G)$	
voloshin-hypergraph_F00251	47	■ 1.000	$G=(X, E), X =n$	$G=(X, E), X =n$	
voloshin-hypergraph_F00252	47	■ 0.997	$G^{\prime}=\left(X, E^{\prime}\right)$	$G^{\prime}=(X, E')$	
voloshin-hypergraph_F00253	48	■ 0.959	$\{1,2\}$	$\{1,2\}$	In Figure 1.29, if we weakly delete edges $\{1,2\}$ and $\{2,3\}$, then we obtain a spanning
voloshin-hypergraph_F00254	48	■ 0.959	$\{2,3\}$	$\{2,3\}$	In Figure 1.29, if we weakly delete edges $\{1,2\}$ and $\{2,3\}$, then we obtain a spanning
voloshin-hypergraph_F00255	48	■ 0.833	$\nu(G)$	$\nu(G)$	$\nu=n/2$.
voloshin-hypergraph_F00256	48	■ 0.955	$\nu=n / 2$	$\nu=n / 2$	
voloshin-hypergraph_F00257	48	■ 1.000	$\{c, d\}$	$\{c, d\}$	However, a maximum matching is formed for example by edges $\{a, c\}$ and $\{a, e\}$, and $\nu(G)=2$
voloshin-hypergraph_F00258	48	■ 1.000	$\{a, c\}$	$\{a, c\}$	$\nu(G)=2$. Since at least one vertex from every edge of any matching must belong to any transver-
voloshin-hypergraph_F00259	48	■ 1.000	$\{d, e\}$	$\{d, e\}$	$\nu(G)=2$. Since at least one vertex from every edge of any matching must belong to any transver-
voloshin-hypergraph_F00260	48	■ 1.000	$\nu\left(G_{2}\right)=2$	$\nu\left(G_{2}\right)=2$	sal,
voloshin-hypergraph_F00261	49	■ 1.000	$P_{6}, C_{3}, C_{4}, C_{5}, W_{4}, W_{5}, K_{2,3}$	$P_6, C_3, C_4, C_5, W_4, W_5, K_{2,3}$	3. Write down the list of all maximal cliques of $P_6, C_3, C_4, C_5, W_4, W_5, K_{2,3}$, cube and Petersen
voloshin-hypergraph_F00262	49	■ 1.000	C_{7}	C_7	 15. Find $\alpha(G)$ for every graph drawn in this section and in F
voloshin-hypergraph_F00263	49	■ 1.000	W_{9}	W_9	 15. Find $\alpha(G)$ for every graph drawn in this section and in F
voloshin-hypergraph_F00264	49	■ 1.000	$K_{4,4}$	$K_{4,4}$	16. Find $\nu(G)$ for every graph drawn in this section and in Figure 1.21.
voloshin-hypergraph_F00265	49	■ 1.000	$K_{6,9}$	$K_{6,9}$	17. Find α, ω, τ , and ν of cube, Petersen graph, K_{999}, K

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voloshin-hypergraph_F00266	49	■ 0.939	$v(G)$	$v(G)$	
voloshin-hypergraph_F00267	49	■ 0.999	α, ω, τ	α, ω, τ	
voloshin-hypergraph_F00268	49	■ 0.999	ν	ν	
voloshin-hypergraph_F00269	49	■ 0.999	$K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$	$K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$	
voloshin-hypergraph_F00270	49	■ 1.000	$n \geq 2$	$n \geq 2$	
voloshin-hypergraph_F00271	49	■ 1.000	W_5	W_5	
voloshin-hypergraph_F00272	50	■ 0.988	$x, y \in X$	$x, y \in X$	only complete graphs do not have separators. For disconnected graphs, a separator is any
voloshin-hypergraph_F00273	50	■ 0.999	$\{x, y\}$	$\{x, y\}$	set of vertices deleting of which increases the number of connected components. So, if S is
voloshin-hypergraph_F00274	50	■ 0.999	$S \subseteq X$	$S \subseteq X$	set of vertices deleting of which increases the number of connected components. So, if S is
voloshin-hypergraph_F00275	50	■ 1.000	S	S	does not contain any other separator as a proper subset, is called a minimal separator . As
voloshin-hypergraph_F00276	50	■ 0.448	$\mathrm{k}(G)$	$\mathrm{k}(G)$	a graph G is k -connected, then it is $(k-1)$ -connected, $(k-2)$ -connected, and so on. But this implication is not working in the opposite way: if a graph is k -connected, it may be not
voloshin-hypergraph_F00277	50	■ 0.976	$\mathrm{K}(K_n) = n - 1$	$\mathrm{K}(K_n) = n - 1$	$(k+1)$ -connected. The inclusion of connectivity classes is shown in Figure 1.32.
voloshin-hypergraph_F00278	50	■ 0.530	$G, \mathrm{k}(G)$	$G, \mathrm{k}(G)$	
voloshin-hypergraph_F00279	50	■ 0.953	$\mathrm{K}(G) \geq k$	$\mathrm{K}(G) \geq k$	
voloshin-hypergraph_F00280	50	■ 0.898	$\mathrm{K}(G) \geq 2$	$\mathrm{K}(G) \geq 2$	
voloshin-hypergraph_F00281	50	■ 1.000	$(k-1)$	$(k-1)$	
voloshin-hypergraph_F00282	50	■ 1.000	$(k-2)$	$(k-2)$	
voloshin-hypergraph_F00283	50	■ 1.000	$(k+1)$	$(k+1)$	
voloshin-hypergraph_F00284	51	■ 0.865	$\{2, 3, 5, 6\}$	$\{2, 3, 5, 6\}$	Graph G in Figure 1.33 is 2-connected. Therefore, it is 1-connected and 0-connected. But it is not 3-connected and it is not k -connected for any $k > 3$.
voloshin-hypergraph_F00285	51	■ 0.826	$\{2, 6\}$	$\{2, 6\}$	
voloshin-hypergraph_F00286	51	■ 0.826	$\{2, 5\}$	$\{2, 5\}$	
voloshin-hypergraph_F00287	51	■ 0.767	$\mathrm{K}(G) = 2$	$\mathrm{K}(G) = 2$	
voloshin-hypergraph_F00288	51	■ 1.000	$X_{\{1\}} \cup S$	$X_1 \cup S$	S has neighbors in both X_1 and X_2 .
voloshin-hypergraph_F00289	51	■ 1.000	$X_{\{2\}} \cup S$	$X_2 \cup S$	Proof. If some $x \in S$ has no vertex in X_1 adjacent to it, then $S - \{x\}$ is also a separator of
voloshin-hypergraph_F00290	51	■ 1.000	$G_{X_{\{1\}} \cup S}$	$G_{X_1 \cup S}$	
voloshin-hypergraph_F00291	51	■ 1.000	$G_{X_{\{2\}} \cup S}$	$G_{X_2 \cup S}$	
voloshin-hypergraph_F00292	51	■ 1.000	$x \in S$	$x \in S$	—
voloshin-hypergraph_F00293	51	■ 1.000	$S - \{x\}$	$S - \{x\}$	—

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voloshin-hypergraph_F00294	52	■ 0.965	$\{3,4\}$	$\{3,4\}$	
voloshin-hypergraph_F00295	52	■ 0.695	$\{1,6\}$	$\{1,6\}$	
voloshin-hypergraph_F00296	52	■ 1.000	$k \geq 2$	$k \geq 2$	
voloshin-hypergraph_F00297	52	■ 1.000	$X_{\{1\}}, X_{\{2\}}, \ldots, X_{\{k\}}$	$X_1, X_2, \dots, X_k$	
voloshin-hypergraph_F00298	52	■ 1.000	$G_{\{1\}}=G_{\{X_{\{1\}} \cup S\}}, G_{\{2\}}=G_{\{X_{\{2\}} \cup S\}}, \ldots$	$G_1 = G_{X_1 \cup S}, G_2 = G_{X_2 \cup S}, \dots$	
voloshin-hypergraph_F00299	52	■ 1.000	$G_{\{k\}}=G_{\{X_{\{k\}} \cup S\}}$	$G_k = G_{X_k \cup S}$	
voloshin-hypergraph_F00300	53	■ 0.822	x, y	x, y	and y , the minimum number of vertices in an (x,y) -separator equals the maximum number of (internally vertex disjoint (x,y) paths.
voloshin-hypergraph_F00301	53	■ 0.783	$\mathbf{K}(G)$	$\mathbf{K}(G)$	7. For separator $\{5,6,7,8,2\}$, construct the derived subgraph
voloshin-hypergraph_F00302	53	■ 1.000	$k \geq 0$	$k \geq 0$	6. Apply Proposition 1.6.1 for separator $\{1,6,3\}$.
voloshin-hypergraph_F00303	53	■ 1.000	$\{1,3,4,7,8\}$	$\{1,3,4,7,8\}$	10. For graph G in Figure 1.36 find a minimal $(1,4)$ -separator.
voloshin-hypergraph_F00304	53	■ 1.000	$\{1,8,3,4,6\}$	$\{1,8,3,4,6\}$	11. For graph G in Figure 1.36 find a minimal $(1,4)$ -edge-separator.
voloshin-hypergraph_F00305	53	■ 1.000	$\{5,6,7,8,2\}$	$\{5,6,7,8,2\}$	12. What is the smallest number of edges that disconnect graph G ?
voloshin-hypergraph_F00306	55	■ 1.000	$m(G)=n(G)-1$	$m(G) = n(G) - 1$	4. There is a unique path connecting any
voloshin-hypergraph_F00307	55	■ 1.000	$1 \Rightarrow \text{Rightarrow } 2 \Rightarrow G$	$1. \Rightarrow 2.G$	2. $\Rightarrow$ 3. Prove that G has no cycles. By contradiction, suppose that G has a cycle. We shall delete an edge $e = \{u,v\}$ of this cycle. $G - e$ is connected since $\{u,v\}$ path remains.
voloshin-hypergraph_F00308	55	■ 1.000	$n=1,2$	$n = 1, 2$	we may delete an edge $e = \{u,v\}$ of any cycle, $G - e$ is connected since $\{u,v\}$ path remains. If $G - e$ contains cycles, repeat the procedure until graph obtained contains no cycles. Since
voloshin-hypergraph_F00309	55	■ 1.000	$n(G)>2$	$n(G) > 2$	we may delete an edge $e = \{u,v\}$ of any cycle, $G - e$ is connected since $\{u,v\}$ path remains. If $G - e$ contains cycles, repeat the procedure until graph obtained contains no cycles. Since
voloshin-hypergraph_F00310	55	■ 1.000	$m\left(G_{\{1\}}\right)=n\left(G_{\{1\}}\right)-1$	$m\left(G_1\right)=n\left(G_1\right)-1$	deleted at least one edge.
voloshin-hypergraph_F00311	55	■ 1.000	$m(G)=m\left(G_{\{1\}}\right)+1$	$m(G) = m\left(G_1\right) + 1$	3. $\Rightarrow$ 4. Let G have no cycles and $m = n - 1$. It may have many components. Let G_1 be a component of G . Then G_1 is a tree. But $m\left(G_1\right) \geq 0$ and $n\left(G_1\right) \geq 1$.
voloshin-hypergraph_F00312	55	■ 1.000	$n(G)=n\left(G_{\{1\}}\right)+1$	$n(G) = n\left(G_1\right) + 1$	3. $\Rightarrow$ 4. Let G have no cycles and $m = n - 1$. It may have many components. Let G_1 be a component of G . Then G_1 is a tree. But $m\left(G_1\right) \geq 0$ and $n\left(G_1\right) \geq 1$.
voloshin-hypergraph_F00313	55	■ 1.000	$e=\{x, y\}$	$e = \{x, y\}$	Therefore $m(G) = m_1 + m_2 + \dots + m_k = (n_1 - k) + (n_2 - k) + \dots + (n_k - k) = n - nk$. So we obtain $m = n - 1 = n - k$ which implies $k = 1$ and G is connected. It means that there
voloshin-hypergraph_F00314	55	■ 1.000	$G-e$	$G - e$	Therefore $m(G) = m_1 + m_2 + \dots + m_k = (n_1 - k) + (n_2 - k) + \dots + (n_k - k) = n - nk$. So we obtain $m = n - 1 = n - k$ which implies $k = 1$ and G is connected. It means that there
voloshin-hypergraph_F00315	55	■ 0.882	$\text{Rightarrow } 2 \Rightarrow m=n-1$	$\Rightarrow 2.m = n - 1$	vertices, then there is a cycle what contradicts to condition 3. Therefore, 4. holds.
voloshin-hypergraph_F00316	55	■ 0.999	$k>1$	$k > 1$	
voloshin-hypergraph_F00317	55	■ 0.999	$G_{\{1\}}, G_{\{2\}}, \ldots, G_{\{k\}}$	$G_1, G_2, \dots, G_k$	
voloshin-hypergraph_F00318	55	■ 0.999	$G_{\{i\}}$	G_i	
voloshin-hypergraph_F00319	55	■ 0.999	$1 \Rightarrow \text{Rightarrow } 2 \Rightarrow m_{\{i\}}=n_{\{i\}}-1$	$1. \Rightarrow 2.m_i = n_i - 1$	
voloshin-hypergraph_F00320	55	■ 0.965	$m(G)=m_{\{1\}}+m_{\{2\}}+\cdots+m_{\{k\}}=\left(n_{\{1\}}-1\right)+\left(n_{\{2\}}-1\right)+\cdots+\left(n_{\{k\}}-1\right)=n-k$	$m(G) = m_1 + m_2 + \dots + m_k = (n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1) = n - k$	

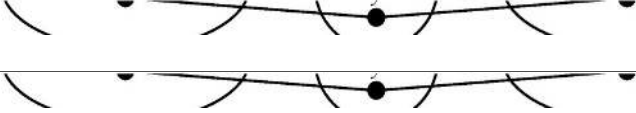
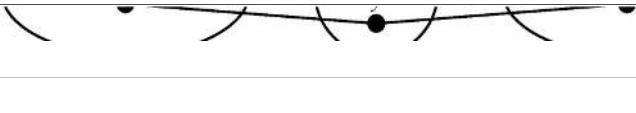
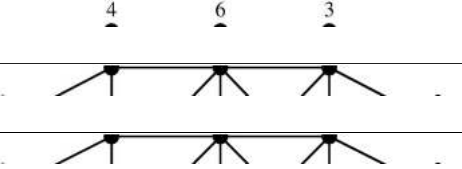
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voloshin-hypergraph_F00321	55	1.000	m=n-1=n-k	$m = n - 1 = n - k$	
voloshin-hypergraph_F00322	55	1.000	k=1	$k = 1$	
voloshin-hypergraph_F00323	56	0.996	C_{l}	C_l	otherwise, if we obtain two cycles C_k and C_l , the subgraph $(C_k \cup C_l) - \{x, y\}$ forms the third
voloshin-hypergraph_F00324	56	0.996	\left(C_{\{k\} \cup C_{\{l\}}\right)-\{x, y\}	$(C_k \cup C_l) - \{x, y\}$	otherwise, if we obtain two cycles C_k and C_l , the subgraph $(C_k \cup C_l) - \{x, y\}$ forms the third
voloshin-hypergraph_F00325	56	1.000	\Lambda(G)=m(G)-	$\Lambda(G) = m(G) -$	$n(G) + k$ is called the cyclomatic number of G . If G is connected, then $k = 1$ and $\Lambda =$
voloshin-hypergraph_F00326	56	0.516	n(G)+k	$n(G) + k$	$m - n + 1 = m - (n - 1)$. Theorem 2.1.1 in fact states that $n - 1$ is the minimum number
voloshin-hypergraph_F00327	56	0.516	\wedge=	$\wedge =$	$m - n + 1 = m - (n - 1)$. Theorem 2.1.1 in fact states that $n - 1$ is the minimum number
voloshin-hypergraph_F00328	56	1.000	m-n+1=m-(n-1)	$m - n + 1 = m - (n - 1)$	of edges for a graph to be connected, or, equivalently, to be a tree. Therefore, the number $m - (n - 1)$ shows how many extra edges graph G has. Starting from any spanning tree, one
voloshin-hypergraph_F00329	56	1.000	m-(n-1)	$m - (n - 1)$	can sequentially add remaining n edges to reconsider G . Every such edge forms previously one cycle with spanning tree. These cycles are called elementary and edges are called
voloshin-hypergraph_F00330	56	0.743	\Lambda	Λ	chords with respect to the spanning tree.
voloshin-hypergraph_F00331	56	0.535	\wedge(G)=0	$\wedge(G) = 0$	
voloshin-hypergraph_F00332	56	0.979	\Lambda(G)=	$\Lambda(G) =$	of elementary cycles and the number of elementary cycles is always the same, namely equal
voloshin-hypergraph_F00333	56	1.000	m-n+1=7-5+1=3	$m - n + 1 = 7 - 5 + 1 = 3$	to the cyclomatic number. Let us agree that two spanning trees of K_7 are considered different if they are formed
voloshin-hypergraph_F00334	56	1.000	n^{n-2}	n^{n-2}	—
voloshin-hypergraph_F00335	57	1.000	n=3	$n = 3$	Since any tree is a connected graph, with the agreement above Cayley's formula implies that there are total n^{n-2} trees for every n . For example, if $n = 3$, then there are three
voloshin-hypergraph_F00336	57	1.000	E_{\{n\}}, C_{\{n\}}, K_{\{n\}}, W_{\{n\}}	E_n, C_n, K_n, W_n	1. Find the cyclomatic number of E_n, C_n, K_n, W_n , prism, cube and the Petersen graph.
voloshin-hypergraph_F00337	57	1.000	C_{\{6\}}, K_{\{4\}}, W_{\{6\}}	C_6, K_4, W_6	4. Describe the procedure for finding all spanning trees in cube and prism.
voloshin-hypergraph_F00338	57	1.000	K_{\{n\}}, W_{\{n\}}, n \geq 5	$K_n, W_n, n \geq 5$	2.2. Trees and Distance
voloshin-hypergraph_F00339	57	1.000	d(x, y)	$d(x, y)$	defined as $d(x, Y) = \min_{y \in Y} d(x, y)$. It is the shortest distance between x and any vertex of
voloshin-hypergraph_F00340	57	0.731	d(x, y)=\infty	$d(x, y) = \infty$	Y .
voloshin-hypergraph_F00341	57	1.000	d(x, Y)=\min_{y \in Y} d(x, y)	$d(x, Y) = \min_{y \in Y} d(x, y)$	$d(x, Y) = d(x, y)$.
voloshin-hypergraph_F00342	57	1.000	Y	Y	
voloshin-hypergraph_F00343	57	1.000	x, y, z \in X	$x, y, z \in X$	3. $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).
voloshin-hypergraph_F00344	57	1.000	d(x, y) \geq 0	$d(x, y) \geq 0$	The diameter of G denotes distance between the farthest ve
voloshin-hypergraph_F00345	57	1.000	d(x, x)=0	$d(x, x) = 0$	The diameter of G denotes distance between the farthest ve
voloshin-hypergraph_F00346	57	1.000	d(x, y)=d(y, x)	$d(x, y) = d(y, x)$	number. $d(x, y)$ – the many diametral
voloshin-hypergraph_F00347	57	1.000	d(x, y) \leq d(x, z) + d(z, y)	$d(x, y) \leq d(x, z) + d(z, y)$	1. It means that if $y \in N_{\infty}(x)$ and $z \notin N_{\infty}(x)$, then $d(x, z) = 1$. Vertex x and set $N_{\infty}(x)$ is called the eccentricity of x .

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voloshin-hypergraph_F08348	57	■ 0.999	$\operatorname{diam}(G)$	$\operatorname{diam}(G)$	of minimum eccentricity. The radius of G is equal to eccentricity of any vertex from the center. At last, a vertex of degree 1 is called pendant vertex or leaf .
voloshin-hypergraph_F08349	57	■ 0.999	$\max_{x,y \in X} d(x,y)$	$\max_{x,y \in X} d(x,y)$	of minimum eccentricity. The radius of G is equal to eccentricity of any vertex from the center. At last, a vertex of degree 1 is called pendant vertex or leaf .
voloshin-hypergraph_F08350	57	■ 0.876	$d(x,y)=\operatorname{diam}(G)$	$d(x,y) = \operatorname{diam}(G)$	Theorem 2.2.1 Let $x \in X$ be an arbitrary vertex in a tree T . Then every vertex $y \in N_\infty(x)$
voloshin-hypergraph_F08351	57	■ 1.000	$N_{\infty}(x)$	$N_\infty(x)$	is pendant.
voloshin-hypergraph_F08352	57	■ 1.000	$y \in N_{\infty}(x)$	$y \in N_\infty(x)$	
voloshin-hypergraph_F08353	57	■ 1.000	$z \notin N_{\infty}(x)$	$z \notin N_\infty(x)$	
voloshin-hypergraph_F08354	57	■ 1.000	$d(x,z) < d(x,y)$	$d(x,z) < d(x,y)$	
voloshin-hypergraph_F08355	58	■ 1.000	z	z	Figure 2.2. By contradiction, assume there is another vertex z' adjacent to y . The shortest (x,z') path cannot use y because it would be longer than $d(x,y)$. Therefore, it must use
voloshin-hypergraph_F08356	58	■ 1.000	$z^{\prime}$	z'	(x,z') -path cannot use y because it would be longer than $d(x,y)$. Therefore, it must use vertex z or any other vertex from (x,y) -path. In any case we obtain a cycle (at least triangle)
voloshin-hypergraph_F08357	58	■ 0.999	$\left(x,z^{\prime}\right)$	(x,z')	what contradicts that T is a tree. □
voloshin-hypergraph_F08358	58	■ 0.570	$d(x,y)=\operatorname{diam}(T)$	$d(x,y) = \operatorname{diam}(T)$	Theorem 2.2.2 (Jordan, 1869) The center of a tree is either K_1 or K_2 .
voloshin-hypergraph_F08359	58	■ 0.570	$x \in N_{\infty}(y)$	$x \in N_\infty(y)$	Theorem 2.2.2 (Jordan, 1869) The center of a tree is either K_1 or K_2 .
voloshin-hypergraph_F08360	58	■ 1.000	K_1	K_1	Since we delete all pendant vertices, by Theorem 2.2.1, the eccentricity of ea
voloshin-hypergraph_F08361	58	■ 0.998	$T_1, T_2, T_3, \ldots$	$T_1, T_2, T_3, \ldots$	Definition 2.3
voloshin-hypergraph_F08362	58	■ 0.941	$\operatorname{diam}(T)$	$\operatorname{diam}(T)$	1. For the tree in Figure 2.3, find the diameter, radius and center.
voloshin-hypergraph_F08363	58	■ 0.991	$x \in N_{\infty}(z)$	$x \in N_\infty(z)$	1. For the tree in Figure 2.3, find the diameter, radius and center. 2. For the tree in Figure 2.3, find the diameter, radius and center.
voloshin-hypergraph_F08364	58	■ 0.991	$y \in N_{\infty}(x): \operatorname{diam}(T)=d(x,y)$	$y \in N_\infty(x): \operatorname{diam}(T) = d(x,y)$	1. For the tree in Figure 2.3, find the diameter, radius and center. 2. For the tree in Figure 2.3, find the diameter, radius and center.
voloshin-hypergraph_F08365	59	■ 1.000	$\mathcal{D}$	$\mathcal{D}$	If each edge $D \in \mathcal{D}$ is assigned a positive real number $w(D)$ called the weight of edge D . Usually, in many practical applications, the weight represents a distance, time, cost.
voloshin-hypergraph_F08366	59	■ 1.000	D	D	If each edge $D \in \mathcal{D}$ is assigned a positive real number $w(D)$ called the weight of edge D . Usually, in many practical applications, the weight represents a distance, time, cost.
voloshin-hypergraph_F08367	59	■ 1.000	$G=(X,\mathcal{D})$	$G=(X,\mathcal{D})$	capacity, resistance, probability, etc. Consider any spanning tree $T=(X,E)$ of G . The
voloshin-hypergraph_F08368	59	■ 1.000	$D \in \mathcal{D}$	$D \in \mathcal{D}$	weight $w(T)$ of tree T is the sum of weights of all edges of T . Different spanning trees may
voloshin-hypergraph_F08369	59	■ 1.000	$w(D)$	$w(D)$	weight $w(T)$ of tree T is the sum of weights of all edges of T . Different spanning trees may
voloshin-hypergraph_F08370	59	■ 1.000	$T=(X,E)$	$T=(X,E)$	weight is called the minimum (maximum) spanning tree problem . It is one of the few optimization problems that allows an efficient algorithm for any graph.
voloshin-hypergraph_F08371	59	■ 1.000	$w(T)$	$w(T)$	optimization problems that allows an efficient algorithm for any graph.
voloshin-hypergraph_F08372	60	■ 0.991	$T^* \neq T$	$T^* \neq T$	Suppose T is not an optimal spanning tree, i.e. there exist a spanning tree $T^* \neq T$ of
voloshin-hypergraph_F08373	60	■ 1.000	$w(T^*) < w(T)$	$w(T^*) < w(T)$	Suppose T is not an optimal spanning tree, i.e. there exist a spanning tree $T^* \neq T$ of minimum weight such that $w(T^*) < w(T)$. Let D be the first edge in the ordering that was
voloshin-hypergraph_F08374	60	■ 1.000	T^*	T^*	chosen for T and is missing in T^* . Adding D to T^* creates a unique cycle. Since T has

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voloshin-hypergraph_F00375	60	■ 1.000	$D^{\{\prime\}}$	D'	no cycles, this cycle contains an edge D' which is not in T . Construct the spanning tree
voloshin-hypergraph_F00376	60	■ 1.000	$T^{\{*\}}+D-D^{\{\prime\}}$	$T^* + D - D'$	$T^* + D - D'$. Now notice that $w(D) \leq w(D')$, and therefore $w(T^* + D - D') = w(T^*) +$
voloshin-hypergraph_F00377	60	■ 1.000	$w(D) \leq w(D^{\{\prime\}}$	$w(D) \leq w(D')$	$T^* + D - D'$. Now notice that $w(D) \leq w(D')$, and therefore $w(T^* + D - D') = w(T^*) +$
voloshin-hypergraph_F00378	60	■ 1.000	$w(D^{\{\prime\}}+D-D^{\{\prime\}})=w(D^{\{\prime\}})+$	$w(T^* + D - D') = w(T^*) +$	$T^* + D - D'$. Now notice that $w(D) \leq w(D')$, and therefore $w(T^* + D - D') = w(T^*) +$
voloshin-hypergraph_F00379	60	■ 1.000	$w(D)-w(D^{\{\prime\}}) \leq w(D^{\{\prime\}})$	$w(D) - w(D') \leq w(T^*)$	$w(D) - w(D') \leq w(T^*)$. We obtain a new spanning tree of the minimum weight which has one edge more in common with T . Repeating this procedure as many times as necessary
voloshin-hypergraph_F00380	60	■ 1.000	$w>0$	$w > 0$	1. Given a weighted graph G , find a spanning tree and compute its weight.
voloshin-hypergraph_F00381	61	■ 0.973	B	B	Proof. $\Rightarrow$ Let G be a bipartite graph with parts A and B , and C_k be any cycle in it with a vertex $x \in A$. Traverse C_k starting at x in any direction. Since C_k is bipartite, each time we
voloshin-hypergraph_F00382	61	■ 1.000	$x \in A$	$x \in A$	vertex $x \in A$. Traverse C_k starting at x in any direction. Since C_k is bipartite, each time we alternate parts A and B . Since we end at x , k is even.
voloshin-hypergraph_F00383	61	■ 1.000	x, k	x, k	$\Leftarrow$ Let $G = (X, E)$ be a connected nontrivial ($n >$
voloshin-hypergraph_F00384	61	■ 0.999	$\{x\}$	$\{x\}$	N_2 , vertices at distance 3 from x by N_3 and so on. The last set in this sequence is $N_\infty(x)$
voloshin-hypergraph_F00385	61	■ 0.999	$N_{\{0\}}$	N_0	N_2 , vertices at distance 3 from x by N_3 and so on. The last set in this sequence is $N_\infty(x)$
voloshin-hypergraph_F00386	61	■ 0.999	$N_{\{1\}}$	N_1	N_2 , vertices at distance 3 from x by N_3 and so on. The last set in this sequence is $N_\infty(x)$
voloshin-hypergraph_F00387	61	■ 0.994	$N_{\{2\}}$	N_2	denoted by N_k , we obtain the following partition of X :
voloshin-hypergraph_F00388	61	■ 0.994	$N_{\{3\}}$	N_3	denoted by N_k , we obtain the following partition of X :
voloshin-hypergraph_F00389	61	■ 1.000	$N_{\{k\}}$	N_k	$X = N_0 \cup N_1 \cup N_2 \cup N_3 \cup \dots$
voloshin-hypergraph_F00390	61	■ 0.979	$N_{\{i\}} \cap N_{\{j\}} = \emptyset, i \neq j$	$N_i \cap N_j = \emptyset, i \neq j$	
voloshin-hypergraph_F00391	61	■ 1.000	$N_{\{i\}}$	N_i	
voloshin-hypergraph_F00392	61	■ 1.000	$X=A \cup B$	$X = A \cup B$	have the same length, say l , and no common vertices other than x' . Therefore, these paths
voloshin-hypergraph_F00393	61	■ 1.000	$e=\{y, z\}$	$e = \{y, z\}$	and edge e form a cycle C_{2l+1} , a contradiction. $\square$
voloshin-hypergraph_F00394	61	■ 1.000	$y, z \in N_{\{i\}}$	$y, z \in N_i$	and edge e form a cycle C_{2l+1} , a contradiction. $\square$
voloshin-hypergraph_F00395	61	■ 0.926	$i>0$	$i > 0$	The idea of the proof of König's theorem allows to suggest a simple algorithm which not
voloshin-hypergraph_F00396	61	■ 0.926	(x, z)	(x, z)	The idea of the proof of König's theorem allows to suggest a simple algorithm which not
voloshin-hypergraph_F00397	61	■ 1.000	$x^{\{\prime\}}$	x'	or many other problems.
voloshin-hypergraph_F00398	61	■ 1.000	$\left(x^{\{\prime\}}, y\right)$	(x', y)	or many other problems.
voloshin-hypergraph_F00399	61	■ 1.000	$\left(x^{\{\prime\}}, z\right)$	(x', z)	or many other problems.
voloshin-hypergraph_F00400	61	■ 1.000	l	l	Algorithm 2.4.1 (Breadth-first search)
voloshin-hypergraph_F00401	61	■ 1.000	$C_{\{2\ l+1\}}$	C_{2l+1}	<i>INPUT:</i> A graph $G = (X, E)$

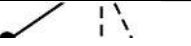
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00402	61	■ 0.993	$i=0$	$i = 0$	
voloshin-hypergraph_F00403	61	■ 0.992	$i+1$	$i + 1$	—
voloshin-hypergraph_F00404	62	■ 1.000	$i:=i+1$	$i := i + 1$	$2 \leq X \leq k$ is a nonempty subset of X and $Y \leq k$. We shall show that $N_G(S) \geq S$ for every such subset S. We shall prove this by induction on S. The base case is $S = 1$. Let $S = \{x\}$. Then $N_G(S) = N_G(x) = k \geq 1 = S$. The induction hypothesis is that for every nonempty subset $S' \subset X$ with $S' < S$, we have $N_G(S') \geq S'$. We now prove the induction step. Let $S \subset X$ with $S \geq 2$. We shall show that $N_G(S) \geq S$. Let $x \in S$. Then $N_G(x) = k$. Let $S' = S - \{x\}$. Then $S' = S - 1$. By the induction hypothesis, $N_G(S') \geq S'$. We have $N_G(S) = N_G(x) \cup N_G(S')$. Since $N_G(x) \cap N_G(S') \subset N_G(x)$, we have $N_G(S) \geq N_G(x) + N_G(S') - N_G(x) \cap N_G(S') \geq k + S' - N_G(x) \cap N_G(S')$. Since $N_G(x) \cap N_G(S') \leq k$, we have $N_G(S) \geq k + S' - k = S' = S - 1$. Since $N_G(x) = k$, we have $N_G(S) \geq S - 1 + 1 = S$. This completes the induction step. Therefore, $N_G(S) \geq S$ for every nonempty subset $S \subset X$. This completes the proof of the theorem.
voloshin-hypergraph_F00405	62	■ 1.000	$N(S)$	$N(S)$	emphasize that the neighborhood is considered in a graph G.
voloshin-hypergraph_F00406	62	■ 1.000	$G=(X, Y ; E)$	$G = (X, Y; E)$	
voloshin-hypergraph_F00407	62	■ 1.000	$N_{\{G\}}(S)$	$N_G(S)$	
voloshin-hypergraph_F00408	63	■ 1.000	$ S $	$ S $	$\Leftarrow$ Let $G = (X, Y; E)$ be a bipartite graph. We
voloshin-hypergraph_F00409	63	■ 1.000	$\left N_{\{G\}}(S)\right \geq S $	$ N_G(S) \geq S $	$\Leftarrow$ Let $G = (X, Y; E)$ be a bipartite graph. We
voloshin-hypergraph_F00410	63	■ 1.000	$ X $	$ X $	x has at least one neighbor y in Y , thus the edge xy is the required matching. Let now $ X > 1$ and the theorem be true for all bipartite graphs with the first part on
voloshin-hypergraph_F00411	63	■ 1.000	$ X =1$	$ X = 1$	x has at least one neighbor y in Y , thus the edge xy is the required matching. Let now $ X > 1$ and the theorem be true for all bipartite graphs with the first part on
voloshin-hypergraph_F00412	63	■ 1.000	$ X >1$	$ X > 1$	Case 1: for any subset $S \subset X, S \neq X$,
voloshin-hypergraph_F00413	63	■ 1.000	$< X $	$< X $	
voloshin-hypergraph_F00414	63	■ 1.000	$S \subsetneq X, S \neq X$	$S \subset X, S \neq X$	$ S $
voloshin-hypergraph_F00415	63	■ 1.000	$G^{\prime}=\left(X^{\prime}, Y^{\prime} ; E^{\prime}\right)$	$G' = (X', Y'; E')$	$ E' \leq X' + Y' $
voloshin-hypergraph_F00416	63	■ 1.000	$\left X^{\prime}\right = X -1< X $	$ X' = X - 1 < X $	$ E' \leq X' + Y' $
voloshin-hypergraph_F00417	63	■ 1.000	$S^{\prime} \subseteq X^{\prime}$	$S' \subseteq X'$	By the induction hypc
voloshin-hypergraph_F00418	63	■ 1.000	$S_{\{0\}} \subsetneq X, S_{\{0\}} \neq X$	$S_0 \subset X, S_0 \neq X$	split the vertex set of G into two subsets:
voloshin-hypergraph_F00419	63	■ 1.000	$G_{\{A\}}$	G_A	see Figure 2.6. Therefore,
voloshin-hypergraph_F00420	63	■ 1.000	$G_{\{B\}}$	G_B	see Figure 2.6. Therefore,
voloshin-hypergraph_F00421	63	■ 1.000	$S \subseteqq S_{\{0\}}$	$S \subseteq S_0$	$ S + S_0 = S \cup S_0 \leq X + Y = X + Y $
voloshin-hypergraph_F00422	63	■ 1.000	$N_{\{G_{\{A\}}\}}(S)=N_{\{G\}}(S)$	$N_{G_A}(S) = N_G(S)$	$ S + S_0 = S \cup S_0 \leq X + Y = X + Y $
voloshin-hypergraph_F00423	63	■ 1.000	$ S \leq\left N_{\{G_{\{A\}}\}}(S)\right $	$ S \leq N_{G_A}(S) $	$ S + S_0 = S \cup S_0 \leq X + Y = X + Y $
voloshin-hypergraph_F00424	63	■ 1.000	$S_{\{0\}}$	S_0	$ S_0 + S = S_0 \cup S \leq X + Y = X + Y $
voloshin-hypergraph_F00425	63	■ 1.000	$S \subseteqq X-S_{\{0\}}$	$S \subseteq X - S_0$	
voloshin-hypergraph_F00426	64	■ 1.000	$\left S_{\{0\}}\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right $	$ S_0 = N_G(S_0) $	Corollary 2.4.1 Every regular bipartite graph has a perfect matching.
voloshin-hypergraph_F00427	64	■ 1.000	$ S \leq\left N_{\{G_{\{B\}}\}}(S)\right $	$ S \leq N_{G_B}(S) $	Corollary 2.4.1 Every regular bipartite graph has a perfect matching.
voloshin-hypergraph_F00428	64	■ 1.000	$X-S_{\{0\}}$	$X - S_0$	Proof. Let $G = (X, Y; E)$ be a k -regular ($k \geq 1$) bipart

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00429	64	0.746	$k \geq 1$	$k \geq 1$	Recall that $\tau(G)$ is the cardinality of a minimum transversal of a graph G , i.e. the
voloshin-hypergraph_F00430	64	1.000	$k X =k Y $	$k X = k Y $	minimum number of vertices that “touch” all edges. As we mentioned in Section 1.7., for
voloshin-hypergraph_F00431	64	1.000	$ X = Y $	$ X = Y $	minimum number of vertices that “touch” all edges. As we mentioned in Section 1.7., for
voloshin-hypergraph_F00432	64	1.000	$i=k S $	$i = k S $	Theorem 2.4.3 (König, 1931) For any bipartite graph G ,
voloshin-hypergraph_F00433	64	1.000	$k, i \leq k N_G(S) $	$k, i \leq k N_G(S) $	$\tau(G) = \nu(G)$
voloshin-hypergraph_F00434	64	1.000	$S \subseteq X, N_G(S) \geq S $	$S \subseteq X, N_G(S) \geq S $	$\tau(G) = \nu(G)$
voloshin-hypergraph_F00435	64	0.987	$G, \tau(G) \geq v(G)$	$G, \tau(G) \geq v(G)$	
voloshin-hypergraph_F00436	65	1.000	$T \subseteq (X \cup Y)$	$T \subseteq (X \cup Y)$	Since $A \cup B$ is a transversal, there are no edges between $Y \setminus B$ and $X \setminus A$. For each $S \subseteq A$,
voloshin-hypergraph_F00437	65	1.000	$\tau(G)= T $	$\tau(G) = T $	consider $N_{G_1}(S)$. If $ N_{G_1}(S) < S $, then, because $N_{G_1}(S)$ “touches” all edges incident to
voloshin-hypergraph_F00438	65	0.999	$X \cap T=A$	$X \cap T = A$	that are not “touched” by B , we could replace S by $N_{G_1}(S)$ and obtain a smaller transversal of
voloshin-hypergraph_F00439	65	0.999	$Y \cap T=B$	$Y \cap T = B$	that are not “touched” by B , we could replace S by $N_{G_1}(S)$ and obtain a smaller transversal of
voloshin-hypergraph_F00440	65	0.999	$G_{\{1\}}=G_{\{A \cup (Y \backslash B)\}}$	$G_1 = G_{A \cup (Y \setminus B)}$	that are not “touched” by B , we could replace S by $N_{G_1}(S)$ and obtain a smaller transversal of
voloshin-hypergraph_F00441	65	0.999	$G_{\{2\}}=G_{\{B \cup (X \backslash A)\}}$	$G_2 = G_{B \cup (X \setminus A)}$	that are not “touched” by B , we could replace S by $N_{G_1}(S)$ and obtain a smaller transversal of
voloshin-hypergraph_F00442	65	1.000	$A \cup B$	$A \cup B$	G than T . Since T is a minimum transversal, this is impossible, and therefore, $ N_{G_1}(S) \geq S $
voloshin-hypergraph_F00443	65	1.000	$Y \backslash B$	$Y \setminus B$	G than T . Since T is a minimum transversal, this is impossible, and therefore, $ N_{G_1}(S) \geq S $
voloshin-hypergraph_F00444	65	1.000	$X \backslash A$	$X \setminus A$	G than T . Since T is a minimum transversal, this is impossible, and therefore, $ N_{G_1}(S) \geq S $
voloshin-hypergraph_F00445	65	1.000	$S \subseteq A$	$S \subseteq A$	G than T . Since T is a minimum transversal, this is impossible, and therefore, $ N_{G_1}(S) \geq S $
voloshin-hypergraph_F00446	65	1.000	$N_{\{G_{\{1\}}\}}(S)$	$N_{G_1}(S)$	for any subset $S \subseteq A$. By Theorem 2.4.2, graph G_1 has a matching that covers A . Applying the same reasoning to graph G_2 , we obtain a matching in G_2 that covers B . Since graphs
voloshin-hypergraph_F00447	65	1.000	$ \left N_{\{G_{\{1\}}\}}(S)\right < S $	$ N_{G_1}(S) < S $	for any subset $S \subseteq A$. By Theorem 2.4.2, graph G_1 has a matching that covers A . Applying the same reasoning to graph G_2 , we obtain a matching in G_2 that covers B . Since graphs
voloshin-hypergraph_F00448	65	1.000	$ \left N_{\{G_{\{1\}}\}}(S)\right \geq S $	$ N_{G_1}(S) \geq S $	graph G that has $ A + B = T = \tau(G)$ edges. Hence $\tau(G) = \nu(G)$. $\square$
voloshin-hypergraph_F00449	65	0.887	$ A + B = T =\tau(G)$	$ A + B = T = \tau(G)$	1. Run the breadth-first search algorithm for prism, cube and Petersen graph bitrary vertex.
voloshin-hypergraph_F00450	65	0.887	$\tau(G)=v(G)$	$\tau(G) = v(G)$	1. Run the breadth-first search algorithm for prism, cube and Petersen graph bitrary vertex.
voloshin-hypergraph_F00451	65	1.000	$m, n \geq 1$	$m, n \geq 1$	
voloshin-hypergraph_F00452	65	1.000	$K_{\{m, n\}}$	$K_{m,n}$	
voloshin-hypergraph_F00453	65	0.999	$\tau\left(K_{\{m, n\}}\right)$	$\tau\left(K_{m,n}\right)$	following algorithmic problems.
voloshin-hypergraph_F00454	65	0.999	$v\left(K_{\{m, n\}}\right)$	$v\left(K_{m,n}\right)$	following algorithmic problems.
voloshin-hypergraph_F00455	65	0.637	τ	τ	100.

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voloshin-hypergraph_F00456	67	■ 1.000	≥ 4	≥ 4	of them is induced. Notice that cycles C_1, C_2, and C_3 do not have non-consecutive vertices and therefore cannot have chords: it explains why “chordality” begins with the cycles of cycles is not a triangle, then it has another chord, and so on. Eventually, every cycle is split 
voloshin-hypergraph_F00457	67	■ 1.000	$C_{\{1\}}, C_{\{2\}}$	C_1, C_2	
voloshin-hypergraph_F00458	68	■ 1.000	$G_{\{1\}}=G_{\{X_{\{1\}} \cup S\}}$	$G_1 = G_{X_1 \cup S}$	
voloshin-hypergraph_F00459	68	■ 1.000	$G_{\{2\}}=G_{\{X_{\{2\}} \cup S\}}$	$G_2 = G_{X_2 \cup S}$	
voloshin-hypergraph_F00460	68	■ 1.000	$G_{\{S\}}$	G_S	
voloshin-hypergraph_F00461	68	■ 1.000	$x, y \in S$	$x, y \in S$	
voloshin-hypergraph_F00462	69	■ 0.997	$y \in C_{\{k\}}$	$y \in C_k$	Since $k \geq 4$, each such path contains at least one internal vertex. Since C_k is induced, inter-
voloshin-hypergraph_F00463	69	■ 1.000	$k \geq 4$	$k \geq 4$	nal vertices of the first path are not adjacent to the internal vertices of the second path. Now
voloshin-hypergraph_F00464	69	■ 1.000	$ X =$	$ X =$	the theorem is due for an noncomplete connected chordal graphs on less than n vertices and prove it for G .
voloshin-hypergraph_F00465	69	■ 1.000	$X - \{x, y\}$	$X - \{x, y\}$	$G_1 = G_{X_1 \cup S}$ and $G_2 = G_{X_2 \cup S}$ be the derived subgraphs with respect to S , see Figure 211. Graphs G_1 and G_2 both are chordal as subgraphs of G .
voloshin-hypergraph_F00466	69	■ 1.000	$G_{\{X_{\{1\}}\}}$	G_{X_1}	is simplicial in G because there are no edges between X_1 and X_2 . Suppose G_1 is not a clique.
voloshin-hypergraph_F00467	69	■ 1.000	$G_{\{X_{\{2\}}\}}$	G_{X_2}	is simplicial in G because there are no edges between X_1 and X_2 . Suppose G_1 is not a clique.
voloshin-hypergraph_F00468	69	■ 1.000	$< n$	$< n$	Consider subgraph G_2 and apply the same reasoning. We obtain that in G there are two
voloshin-hypergraph_F00469	69	■ 0.549	$X_{\{2\}} \neq \emptyset$	$X_2 \neq \emptyset$	simplicial vertices which are not adjacent.
voloshin-hypergraph_F00470	69	■ 1.000	$K_{\{n\}}, n \geq 2$	$K_n, n \geq 2$	
voloshin-hypergraph_F00471	70	■ 1.000	$G_{\{1\}}=(X, E)$	$G_1 = (X, E)$	
voloshin-hypergraph_F00472	70	■ 0.875	$G_{\{n+1\}}=\emptyset$	$G_{n+1} = \emptyset$	step in this procedure will occur when we delete the last vertex x_n and arrive to $G_{n+1} = \emptyset$. We obtain the ordering of vertices $x_1, x_2, \dots, x_n$. Denote it by σ so $\sigma = (x_1, x_2, \dots, x_n)$. The
voloshin-hypergraph_F00473	70	■ 0.999	$\boldsymbol{\sigma} = \left(x_{\{1\}}, x_{\{2\}}, \dots, x_{\{n\}} \right)$	$\boldsymbol{\sigma} = (x_1, x_2, \dots, x_n)$	we obtain the ordering of vertices $x_1, x_2, \dots, x_n$. Denote it by σ , so $\sigma = (x_1, x_2, \dots, x_n)$. The ordering σ is called a simplicial elimination ordering or a <i>perfect</i> elimination ordering.
voloshin-hypergraph_F00474	70	■ 1.000	$x_{\{i\}}, x_{\{i+1\}}, \dots, x_{\{n\}}$	$x_i, x_{i+1}, \dots, x_n$	It follows from the observations above that G is chordal if and only if it has a simplicial decomposition. But if a
voloshin-hypergraph_F00475	70	■ 1.000	$G_{\{i+1\}}=G_{\{i\}}-x_{\{i\}}, i=1,2, \dots, n$	$G_{i+1} = G_i - x_i, i = 1, 2, \dots, n$	It follows from the observations above that G is chordal if and only if it has a simplicial decomposition. But if a
voloshin-hypergraph_F00476	70	■ 1.000	$G_{\{2\}}, G_{\{3\}}, \dots, G_{\{8\}}, G_{\{9\}}=\emptyset$	$G_2, G_3, \dots, G_8, G_9 = \emptyset$	
voloshin-hypergraph_F00477	70	■ 0.846	$1, 2, 3, \dots, 8$	$1, 2, 3, \dots, 8$	
voloshin-hypergraph_F00478	70	■ 0.846	$\boldsymbol{\sigma} = (1, 2, 3, 4, 5, 6, 7, 8)$	$\boldsymbol{\sigma} = (1, 2, 3, 4, 5, 6, 7, 8)$	
voloshin-hypergraph_F00479	70	■ 0.907	$\binom{8}{4}$	$\binom{8}{4}$	
voloshin-hypergraph_F00480	70	■ 0.943	$\binom{8}{5}$	$\binom{8}{5}$	
voloshin-hypergraph_F00481	70	■ 0.668	$\binom{8}{6}$	$\binom{8}{6}$	
voloshin-hypergraph_F00482	70	■ 0.668	$C_{\{6\}}, \binom{8}{7}$	$C_6, \binom{8}{7}$	

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voloshin-hypergraph_F00483	70	0.668	C_8	C_8	
voloshin-hypergraph_F00484	70	0.668	$\binom{8}{8}$	$\binom{8}{8}$	
voloshin-hypergraph_F00485	71	1.000	$\sigma^{\prime}=(2,3,8,7,6,5,4,1)$	$\sigma' = (2, 3, 8, 7, 6, 5, 4, 1)$	to z may be the last but one. deleted from the end of the
voloshin-hypergraph_F00486	71	1.000	$S^{\prime}$	S'	Notice that the theorem above holds for any graph, not necessarily chordal. However,
voloshin-hypergraph_F00487	71	1.000	$\{x\} \cup N(x)$	$\{x\} \cup N(x)$	the existence of a simplicial elimination ordering allows to suggest a simple algorithm for finding the independence number and the respective maximum stable set of a chordal graph.
voloshin-hypergraph_F00488	71	1.000	$S^{\prime} \cup \{x\}$	$S' \cup \{x\}$	the existence of a simplicial elimination ordering allows to suggest a simple algorithm for finding the independence number and the respective maximum stable set of a chordal graph.
voloshin-hypergraph_F00489	71	1.000	$y \in N(x)$	$y \in N(x)$	finding the independence number and the respective maximum stable set of a chordal graph.
voloshin-hypergraph_F00490	71	1.000	$y \in S^{\prime}$	$y \in S'$	finding the independence number and the respective maximum stable set of a chordal graph.
voloshin-hypergraph_F00491	71	1.000	$S=S^{\prime} \backslash\{y\} \cup\{x\}$	$S = S' \setminus \{y\} \cup \{x\}$	Algorithm 3.2.1 (Finding
voloshin-hypergraph_F00492	71	0.995	$ S =\alpha(G)$	$ S = \alpha(G)$	4. End.
voloshin-hypergraph_F00493	71	0.872	$S=\emptyset$	$S = \emptyset$	
voloshin-hypergraph_F00494	72	0.813	$\{1,4,5\}$	$\{1, 4, 5\}$	If we run the algorithm for graph G shown in Figure 3.3, then on the first step, vertex 1 is included in S and set $\{1, 4, 5\}$ is strongly deleted; on the second step, vertex 2 is
voloshin-hypergraph_F00495	72	1.000	$\{2,3,8\}$	$\{2, 3, 8\}$	included in S , and set $\{2, 3, 8\}$ is strongly deleted; at last, on the third step, vertex 6 is
voloshin-hypergraph_F00496	72	0.968	$S=\{1,2,6\}$	$S = \{1, 2, 6\}$	conclude that one of the maximum stable sets of graph G is $S = \{1, 2, 6\}$, and $\alpha(G) = 3$.
voloshin-hypergraph_F00497	72	0.968	$\alpha(G)=3$	$\alpha(G) = 3$	conclude that one of the maximum stable sets of graph G is $S = \{1, 2, 6\}$, and $\alpha(G) = 3$.
voloshin-hypergraph_F00498	72	1.000	$\{3,4,5,7,8\}$	$\{3, 4, 5, 7, 8\}$	Respectively, vertices $\{3, 4, 5, 7, 8\}$ form a minimum transversal and $\tau(G) = 5$. Depending on the simplicial vertices chosen at each step, there are other maximum stable sets and
voloshin-hypergraph_F00499	72	1.000	$\tau(G)=5$	$\tau(G) = 5$	Respectively, vertices $\{3, 4, 5, 7, 8\}$ form a minimum transversal and $\tau(G) = 5$. Depending on the simplicial vertices chosen at each step, there are other maximum stable sets and
voloshin-hypergraph_F00500	72	0.966	$\{1,4,5\},\{2,3,8\}$	$\{1, 4, 5\}, \{2, 3, 8\}$	and $\{6, 7\}$ have the property that each vertex of G belongs to one of them, and the number of such cliques is the minimum
voloshin-hypergraph_F00501	72	0.999	$\theta(G)$	$\theta(G)$	cliques cover the vertex set of G . As we will see, clique coverings play an important role
voloshin-hypergraph_F00502	72	0.986	$\theta(G)=3$	$\theta(G) = 3$	
voloshin-hypergraph_F00503	73	1.000	$M(G)$	$M(G)$	Indeed, on one hand, $M(G)$ cannot be less than t since t is the minimum degree in
voloshin-hypergraph_F00504	73	1.000	t	t	Generally, notice that the degree of any vertex in sub
voloshin-hypergraph_F00505	73	1.000	$M(G)>t$	$M(G) > t$	Theorem 3.3.1 For every graph G with $M(G) > \alpha(G) + 1$,
voloshin-hypergraph_F00506	73	1.000	$M(G)=t$	$M(G) = t$	in G . Therefore, there is at least one induced complete subgraph of G with minimum degree
voloshin-hypergraph_F00507	73	1.000	$G, M(G) \geq \omega(G)-1$	$G, M(G) \geq \omega(G) - 1$	
voloshin-hypergraph_F00508	73	0.999	$\omega(G)-1$	$\omega(G) - 1$	
voloshin-hypergraph_F00509	74	0.969	$\boldsymbol{\omega}(G)-1$	$\omega(G) - 1$	next theorem shows that chordal graphs realize the absolute minimum for $M(G)$:

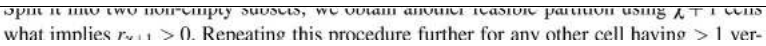
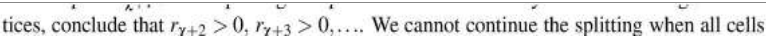
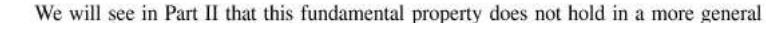
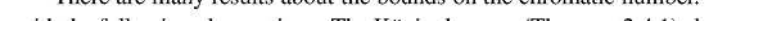
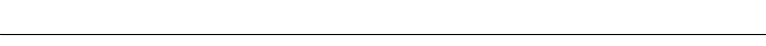
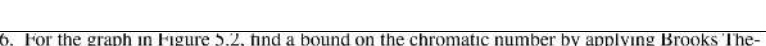
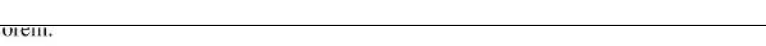
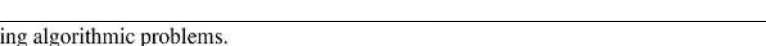
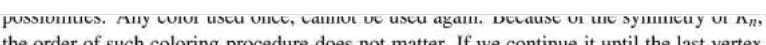
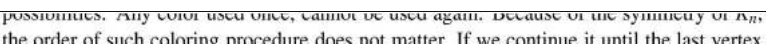
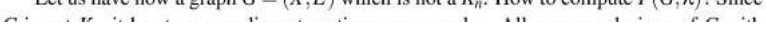
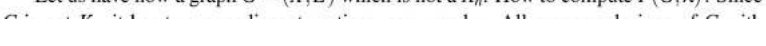
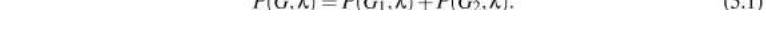
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F08510	74	■ 1.000	$M(\text{G}^{\{\prime\}})=\omega(\text{G}^{\{\prime\}})-1$	$M(G') = \omega(G') - 1$	
voloshin-hypergraph_F08511	74	■ 1.000	$M(\text{C}_{\{k\}})=2=\omega(\text{C}_{\{k\}})$	$M(C_k) = 2 = \omega(C_k)$	
voloshin-hypergraph_F08512	74	■ 1.000	$M(\text{G})\geq \omega(\text{G})-1$	$M(G) \geq \omega(G) - 1$	
voloshin-hypergraph_F08513	74	■ 1.000	$\omega(\text{G})=t+1$	$\omega(G) = t + 1$	
voloshin-hypergraph_F08514	74	■ 1.000	$M(\text{G})\leq t=\omega(\text{G})-1$	$M(G) \leq t = \omega(G) - 1$	
voloshin-hypergraph_F08515	74	■ 1.000	$M(\text{G})=\omega(\text{G})-1$	$M(G) = \omega(G) - 1$	
voloshin-hypergraph_F08516	74	■ 1.000	$K_{\{n,\ n\}}$	$K_{n,n}$	
voloshin-hypergraph_F08517	74	■ 1.000	$M(\text{K}_{\{n,\ n\}})=n,\ \omega(\text{K}_{\{n,\ n\}})=2$	$M(K_{n,n}) = n, \omega(K_{n,n}) = 2$	
voloshin-hypergraph_F08518	74	■ 1.000	$M(\text{G}^{\{\prime\}})>\omega(\text{G}^{\{\prime\}})-1$	$M(G') > \omega(G') - 1$	
voloshin-hypergraph_F08519	74	■ 1.000	$M(\text{C}_{\{n\}}),\ M(\text{K}_{\{n\}}),\ M(\text{W}_{\{n\}})$	$M(C_n), M(K_n), M(W_n)$	
voloshin-hypergraph_F08520	74	■ 1.000	$\text{G},\ M(\text{G})=k$	$G, M(G) = k$	
voloshin-hypergraph_F08521	75	■ 1.000	$\text{G} =$	$G =$	
voloshin-hypergraph_F08522	75	■ 0.998	$(\text{X},\ \text{E})$	(X, E)	
voloshin-hypergraph_F08523	75	■ 0.998	$n=2$	$n = 2$	
voloshin-hypergraph_F08524	75	■ 0.999	$\text{N}_{\{k\}}=\text{N}_{\{\infty\}}(\text{x})$	$N_k = N_{\infty}(x)$	
voloshin-hypergraph_F08525	76	■ 1.000	$\text{N}_{\{\infty\}}(\text{x})=\text{X}-\{\text{x}\}$	$N_{\infty}(x) = X - \{x\}$	
voloshin-hypergraph_F08526	76	■ 1.000	$\text{G}-\text{x}$	$G - x$	
voloshin-hypergraph_F08527	76	■ 1.000	$\text{N}_{\{k-1\}}$	N_{k-1}	
voloshin-hypergraph_F08528	76	■ 1.000	$\text{G}_{\{\text{N}_{\{k\}}\}}$	G_{N_k}	
voloshin-hypergraph_F08529	76	■ 1.000	$\text{G}_{\{\text{N}_{\{k-1\}}\}}$	$G_{N_{k-1}}$	
voloshin-hypergraph_F08530	76	■ 1.000	$\text{G}_{\{\text{N}_{\{k-1\}}\}\cup\text{N}_{\{k\}}\}}$	$G_{N_{k-1} \cup N_k}$	
voloshin-hypergraph_F08531	76	■ 1.000	$y\in\text{N}_{\{k\}}$	$y \in N_k$	
voloshin-hypergraph_F08532	76	■ 0.976	$y^{\{\prime\}}\in\text{N}_{\{\infty\}}(\text{x})$	$y' \in N_{\infty}(x)$	
voloshin-hypergraph_F08533	76	■ 0.986	$d(\text{x},\ y^{\{\prime\}})=d(\text{x},\ y)=\operatorname{diam}(\text{G})$	$d(x, y') = d(x, y) = \text{diam}(G)$	
voloshin-hypergraph_F08534	76	■ 0.934	$y^{\{\prime\}}$	y'	
voloshin-hypergraph_F08535	76	■ 0.934	$x^{\{\prime\}}\in\text{N}_{\{\infty\}}(y^{\{\prime\}})$	$x' \in N_{\infty}(y')$	
voloshin-hypergraph_F08536	76	■ 0.934	$d(y^{\{\prime\}},\ x^{\{\prime\}})=d(y^{\{\prime\}},\ x)=\operatorname{diam}(\text{G})$	$d(y', x') = d(y', x) = \text{diam}(G)$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F08537	76	■ 0.997	$\operatorname{diam}(G)=4$	$\operatorname{diam}(G) = 4$	
voloshin-hypergraph_F08538	76	■ 0.997	$N_{\{0\}}=\{2\}$	$N_0 = \{2\}$	
voloshin-hypergraph_F08539	76	■ 0.997	$N_{\{1\}}(2)=\{1,12,10,3\}$	$N_1(2) = \{1,12,10,3\}$	
voloshin-hypergraph_F08540	76	■ 1.000	$N_{\{2\}}(2)=\{11,4,8,9\}$	$N_2(2) = \{11,4,8,9\}$	
voloshin-hypergraph_F08541	76	■ 1.000	$N_{\{3\}}(2)=N_{\{\infty\}}(2)=\{5,6,7\}$	$N_3(2) = N_\infty(2) = \{5,6,7\}$	
voloshin-hypergraph_F08542	77	■ 0.997	$N_{\{0\}}(5)=\{5\}, N_{\{1\}}(5)=\{4,6\}$	$N_0(5) = \{5\}, N_1(5) = \{4,6\}$	We start with the same procedure starting at $x = 5$. $N_0(5) = \{5\}$, $N_1(5) = \{4,6\}$.
voloshin-hypergraph_F08543	77	■ 1.000	$N_{\{2\}}(5)=\{7,8,9,3\}, N_{\{3\}}(5)=\{2,10\}, N_{\{4\}}(5)=N_{\{\infty\}}(5)=\{1,12,11\}$	$N_2(5) = \{7,8,9,3\}, N_3(5) = \{2,10\}, N_4(5) = N_\infty(5) = \{1,12,11\}$	We now run the same procedure starting at, say, $y = 11$. $N_0(11) = \{11\}$, $N_1(11) = \{5,6,7\}$, $N_2(11) = \{7,8,9,3\}$, $N_3(11) = \{2,10\}$, $N_4(11) = N_\infty(11) = \{1,12,11\}$. We find two simplicial
voloshin-hypergraph_F08544	77	■ 0.993	$d(11,5)=\operatorname{diam}(G)=4$	$d(11,5) = \operatorname{diam}(G) = 4$	vertices, 11 and 5 such that $d(11,5) = \operatorname{diam}(G) = 4$. There are two diametral (11,5)-paths:
voloshin-hypergraph_F08545	78	■ 1.000	$\overline{C_l}, l \geq 4$	$\overline{C_l}, l \geq 4$	as on graphs with all vertices being strongly cyclic. All cycles $C_k, k \geq 0$ are induced. Each vertex belongs to C_k and $\overline{C_k}$, see Figure 3.11.
voloshin-hypergraph_F08546	78	■ 1.000	$C_{\{k\}}, k \geq 6$	$C_k, k \geq 6$	
voloshin-hypergraph_F08547	78	■ 1.000	$\overline{C_4}$	$\overline{C_4}$	
voloshin-hypergraph_F08548	79	■ 0.971	$G-S$	$G - S$	assumption on x . Therefore, y and z must be adjacent, i.e. S induces a clique. $\square$
voloshin-hypergraph_F08549	79	■ 1.000	$y, z \in S$	$y, z \in S$	Theorem 3.5.1 (Characterization theorem, Caracciolo, 1994). Every graph G that follows
voloshin-hypergraph_F08550	80	■ 1.000	$n(G)$	$n(G)$	We prove (iii) $\Rightarrow$ (iv) by induction on the number of vertices $n(G)$. Let $G = (V, E)$ be a
voloshin-hypergraph_F08551	80	■ 1.000	$C_l, l \geq 4$	$C_l, l \geq 4$	form a latticed subgraph because each vertex belongs to some cycle ($C_k, k \geq 4$ or $C_l, l \geq 4$)
voloshin-hypergraph_F08552	80	■ 0.989	$X^{\prime} \neq \emptyset$	$X' \neq \emptyset$	The graph G' is connected by Lemma 3.5.1, and therefore it is nonempty. We denote it by y .
voloshin-hypergraph_F08553	80	■ 1.000	$G^{\prime}=G-X^{\prime}$	$G' = G - X'$	Since every vertex of G' lies in some $C_k, k \geq 4$, y is co-simplicial. We will prove that
voloshin-hypergraph_F08554	80	■ 1.000	$X^{\prime}=X$	$X' = X$	Since every vertex of G' lies in some $C_k, k \geq 4$, y is co-simplicial. We will prove that
voloshin-hypergraph_F08555	80	■ 1.000	$n\left(G^{\prime}\right)<n(G)$	$n\left(G'\right) < n(G)$	contradiction. Step 2. Proof that x is co-simplicial in G . Let x be any vertex in V . Since G is
voloshin-hypergraph_F08556	80	■ 1.000	$C_{\{k\}}, k \geq 4, y$	$C_k, k \geq 4, y$	Let G_1, G_2 be induced subgraphs of G such that $G = G_1 \cup G_2, G_1 \cap G_2 = S$, and there is
voloshin-hypergraph_F08557	80	■ 1.000	$G=G_{\{1\}} \cup G_{\{2\}}, G_{\{1\}} \cap G_{\{2\}}=S$	$G = G_1 \cup G_2, G_1 \cap G_2 = S$	
voloshin-hypergraph_F08558	80	■ 1.000	$G_{\{1\}}-S$	$G_1 - S$	
voloshin-hypergraph_F08559	80	■ 1.000	$G_{\{2\}}-S$	$G_2 - S$	
voloshin-hypergraph_F08560	80	■ 1.000	$e_{\{1\}}, e_{\{2\}}$	e_1, e_2	
voloshin-hypergraph_F08561	81	■ 1.000	$S \subseteq N(x)$	$S \subseteq N(x)$	
voloshin-hypergraph_F08562	81	■ 1.000	$X^{\prime}, y$	X', y	Exercises 3.5.
voloshin-hypergraph_F08563	82	■ 1.000	$C_{\{n\}}, K_{\{n\}}, W_{\{n\}}, n \geq 3$	$C_n, K_n, W_n, n \geq 3$	$\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ are isomorphic to $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$.
voloshin-hypergraph_F08564	83	■ 1.000	$f(G)$	$f(G)$	

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voloshin-hypergraph_F00565	83	■ 1.000	$f_{\{1\}}$	f_1	
voloshin-hypergraph_F00566	83	■ 1.000	$f_{\{4\}}$	f_4	—
voloshin-hypergraph_F00567	85	■ 1.000	$K_{\{2,3\}}$	$K_{2,3}$	G embed at least two different plane embeddings of the same order $ V $ and $ W $ ($n \leq 4$)
voloshin-hypergraph_F00568	85	■ 1.000	$W_{\{n\}}(n \geq 4)$	$W_n(n \geq 4)$	G embed at least two different plane embeddings of the same order $ V $ and $ W $ ($n \leq 4$)
voloshin-hypergraph_F00569	85	■ 1.000	$m(T)=n-1, f(T)=1$	$m(T) = n - 1, f(T) = 1$	number of edges and the number of faces by 1. Since in the formula m and f are of the opposite signs, the equality holds for G itself. $\square$
voloshin-hypergraph_F00570	85	■ 1.000	$n(T)-$	$n(T)-$	number of edges and the number of faces by 1. Since in the formula m and f are of the opposite signs, the equality holds for G itself. $\square$
voloshin-hypergraph_F00571	85	■ 1.000	$m(T)+f(T)=n-(n-1)+1=2$	$m(T) + f(T) = n - (n - 1) + 1 = 2$	opposite signs, the equality holds for G itself.
voloshin-hypergraph_F00572	85	■ 1.000	$m-n+2$	$m - n + 2$	Corollary 4.2.2 If G is a connect
voloshin-hypergraph_F00573	85	■ 1.000	$m(G)$	$m(G)$	$m(G) \leq 3n(G) - 6.$
voloshin-hypergraph_F00574	85	■ 1.000	$f(G)=$	$f(G) =$	$m(G) \leq 3n(G) - 6.$
voloshin-hypergraph_F00575	85	■ 1.000	$3 f$	$3f$	
voloshin-hypergraph_F00576	85	■ 1.000	$2 m$	$2m$	
voloshin-hypergraph_F00577	85	■ 1.000	$3 f \leq 2 m$	$3f \leq 2m$	
voloshin-hypergraph_F00578	85	■ 1.000	$f=2-n+m$	$f = 2 - n + m$	
voloshin-hypergraph_F00579	85	■ 1.000	$3(2-n+m) \leq 2 m$	$3(2 - n + m) \leq 2m$	
voloshin-hypergraph_F00580	85	■ 1.000	$m \leq 3 n-6$	$m \leq 3n - 6$	
voloshin-hypergraph_F00581	86	■ 0.615	$6 n \leq 2 m$	$6n \leq 2m$	For any plane graph G and every its bounded face f , there exists such a plane embedding
voloshin-hypergraph_F00582	86	■ 1.000	$3 n \leq m$	$3n \leq m$	For any plane graph G and every its bounded face f , there exists such a plane embedding of G that face f becomes unbounded. This can be shown using the so called stereographic
voloshin-hypergraph_F00583	86	■ 1.000	$3 n \leq m \leq 3 n-6$	$3n \leq m \leq 3n - 6$	projection.
voloshin-hypergraph_F00584	86	■ 1.000	$0 \leq -6$	$0 \leq -6$	projection.
voloshin-hypergraph_F00585	86	■ 0.997	$n-m+$	$n - m +$	correspond to the vertices, edges and faces of the initial polyhedron. $\square$
voloshin-hypergraph_F00586	86	■ 1.000	$f=2$	$f = 2$	
voloshin-hypergraph_F00587	87	■ 1.000	$n=8, m=12, f=6$	$n = 8, m = 12, f = 6$	For example, consider the cube $n = 8, m = 12, f = 6$ and substitute the formula (4.14)
voloshin-hypergraph_F00588	87	■ 1.000	$K_{\{n\}}, W_{\{n\}}, K_{\{m, n\}}(m \geq 1, n \geq 3)$	$K_n, W_n, K_{m,n}(m \geq 1, n \geq 3)$	
voloshin-hypergraph_F00589	89	■ 1.000	$X=\{1,2,3,4,5,6\}$	$X = \{1, 2, 3, 4, 5, 6\}$	Many examples K_n in planes and realize set $V = \{1, 2, 3, 4, 5, 6\}$. Since K_n is bipartite
voloshin-hypergraph_F00590	90	■ 1.000	$k, l \geq 3$	$k, l \geq 3$	For example, any two cycles C_k and C_l , with $k, l \geq 3$ are homeomorphic because they both
voloshin-hypergraph_F00591	90	■ 0.995	$G_{\{i\}}^{\prime \prime}=\overline{G_{\{i\}}} \cup\{x, y\}$	$G'_i = G_i \cup \{x, y\}$	or. The last implies that any minimal non planar graph must be 3-connected. In other words, it is sufficient to consider further 3-connected graphs only.

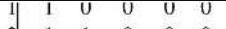
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voloshin-hypergraph_F00592	90	■ 1.000	$G_{1}^{\prime}, G_{2}^{\prime}$	G_1', G_2'	At this point one uses the lemma stating that any 2 connected graph
voloshin-hypergraph_F00593	90	■ 1.000	≥ 5	≥ 5	$K_{3,3}$, then the initial graph G also contains subgraphs homeomorphic to K_5 or $K_{3,3}$.
voloshin-hypergraph_F00594	91	■ 1.000	$G-V\left(G^{\prime}\right)$	$G-V\left(G'\right)$	Z) a connected component of $G-V\left(G'\right)$ together with edges connecting it to G' (vertices of attachment from G' included).
voloshin-hypergraph_F00595	91	■ 1.000	$F(A)$	$F(A)$	vertices of attachment. If $F(A)=\emptyset$ for some A , then G is not planar. If $ F(A) =1$ for some
voloshin-hypergraph_F00596	91	■ 1.000	$F(A)=\emptyset$	$F(A)=\emptyset$	A , then select A for the next step. If $ F(A) >1$ for all A , then select any fragment A . Next, choose any path connecting two vertices of attachment of the selected fragment A .
voloshin-hypergraph_F00597	91	■ 1.000	$ F(A) =1$	$ F(A) =1$	A , then select A for the next step. If $ F(A) >1$ for all A , then select any fragment A . Next, choose any path connecting two vertices of attachment of the selected fragment A .
voloshin-hypergraph_F00598	91	■ 1.000	$ F(A) >1$	$ F(A) >1$	Next, choose any path connecting two vertices of attachment of the selected fragment A . Embed the path inside a face from $F(A)$. Call the resulting plane graph G' and repeat the
voloshin-hypergraph_F00599	91	■ 1.000	$G^{\prime} \neq G$	$G' \neq G$	Exercises 4.4.
voloshin-hypergraph_F00600	91	■ 1.000	$m \geq 1$	$m \geq 1$	make it planar :
voloshin-hypergraph_F00601	91	■ 1.000	$n \geq 3$	$n \geq 3$	make it planar :
voloshin-hypergraph_F00602	91	■ 1.000	$K_{\{n\}}, K_{\{m, n\}}, W_{\{n\}}$	$K_n, K_{m,n}, W_n$	make it planar :
voloshin-hypergraph_F00603	92	■ 1.000	$n \geq 4$	$n \geq 4$	5. Given a graph G and $n > 4$, determine if G is homeomorphic to K_n .
voloshin-hypergraph_F00604	92	■ 1.000	$m=3n-6$	$m=3n-6$	
voloshin-hypergraph_F00605	92	■ 1.000	$3f=2m$	$3f=2m$	
voloshin-hypergraph_F00606	93	■ 1.000	$m=3n-6=15-6=9$	$m=3n-6=15-6=9$	$m=3n-6=15-6=9$ edges. Thus any plane graph on five vertices is a subgraph of
voloshin-hypergraph_F00607	93	■ 1.000	$G^{\{*\}}$	G^*	
voloshin-hypergraph_F00608	93	■ 1.000	$f_{\{i\}}$	f_i	vertices f_i and f_j .
voloshin-hypergraph_F00609	93	■ 1.000	$f_{\{j\}}$	f_j	vertices f_i and f_j .
voloshin-hypergraph_F00610	93	■ 1.000	$n\left(G^{\{*\}}\right)=f, m\left(G^{\{*\}}\right)=m$	$n\left(G^*\right)=f, m\left(G^*\right)=m$	same rules. $\square$
voloshin-hypergraph_F00611	93	■ 1.000	$f\left(G^{\{*\}}\right)=n$	$f\left(G^*\right)=n$	same rules. $\square$
voloshin-hypergraph_F00612	93	■ 0.992	$\left(G^{\{*\}}\right)^{\{*\}}$	$\left(G^*\right)^*$	
voloshin-hypergraph_F00613	94	■ 1.000	$W_{\{n\}}, n \geq 4$	$W_n, n \geq 4$	
voloshin-hypergraph_F00614	96	■ 0.989	$1,2, \ldots, \lambda$	$1,2, \ldots, \lambda$	coloring the vertices of a graph, we just label them by numbers $1,2, \ldots, \lambda$ where λ is the
voloshin-hypergraph_F00615	96	■ 0.989	λ	λ	coloring the vertices of a graph, we just label them by numbers $1,2, \ldots, \lambda$ where λ is the
voloshin-hypergraph_F00616	96	■ 1.000	$\{1,2, \ldots, \lambda\}$	$\{1,2, \ldots, \lambda\}$	labeling of vertices of graph G by the numbers from $\{1,2, \ldots, \lambda\}$ is called a coloring . Each vertex is assigned precisely one color.
voloshin-hypergraph_F00617	96	■ 1.000	$P(G, \lambda)$	$P(G, \lambda)$	colors in a proper λ -coloring may be strictly less than λ ; but it can never be greater than
voloshin-hypergraph_F00618	96	■ 1.000	$\chi(G)$	$\chi(G)$	time a proper λ -coloring for any $\lambda \geq \chi(G)$, we conclude that $P(G, \lambda) \geq 1$ for any integer $\lambda \geq \chi(G)$. Since λ is the number of colors, in our discussions it is always an integer variable

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00619	96	■ 1.000	$\min\{n(G), \lambda\}$	$\min\{n(G), \lambda\}$	$\lambda \leq \chi(G)$ Con:
voloshin-hypergraph_F00620	96	■ 1.000	$P(G, \lambda)=0$	$P(G, \lambda) = 0$	$\{y, z\}$, see Figure 5.1. There is no coloring with one color, so $\chi(G) > 1$. On the other
voloshin-hypergraph_F00621	96	■ 1.000	$1 \leq \lambda \leq \chi(G) - 1$	$1 \leq \lambda \leq \chi(G) - 1$	hand, it is easy to see a coloring with two colors, say 1-2-1, so $\chi(G) = 2$. Suppose the set
voloshin-hypergraph_F00622	96	■ 1.000	$\lambda \geq \chi(G)$	$\lambda \geq \chi(G)$	of available colors is $\{1, 2, 3\}$, consequently, $\lambda = 3$. Having totally three colors and keep-
voloshin-hypergraph_F00623	96	■ 1.000	$P(G, \lambda) \geq 1$	$P(G, \lambda) \geq 1$	of available colors is $\{1, 2, 3\}$, consequently, $\lambda = 3$. Having totally three colors and keep-
voloshin-hypergraph_F00624	96	■ 1.000	$X=\{x, y, z\}$	$X = \{x, y, z\}$	add color 1 = and 1 = if we use only 2 colors, then we have to choose them from $\{1, 2, 3\}$. Thus we have the choices: colors 1,2, colors 2,3, and colors 1,3. The total number
voloshin-hypergraph_F00625	96	■ 1.000	$E=\{x, y\}$	$E = \{x, y\}$	add color 1 = and 1 = if we use only 2 colors, then we have to choose them from $\{1, 2, 3\}$. Thus we have the choices: colors 1,2, colors 2,3, and colors 1,3. The total number
voloshin-hypergraph_F00626	96	■ 0.997	$\{y, z\}$	$\{y, z\}$	of choices equals $\binom{2}{1} = \binom{3}{2} = 3$. For the first choice, the colorings are 1-2-1 and 2-1-2, see
voloshin-hypergraph_F00627	96	■ 0.997	$\chi(G)>1$	$\chi(G) > 1$	of choices equals $\binom{2}{1} = \binom{3}{2} = 3$. For the first choice, the colorings are 1-2-1 and 2-1-2, see
voloshin-hypergraph_F00628	96	■ 0.998	$\chi(G)=2$	$\chi(G) = 2$	the first two colorings in Figure 5.1, column $i = 2$. For the second choice, the colorings are
voloshin-hypergraph_F00629	96	■ 0.998	$\{1, 2, 3\}$	$\{1, 2, 3\}$	2-3-2 and 3-2-3. For the last choice, the colorings are 1-3-1 and 3-1-3. Notice that in this case the colorings come in pairs, and in each pair we just permute the colors.
voloshin-hypergraph_F00630	96	■ 0.998	$\lambda=3$	$\lambda = 3$	2-3-2 and 3-2-3. For the last choice, the colorings are 1-3-1 and 3-1-3. Notice that in this case the colorings come in pairs, and in each pair we just permute the colors.
voloshin-hypergraph_F00631	96	■ 1.000	$i=2$	$i = 2$	if $i = 2$, we observe that any coloring will be proper because all the vertices are of different colors. To construct all colorings, fix color 1 for x and obtain two colorings by
voloshin-hypergraph_F00632	96	■ 1.000	$i=3$	$i = 3$	if $i = 3$, we observe that any coloring will be proper because all the vertices are of different colors. To construct all colorings, fix color 1 for x and obtain two colorings by
voloshin-hypergraph_F00633	96	■ 0.523	$\binom{\lambda}{i}=\binom{3}{2}=3$	$\binom{\lambda}{i} = \binom{3}{2} = 3$	
voloshin-hypergraph_F00634	97	■ 1.000	$P(G, 3)=12$	$P(G, 3) = 12$	that there are general procedures for computation of $P(G, \kappa)$ for an arbitrary graph G . Before proceeding to the next section, let us consider one more concept. Let $G = (X, E)$
voloshin-hypergraph_F00635	97	■ 1.000	$1 \leq i \leq n(G)$	$1 \leq i \leq n(G)$	
voloshin-hypergraph_F00636	97	■ 1.000	$r_i(G)$	$r_i(G)$	
voloshin-hypergraph_F00637	98	■ 0.927	$X: X_1=\{x, z\}$	$X : X_1 = \{x, z\}$	there is only one feasible partition of X : $X_1 = \{x, z\}$ (first cell) and $X_2 = \{y\}$ (second cell).
voloshin-hypergraph_F00638	98	■ 0.927	$X_2=\{y\}$	$X_2 = \{y\}$	there is only one feasible partition of X : $X_1 = \{x, z\}$ (first cell) and $X_2 = \{y\}$ (second cell).
voloshin-hypergraph_F00639	98	■ 1.000	$r_2(G)=1$	$r_2(G) = 1$	Therefore, $r_2(G)$
voloshin-hypergraph_F00640	98	■ 1.000	$X_1=\{x\}, X_2=\{y\}$	$X_1 = \{x\}, X_2 = \{y\}$	partition with cells $\alpha_1 = \{x\}$, $\alpha_2 = \{y\}$, and $\alpha_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently, $r_1(G) = 0$, we obtain that the chromatic spectrum of G is:
voloshin-hypergraph_F00641	98	■ 1.000	$X_3=\{z\}$	$X_3 = \{z\}$	partition with cells $\alpha_1 = \{x\}$, $\alpha_2 = \{y\}$, and $\alpha_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently, $r_1(G) = 0$, we obtain that the chromatic spectrum of G is:
voloshin-hypergraph_F00642	98	■ 1.000	$r_3(G)=1$	$r_3(G) = 1$	partition with cells $\alpha_1 = \{x\}$, $\alpha_2 = \{y\}$, and $\alpha_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently, $r_1(G) = 0$, we obtain that the chromatic spectrum of G is:
voloshin-hypergraph_F00643	98	■ 1.000	$r_1(G)=0$	$r_1(G) = 0$	
voloshin-hypergraph_F00644	98	■ 1.000	$R(G)$	$R(G)$	zeros between positive components. Otherwise it is called broken (has gaps) .
voloshin-hypergraph_F00645	98	■ 0.850	$R(G)=\left(0, 0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n\right)$	$R(G) = (0, 0, \dots, 0, r_{\chi}, r_{\chi+1}, \dots, r_n)$	split it into two non-empty subsets, we obtain another feasible partition using $\chi + 1$ cells what implies $r_{\chi+1} > 0$. Repeating this procedure further for any other cell having > 1 ver-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00646	98	0.850	$r_{\chi}>0$	$r_{\chi} > 0$	 Split it into two non-empty subsets, we obtain another feasible partition using $\chi + 1$ cells, what implies $r_{\chi+1} > 0$. Repeating this procedure further for any other cell having $r_{\chi} > 0$ ver-
voloshin-hypergraph_F00647	98	0.476	χ	χ	 tices, conclude that $r_{\chi+2} > 0, r_{\chi+3} > 0, \dots$. We cannot continue the splitting when all cells
voloshin-hypergraph_F00648	98	0.987	$\chi+1$	$\chi + 1$	 have precisely one vertex each, and that corresponds to $r_n = 1 > 0$. $\square$
voloshin-hypergraph_F00649	98	0.360	$r_{\chi+1}>0$	$r_{\chi+1} > 0$	 We will see in Part II that this fundamental property does not hold in a more general
voloshin-hypergraph_F00650	98	1.000	$r_{\chi+2}>0, r_{\chi+3}>0, \ldots$	$r_{\chi+2} > 0, r_{\chi+3} > 0, \dots$	 case of hypergraph coloring.
voloshin-hypergraph_F00651	98	1.000	$r_n=1>0$	$r_n = 1 > 0$	 There are many results about the bounds on the chromatic number.
voloshin-hypergraph_F00652	98	1.000	$G, \chi(G) \leq 2$	$G, \chi(G) \leq 2$	 Theorem 5.2.5 (Brooks, 1941) <i>If G is a connected graph different from a clique and $\Delta(G) \leq \chi(G)$,</i>
voloshin-hypergraph_F00653	98	1.000	$\chi(G) \leq \Delta(G)$	$\chi(G) \leq \Delta(G)$	 graph.
voloshin-hypergraph_F00654	98	1.000	$E_n, K_n, K_{m,n}$	$E_n, K_n, K_{m,n}$	
voloshin-hypergraph_F00655	98	1.000	$T_n, C_{2n}, C_{2n+1}, W_n$	$T_n, C_{2n}, C_{2n+1}, W_n$	
voloshin-hypergraph_F00656	98	1.000	$P(G^{\prime}, 3)$	$P(G', 3)$	
voloshin-hypergraph_F00657	98	1.000	$R(G^{\prime})$	$R(G')$	
voloshin-hypergraph_F00658	99	1.000	$i=6$	$i = 6$	 6. For the graph in Figure 5.2, find a bound on the chromatic number by applying Brooks The-
voloshin-hypergraph_F00659	99	1.000	$i+1, i+2, \ldots, n=10$	$i + 1, i + 2, \dots, n = 10$	 orem.
voloshin-hypergraph_F00660	99	0.904	$\chi, \chi+1, \chi+2, \ldots, n=10$	$\chi, \chi + 1, \chi + 2, \dots, n = 10$	 following algorithmic problems.
voloshin-hypergraph_F00661	99	1.000	$\lambda \geq 1$	$\lambda \geq 1$	 possibilities. Any color used once, cannot be used again. Because of the symmetry of K_n, the order of such coloring procedure does not matter. If we continue it until the last vertex
voloshin-hypergraph_F00662	99	1.000	$\lambda < n$	$\lambda < n$	 possibilities. Any color used once, cannot be used again. Because of the symmetry of K_n, the order of such coloring procedure does not matter. If we continue it until the last vertex
voloshin-hypergraph_F00663	99	1.000	$\lambda \geq n$	$\lambda \geq n$	
voloshin-hypergraph_F00664	99	1.000	$\chi(K_n)=n$	$\chi(K_n) = n$	
voloshin-hypergraph_F00665	99	1.000	$P(K_n, \lambda)$	$P(K_n, \lambda)$	
voloshin-hypergraph_F00666	99	0.647	$\lambda-1$	$\lambda - 1$	
voloshin-hypergraph_F00667	99	0.647	$\lambda-2$	$\lambda - 2$	
voloshin-hypergraph_F00668	100	0.998	K_1, K_2, K_3, K_4	K_1, K_2, K_3, K_4	
voloshin-hypergraph_F00669	100	1.000	$\lambda(\lambda-1)(\lambda-2) \cdots (\lambda-n+1)$	$\lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - n + 1)$	 Let us have now a graph $G = (X, E)$ which is not a K_n. How to compute $P(G, \lambda)$? Since
voloshin-hypergraph_F00670	100	1.000	$\lambda^{\{n\}}$	$\lambda^{(n)}$	 Let us have now a graph $G = (X, E)$ which is not a K_n. How to compute $P(G, \lambda)$? Since
voloshin-hypergraph_F00671	100	0.998	$G_1=G \cup \{x, y\}$	$G_1 = G \cup \{x, y\}$	 Therefore,
voloshin-hypergraph_F00672	100	1.000	$G_2=G_1 \cdot \{x, y\}$	$G_2 = G_1 \cdot \{x, y\}$	 $P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda).$ (5.1)

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voloshin-hypergraph_F00673	100	■ 1.000	$G \cdot e$	$G \cdot e$	Notice that compared to G , graph G' has the same vertices and edges except a newly
voloshin-hypergraph_F00674	100	■ 1.000	$P(\text{G}_{\{1\}}, \lambda) = P(\text{G}_{\{3\}}, \lambda) + P(\text{G}_{\{4\}}, \lambda)$	$P(G_1, \lambda) = P(G_3, \lambda) + P(G_4, \lambda)$	
voloshin-hypergraph_F00675	100	■ 1.000	$P(\text{G}_{\{2\}}, \lambda) = P(\text{G}_{\{5\}}, \lambda) + P(\text{G}_{\{6\}}, \lambda)$	$P(G_2, \lambda) = P(G_5, \lambda) + P(G_6, \lambda)$	
voloshin-hypergraph_F00676	100	■ 0.950	$P(G, \lambda) = P(\text{G}_{\{3\}}, \lambda) + P(\text{G}_{\{4\}}, \lambda) + P(\text{G}_{\{5\}}, \lambda) + P(\text{G}_{\{6\}}, \lambda)$	$P(G, \lambda) = P(G_3, \lambda) + P(G_4, \lambda) + P(G_5, \lambda) + P(G_6, \lambda)$	
voloshin-hypergraph_F00677	101	■ 1.000	$s \geq 1$	$s \geq 1$	graphs $K_i, K_{i+1}, \dots$
voloshin-hypergraph_F00678	101	■ 1.000	$P(K_{\{i_j\}}, \lambda)$	$P(K_{i_j}, \lambda)$	connections only. All the other complete graphs have $< n$ vertices because they are obtained
voloshin-hypergraph_F00679	101	■ 1.000	$K_{\{i_1\}}, K_{\{i_2\}}, \dots, K_{\{i_s\}}$	$K_{i_1}, K_{i_2}, \dots, K_{i_s}$	$2) \cdots (\lambda - n + 1)$, our next conclusion is that the degree of $P(G, \lambda)$ is n and the major coefficient is 1
voloshin-hypergraph_F00680	101	■ 1.000	$K_{\{i\}}$	K_i	name corresponds to all vertices of G colored with that color.
voloshin-hypergraph_F00681	101	■ 0.996	$x \cdot y \cdot z$	xyz	the composite names of vertices. It is so because each K_i was obtained by a unique way from G . Therefore, the number of all complete graphs having i vertices equals the number
voloshin-hypergraph_F00682	102	■ 0.998	$r_1=r_2=\cdots=r_{\chi-1}=0$	$r_1 = r_2 = \cdots = r_{\chi-1} = 0$	Since $r_1 = r_2 = \cdots = r_{\chi-1} = 0$ and $P(K_i, \lambda) = \lambda^{(i)}$, we arrive at last to the final funda-
voloshin-hypergraph_F00683	102	■ 0.998	$P(K_{\{i\}}, \lambda) = \lambda^{(i)}$	$P(K_i, \lambda) = \lambda^{(i)}$	Since $r_1 = r_2 = \cdots = r_{\chi-1} = 0$ and $P(K_i, \lambda) = \lambda^{(i)}$, we arrive at last to the final funda-
voloshin-hypergraph_F00684	102	■ 0.998	$\left(\lambda^{(i)}\right)$	$\left(\lambda^{(i)}\right)$	partitions $r_i(G/\lambda^{(i)})$, and at last count that for an i (obtain $\sum_{i=\chi} r_i(G/\lambda^{(i)})$). Connection-contraction algorithm not only shows this structure, it also shows the way how to obtain
voloshin-hypergraph_F00685	102	■ 0.875	$\left(r_i(G)\lambda^{(i)}\right)$	$\left(r_i(G)\lambda^{(i)}\right)$	connection algorithm not only shows this structure, it also shows the way how to obtain $P(G, \lambda)$ for any graph G .
voloshin-hypergraph_F00686	102	■ 0.875	$\sum_{i=\chi}^n r_i(G)\lambda^{(i)}$	$\sum_{i=\chi}^n r_i(G)\lambda^{(i)}$	connection algorithm not only shows this structure, it also shows the way how to obtain $P(G, \lambda)$ for any graph G .
voloshin-hypergraph_F00687	102	■ 1.000	$P(G, 3)=3(3-1)^2=12$	$P(G, 3) = 3(3 - 1)^2 = 12$	The connection-contraction algorithm is good for small graphs but it is not efficient for
voloshin-hypergraph_F00688	102	■ 1.000	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = 0 \cdot P(K_1, \lambda) +$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = 0 \cdot P(K_1, \lambda) +$	The connection-contraction algorithm is good for small graphs but it is not efficient for
voloshin-hypergraph_F00689	102	■ 1.000	$1 \cdot P(K_2, \lambda) + 1 \cdot P(K_3, \lambda)$	$1 \cdot P(K_2, \lambda) + 1 \cdot P(K_3, \lambda)$	range n . Since every graph produces two new graphs, the number of graphs is doubling at each step and is power of 2. The whole procedure can be denoted itself as a graph, which
voloshin-hypergraph_F00690	102	■ 1.000	$R(G)=(0,1,1)$	$R(G) = (0, 1, 1)$	range n . Since every graph produces two new graphs, the number of graphs is doubling at each step and is power of 2. The whole procedure can be denoted itself as a graph, which
voloshin-hypergraph_F00691	102	■ 1.000	$G_{\{3\}}, G_{\{4\}}, G_{\{5\}}$	G_3, G_4, G_5	—
voloshin-hypergraph_F00692	102	■ 1.000	$G_{\{6\}}$	G_6	—
voloshin-hypergraph_F00693	103	■ 1.000	$K_{\{i_j\}}$	K_{i_j}	of G into combination of complete graphs as described in equality (5.2). This observation
voloshin-hypergraph_F00694	103	■ 1.000	$P(C_{\{4\}}, 10) = 10 \cdot 9 \cdot (10^2 - 3 \cdot 10 + 3) = 6570$	$P(C_4, 10) = 10 \cdot 9 \cdot (10^2 - 3 \cdot 10 + 3) = 6570$	
voloshin-hypergraph_F00695	104	■ 1.000	$E_{\{4\}}, E_{\{5\}}, C_{\{5\}}, C_{\{6\}}, W_{\{4\}}, W_{\{5\}}, P_{\{4\}}, P_{\{5\}}, P_{\{6\}}$	$E_4, E_5, C_5, C_6, W_4, W_5, P_4, P_5, P_6$	4. A manager needs to arrange five people A, B, C, D and E in the C conflict with B and E; person B is in conflict with A and C; person C is
voloshin-hypergraph_F00696	104	■ 1.000	$n \geq 6$	$n \geq 6$	4. A manager needs to arrange five people A, B, C, D and E in the C conflict with B and E; person B is in conflict with A and C; person C is
voloshin-hypergraph_F00697	104	■ 1.000	$P_{\{4\}}, P_{\{5\}}, P_{\{6\}}, C_{\{5\}}$	P_4, P_5, P_6, C_5	D; person D is in conflict with A, B, C, E; person E is in conflict with A and D. What is the
voloshin-hypergraph_F00698	105	■ 1.000	$P(\text{G}_{\{1\}}, \lambda)$	$P(G_1, \lambda)$	$P(E_0, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$

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voloshin-hypergraph_F00699	105	■ 1.000	$P(\text{left}(G_2), \lambda)$	$P(G_2, \lambda)$	$P(E_0, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n$
voloshin-hypergraph_F00700	105	■ 1.000	$E_n = n K_1$	$E_n = nK_1$	
voloshin-hypergraph_F00701	105	■ 1.000	$P(\text{left}(K_1), \lambda) = \lambda$	$P(K_1, \lambda) = \lambda$	
voloshin-hypergraph_F00702	105	■ 1.000	$S(n, i) = r_i(\text{left}(E_n))$	$S(n, i) = r_i(E_n)$	
voloshin-hypergraph_F00703	105	■ 1.000	$S(n, i)$	$S(n, i)$	
voloshin-hypergraph_F00704	106	■ 1.000	$R(\text{left}(E_1)) = (1), R(\text{left}(E_2)) = (1, 1), R(\text{left}(E_3)) = (1, 3, 1)$	$R(E_1) = (1), R(E_2) = (1, 1), R(E_3) = (1, 3, 1)$	One can check, for example, that $R(E_1) = (1), R(E_2) = (1, 1), R(E_3) = (1, 3, 1)$,
voloshin-hypergraph_F00705	106	■ 1.000	$R(\text{left}(E_4)) = (1, 7, 6, 1)$	$R(E_4) = (1, 7, 6, 1)$	$R(E_4) = (1, 7, 6, 1)$ and so (
voloshin-hypergraph_F00706	106	■ 0.890	$\lambda^{\text{left}(4)} = -6 \lambda + 11 \lambda^2 -$	$\lambda^{(4)} = -6\lambda + 11\lambda^2 -$	$6\lambda^3 + \lambda^4$, and therefore $s(4, 1) = -6, s(4, 2) = 11, s(4, 3) = -6$, and $s(4, 4) = 1$.
voloshin-hypergraph_F00707	106	■ 1.000	$6 \lambda^{\text{left}(3)} + \lambda^{\text{left}(4)}$	$6\lambda^3 + \lambda^4$	Generally, Stirling numbers serve as the coefficients to express κ_n (or, equivalently, z in terms of $\lambda^{(i)}$ (or, equivalently, K_i) and $\lambda^{(n)}$ in terms of $\lambda^i, i = 1, 2, \dots, n$. The latter ir
voloshin-hypergraph_F00708	106	■ 1.000	$s(4, 1) = -6, s(4, 2) = 11, s(4, 3) = -6$	$s(4, 1) = -6, s(4, 2) = 11, s(4, 3) = -6$	Generally, Stirling numbers serve as the coefficients to express κ_n (or, equivalently, z in terms of $\lambda^{(i)}$ (or, equivalently, K_i) and $\lambda^{(n)}$ in terms of $\lambda^i, i = 1, 2, \dots, n$. The latter ir
voloshin-hypergraph_F00709	106	■ 1.000	$s(4, 4) = 1$	$s(4, 4) = 1$	Generally, Stirling numbers serve as the coefficients to express κ_n (or, equivalently, z in terms of $\lambda^{(i)}$ (or, equivalently, K_i) and $\lambda^{(n)}$ in terms of $\lambda^i, i = 1, 2, \dots, n$. The latter ir
voloshin-hypergraph_F00710	106	■ 1.000	$\lambda^{\text{left}(n)}$	λ^n	more general setting can be expressed as the disconnection-contraction algorithm .
voloshin-hypergraph_F00711	106	■ 1.000	$\lambda^{\text{left}(i)}$	$\lambda^{(i)}$	The idea of it consists in the following. Recall that in the connection-contraction we
voloshin-hypergraph_F00712	106	■ 1.000	$\lambda^{\text{left}(i)}, i = 1, 2, \dots, n$	$\lambda^i, i = 1, 2, \dots, n$	The idea of it consists in the following. Recall that in the connection-contraction we
voloshin-hypergraph_F00713	106	■ 0.736	$1, -m(G), \dots$	$1, -m(G), \dots$	and $\{b_i\}$ such that
voloshin-hypergraph_F00714	106	■ 0.998	$m = 0$	$m = 0$	
voloshin-hypergraph_F00715	106	■ 1.000	$G = E_n$	$G = E_n$	
voloshin-hypergraph_F00716	106	■ 1.000	$P(G, \lambda) = \lambda^{\text{left}(n)}$	$P(G, \lambda) = \lambda^n$	
voloshin-hypergraph_F00717	106	■ 1.000	$m > 1$	$m > 1$	
voloshin-hypergraph_F00718	106	■ 1.000	$\text{left}\{a_i\}$	$\{a_i\}$	
voloshin-hypergraph_F00719	106	■ 1.000	$\text{left}\{b_i\}$	$\{b_i\}$	
voloshin-hypergraph_F00720	107	■ 1.000	$\lambda^{\text{left}(k)}$	$\lambda^{(k)}$	and consider all proper κ -colorings of G that can be obtained by the extension of a coloring of S . Denote the number of such colorings by N_2 . We claim that $N_1 = N_2$.
voloshin-hypergraph_F00721	107	■ 1.000	$P(G, \lambda) / \lambda^{\text{left}(k)}$	$P(G, \lambda) / \lambda^{(k)}$	and consider all proper κ -colorings of G that can be obtained by the extension of a coloring of S . Denote the number of such colorings by N_2 . We claim that $N_1 = N_2$.
voloshin-hypergraph_F00722	107	■ 1.000	$N_1 = N_2$	$N_1 = N_2$	
voloshin-hypergraph_F00723	107	■ 1.000	j	j	
voloshin-hypergraph_F00724	108	■ 1.000	$t = P(\text{left}(G_S), \lambda) = P(\text{left}(K_k), \lambda)$	$t = P(G_S, \lambda) = P(K_k, \lambda) = \lambda^{(k)}$	by a permutation of colors. Let $t = P(G_S, \lambda) = P(K_k, \lambda) = \lambda^{(k)}$. Repeating the reasoning above for each of t color-

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voloshin-hypergraph_F00725	108	■ 1.000	$N_{\{1\}}=N_{\{2\}}=N_{\{3\}}=\cdots=N_{\{t\}}$	$N_1 = N_2 = N_3 = \cdots = N_t$	ings of S , we arrive to the conclusion that $N_1 = N_2 = N_3 = \cdots = N_t$. Hence the set of all
voloshin-hypergraph_F00726	108	■ 1.000	$k-1$	$k - 1$	These colorings cannot be obtained from each other by a permutation of colors. When S
voloshin-hypergraph_F00727	108	■ 1.000	$P\left(G_{X_{\{1\}} \cup S}, \lambda\right)$	$P\left(G_{X_1 \cup S}, \lambda\right)$	into $\lambda^{(k)}$ equal classes, and each such class contains exactly $P\left(G_{X_2 \cup S}, \lambda\right) / \lambda^{(k)}$ colorings.
voloshin-hypergraph_F00728	108	■ 1.000	$P\left(G_{\{S\}}, \lambda\right)=\lambda^{\{k\}}$	$P\left(G_S, \lambda\right)=\lambda^{(k)}$	
voloshin-hypergraph_F00729	108	■ 1.000	$P\left(G_{X_{\{1\}} \cup S}, \lambda\right) / \lambda^{\{k\}}$	$P\left(G_{X_1 \cup S}, \lambda\right) / \lambda^{(k)}$	
voloshin-hypergraph_F00730	108	■ 1.000	$P\left(G_{X_{\{2\}} \cup S}, \lambda\right)$	$P\left(G_{X_2 \cup S}, \lambda\right)$	
voloshin-hypergraph_F00731	108	■ 1.000	$P\left(G_{X_{\{2\}} \cup S}, \lambda\right) / \lambda^{\{k\}}$	$P\left(G_{X_2 \cup S}, \lambda\right) / \lambda^{(k)}$	
voloshin-hypergraph_F00732	109	■ 0.947	$G_{\{1\}}, G_{\{2\}}, \ldots G_{\{l\}}$	$G_1, G_2, \ldots G_l$	Corollary 5.4.1 Let $G=(V, E)$ be a connected graph having a separator S which is a clique of size k . Suppose $G_1, G_2, \ldots, G_l$ are the derived subgraphs with respect to the separator S . Then the chromatic polynomial of G can be written as a sum of chromatic polynomials of G_i for $i=1, 2, \ldots, l$.
voloshin-hypergraph_F00733	109	■ 0.996	$G_{\{i\}}, i=1,2, \ldots, l$	$G_i, i=1,2, \ldots, l$	by any single coloring of S . Since the number of classes is $\lambda^{(k)}$, the formula follows. $\square$
voloshin-hypergraph_F00734	109	■ 0.733	$P(\emptyset, \lambda)=1$	$P(\emptyset, \lambda)=1$	
voloshin-hypergraph_F00735	109	■ 1.000	$G_{\{1\}}=C_{\{4\}}, G_{\{2\}}=K_{\{3\}}$	$G_1 = C_4, G_2 = K_3$	
voloshin-hypergraph_F00736	109	■ 1.000	$ S =2$	$ S = 2$	$\frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)][\lambda(\lambda-1)(\lambda-2)]}{\lambda^{(2)}} =$
voloshin-hypergraph_F00737	109	■ 1.000	$P\left(C_{\{4\}}, \lambda\right)$	$P\left(C_4, \lambda\right)$	$\frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)][\lambda(\lambda-1)(\lambda-2)]}{\lambda^{(2)}} =$
voloshin-hypergraph_F00738	109	■ 1.000	$P\left(K_{\{3\}}, \lambda\right)=\lambda(\lambda-1)(\lambda-2)$	$P\left(K_3, \lambda\right)=\lambda(\lambda-1)(\lambda-2)$	$\frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)][\lambda(\lambda-1)(\lambda-2)]}{\lambda^{(2)}} =$
voloshin-hypergraph_F00739	109	■ 0.986	$X_{\{2\}}=\{x\}$	$X_2 = \{x\}$	
voloshin-hypergraph_F00740	109	■ 1.000	$G_{\{1\}}=G-x, G_{\{2\}}=K_{\{k+1\}}$	$G_1 = G - x, G_2 = K_{k+1}$	
voloshin-hypergraph_F00741	109	■ 1.000	$G_{\{S\}}=K_{\{k\}}$	$G_S = K_k$	
voloshin-hypergraph_F00742	110	■ 1.000	$G=K_{\{k+1\}}, G-x=K_{\{k\}}$	$G = K_{k+1}, G - x = K_k$	If G is a complete graph, then $G = K_{k+1}$, $G - x = K_k$ and the formula follows directly.
voloshin-hypergraph_F00743	110	■ 1.000	$G_{\{1\}}=G-x=C_{\{4\}}$	$G_1 = G - x = C_4$	and $G_1 = G - x = C_4$, compute the chromatic polynomial of G directly.
voloshin-hypergraph_F00744	110	■ 0.999	$(i-k) r_{\{i\}}(G-x)$	$(i-k)r_i(G-x)$	
voloshin-hypergraph_F00745	110	■ 1.000	$r_{\{i-1\}}(G-x)$	$r_{i-1}(G-x)$	Corollary 5.4.3 If $S(n, i)$ is the Stirling number of the second kind, then
voloshin-hypergraph_F00746	110	■ 1.000	$E_{\{n\}}: r_{\{i\}}\left(E_{\{n\}}\right)=S(n, i)$	$E_n: r_i\left(E_n\right)=S(n, i)$	
voloshin-hypergraph_F00747	110	■ 1.000	$r_{\{1\}}\left(E_{\{1\}}\right)=S(1,1)=1$	$r_1\left(E_1\right)=S(1,1)=1$	
voloshin-hypergraph_F00748	110	■ 1.000	$P_{\{3\}}, P_{\{4\}}, P_{\{5\}}, P_{\{n\}}$	P_3, P_4, P_5, P_n	—
voloshin-hypergraph_F00749	110	■ 1.000	$C_{\{4\}}, C_{\{5\}}, C_{\{6\}}, C_{\{7\}}$	C_4, C_5, C_6, C_7	—
voloshin-hypergraph_F00750	110	■ 1.000	$W_{\{4\}}, W_{\{5\}}, W_{\{6\}}$	W_4, W_5, W_6	—
voloshin-hypergraph_F00751	111	■ 1.000	$E_{\{6\}}, E_{\{7\}}$	E_6, E_7	3. For each graph in Figure 5.12, by finding a simplicial vertex or a complete separator compute
voloshin-hypergraph_F00752	111	■ 1.000	$E_{\{8\}}$	E_8	3. For each graph in Figure 5.12, by finding a simplicial vertex or a complete separator compute

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voloshin-hypergraph_F00753	111	1.000	$K_{\{3\}}, K_{\{4\}}, K_{\{5\}}$	K_3, K_4, K_5	the chromatic polynomial.
voloshin-hypergraph_F00754	111	1.000	$i \leq n$	$i \leq n$	
voloshin-hypergraph_F00755	111	0.994	$G_{\{1\}}=(X, E), X =n$	$G_1 = (X, E), X = n$	$P(G_1, \lambda) = (\lambda - k_1)P(G_2, \lambda) = (\lambda - k_1)(\lambda - k_2)P(G_3, \lambda) \cdots =$
voloshin-hypergraph_F00756	111	0.999	$G_{\{2\}}=G_{\{1\}}-x_{\{1\}}, G_{\{3\}}=G_{\{2\}}-x_{\{2\}}, \ldots, G_{\{n\}}=G_{\{n-1\}}-x_{\{n-1\}}=\text{\texttt{\textbackslash left\{x_{\{n\}\right\}}}$	$G_2 = G_1 - x_1, G_3 = G_2 - x_2, \dots, G_n = G_{n-1} - x_{n-1} = \{x_n\}$	$(\lambda - k_1)(\lambda - k_2) \cdots (\lambda - k_{n-1})(\lambda - k_n)$
voloshin-hypergraph_F00757	111	0.858	$G_{\{n+1\}}=G_{\{n\}}-x_{\{n\}}=\text{\texttt{\textbackslash emptyset}}$	$G_{n+1} = G_n - x_n = \emptyset$	$(\lambda - k_1)(\lambda - k_2) \cdots (\lambda - k_{n-1})(\lambda - k_n)$
voloshin-hypergraph_F00758	111	0.858	$k_{\{i\}}$	k_i	$(\lambda - k_1)(\lambda - k_2) \cdots (\lambda - k_{n-1})(\lambda - k_n)$
voloshin-hypergraph_F00759	111	1.000	$G_{\{i\}}, i=1, \ldots, n$	$G_i, i = 1, \dots, n$	We see that all the roots of $P(G_1, \lambda)$ are integer numbers equal to the degrees of the vertices of G_1 . Since G_1 contains each vertex v , we conclude that $k_v = 0$.
voloshin-hypergraph_F00760	111	1.000	$G_{\{n\}}$	G_n	
voloshin-hypergraph_F00761	111	1.000	$k_{\{n\}}=0$	$k_n = 0$	
voloshin-hypergraph_F00762	111	1.000	$\lambda=0$	$\lambda = 0$	
voloshin-hypergraph_F00763	112	1.000	$\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}}}$	$\omega(G_1)$	By the definition of clique number, G_1 contains a complete subgraph on $\omega(G_1)$ vertices.
voloshin-hypergraph_F00764	112	1.000	$\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}}}-1$	$\omega(G_1) - 1$	By the definition of clique number, G_1 contains a complete subgraph on $\omega(G_1)$ vertices, so maximum among all k_i 's is $\omega(G_1) - 1$. We now fix the second root of the chromatic
voloshin-hypergraph_F00765	112	1.000	$\lambda=\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}}}-1$	$\lambda = \omega(G_1) - 1$	polynomial, namely, $\lambda = \omega(G_1) - 1$. Recall that $\omega(G_1) \leq \chi(G_1)$ what implies that there is
voloshin-hypergraph_F00766	112	1.000	$\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}}) \leq \chi\text{\texttt{\textbackslash left(G_{\{1\}\right}}}$	$\omega(G_1) \leq \chi(G_1)$	polynomial, namely, $\lambda = \omega(G_1) - 1$. Recall that $\omega(G_1) \leq \chi(G_1)$ what implies that there is
voloshin-hypergraph_F00767	112	0.999	ω	ω	no coloring using less than ω colors. Consequently, every integer on the closed interval
voloshin-hypergraph_F00768	112	0.987	$[0, \omega-1]$	$[0, \omega - 1]$	$[0, \omega - 1]$ is a root of $P(G_1, \lambda)$. Therefore, all roots k_i are the integer numbers from the
voloshin-hypergraph_F00769	112	1.000	$\chi\text{\texttt{\textbackslash left(G_{\{1\}\right}})=\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}}}$	$\chi(G_1) = \omega(G_1)$	to σ , i.e. $x_n, x_{n-1}, x_{n-2}, \dots, x_2, x_1$. In general case this procedure is called online coloring
voloshin-hypergraph_F00770	112	0.866	$x_{\{n\}}, x_{\{n-1\}}, x_{\{n-2\}}, \ldots, x_{\{2\}}, x_{\{1\}}$	$x_n, x_{n-1}, x_{n-2}, \dots, x_2, x_1$	because the vertices of a graph can imaginably be put on the real line and colored from left
voloshin-hypergraph_F00771	112	0.993	$x_{\{n-1\}}$	x_{n-1}	try to use color 2, if not possible use the color 3. Continue this procedure each time trying
voloshin-hypergraph_F00772	112	0.699	$x_{\{n-2\}}$	x_{n-2}	to use the smallest possible color. If none of the used colors fits, use the new color. This is
voloshin-hypergraph_F00773	112	0.982	$\omega-1$	$\omega - 1$	Theorem 5.5.1 If G is a chordal graph, then the chromatic polynomial has the following
voloshin-hypergraph_F00774	112	1.000	$\omega(G)=\chi(G)$	$\omega(G) = \chi(G)$	form:
voloshin-hypergraph_F00775	112	0.995	$s_{\{i\}} \geq 1(i=0,1, \ldots, \chi-1)$	$s_i \geq 1(i = 0, 1, \dots, \chi - 1)$	
voloshin-hypergraph_F00776	112	0.531	$2, 2, 1, 0$	$2, 2, 1, 0$	
voloshin-hypergraph_F00777	113	1.000	$\omega\text{\texttt{\textbackslash left(G_{\{1\}\right}})=\chi\text{\texttt{\textbackslash left(G_{\{1\}\right}}}=4$	$\omega(G_1) = \chi(G_1) = 4$	Verifying 2, 2, 2 and 3 from the previous proposition allows us to get $P(G_1) = \chi(G_1) = 4 \cdot \omega(G_1)$
voloshin-hypergraph_F00778	113	1.000	$k_{\{1\}}=k_{\{2\}}=\cdots=k_{\{n-1\}}=1, k_{\{n\}}=0$	$k_1 = k_2 = \dots = k_{n-1} = 1, k_n = 0$	Proof. Induction on the number of
voloshin-hypergraph_F00779	113	1.000	$C_{\{n\}}, n \geq 1$	$C_n, n \geq 1$	$< n$ vertices. Choose any edge of C_n an

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voloshin-hypergraph_F00780	113	1.000	$n=1,2,3$	$n = 1, 2, 3$	$\lambda(\lambda-1)^{n-1}-[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)]=$
voloshin-hypergraph_F00781	114	0.987	$\chi^{\{\prime\}}=\chi\left(G^{\{\prime\}}\right)$	$\chi' = \chi(G')$	where $\chi' = \chi(G')$, and $s'_i \geq 1$ ($i = 0, 1, \dots, \chi' - 1$) is the number of simplicial vertices of G' .
voloshin-hypergraph_F00782	114	0.987	$s_{\{i\}^{\{\prime\}}}\geq 1\left(i=0,1,\ldots,\chi^{\{\prime\}}-1\right)$	$s'_i \geq 1$ ($i = 0, 1, \dots, \chi' - 1$)	where $\chi' = \chi(G')$, and $s'_i \geq 1$ ($i = 0, 1, \dots, \chi' - 1$) is the number of simplicial vertices of G' .
voloshin-hypergraph_F00783	114	1.000	$\chi\left(C_{\{k\}}\right)=2$	$\chi(C_k) = 2$	of $P(C_k, \lambda)$ should be from the set $\{0, 1, 2\}$. But by Lemma 5.5.1 $P(C_k, \lambda) = (\lambda - 1) + (-1)^k(\lambda - 1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k > 4$ has at least
voloshin-hypergraph_F00784	114	1.000	$\chi\left(C_{\{k\}}\right)=3$	$\chi(C_k) = 3$	of $P(C_k, \lambda)$ should be from the set $\{0, 1, 2\}$. But by Lemma 5.5.1 $P(C_k, \lambda) = (\lambda - 1) + (-1)^k(\lambda - 1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k > 4$ has at least
voloshin-hypergraph_F00785	114	1.000	$P\left(C_{\{k\}},\lambda\right)$	$P(C_k, \lambda)$	$(-1)^k(\lambda - 1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k \geq 4$ has at least one complex root; a contradiction. $\square$
voloshin-hypergraph_F00786	114	1.000	$\{0,1,2\}$	$\{0, 1, 2\}$	$(-1)^k(\lambda - 1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k \geq 4$ has at least one complex root; a contradiction. $\square$
voloshin-hypergraph_F00787	114	1.000	$P\left(C_{\{k\}},\lambda\right)=(\lambda-1)^k+$	$P(C_k, \lambda) = (\lambda - 1)^k +$	$(-1)^k(\lambda - 1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k \geq 4$ has at least one complex root; a contradiction. $\square$
voloshin-hypergraph_F00788	114	1.000	$(-1)^{\{k\}}(\lambda-1)$	$(-1)^k(\lambda - 1)$	Let $G = (X, E)$ be a graph. $A \subset X$ be an arbitrary separator. $G_1^* = (X_1, E_1)$, $G_2^* =$
voloshin-hypergraph_F00789	114	0.997	$A\subset X$	$A \subset X$	vertex set A together with all incident edges from G . If G is a disconnected graph having l connected components, then we assume that $k \leq l$. So, we have $X_1 \cup X_2 \cup \dots \cup X_k \cup A = X$.
voloshin-hypergraph_F00790	114	0.997	$G_{\{1\}^{\{\ast\}}}=\left(X_{\{1\}},E_{\{1\}}\right),G_{\{2\}^{\{\ast\}}}=\left(X_{\{2\}},E_{\{2\}}\right),\ldots,G_{\{k\}^{\{\ast\}}}=\left(X_{\{k\}},E_{\{k\}}\right),k\geq 2$	$G_1^* = (X_1, E_1), G_2^* =$	vertex set A together with all incident edges from G . If G is a disconnected graph having l connected components, then we assume that $k \leq l$. So, we have $X_1 \cup X_2 \cup \dots \cup X_k \cup A = X$.
voloshin-hypergraph_F00791	114	1.000	$\left(X_{\{2\}},E_{\{2\}}\right),\ldots,G_{\{k\}^{\{\ast\}}}=\left(X_{\{k\}},E_{\{k\}}\right),k\geq 2$	$(X_2, E_2), \dots, G_k^* = (X_k, E_k), k \geq 2$	connected components, then we assume that $k \geq 1$. So, we have $X_1 \cup X_2 \cup \dots \cup X_k \cup A = X$, $X_i \cap X_j = \emptyset, i \neq j$. As usually, denote the derived induced subgraphs in the following way:
voloshin-hypergraph_F00792	114	1.000	$k>l$	$k > l$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0.$
voloshin-hypergraph_F00793	114	1.000	$X_{\{1\}}\cup X_{\{2\}}\cup\ldots\cup X_{\{k\}}\cup A=X$	$X_1 \cup X_2 \cup \dots \cup X_k \cup A = X$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0.$
voloshin-hypergraph_F00794	114	0.753	$X_{\{i\}}\cap X_{\{j\}}=\emptyset,i\neq j$	$X_i \cap X_j = \emptyset, i \neq j$	Theorem 5.5.3 A graph $G = (X, E)$ is chordal if and only if, for any induced subgraph
voloshin-hypergraph_F00795	114	1.000	$A^{\{\prime\}}\subset X^{\{\prime\}}$	$A' \subset X'$	Proof. $\Rightarrow$ Since every induced subgraph of a chordal graph is chord;
voloshin-hypergraph_F00796	114	1.000	$G^{\{\prime\}}=G$	$G' = G$	the neighborhood $N(x_0) \subseteq X_1 \cup A$. Therefore, x_0 is a simplicial vertex in G_1 . Applying
voloshin-hypergraph_F00797	114	1.000	$n=2,3,4$	$n = 2, 3, 4$	the neighborhood $N(x_0) \subseteq X_1 \cup A$. Therefore, x_0 is a simplicial vertex in G_1 . Applying
voloshin-hypergraph_F00798	114	1.000	$n>4$	$n > 4$	$P(G_1, \lambda) = (\lambda - p)P(G_1 - x_0, \lambda).$
voloshin-hypergraph_F00799	114	1.000	$n(G)=n$	$n(G) = n$	$P(G_1, \lambda) = (\lambda - p)P(G_1 - x_0, \lambda).$
voloshin-hypergraph_F00800	114	1.000	$x_{\{0\}}\in X$	$x_0 \in X$	$P(G_1, \lambda) = (\lambda - p)P(G_1 - x_0, \lambda).$
voloshin-hypergraph_F00801	114	1.000	p	p	$P(G_1, \lambda) = (\lambda - p)P(G_1 - x_0, \lambda).$
voloshin-hypergraph_F00802	114	1.000	$x_{\{0\}}\notin A$	$x_0 \notin A$	
voloshin-hypergraph_F00803	114	1.000	$x_{\{0\}}\in X_{\{1\}}$	$x_0 \in X_1$	
voloshin-hypergraph_F00804	114	1.000	$x_{\{0\}}$	x_0	
voloshin-hypergraph_F00805	114	1.000	$N\left(x_{\{0\}}\right)\subseteq X_{\{1\}}\cup A$	$N(x_0) \subseteq X_1 \cup A$	$P(G, \lambda) = (\lambda - p)P(G - x_0, \lambda).$
voloshin-hypergraph_F00806	114	1.000	$G-x_{\{0\}}$	$G - x_0$	

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voloshin-hypergraph_F00807	114	■ 1.000	$n\left(G-x_{\{0\}}\right)<n$	$n\left(G-x_0\right)<n$	
voloshin-hypergraph_F00808	115	■ 0.958	$\lambda-p$	$\lambda-p$	Multiplying both sides of (5.16) with the factor $(\lambda-p)$, we obtain
voloshin-hypergraph_F00809	115	■ 1.000	$x_{\{0\}} \in A$	$x_0 \in A$	Case 2. $x_0 \in A$, see Figure 5.15. Let $N(x_0) \cap A = p_1$. Since $N(x_0)$ induces a complete
voloshin-hypergraph_F00810	115	■ 1.000	$\left N\left(x_{\{0\}}\right) \cap A\right =p_{\{1\}}$	$ N\left(x_0\right) \cap A =p_1$	Case 2. $x_0 \in A$, see Figure 5.15. Let $N(x_0) \cap A = p_1$. Since $N(x_0)$ induces a complete
voloshin-hypergraph_F00811	115	■ 1.000	$N\left(x_{\{0\}}\right)$	$N\left(x_0\right)$	Case 2. $x_0 \in A$, see Figure 5.15. Let $N(x_0) \cap A = p_1$. Since $N(x_0)$ induces a complete
voloshin-hypergraph_F00812	115	■ 1.000	$p-p_{\{1\}}=p_{\{2\}}$	$p-p_1=p_2$	graph, the remaining $p-p_1=p_2$ vertices from $N(x_0)$ belong to at most one, say A_1, of the sets X, $1 \leq i \leq n$. Then
voloshin-hypergraph_F00813	115	■ 1.000	$X_{\{i\}}, 1 \leq i \leq n$	$X_i, 1 \leq i \leq n$	$p-p_1=p_2=\sum_{i=1}^n\left X_i\right$
voloshin-hypergraph_F00814	115	■ 1.000	$G_{\{i\}}, i=$	$G_i, i=$	$0, 2, 3, \dots, k$ with degree p_1. Apply formula (3.10) to G_i.
voloshin-hypergraph_F00815	115	■ 1.000	$0, 2, 3, \ldots, k$	$0, 2, 3, \dots, k$	$P(G, \lambda)=(\lambda-p) P\left(G-x_0,\right.$
voloshin-hypergraph_F00816	115	■ 1.000	$p_{\{1\}}$	p_1	$P(G, \lambda)=(\lambda-p) P\left(G-x_0,\right.$
voloshin-hypergraph_F00817	115	■ 1.000	$A-x_{\{0\}}$	$A-x_0$	On the other hand,
voloshin-hypergraph_F00818	115	■ 1.000	$G_{\{i\}}^{\prime}$	$G_i^{\prime}$	
voloshin-hypergraph_F00819	115	■ 1.000	$x_{\{0\}}, i=0, 1, \ldots, k$	$x_0, i=0, 1, \dots, k$	
voloshin-hypergraph_F00820	115	■ 1.000	$i=0, 2, \ldots, k$	$i=0, 2, \dots, k$	—
voloshin-hypergraph_F00821	116	■ 1.000	$G_{\{1\}}, \ldots, G_{\{k\}}$	$G_1, \dots, G_k$	From this point of view, chordal graphs have so nice structure or colorings that they mysteriously look like “complete graphs having separators”. However, (5.15) has an addi-
voloshin-hypergraph_F00822	116	■ 1.000	$A=\emptyset$	$A=\emptyset$	tional 1
voloshin-hypergraph_F00823	116	■ 0.999	$x \in X^{\prime}$	$x \in X^{\prime}$	is chordal, we prove the necessity for $G^{\prime}=G$. Hence, assume that $G=(X, E)$ is a connected chordal graph and $x \in X$ is an arbitrary vertex
voloshin-hypergraph_F00824	116	■ 0.982	$X=\{x\} \cup N(x)$	$X=\{x\} \cup N(x)$	Case 2. $X \neq\{x\} \cup N(x)$. Since G is connected, the subgraph $G_{N(x)}$ is a separator. Apply
voloshin-hypergraph_F00825	116	■ 1.000	$X \neq\{x\} \cup N(x)$	$X \neq\{x\} \cup N(x)$	
voloshin-hypergraph_F00826	116	■ 1.000	$G_{\{N(x)\}}$	$G_{N(x)}$	
voloshin-hypergraph_F00827	117	■ 1.000	C	C	denoted by just C. Observe that $C-x$ is a tree on $k-1$ vertices, and $C_{N(x)}=E_2$. We see that $P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W(C-x, \lambda)=\lambda^{-1}(\lambda-1)^2$ and by the formula
voloshin-hypergraph_F00828	117	■ 1.000	$C-x$	$C-x$	denoted by just C. Observe that $C-x$ is a tree on $k-1$ vertices, and $C_{N(x)}=E_2$. We see that $P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W(C-x, \lambda)=\lambda^{-1}(\lambda-1)^2$ and by the formula
voloshin-hypergraph_F00829	117	■ 1.000	$C_{\{N(x)\}}=E_{\{2\}}$	$C_{N(x)}=E_2$	denoted by just C. Observe that $C-x$ is a tree on $k-1$ vertices, and $C_{N(x)}=E_2$. We see that $P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W(C-x, \lambda)=\lambda^{-1}(\lambda-1)^2$ and by the formula
voloshin-hypergraph_F00830	117	■ 0.996	$P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W\left(C_{\{N(x)\}}, \lambda\right)=\lambda^{-1}(\lambda-1)^2$	$P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W\left(C_{N(x)}, \lambda\right)=\lambda^{-1}(\lambda-1)^2$	that $P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W\left(C_{N(x)}, \lambda\right)=\lambda^{-1}(\lambda-1)^2$, and by the formula
voloshin-hypergraph_F00831	117	■ 1.000	$N(x)=K_{\{k\}}$	$N(x)=K_k$	$W\left(G_{N(x)}\right.$
voloshin-hypergraph_F00832	117	■ 1.000	$W\left(G_{\{N(4)\}}, \lambda\right)$	$W\left(G_{N(4)}, \lambda\right)$	$P(G-x, \lambda)=\lambda(\lambda-1)^2, W(G-x, \lambda)=\lambda^{-1}(\lambda-1)^2$

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voloshin-hypergraph_F00833	117	■ 1.000	$N(4)=\{1,5,6\}$	$N(4) = \{1, 5, 6\}$	
voloshin-hypergraph_F00834	118	■ 1.000	$\{5,6,7,8\}$	$\{5, 6, 7, 8\}$	and respective derived subgraphs.
voloshin-hypergraph_F00835	118	■ 0.975	$G, \chi(G) \leq M(G)+1$	$G, \chi(G) \leq M(G) + 1$	structing it in ordering $x_n, x_{n-1}, \dots, x_2, x_1$. That me
voloshin-hypergraph_F00836	118	■ 1.000	$G_{\{1\}}=G, G_{\{2\}}, G_{\{3\}}, \ldots, G_{\{n\}}$	$G_1 = G, G_2, G_3, \dots, G_n$	The worst case occurs when all $M(G)$ neighbors of x_i have distinct colors (other-
voloshin-hypergraph_F00837	118	■ 1.000	$x_{\{n\}}, x_{\{n-1\}}, \ldots, x_{\{2\}}, x_{\{1\}}$	$x_n, x_{n-1}, \dots, x_2, x_1$	$M(G)+1$. $\square$
voloshin-hypergraph_F00838	118	■ 0.988	$(M(G)+1)$	$(M(G) + 1)$	
voloshin-hypergraph_F00839	118	■ 1.000	$M(G)+1$	$M(G) + 1$	
voloshin-hypergraph_F00840	119	■ 1.000	$G_{\{i+1\}}$	G_{i+1}	
voloshin-hypergraph_F00841	119	■ 0.999	$\chi(G) \leq 6$	$\chi(G) \leq 6$	Proof. Recall that by Corollary 4.2.3, G contains a vi
voloshin-hypergraph_F00842	119	■ 1.000	$M(G) \leq 5$	$M(G) \leq 5$	Theorem 5.6.2 (Five Color Theorem, Heawood, 1890) If $G = (X, E)$ is a planar graph,
voloshin-hypergraph_F00843	119	■ 1.000	$\chi(G) \leq 5$	$\chi(G) \leq 5$	Assu
voloshin-hypergraph_F00844	119	■ 1.000	≤ 5	≤ 5	By this, all the vertices of $N(x)$ get some colors. We will show that
voloshin-hypergraph_F00845	119	■ 0.998	$d(x) \leq 5$	$d(x) \leq 5$	assign a color to vertex x such that G is 5-colorable
voloshin-hypergraph_F00846	119	■ 0.998	$\chi(G-x) \leq 5$	$\chi(G - x) \leq 5$	obtain a 5-coloring of G .
voloshin-hypergraph_F00847	119	■ 0.693	≤ 4	≤ 4	obtain a 5-coloring of G .
voloshin-hypergraph_F00848	119	■ 0.988	$d(x)=5$	$d(x) = 5$	The situation is equivalent to that shown in Figure 5.17 for $G_i = G$, but now it can
voloshin-hypergraph_F00849	119	■ 1.000	$G_{\{i\}}=G$	$G_i = G$	Subcase 2.1: in the 5-coloring of $G - x$ there exists an (a, c) -path P with vertices
voloshin-hypergraph_F00850	119	■ 0.999	$N(x)=$	$N(x) =$	and 3, and vertex x uncolored. Switch the colors 2 and 4 on vertices located in the plane
voloshin-hypergraph_F00851	119	■ 0.812	(a, c)	(a, c)	
voloshin-hypergraph_F00852	119	■ 0.812	P	P	
voloshin-hypergraph_F00853	119	■ 1.000	$a \ x$	ax	
voloshin-hypergraph_F00854	119	■ 1.000	$x \ c$	xc	
voloshin-hypergraph_F00855	120	■ 1.000	$G_{\{13\}}$	G_{13}	obtain a proper 5-coloring of G , what proves the theorem. $\square$
voloshin-hypergraph_F00856	120	■ 0.725	c	c	Theorem 5.6.3 (Four Color Theorem. Appel. Haken. Koch. 1977) If G is a planar
voloshin-hypergraph_F00857	120	■ 0.994	$\chi(G) \leq 4$	$\chi(G) \leq 4$	that under this as:
voloshin-hypergraph_F00858	120	■ 1.000	$\chi(G-x) \leq 4$	$\chi(G - x) \leq 4$	
voloshin-hypergraph_F00859	121	■ 0.912	≤ 3	≤ 3	Case 1: the number of colors used in $M(x)$ is ≤ 2 . We assign the 4th color to x and

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voloshin-hypergraph_F00860	121	■ 0.997	$d(x)=4$	$d(x) = 4$	Case 2: the number of colors used in $N(x)$ is 4. It means that $d(x) = 4$ or $d(x) = 5$.
voloshin-hypergraph_F00861	121	■ 0.999	$G^{\{\prime\}}=G-e$	$G' = G - e$	Subcase 2.1: $d(x) = 4$. Apply the reasoning from Theorem 5.6.2 for $G' = G - e$ (in
voloshin-hypergraph_F00862	121	■ 0.973	$N(x)=\{a, b, c, d, e\}$	$N(x) = \{a, b, c, d, e\}$	Subcase 2.2.1: in the 4-coloring of $G - x$, there is no (a, c) -path P_{13} or there is no
voloshin-hypergraph_F00863	121	■ 1.000	a, b, c, d, e	a, b, c, d, e	is disconnected, and vertices a and c are in different components. Switching the colors 1
voloshin-hypergraph_F00864	121	■ 1.000	$G_{\{i\ j\}}$	G_{ij}	and 3 in the component containing vertex a eliminates color 1 from $N(x)$; we then assign
voloshin-hypergraph_F00865	121	■ 1.000	$P_{\{i\ j\}}$	P_{ij}	and 3 in the component containing vertex a eliminates color 1 from $N(x)$; we then assign
voloshin-hypergraph_F00866	121	■ 0.936	$P_{\{13\}}$	P_{13}	Subcase 2.2.2: in the 4-coloring of $G - x$, both (a, c) -path P_{13} and (a, d) -path P_{14} exist,
voloshin-hypergraph_F00867	121	■ 1.000	(a, d)	(a, d)	see Figure 5.19. Observe that the component G' of G_{24} containing vertex b is separated from
voloshin-hypergraph_F00868	121	■ 1.000	$P_{\{14\}}$	P_{14}	see Figure 5.19. Observe that the component G' of G_{24} containing vertex b is separated from
voloshin-hypergraph_F00869	121	■ 1.000	$G_{\{24\}}$	G_{24}	
voloshin-hypergraph_F00870	121	■ 1.000	$G^{\{\prime\prime\}}\prime\prime\}$	G''	—
voloshin-hypergraph_F00871	121	■ 1.000	$G_{\{23\}}$	G_{23}	—
voloshin-hypergraph_F00872	122	■ 1.000	$x\ d$	xd	separated by (a, d) with P_{14} and edges ac and cd . We now permute color 2 with 4 in G'
voloshin-hypergraph_F00873	123	■ 1.000	$\chi\left(W_{\{6\}}\right)=4$	$\chi(W_6) = 4$	—
voloshin-hypergraph_F00874	124	■ 1.000	$W_{\{8\}}$	W_8	four colors is the minimum for the entire map of USA.
voloshin-hypergraph_F00875	124	■ 1.000	$P_{\{n\}}, K_{\{n\}}, K_{\{m, n\}}, C_{\{n\}}, W_{\{n\}}, m, n \geq 4$	$P_n, K_n, K_{m,n}, C_n, W_n, m, n \geq 4$	prove that the obtained regions can be colored with two colors in such a way that no two of
voloshin-hypergraph_F00876	124	■ 1.000	$\chi(G)=4$	$\chi(G) = 4$	1. Given a graph G , find $M(G)$, bound on the chromatic number, and the respective coloring.
voloshin-hypergraph_F00877	124	■ 0.830	$\chi(G)=\omega(G)$	$\chi(G) = \omega(G)$	
voloshin-hypergraph_F00878	124	■ 0.830	$\chi(G)>\omega(G)$	$\chi(G) > \omega(G)$	
voloshin-hypergraph_F00879	124	■ 0.998	$\chi\left(C_{\{4\}}\right)=\omega\left(C_{\{4\}}\right)=2$	$\chi(C_4) = \omega(C_4) = 2$	
voloshin-hypergraph_F00880	124	■ 0.998	$\chi\left(C_{\{5\}}\right)=3>\omega\left(C_{\{5\}}\right)=2$	$\chi(C_5) = 3 > \omega(C_5) = 2$	
voloshin-hypergraph_F00881	124	■ 1.000	$\chi(G)-\omega(G)$	$\chi(G) - \omega(G)$	
voloshin-hypergraph_F00882	125	■ 0.997	$\chi=4$	$\chi = 4$	
voloshin-hypergraph_F00883	125	■ 1.000	$\omega=2$	$\omega = 2$	
voloshin-hypergraph_F00884	125	■ 0.970	$\boldsymbol{\chi}-\boldsymbol{\omega}=k$	$\chi - \omega = k$	
voloshin-hypergraph_F00885	125	■ 0.838	$\boldsymbol{\chi}-\boldsymbol{\omega}$	$\chi - \omega$	
voloshin-hypergraph_F00886	125	■ 0.838	$\boldsymbol{\omega}(G)$	$\omega(G)$	
voloshin-hypergraph_F00887	125	■ 0.838	$\boldsymbol{\chi}(G)$	$\chi(G)$	

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voloshin-hypergraph_F08888	126	■ 1.000	$\chi(G^{\prime})=\omega(G^{\prime})$	$\chi(G') = \omega(G')$	Definition 5.7.1 A graph G is perfect if $\chi(G') = \omega(G')$ for any induced subgraph G' in-
voloshin-hypergraph_F08889	126	■ 0.998	$\chi(G)=\theta(\bar{G})$	$\chi(G) = \theta(\bar{G})$	
voloshin-hypergraph_F08890	126	■ 0.998	$\theta(G^{\prime})=\alpha(G^{\prime})$	$\theta(G') = \alpha(G')$	
voloshin-hypergraph_F08891	126	■ 0.999	$\chi(G)=3, \omega(G)=3$	$\chi(G) = 3, \omega(G) = 3$	perfect graphs. An example of an imperfect graph is C_5 : $\chi(C_5) = 3, \omega(C_5) = 2, \alpha(C_5) = 2$ and $\theta(C_5) = 2$ (recall that C_5 is isomorphic to $\bar{C}_5$). In these examples graphs and their
voloshin-hypergraph_F08892	126	■ 1.000	$\theta(G)=2$	$\theta(G) = 2$	and $\theta(C_5) = 2$ (recall that C_5 is isomorphic to $\bar{C}_5$). In these examples graphs and their complements behave similarly with respect to graph perfection. Is this the rule?
voloshin-hypergraph_F08893	126	■ 1.000	$\chi(G)=2, \omega(G)=2, \alpha(G)=3$	$\chi(G) = 2, \omega(G) = 2, \alpha(G) = 3$	May we have the equality $\chi(G') = \omega(G')$ for all induced subgraphs G' and inequality
voloshin-hypergraph_F08894	126	■ 1.000	$C_5: \chi(C_5)=3, \omega(C_5)=2, \alpha(C_5)=2$	$C_5: \chi(C_5) = 3, \omega(C_5) = 2, \alpha(C_5) = 2$	Perfect Graph Conjecture by Berge in 1960 and proved by Lovasz in 1972 (the first proof used hypergraph approach).
voloshin-hypergraph_F08895	126	■ 0.972	$\theta(C_5)=3$	$\theta(C_5) = 3$	used hypergraph approach).
voloshin-hypergraph_F08896	126	■ 1.000	$\theta(G^{\prime}) \neq \alpha(G^{\prime})$	$\theta(G') \neq \alpha(G')$	
voloshin-hypergraph_F08897	127	■ 1.000	$n=3,4,5$	$n = 3, 4, 5$	$\omega(G-x)$ colors. If $\omega(G) = \omega(G-x)$, then $ N(x) \leq \omega(G-x) - 1$ and we can use a color missing in $N(x)$ to color vertex x . If $\omega(G) = \omega(G-x) + 1$ then $ N(x) = \omega(G-x)$ and via
voloshin-hypergraph_F08898	127	■ 1.000	$\chi(G-x)=$	$\chi(G-x) =$	colors. Therefore, G is perfect.
voloshin-hypergraph_F08899	127	■ 1.000	$\omega(G-x)$	$\omega(G-x)$	Case 2: x is a simplicial vertex in $\bar{G}$. Again, graph $G-x$ is quasi-triangulated, has
voloshin-hypergraph_F08900	127	■ 1.000	$\omega(G)=\omega(G-x)$	$\omega(G) = \omega(G-x)$	Case 2: x is a simplicial vertex in $\bar{G}$. Again, graph $G-x$ is quasi-triangulated, has
voloshin-hypergraph_F08901	127	■ 1.000	$ N(x) \leq \omega(G-x)-1$	$ N(x) \leq \omega(G-x) - 1$	Case 2: x is a simplicial vertex in $\bar{G}$. Again, graph $G-x$ is quasi-triangulated, has
voloshin-hypergraph_F08902	127	■ 0.953	$\omega(G)=\omega(G-x)+1$	$\omega(G) = \omega(G-x) + 1$	$\leq n$ vertices, and by the induction hypothesis is perfect. By Theorem 5.7.1, graph $G-x$ is
voloshin-hypergraph_F08903	127	■ 0.953	$ N(x) =\omega(G-x)$	$ N(x) = \omega(G-x)$	$\leq n$ vertices, and by the induction hypothesis is perfect. By Theorem 5.7.1, graph $G-x$ is
voloshin-hypergraph_F08904	127	■ 0.902	$\overline{G-x}$	$\overline{G-x}$	chordal graphs (containing no induced cycles of length ≥ 3 in G and in $\bar{G}$), strongly perfect (every induced subgraph has an independent set intersecting all maximal cliques). Meyniel
voloshin-hypergraph_F08905	127	■ 0.993	C_{2k+1}	C_{2k+1}	
voloshin-hypergraph_F08906	127	■ 0.993	$\bar{C}_{2k+1}$	$\bar{C}_{2k+1}$	
voloshin-hypergraph_F08907	128	■ 0.999	P_5, C_4, C_6, K_4	P_5, C_4, C_6, K_4	
voloshin-hypergraph_F08908	128	■ 0.999	$\theta(C_7)>\alpha(C_7)$	$\theta(C_7) > \alpha(C_7)$	5. Are cube and prism per
voloshin-hypergraph_F08909	128	■ 0.829	$\Theta(G)$	$\Theta(G)$	5.8 Edge Coloring and Vizing's Theorem
voloshin-hypergraph_F08910	128	■ 1.000	$\lambda \geq 0$	$\lambda \geq 0$	vertices colored. A graph is edge is colorable by n new colors.
voloshin-hypergraph_F08911	128	■ 1.000	$ E $	$ E $	matchings. The chromatic index $\chi'(G)$ is the minimum number of distinct matchings over
voloshin-hypergraph_F08912	128	■ 1.000	$\chi^{\prime}(G)$	$\chi'(G)$	vertices representing (X', E') such that X'
voloshin-hypergraph_F08913	128	■ 1.000	$G^{\prime}=$	$G' =$	
voloshin-hypergraph_F08914	128	■ 1.000	$\left(X^{\prime}, E^{\prime}\right)$	(X', E')	

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voloshin-hypergraph_F00915	128	■ 1.000	$X^{\{\prime\}}=E$	$X' = E$	
voloshin-hypergraph_F00916	129	■ 1.000	$G^{\{\prime\}}=L(G)$	$G' = L(G)$	simplicial decomposition. Notice that every non pendant vertex of G forms a clique in $L(G)$
voloshin-hypergraph_F00917	129	■ 1.000	$P(L(G), \lambda)=\lambda(\lambda-1)(\lambda-2)^2(\lambda-3)$	$P(L(G), \lambda) = \lambda(\lambda - 1)(\lambda - 2)^2(\lambda - 3)$	for any graph G without loops. As usually, let $\Delta(G)$ denote the maximum vertex degree of G . Evidently, $\chi'(G) \geq \Delta(G)$
voloshin-hypergraph_F00918	129	■ 1.000	$\{b, c, d, e\}$	$\{b, c, d, e\}$	plays for the chromatic number χ in vertex coloring where $\chi \geq \omega$. But how far may be the
voloshin-hypergraph_F00919	129	■ 1.000	$\chi^{\{\prime\}}(G)=4=\chi(L(G))$	$\chi'(G) = 4 = \chi(L(G))$	value $\chi'(G)$ from $\Delta(G)$? Surprisingly, it must be very close.
voloshin-hypergraph_F00920	129	■ 1.000	$\chi^{\{\prime\}}(G)=\chi(L(G))$	$\chi'(G) = \chi(L(G))$	value $\chi'(G)$ from $\Delta(G)$? Surprisingly, it must be very close.
voloshin-hypergraph_F00921	129	■ 1.000	$\chi^{\{\prime\}}(G) \geq \Delta(G)$	$\chi'(G) \geq \Delta(G)$	THEOREM 5.9.1 (VIZING, 1964) If G is a simple graph, then either
voloshin-hypergraph_F00922	129	■ 1.000	Δ	Δ	$\chi'(G) = \Delta(G)$, or $\chi'(G) = \Delta(G) + 1$.
voloshin-hypergraph_F00923	129	■ 1.000	$\chi^{\{\prime\}}$	χ'	$\chi'(G) = \Delta(G)$, or $\chi'(G) = \Delta(G) + 1$.
voloshin-hypergraph_F00924	129	■ 0.996	$\boldsymbol{\chi}$	χ	Proof Simple examples (like C_{2k} and C_{2k+1}) show that there are graphs with both values
voloshin-hypergraph_F00925	129	■ 0.996	$\boldsymbol{\chi} \geq \omega$	$\chi \geq \omega$	Proof Simple examples (like C_{2k} and C_{2k+1}) show that there are graphs with both values
voloshin-hypergraph_F00926	129	■ 1.000	C_{2k}	C_{2k}	coloring of G will be obtained. Since we use $\Delta + 1$ colors and Δ is the maximum degree,
voloshin-hypergraph_F00927	129	■ 0.990	$\Delta+1$	$\Delta + 1$	
voloshin-hypergraph_F00928	129	■ 1.000	$y_{\{0\}}$	y_0	
voloshin-hypergraph_F00929	129	■ 1.000	$x y_{\{0\}}$	xy_0	
voloshin-hypergraph_F00930	130	■ 1.000	$c_{\{0\}}$	c_0	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00931	130	■ 1.000	$c_{\{1\}}$	c_1	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00932	130	■ 1.000	$x y_{\{k\}}$	xy_k	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00933	130	■ 1.000	$c_{\{i\}}$	c_i	ends missing a color c_i , then re-color xy_k with c_i and color xy_0 with c_1 . The bad case: there is no edge xy_k colored c_1 with both ends missing a color.
voloshin-hypergraph_F00934	130	■ 1.000	$x y_{\{1\}}$	xy_1	The bad case: there is no edge xy_k colored c_1 with both ends missing a color. Edge xy_1 is colored c_1 ; a color c_2 is missing at v_1 . Edge xy_2 is colored c_2 ; a color c_3 is missing at
voloshin-hypergraph_F00935	130	■ 1.000	$c_{\{2\}}$	c_2	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00936	130	■ 1.000	$y_{\{1\}}$	y_1	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00937	130	■ 1.000	$x y_{\{2\}}$	xy_2	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00938	130	■ 1.000	$c_{\{3\}}$	c_3	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00939	130	■ 1.000	$y_{\{2\}}$	y_2	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at
voloshin-hypergraph_F00940	130	■ 1.000	$x y_{\{3\}}$	xy_3	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at
voloshin-hypergraph_F00941	130	■ 1.000	$c_{\{4\}}$	c_4	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at

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voloshin-hypergraph_F00942	130	■ 1.000	$y_{\{3\}}$	y_3	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at
voloshin-hypergraph_F00943	130	■ 0.999	$\bar{c}_{\{i\}}$	$\bar{c}_i$	both vertex x and y_i . Then we re-color each edge xy_i from c_i to c_{l+1} , $i = 1, 2, \dots, l$. Call this
voloshin-hypergraph_F00944	130	■ 0.999	$c_{\{l+1\}}$	c_{l+1}	both vertex x and y_l . Then we re-color each edge xy_i from c_i to c_{l+1} , $i = 1, 2, \dots, l$. Call this
voloshin-hypergraph_F00945	130	■ 1.000	$y_{\{l\}}$	y_l	re-coloring downshifting from y_l , we arrive to a situation when color c_1 is missing at x . Since it is missing at x , too, color edge xy_0 with c_1 .
voloshin-hypergraph_F00946	130	■ 1.000	$x \times y_{\{i\}}$	xy_i	re-coloring downshifting from y_l , we arrive to a situation when color c_1 is missing at x . Since it is missing at x , too, color edge xy_0 with c_1 .
voloshin-hypergraph_F00947	130	■ 1.000	$c_{\{i+1\}}$, $i=1,2, \ldots, l$	$c_{i+1}, i = 1, 2, \dots, l$	re-coloring downshifting from y_l , we arrive to a situation when color c_1 is missing at x . Since it is missing at x , too, color edge xy_0 with c_1 .
voloshin-hypergraph_F00948	130	■ 1.000	$c_{\{1\}}$, $c_{\{2\}}$, $c_{\{3\}}$, $\ldots$	$c_1, c_2, c_3, \dots$	
voloshin-hypergraph_F00949	130	■ 1.000	$c_{\{l+1\}}=c_{\{k\}}$	$c_{l+1} = c_k$	
voloshin-hypergraph_F00950	130	■ 1.000	$1 \leq k \leq l$	$1 \leq k \leq l$	
voloshin-hypergraph_F00951	131	■ 1.000	$c_{\{k\}}$	c_k	two alternating colors c_0 and c_k beginning at vertex x . Since we color the edges xy_k
voloshin-hypergraph_F00952	131	■ 1.000	$y_{\{k\}}$	y_k	Subcase 1: P reaches vertex y_k and therefore ends at x . Switch colors c_0 and c_k along
voloshin-hypergraph_F00953	131	■ 1.000	$y_{\{k-1\}}$	y_{k-1}	Subcase 2: P reaches vertex y_{k-1} and therefore ends at y_{k-1} . Again, switch colors c_0 and c_k along P . As a result, color c_0 is missing at y_{k-1} . Since it is also missing at x , we
voloshin-hypergraph_F00954	131	■ 1.000	$x \times y_{\{k-1\}}$	xy_{k-1}	again, release color c_1 to color edge xy_0 .
voloshin-hypergraph_F00955	131	■ 1.000	x, P	x, P	Switch the colors c_0 and c_k along P , color c_0 becomes missing at y_l . Perform downshifting from y_l and release color c_1 to color edge xy_0 . $\square$
voloshin-hypergraph_F00956	131	■ 1.000	$\chi^{\{\prime\}}(G)=$	$\chi'(G) =$	Class 2 is a hard problem.
voloshin-hypergraph_F00957	131	■ 0.998	$\chi^{\{\prime\}}(G)=\Delta(G)+1$	$\chi'(G) = \Delta(G) + 1$	If a graph G has multiple (parallel) edges (but not loops), the multiplicity of G , denoted by $\mu(G)$, is the maximum number of parallel edges connecting a pair of vertices. Theorem
voloshin-hypergraph_F00958	131	■ 1.000	$\mu(G)$	$\mu(G)$	Theorem 5.8.2 (Vizing, 1964) If G is a loop-less multigraph with multiplicity $\mu(G)$, then
voloshin-hypergraph_F00959	131	■ 1.000	$\Delta(G) \leq \chi^{\{\prime\}}(G) \leq \Delta(G)+\mu(G)$	$\Delta(G) \leq \chi'(G) \leq \Delta(G) + \mu(G)$	
voloshin-hypergraph_F00960	131	■ 1.000	$\Delta+\mu$	$\Delta + \mu$	(König, 1916) If G is a bipartite graph, then $\chi'(G) = \Delta(G)$
voloshin-hypergraph_F00961	131	■ 1.000	$\chi^{\{\prime\}}(G)=\Delta(G)+\mu(G)$	$\chi'(G) = \Delta(G) + \mu(G)$	
voloshin-hypergraph_F00962	131	■ 1.000	$\mu=1$	$\mu = 1$	1. Find the chromatic index and an optimal edge coloring of: $K_n, C_n, W_n,$
voloshin-hypergraph_F00963	131	■ 1.000	$\chi^{\{\prime\}}(G)=\Delta(G)$	$\chi'(G) = \Delta(G)$	
voloshin-hypergraph_F00964	131	■ 1.000	$K_{\{n\}}$, $C_{\{n\}}$, $W_{\{n\}}$, $K_{\{m, n\}}$	$K_n, C_n, W_n, K_{m,n}$	—
voloshin-hypergraph_F00965	131	■ 1.000	$m, n \geq 3$	$m, n \geq 3$	—
voloshin-hypergraph_F00966	132	■ 1.000	$\chi^{\{\prime\}}=4$	$\chi' = 4$	3. Prove that for the Petersen graph $\chi' = 4$.
voloshin-hypergraph_F00967	132	■ 0.988	$k \leq \lambda$	$k \leq \lambda$	An example of a multigraph G and its edge 3-coloring is shown in Figure 5.28. We will
voloshin-hypergraph_F00968	132	■ 0.881	$\boldsymbol{k}$	$\boldsymbol{k}$	An example of a multigraph G and its edge 3-coloring is shown in Figure 5.28. We will show that it uses the maximum number of colors.
voloshin-hypergraph_F00969	132	■ 1.000	$\bar{\chi}^{\{\prime\}}(G)$	$\bar{\chi}'(G)$	only if the maximum degree $\Delta(G) \leq 2$.

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voloshin-hypergraph_F00970	132	0.988	$\bar{\chi}^{\prime}(G)=3$	$\bar{\chi}^{\prime}(G)=3$	a tree, color any maximal path with color 1 and all other edges with c
voloshin-hypergraph_F00971	132	0.990	$\bar{\chi}^{\prime}(G)=1$	$\bar{\chi}^{\prime}(G)=1$	$\Leftarrow$ The converse is evident since $\Delta(G) \leq 2$ implies that G is either a cycle or a path. $\square$
voloshin-hypergraph_F00972	132	1.000	$\Delta(G) \leq 2$	$\Delta(G) \leq 2$	
voloshin-hypergraph_F00973	132	0.876	$\Delta(G) \geq 3$	$\Delta(G) \geq 3$	
voloshin-hypergraph_F00974	133	0.999	$\bar{\chi}^{\prime}(G) \geq 2$	$\bar{\chi}^{\prime}(G) \geq 2$	Consider a proper edge k -coloring of $G=(X, E)$; it partitions E into k color classes
voloshin-hypergraph_F00975	133	0.996	$\left\{C_1, C_2, \ldots, C_k\right\}$	$\left\{C_1, C_2, \ldots, C_k\right\}$	Let $G_i=\left(X_i, C_i\right)$ be the subgraph of G with a set of vertices X_i determined by endpoints of the edges of C_i . Notice that subgraphs G_i are not necessarily connected multigraphs and
voloshin-hypergraph_F00976	133	0.996	C_i	C_i	Let $G_i=\left(X_i, C_i\right)$ be the subgraph of G with a set of vertices X_i determined by endpoints of the edges of C_i . Notice that subgraphs G_i are not necessarily connected multigraphs and
voloshin-hypergraph_F00977	133	1.000	$G_i=\left(X_i, C_i\right)$	$G_i=\left(X_i, C_i\right)$	A vertex x of a subgraph G_i is said to be satisfied by subgraph G_i if it is not a pendant
voloshin-hypergraph_F00978	133	1.000	X_i	X_i	A vertex x of a subgraph G_i is said to be satisfied by subgraph G_i if it is not a pendant
voloshin-hypergraph_F00979	133	0.839	$\left(\bar{\chi}^{\prime}(G)+1\right)$	$\left(\bar{\chi}^{\prime}(G)+1\right)$	Proof. Indeed, in f , every G_i must be connected because otherwise we could increase the
voloshin-hypergraph_F00980	133	0.996	$1 \leq i \leq$	$1 \leq i \leq$	
voloshin-hypergraph_F00981	133	0.939	G_j	G_j	
voloshin-hypergraph_F00982	134	1.000	G_k	G_k	G_i of G into the following three classes,
voloshin-hypergraph_F00983	134	1.000	G_l s	G_l s	G_i of G into the following three classes,
voloshin-hypergraph_F00984	134	0.998	$c^{\prime}$	$c^{\prime}$	
voloshin-hypergraph_F00985	134	0.998	$c^{\prime} \leq c$	$c^{\prime} \leq c$	
voloshin-hypergraph_F00986	134	1.000	$ X =n, E =m$	$ X =n, E =m$	
voloshin-hypergraph_F00987	135	1.000	$\bar{\chi}^{\prime}(G) \leq c+m-n+p$	$\bar{\chi}^{\prime}(G) \leq c+m-n+p$	Proof. We prove first that $\bar{\chi}(G) \leq c+m-n+p$. By Corollary 3.9.1, if a coloring f uses
voloshin-hypergraph_F00988	135	0.999	$G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$	$G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$	Proof. we prove first that $\bar{\chi}(G) \leq c+m-n+p$. By Corollary 3.9.1, if a coloring f uses $\bar{\chi}^{\prime}(G)$ colors, then subgraphs $G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$, are either cycles or paths, while Corollary
voloshin-hypergraph_F00989	135	1.000	$v^{\prime}$	$v^{\prime}$	Let $c^{\prime}$ be the number of subgraphs, and $v^{\prime}$ be the number of satisfied vertices in class
voloshin-hypergraph_F00990	135	1.000	G_l	G_l	disjoint. Let x be a satisfied vertex in G . If x is satisfied by some G_l from class A, then it is possible to look at all the other G_i s which satisfy x , like paths with endpoints x .
voloshin-hypergraph_F00991	135	0.419	G_k s	G_k s	If x is not satisfied by any subgraph from class A, then it is satisfied by a subgraph fr
voloshin-hypergraph_F00992	135	0.915	$n-v^{\prime}-p$	$n-v^{\prime}-p$	as paths with coinciding endpoints. Since the number of edges in each of them equals the
voloshin-hypergraph_F00993	135	0.973	$m^{\prime}$	$m^{\prime}$	Since class C contains $m-m^{\prime}$ edges, the number of colors in f is:
voloshin-hypergraph_F00994	135	1.000	$m-m^{\prime}$	$m-m^{\prime}$	and therefore we obtain
voloshin-hypergraph_F00995	135	0.995	$\bar{\chi}^{\prime}(G) \geq c+m-n+p$	$\bar{\chi}^{\prime}(G) \geq c+m-n+p$	the inequality holds.
voloshin-hypergraph_F00996	135	1.000	$C_1, \ldots, C_c$	$C_1, \ldots, C_c$	Now we implement the following coloring augmenting procedure. Choose any satisfied vertex x , say x incident to an uncolored edge and construct a path along uncolored edges

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00997	135	■ 1.000	$A_{\{0\}}$	A_0	Now we implement the following coloring augmenting procedure. Choose any satisfied vertex x say x incident to an uncolored edge and construct a path along uncolored edges
voloshin-hypergraph_F00998	135	■ 0.999	$1, \ldots, c$	$1, \ldots, c$	vertex say x incident to an uncolored edge and construct a path along uncolored edges until the first satisfied vertex y is reached or we get stuck at a pendant vertex y . Since A_0
voloshin-hypergraph_F00999	135	■ 0.999	$m\left(A_{\{0\}\right)=$	$m\left(A_0\right)=$	vertex say x incident to an uncolored edge and construct a path along uncolored edges until the first satisfied vertex y is reached or we get stuck at a pendant vertex y . Since A_0
voloshin-hypergraph_F01000	135	■ 1.000	$n\left(A_{\{0\}\right)}$	$n\left(A_0\right)$	constitutes the maximum number of disjoint cycles, the new vertices and edges represent
voloshin-hypergraph_F01001	135	■ 1.000	$p\left(A_{\{0\}\right)=0$	$p\left(A_0\right)=0$	constitutes the maximum number of disjoint cycles, the new vertices and edges represent
voloshin-hypergraph_F01002	135	■ 1.000	$c+m\left(A_{\{0\}\right)-n\left(A_{\{0\}\right)+p\left(A_{\{0\}\right)}$	$c+m\left(A_0\right)-n\left(A_0\right)+p\left(A_0\right)$	either a cycle or a path (if they form a cycle, then $x=y$).
voloshin-hypergraph_F01003	135	■ 1.000	$\bar{\chi}^{\prime}\left(A_{\{0\}\right) \geq c+m\left(A_{\{0\}\right)-n\left(A_{\{0\}\right)+p\left(A_{\{0\}\right)}$	$\bar{\chi}^{\prime}\left(A_0\right) \geq c+m\left(A_0\right)-n\left(A_0\right)+p\left(A_0\right)$	either a cycle or a path (if they form a cycle, then $x=y$).
voloshin-hypergraph_F01004	135	■ 1.000	$x=y$	$x=y$	
voloshin-hypergraph_F01005	135	■ 1.000	$A_{\{1\}}$	A_1	
voloshin-hypergraph_F01006	136	■ 1.000	$A_{\{2\}}, A_{\{3\}}, \ldots$	$A_2, A_3, \ldots$	and therefore the inequality holds. We repeat this coloring procedure by constructing subgraphs $A_2, A_3, \ldots$ until all the edges of G are colored and all non-pendant vertices satisfied
voloshin-hypergraph_F01007	136	■ 1.000	$c=1, m=5, n=4$	$c=1, m=5, n=4$	For our example, see Figure 5.28, it is easy to see that $c=1, m=5, n=4$ and $p=1$, and the formula holds.
voloshin-hypergraph_F01008	136	■ 1.000	$p=1$	$p=1$	For our example, see Figure 5.28, it is easy to see that $c=1, m=5, n=4$ and $p=1$, and the formula holds.
voloshin-hypergraph_F01009	136	■ 1.000	$c=0$	$c=0$	EXERCISES 5.27.
voloshin-hypergraph_F01010	136	■ 1.000	$\bar{\chi}^{\prime}(G)=c+m-n+p$	$\bar{\chi}^{\prime}(G)=c+m-n+p$	
voloshin-hypergraph_F01011	137	■ 1.000	$C_{\{n\}}, W_{\{n\}}, K_{\{n\}}, K_{\{m, n\}}$	$C_n, W_n, K_n, K_{m, n}$	5. For a graph G with n vertices and m edges, let $C_n, W_n, K_n, K_{m, n}$ denote the cycle, wheel, complete graph, and complete bipartite graph, respectively.
voloshin-hypergraph_F01012	137	■ 1.000	$m, n \geq 2$	$m, n \geq 2$	5. For $m, n \geq 2$, let $C_n, W_n, K_n, K_{m, n}$ denote the cycle, wheel, complete graph, and complete bipartite graph, respectively.
voloshin-hypergraph_F01013	137	■ 1.000	$K_{\{n\}}, n \geq 5$	$K_n, n \geq 5$	
voloshin-hypergraph_F01014	139	■ 1.000	$x_{\{0\}}, e_{\{1\}}, x_{\{1\}}, e_{\{2\}}, \ldots, e_{\{k\}}, x_{\{k\}}$	$x_0, e_1, x_1, e_2, \ldots, e_k, x_k$	is called closed . If no edge is repeating, then a walk is called a (x_0, x_k) - trail . A trail with no
voloshin-hypergraph_F01015	139	■ 0.999	$x_{\{k\}}$	x_k	the well known concepts as the special cases of a walk and trail. A connected graph G is called Eulerian if it has a closed trail containing all edges of
voloshin-hypergraph_F01016	139	■ 0.999	$\left(x_{\{0\}}, x_{\{k\}\right)}$	$\left(x_0, x_k\right)$	the well known concepts as the special cases of a walk and trail. A connected graph G is called Eulerian if it has a closed trail containing all edges of
voloshin-hypergraph_F01017	139	■ 0.999	$x_{\{0\}}=x_{\{k\}}$	$x_0=x_k$	the well known concepts as the special cases of a walk and trail. A connected graph G is called Eulerian if it has a closed trail containing all edges of
voloshin-hypergraph_F01018	140	■ 1.000	$m\left(G^{\prime}\right)<m(G)$	$m\left(G^{\prime}\right)<m(G)$	way: traverse C and make a detour at the very first vertex of each component. Since every
voloshin-hypergraph_F01019	140	■ 0.562	$\sum_{i=1}^n d\left(x_i\right)=2 m$	$\sum_{i=1}^n d\left(x_i\right)=2 m$	pen-strokes are sufficient to draw G in the plane.
voloshin-hypergraph_F01020	140	■ 1.000	$k / 2$	$k / 2$	1. Given a graph G , recognize if it is Eul
voloshin-hypergraph_F01021	141	■ 1.000	$ X =n \geq 3$	$ X =n \geq 3$	5. For $n \geq 3$, let X be a set of vertices of G such that $d(x)+d(y) \geq n$,
voloshin-hypergraph_F01022	141	■ 1.000	$d\left(x_{\{1\}\right)+d\left(x_{\{n\}\right) \geq n$	$d\left(x_1\right)+d\left(x_n\right) \geq n$	
voloshin-hypergraph_F01023	141	■ 1.000	$d\left(x_{\{1\}\right) \leq n-2$	$d\left(x_1\right) \leq n-2$	

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voloshin-hypergraph_F01024	141	■ 1.000	$d\left(x_n\right) \leq$	$d\left(x_n\right) \leq$	
voloshin-hypergraph_F01025	141	■ 1.000	$n-3$	$n-3$	
voloshin-hypergraph_F01026	142	■ 0.984	$\left(x_i, x_{i+1}\right)$	$\left(x_i, x_{i+1}\right)$	what contradicts the condition of theorem. The last implies that among $n-3$ internal eq of P there exists one, say $\left(x_i, x_{i+1}\right)$ with x_i adjacent to x_{i-1} and x_{i+1} adjacent to x_{i+2} .
voloshin-hypergraph_F01027	142	■ 0.984	x_{i+1}	x_{i+1}	what contradicts the condition of theorem. The last implies that among $n-3$ internal eq of P there exists one, say $\left(x_i, x_{i+1}\right)$ with x_i adjacent to x_{i-1} and x_{i+1} adjacent to x_{i+2} .
voloshin-hypergraph_F01028	142	■ 1.000	$n / 2$	$n / 2$	Proof. The statement follows directly from the theorem.
voloshin-hypergraph_F01029	142	■ 1.000	$d(x)+d(y) \geq n$	$d(x)+d(y) \geq n$	for every pair of vertices in G . □
voloshin-hypergraph_F01030	142	■ 1.000	$\overline{C_7}$	$\overline{C_7}$	following algorithmic problems.
voloshin-hypergraph_F01031	142	■ 1.000	$\overline{C_8}$	$\overline{C_8}$	following algorithmic problems.
voloshin-hypergraph_F01032	143	■ 1.000	$N=(X, A)$	$N=(X, A)$	is called a digraph. A digraph $N=(X, A)$ is called a network , if X is a set of vertices (sometimes called nodes), A is a set of arcs, and to each arc $a \in A$ a non-negative real
voloshin-hypergraph_F01033	143	■ 1.000	$a \in A$	$a \in A$	number $c(a)$ is assigned which is called the capacity of arc a . For any vertex $y \in X$, any
voloshin-hypergraph_F01034	143	■ 1.000	$c(a)$	$c(a)$	arc of type (x, y) is called incoming , and every arc of type (y, z) is called outcoming . In N ,
voloshin-hypergraph_F01035	143	■ 1.000	$y \in X$	$y \in X$	arc of type (x, y) is called incoming , and every arc of type (y, z) is called outcoming . In N ,
voloshin-hypergraph_F01036	143	■ 0.666	(y, z)	(y, z)	there are two special vertices: $u \in X$, which is not incident to any incoming arc and is called
voloshin-hypergraph_F01037	143	■ 0.666	N	N	there are two special vertices: $u \in X$, which is not incident to any incoming arc and is called
voloshin-hypergraph_F01038	143	■ 1.000	$u \in X$	$u \in X$	the source , and $v \in X$, which is not incident to any outcoming arc and is called the sink . A network flow F is an assignment to each arc $a \in A$ a non-negative real number $f(a)$
voloshin-hypergraph_F01039	143	■ 1.000	$v \in X$	$v \in X$	A network flow F is an assignment to each arc $a \in A$ a non-negative real number $f(a)$ (called the flow in a) such that:
voloshin-hypergraph_F01040	143	■ 0.958	F	F	for every vertex $x \in X$
voloshin-hypergraph_F01041	143	■ 0.958	$f(a)$	$f(a)$	for every arc $a \in A$
voloshin-hypergraph_F01042	143	■ 0.991	$f(a) \leq c(a)$	$f(a) \leq c(a)$	$\sum_j f_j$
voloshin-hypergraph_F01043	143	■ 0.999	u	u	An arc $a \in A$ for which $f(a) = c(a)$ is called saturated ; if $f(a) < c(a)$, then arc a is
voloshin-hypergraph_F01044	143	■ 0.999	v	v	An arc $a \in A$ for which $f(a) = c(a)$ is called saturated ; if $f(a) < c(a)$, then arc a is
voloshin-hypergraph_F01045	143	■ 1.000	$f(a)=c(a)$	$f(a)=c(a)$	(x, v) . Given a network $N=(X, A)$ how can we find a maximum flow? Observe that there
voloshin-hypergraph_F01046	143	■ 1.000	$f(a)<c(a)$	$f(a)<c(a)$	(x, v) . Given a network $N=(X, A)$ how can we find a maximum flow? Observe that there
voloshin-hypergraph_F01047	143	■ 1.000	(u, x)	(u, x)	is no other concept as the conservation law for the experiment. Otherwise we could find the maximum flow right from the source. The answer to this question is closely related to the
voloshin-hypergraph_F01048	143	■ 1.000	(x, v)	(x, v)	
voloshin-hypergraph_F01049	143	■ 0.798	$u=x_{\{0\}} \rightarrow x_{\{1\}} \rightarrow x_{\{2\}} \rightarrow \cdots \rightarrow x_{\{k\}}$	$u=x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_k$	
voloshin-hypergraph_F01050	143	■ 0.798	$\left(x_{i+1}, x_i\right)$	$\left(x_{i+1}, x_i\right)$	

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voloshin-hypergraph_F01051	143	■ 0.982	$i=0,1, \ldots, k-1$	$i = 0, 1, \dots, k - 1$	
voloshin-hypergraph_F01052	143	■ 1.000	$u \in Y$	$u \in Y$	
voloshin-hypergraph_F01053	143	■ 1.000	$Z=X \backslash Y$	$Z = X \backslash Y$	
voloshin-hypergraph_F01054	143	■ 1.000	$v \in Z$	$v \in Z$	
voloshin-hypergraph_F01055	143	■ 1.000	$v \in Y$	$v \in Y$	
voloshin-hypergraph_F01056	143	■ 1.000	$u=x_{\{0\}} \rightarrow x_{\{1\}} \rightarrow$	$u = x_0 \rightarrow x_1 \rightarrow$	
voloshin-hypergraph_F01057	143	■ 0.975	$x_{\{2\}} \rightarrow \cdots \rightarrow x_{\{k\}}=v$	$x_2 \rightarrow \cdots \rightarrow x_k = v$	—
voloshin-hypergraph_F01058	143	■ 0.975	$\delta>0$	$\delta > 0$	—
voloshin-hypergraph_F01059	144	■ 0.777	$x_{\{i\}}, x_{\{i+1\}}$	x_i, x_{i+1}	the value needed to estimate any unextracted arc of flow ($x_i \rightarrow x_{i+1}$) and does not exceed the
voloshin-hypergraph_F01060	144	■ 0.997	δ	δ	and decrease it in all arcs of the second type by the same value δ . It is clear that the flow
voloshin-hypergraph_F01061	144	■ 1.000	Z	Z	Now, let S be the set of all arcs with initial vertex in T and terminal vertex in Z . Evi-
voloshin-hypergraph_F01062	144	■ 0.980	$(0,3)$	$(0, 3)$	every arc in each flow. As one can see, the initial flow F_0 is obtained by running one unit of
voloshin-hypergraph_F01063	144	■ 0.980	$\left(u, x_{\{1\}}\right)$	(u, x_1)	every arc in each flow. As one can see, the initial flow F_0 is obtained by running one unit of
voloshin-hypergraph_F01064	144	■ 0.980	$F_{\{0\}}$	F_0	every arc in each flow. As one can see, the initial flow F_0 is obtained by running one unit of
voloshin-hypergraph_F01065	144	■ 0.922	$\left(x_{\{3\}}, x_{\{1\}}\right)$	(x_3, x_1)	backward arc (x_1, x_3) , and adding it in arc the (x_3, v) .
voloshin-hypergraph_F01066	144	■ 1.000	$(1,1)$	$(1, 1)$	Flow F_1 (see the next figure) is obtained from the F_0
voloshin-hypergraph_F01067	144	■ 0.815	$F_{\{1\}}$	F_1	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v.$
voloshin-hypergraph_F01068	144	■ 0.815	$u, x_{\{1\}}$	u, x_1	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v.$
voloshin-hypergraph_F01069	144	■ 0.971	$\left(x_{\{1\}}, x_{\{3\}}\right)$	(x_1, x_3)	Flow F_2 (see the next figure) is obtained from the F_1
voloshin-hypergraph_F01070	144	■ 0.971	$\left(x_{\{3\}}, v\right)$	(x_3, v)	Flow F_3 (see the next figure) is obtained from the F_2
voloshin-hypergraph_F01071	144	■ 1.000	$F_{\{2\}}$	F_2	Flow F_3 (see the next figure) is obtained from the flow F_2 by adding one unit a
voloshin-hypergraph_F01072	144	■ 1.000	$F_{\{3\}}$	F_3	$u \rightarrow v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v$
voloshin-hypergraph_F01073	144	■ 1.000	$F_{\{4\}}$	F_4	augmented the flow. One can easily see that there is no further flow-augmenting path at this
voloshin-hypergraph_F01074	144	■ 0.522	$\left(u, x_{\{1\}}\right), \left(x_{\{3\}}, x_{\{1\}}\right)$	$(u, x_1), (x_3, x_1)$	Exercises 6.3.
voloshin-hypergraph_F01075	144	■ 0.546	$x_{\{4\}}, x_{\{2\}}$	x_4, x_2	
voloshin-hypergraph_F01076	146	■ 1.000	$W_{\{7\}}$	W_7	1. Given a network, find a flow augmenting path
voloshin-hypergraph_F01077	151	■ 0.967	$\mathcal{D}=\left\{D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{m\}}\right\}$	$\mathcal{D} = \{D_1, D_2, \dots, D_m\}$	

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voloshin-hypergraph_F01078	151	■ 1.000	$\mathcal{H}=(X, \mathcal{D})$	$\mathcal{H}=(X, \mathcal{D})$	
voloshin-hypergraph_F01079	151	■ 1.000	$V(\mathcal{H})$	$V(\mathcal{H})$	
voloshin-hypergraph_F01080	151	■ 1.000	$\mathcal{D}(\mathcal{H})$	$\mathcal{D}(\mathcal{H})$	
voloshin-hypergraph_F01081	152	■ 1.000	$n(\mathcal{H})$	$n(\mathcal{H})$	$n(\mathcal{H})$ is called the order of the hypergraph, written also as n or $n(\mathcal{H})$. The elements
voloshin-hypergraph_F01082	152	■ 0.999	$D_1, D_2, \ldots, D_m$	$D_1, D_2, \ldots, D_m$	$ x =n$ is called the order of the hypergraph, which also as n , or $n(\mathcal{H})$. The elements $x_1, x_2, \ldots, x_n$ are called the vertices and the sets $D_1, D_2, \ldots, D_m$ are called the edges (hy-
voloshin-hypergraph_F01083	152	■ 1.000	$m(\mathcal{H})$	$m(\mathcal{H})$	peredges). The number of edges is usually denoted by m or $m(\mathcal{H})$. Sometimes we will
voloshin-hypergraph_F01084	152	■ 1.000	$x_i \in X$	$x_i \in X$	referred to the elements of the same kind (vertices vs vertices, or edges vs edges), while the incidence is referred to the elements of different kind (vertices vs edges).
voloshin-hypergraph_F01085	152	■ 1.000	$D_j \in \mathcal{D}$	$D_j \in \mathcal{D}$	referred to the elements of the same kind (vertices vs vertices, or edges vs edges), while the incidence is referred to the elements of different kind (vertices vs edges).
voloshin-hypergraph_F01086	152	■ 0.996	$\mathcal{D}(x), x \in X$	$\mathcal{D}(x), x \in X$	$\Lambda(\mathcal{H}) = \max \mathcal{D}(x) .$
voloshin-hypergraph_F01087	152	■ 0.996	$ \mathcal{D}(x) $	$ \mathcal{D}(x) $	$\Lambda(\mathcal{H}) = \max \mathcal{D}(x) .$
voloshin-hypergraph_F01088	152	■ 1.000	$ \mathcal{D}(x) $	$ \mathcal{D}(x) $	$\Lambda(\mathcal{H}) = \max_{x \in X} \mathcal{D}(x) .$
voloshin-hypergraph_F01089	152	■ 1.000	D_i	D_i	A hypergraph in which all vertices have the same degree $k \geq 0$ is called k-re-
voloshin-hypergraph_F01090	152	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	A hypergraph in which all vertices have the same degree $k \geq 0$ is called k-re-
voloshin-hypergraph_F01091	152	■ 1.000	$r \geq 0$	$r \geq 0$	degree of an empty edge is trivially 0. A vertex of a hypergraph which is incident to no
voloshin-hypergraph_F01092	152	■ 0.998	$ \mathcal{D}(x) =2$	$ \mathcal{D}(x) =2$	Hypergraph modeling examples. Hypergraphs can model concepts in different settings in a much more general setting than graphs do. In addition, they help to find optimal
voloshin-hypergraph_F01093	152	■ 0.998	$D_i \in \mathcal{D}$	$D_i \in \mathcal{D}$	Hypergraph modeling examples. Hypergraphs can model concepts in different settings in a much more general setting than graphs do. In addition, they help to find optimal
voloshin-hypergraph_F01094	152	■ 0.998	$\mathcal{H}_1$	$\mathcal{H}_1$	the hyperedges represent properties of different subsets, or, even more generally, <i>arbitrary</i>
voloshin-hypergraph_F01095	152	■ 0.998	$\mathcal{H}_2$	$\mathcal{H}_2$	the hyperedges represent properties of different subsets, or, even more generally, <i>arbitrary</i>
voloshin-hypergraph_F01096	155	■ 1.000	$I(\mathcal{H})$	$I(\mathcal{H})$	an example of the incidence matrix is shown in Figure 10.1.1. One can see, in contrast to graphs, any (0,1)-matrix is the incidence matrix of a hypergraph. From this point of view,
voloshin-hypergraph_F01097	156	■ 1.000	$\mathcal{D}=\left\{D_1, D_2, \ldots\right\}.$	$\mathcal{D}=\left\{D_1, D_2, \ldots\right\}.$	$D_m\}$. The dual of the hypergraph $\mathcal{H}$ is a hypergraph $\mathcal{H}^*=(Y, \mathcal{Z})$ whose vertex set is
voloshin-hypergraph_F01098	156	■ 1.000	$\left.D_m\right\}$	$D_m\}$	$Y=\left\{d_1, d_2, \ldots, d_m\right\}$, and the edge set is defined as follows:
voloshin-hypergraph_F01099	156	■ 1.000	$\mathcal{H}^*=(Y, \mathcal{Z})$	$\mathcal{H}^*=(Y, \mathcal{Z})$	$Y=\left\{d_1, d_2, \ldots, d_m\right\}$, and the edge set is defined as follows:
voloshin-hypergraph_F01100	156	■ 1.000	$Y=\left\{d_1, d_2, \ldots, d_m\right\}$	$Y=\left\{d_1, d_2, \ldots, d_m\right\}$	$\mathcal{Z}=\left\{X_1, X_2, \ldots, X_r\right\}$
voloshin-hypergraph_F01101	156	■ 1.000	$I\left(\mathcal{H}^*\right)$	$I\left(\mathcal{H}^*\right)$	same matrix, therefore $\left(\mathcal{H}^*\right)^*=\mathcal{H}$. Following the agreement above this holds even in the
voloshin-hypergraph_F01102	156	■ 1.000	I^*	I^*	same matrix, therefore $\left(\mathcal{H}^*\right)^*=\mathcal{H}$. Following the agreement above this holds even in the
voloshin-hypergraph_F01103	156	■ 1.000	$\mathcal{H}^*$	$\mathcal{H}^*$	case when X or/and $\mathcal{D}$ are empty: the column becomes a row and vice versa. Notice that
voloshin-hypergraph_F01104	156	■ 1.000	$I\left(\mathcal{H}^*\right)=I^*\left(\mathcal{H}\right)$	$I\left(\mathcal{H}^*\right)=I^*\left(\mathcal{H}\right)$	$\Delta(\mathcal{H})=r\left(\mathcal{H}^*\right)$ and the dual of a k -regular hypergraph is k -uniform. The definition above implies that duality dual hypergraph is a finite combinatorial object with a duality of its

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voloshin-hypergraph_F01105	156	■ 1.000	$\left(\mathcal{H}^*\right)^*=\mathcal{H}$	$(\mathcal{H}^*)^* = \mathcal{H}$	implies that drawing dual hypergraph is a simple procedure. begin with a drawing of $\mathcal{H} \rightarrow$ construct incidence matrix $I(\mathcal{H}) \rightarrow$ transpose it to obtain $I^*(\mathcal{H}) \rightarrow$ draw dual $\mathcal{H}^*$
voloshin-hypergraph_F01106	156	■ 1.000	$\Delta(\mathcal{H})=r\left(\mathcal{H}^*\right)$	$\Delta(\mathcal{H}) = r(\mathcal{H}^*)$	bipartite graph $B(\mathcal{H})=(X, \mathcal{D} ; E)$ with the vertex set $X \cup \mathcal{D}$, where X is the left part, $\mathcal{D}$ is
voloshin-hypergraph_F01107	156	■ 0.869	$I(\mathcal{H}) \rightarrow$	$I(\mathcal{H}) \Rightarrow$	and only if vertex $x \in X$ is incident to edge $D \in \mathcal{D}$ in $\mathcal{H}$. In this way, any bipartite graph is a bipartite representation of a hypergraph
voloshin-hypergraph_F01108	156	■ 0.869	$I^*\left(\mathcal{H}\right) \rightarrow$	$I^*(\mathcal{H}) \Rightarrow$	and only if vertex $x \in X$ is incident to edge $D \in \mathcal{D}$ in $\mathcal{H}$. In this way, any bipartite graph is a bipartite representation of a hypergraph
voloshin-hypergraph_F01109	156	■ 1.000	$B(\mathcal{H})=(X, \mathcal{D} ; E)$	$B(\mathcal{H})=(X, \mathcal{D} ; E)$	the left part while preserving all edges.
voloshin-hypergraph_F01110	156	■ 1.000	$X \cup \mathcal{D}$	$X \cup \mathcal{D}$	the left part while preserving all edges.
voloshin-hypergraph_F01111	156	■ 1.000	$B(\mathcal{H})$	$B(\mathcal{H})$	The very same hypergraph $\mathcal{H}$, see Figure 7.1, its incidence matrix $I(\mathcal{H})$, dual hyper-
voloshin-hypergraph_F01112	156	■ 1.000	$B\left(\mathcal{H}^*\right)$	$B\left(\mathcal{H}^*\right)$	
voloshin-hypergraph_F01113	156	■ 1.000	D_1	D_1	
voloshin-hypergraph_F01114	156	■ 1.000	$\left D_1\right =1$	$\left D_1\right =1$	
voloshin-hypergraph_F01115	156	■ 1.000	D_2	D_2	
voloshin-hypergraph_F01116	156	■ 1.000	D_3	D_3	
voloshin-hypergraph_F01117	156	■ 0.817	d_1	d_1	—
voloshin-hypergraph_F01118	157	■ 0.999	$B(\mathcal{H})([6])$	$B(\mathcal{H})([6])$	Every picture of a
voloshin-hypergraph_F01119	158	■ 1.000	$B(\mathcal{H})=$	$B(\mathcal{H})=$	of the equality. Evidently, they coincide because both are equal to the number of edges in
voloshin-hypergraph_F01120	158	■ 1.000	$(X, \mathcal{D} ; E)$	$(X, \mathcal{D} ; E)$	$B(\mathcal{H})$. □
voloshin-hypergraph_F01121	159	■ 1.000	$n \times n$	$n \times n$	$\mathcal{H}$, denoted by $A(\mathcal{H})$, if
voloshin-hypergraph_F01122	159	■ 1.000	$A=\left(a_{i j}\right)$	$A=\left(a_{i j}\right)$	$\mathcal{H}$, denoted by $A(\mathcal{H})$, if
voloshin-hypergraph_F01123	159	■ 1.000	$A(\mathcal{H})$	$A(\mathcal{H})$	
voloshin-hypergraph_F01124	159	■ 1.000	$(0,1)$	$(0,1)$	7. Construct the edge list for graph G and its dual G^* in Figure 7.4.
voloshin-hypergraph_F01125	159	■ 1.000	$B(B(\mathcal{H}))$	$B(B(\mathcal{H}))$	Computer Projects 7.2. Using an appropriate hypergraph representation, write a program for the following algorithmic problems.
voloshin-hypergraph_F01126	159	■ 1.000	$B(G)$	$B(G)$	Computer Projects 7.2. Using an appropriate hypergraph representation, write a program for the following algorithmic problems.
voloshin-hypergraph_F01127	159	■ 1.000	$E_{\{n\}}, K_{\{n\}}, P_{\{n\}}, C_{\{n\}}$	E_n, K_n, P_n, C_n	1. Given the incidence matrix of a hypergraph, construct the edge lis
voloshin-hypergraph_F01128	159	■ 1.000	$W_{\{n\}}, n=3,4,5,6,7$	$W_n, n=3,4,5,6,7$	1. Given the incidence matrix of a hypergraph, construct the edge lis
voloshin-hypergraph_F01129	160	■ 1.000	$0 \leq r \leq n$	$0 \leq r \leq n$	be the simple hypergraph $\mathcal{H}_n=(X, \mathcal{E})$ such that $ x =n$ and $\mathcal{E}=\left\{x_n\right\}$ consists of all the r -subsets of X . Thus a complete graph on n vertices is a complete 2-uniform hypergraph K_n^2
voloshin-hypergraph_F01130	160	■ 1.000	$K_{\{n\}}^r=(X, \mathcal{D})$	$K_n^r=(X, \mathcal{D})$	also denoted by K_n . The complete r -uniform hypergraph and the family of its edges both
voloshin-hypergraph_F01131	160	■ 1.000	$\mathcal{D}\left(K_{\{n\}}^r\right)$	$\mathcal{D}\left(K_n^r\right)$	also denoted by K_n . The complete r -uniform hypergraph and the family of its edges both

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voloshin-hypergraph_F01132	160	■ 0.999	$K_{\{n\}}^{\{2\}}$	K_n^2	are denoted by K_n^r , i.e. $K_n^r = (X, K_n^r)$, where $ X = n$.
voloshin-hypergraph_F01133	160	■ 1.000	$K_{\{n\}}^{\{r\}}$	K_n^r	draw an empty edge to emphasize the empty set. As r edges is $1 + 4 + 6 + 4 + 1 = 16 = 2^4 = 2^n$ what is the
voloshin-hypergraph_F01134	160	■ 1.000	$K_{\{n\}}^{\{r\}}=\left(X, K_{\{n\}}^{\{r\}}\right)$	$K_n^r = (X, K_n^r)$	draw an empty edge to emphasize the empty set. As r edges is $1 + 4 + 6 + 4 + 1 = 16 = 2^4 = 2^n$ what is the
voloshin-hypergraph_F01135	160	■ 1.000	$K_{\{4\}}^{\{0\}}$	K_4^0	edges is $1 + 4 + 6 + 4 + 1 = 16 = 2^4 = 2^n$ what is the number of all subsets on the set of four vertices. This holds for any $n \geq 0$ and all r such that $0 \leq r \leq n$. It is important to notice
voloshin-hypergraph_F01136	160	■ 1.000	$1+4+6+4+1=16=2^{\{4\}}=2^{\{n\}}$	$1 + 4 + 6 + 4 + 1 = 16 = 2^4 = 2^n$	$\frac{1}{(n)!} = \frac{1}{n!} = \frac{n(n-1)(n-2)\dots(n-r+1)}{n!}$
voloshin-hypergraph_F01137	160	■ 1.000	$n \geq 0$	$n \geq 0$	$ \mathcal{D}(K_n^r) = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n!}{r!} \frac{1}{(n-r)!} = \frac{n!}{r!} \frac{1}{(n-r)!}$.
voloshin-hypergraph_F01138	160	■ 1.000	$K_{\{4\}}^{\{3\}}$	K_4^3	Among all complete hyp
voloshin-hypergraph_F01139	160	■ 1.000	$K_{\{4\}}^{\{2\}}$	K_4^2	of distinct vertices $x_0, x_1, x_2, \dots, x_{t-1}$ and distinct edges $D_0, D_1, D_2, \dots, D_{t-1}$ satisfying
voloshin-hypergraph_F01140	160	■ 1.000	$\binom{n-1}{r-1}$	$\binom{n-1}{r-1}$	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$, is called a path connecting the vertices x_0 and x_t , or, equiv-
voloshin-hypergraph_F01141	160	■ 1.000	$x_{\{0\}}, x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{t-1\}}$	$x_0, x_1, x_2, \dots, x_{t-1}$	Connected hypergraphs. The hypergraph $\mathcal{H} = (X, \mathcal{D})$ is called connected if for any pair of its vertices there is a path connecting them. If $\mathcal{H}$ is not connected, then it consists of
voloshin-hypergraph_F01142	160	■ 1.000	$D_{\{0\}}, D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{t-1\}}$	$D_0, D_1, D_2, \dots, D_{t-1}$	Connected hypergraphs. The hypergraph $\mathcal{H} = (X, \mathcal{D})$ is called connected if for any pair of its vertices there is a path connecting them. If $\mathcal{H}$ is not connected, then it consists of
voloshin-hypergraph_F01143	160	■ 1.000	$x_{\{i\}}, x_{\{i+1\}} \in D_{\{i\}}, i=0,1, \ldots, t-1$	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$	two or more connected components each of which is a connected hypergraph. An isolated
voloshin-hypergraph_F01144	160	■ 1.000	$x_{\{t\}}$	x_t	two or more connected components each of which is a connected hypergraph. An isolated
voloshin-hypergraph_F01145	160	■ 1.000	$\left(x_{\{0\}}, x_{\{t\}}\right)$	(x_0, x_t)	vertex, a vertex incident to loops only and an empty edge are also considered connected
voloshin-hypergraph_F01146	160	■ 1.000	$x_{\{t\}}=x_{\{0\}}$	$x_t = x_0$	vertex, a vertex incident to loops only and an empty edge are also considered connected
voloshin-hypergraph_F01147	160	■ 0.831	$\mu_{\{1\}}=1 D_{\{2\}}^2 D_{\{4\}}^3 D_{\{5\}}^5$	$\mu_1 = 1D_2^2D_4^3D_5^5$	
voloshin-hypergraph_F01148	160	■ 1.000	$\mu_{\{2\}}=1 D_{\{2\}}^2 D_{\{4\}}^3 D_{\{5\}}^4 D_{\{3\}}^1$	$\mu_2 = 1D_2^2D_4^3D_5^4D_3^1$	
voloshin-hypergraph_F01149	160	■ 1.000	$\mathcal{H}, \mathcal{H}^{\{*\}}$	$\mathcal{H}, \mathcal{H}^*$	—
voloshin-hypergraph_F01150	161	■ 1.000	$K_{\{4\}}^{\{1\}}$	K_4^1	X can be partitioned into two disjoint sets X_1 and X_2 (called parts) in such a way that each hyperedge of cardinality ≥ 2 contains vertices from both parts. It means that there is no such
voloshin-hypergraph_F01151	161	■ 1.000	$K_{\{4\}}^{\{2\}}, K_{\{4\}}^{\{3\}}$	K_4^2, K_4^3	hyperedge of cardinality ≥ 2 contains vertices from both parts. It means that there is no such hyperedge inside X_1 and there is no such hyperedge inside X_2 . In other words, the vertices
voloshin-hypergraph_F01152	161	■ 1.000	$K_{\{4\}}^{\{4\}}$	K_4^4	hyperedge of cardinality ≥ 2 contains vertices from both parts. It means that there is no such hyperedge inside X_1 and there is no such hyperedge inside X_2 . In other words, the vertices
voloshin-hypergraph_F01153	161	■ 1.000	≥ 2	≥ 2	no role in bipartition, they are usually ignored when discussing the properties of 2-colorable hypergraphs
voloshin-hypergraph_F01154	161	■ 1.000	$K_{\{n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{r\}}\}}^{\{r\}}$	$K_{n_1, n_2, \dots, n_r}^r$	
voloshin-hypergraph_F01155	161	■ 1.000	$n_{\{i\}}$	n_i	
voloshin-hypergraph_F01156	161	■ 0.992	$\mathcal{H}^{\{\prime\}}=\left(X^{\{\prime\}}, \mathcal{D}^{\{\prime\}}\right)$	$\mathcal{H}' = (X', \mathcal{D}')$	
voloshin-hypergraph_F01157	161	■ 0.992	$\mathcal{H} \cong \mathcal{H}^{\{\prime\}}$	$\mathcal{H} \cong \mathcal{H}'$	
voloshin-hypergraph_F01158	161	■ 0.999	$\mathcal{D}^{\{\prime\}}$	$\mathcal{D}'$	—

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voloshin-hypergraph_F01185	167	■ 0.999	$\mathcal{D}-\mathcal{D}'$	$\mathcal{D}-\mathcal{D}'$	that the order in which the vertices and edges are deleted is not important.
voloshin-hypergraph_F01186	167	■ 0.978	$\mathcal{H}:\mathcal{H}_1$	$\mathcal{H}:\mathcal{H}_1$	Induced subhypergraphs. A hypergraph $\mathcal{H}'=(X',\mathcal{D}')$ is called an induced subhy-
voloshin-hypergraph_F01187	167	■ 0.996	$\mathcal{H}_Y$	$\mathcal{H}_Y$	Stable (independent) sets. Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph. A
voloshin-hypergraph_F01188	167	■ 0.996	$Y=\{2,3,4\}$	$Y=\{2,3,4\}$	Stable (independent) sets. Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph. A
voloshin-hypergraph_F01189	167	■ 0.996	$\mathcal{H}_1=\mathcal{H}_Y$	$\mathcal{H}_1=\mathcal{H}_Y$	Stable (independent) sets. Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph. A
voloshin-hypergraph_F01190	167	■ 1.000	$\mathcal{H}^{\prime}=\left(X,\mathcal{D}^{\prime}\right)$	$\mathcal{H}'=(X,\mathcal{D}')$	maximal by inclusion stable sets. The largest size of a stable set over all maximal by
voloshin-hypergraph_F01191	167	■ 0.743	$\alpha(\mathcal{H})$	$\alpha(\mathcal{H})$	
voloshin-hypergraph_F01192	167	■ 0.946	$1\leq\alpha(\mathcal{H})\leq X $	$1\leq\alpha(\mathcal{H})\leq X $	
voloshin-hypergraph_F01193	167	■ 0.976	$\alpha(\mathcal{H})=3$	$\alpha(\mathcal{H})=3$	
voloshin-hypergraph_F01194	168	■ 1.000	$ S\cap D \leq 1$	$ S\cap D \leq 1$	for a graph G , $\alpha(G)=\bar{\alpha}(G)$.
voloshin-hypergraph_F01195	168	■ 1.000	$D\in$	$D\in$	for a graph G , $\alpha(G)=\bar{\alpha}(G)$.
voloshin-hypergraph_F01196	168	■ 1.000	$\bar{\alpha}(\mathcal{H})$	$\bar{\alpha}(\mathcal{H})$	Transversals. A set $T\subseteq X$ is called a transversal of a hypergraph $\mathcal{H}=(X,\mathcal{D})$ if
voloshin-hypergraph_F01197	168	■ 1.000	$G,\alpha(G)=\bar{\alpha}(G)$	$G,\alpha(G)=\bar{\alpha}(G)$	$ T\cap D \geq 1$ for every edge $D\in\mathcal{D}$. The cardinality of a minimum transversal is denoted by
voloshin-hypergraph_F01198	168	■ 1.000	$ T\cap D \geq 1$	$ T\cap D \geq 1$	$\tau(\mathcal{H})$. Following the defini
voloshin-hypergraph_F01199	168	■ 1.000	$\tau(\mathcal{H})$	$\tau(\mathcal{H})$	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X $.
voloshin-hypergraph_F01200	168	■ 1.000	$S=X-T$	$S=X-T$	In Figure 7.12, vertices 4 and 5 form a minimum transversal, so $\tau=2$. The comple-
voloshin-hypergraph_F01201	168	■ 1.000	$\tau=2$	$\tau=2$	In Figure 7.12, vertices 4 and 5 form a minimum transversal, so $\tau=2$. The comple-
voloshin-hypergraph_F01202	168	■ 0.822	$\alpha(\mathcal{H})+\tau(\mathcal{H})=3+2=5= X $	$\alpha(\mathcal{H})+\tau(\mathcal{H})=3+2=5= X $	hypergraph without edges, i.e. an empty hypergraph. Transversals are sometimes called
voloshin-hypergraph_F01203	168	■ 0.995	$\mathcal{V}(\mathcal{H})$	$\mathcal{V}(\mathcal{H})$	Matchings. In a hypergraph $\mathcal{H}$, a set
voloshin-hypergraph_F01204	168	■ 0.956	$\nu(\mathcal{H})=\bar{\alpha}(\mathcal{H}^*)$	$\nu(\mathcal{H})=\bar{\alpha}(\mathcal{H}^*)$	common is called a matching . A perfect
voloshin-hypergraph_F01205	168	■ 0.940	$\{1,2,5\}$	$\{1,2,5\}$	We say that $\mathcal{H}$ satisfies the König property if
voloshin-hypergraph_F01206	168	■ 0.940	$\mathcal{V}(\mathcal{H})=2$	$\mathcal{V}(\mathcal{H})=2$	we say that $\mathcal{H}$ satisfies the König property if
voloshin-hypergraph_F01207	168	■ 0.983	$\tau(\mathcal{H})\geq 2=v(\mathcal{H})$	$\tau(\mathcal{H})\geq 2=v(\mathcal{H})$	
voloshin-hypergraph_F01208	168	■ 0.983	$\tau(\mathcal{H})=v(\mathcal{H}),\mathcal{H}$	$\tau(\mathcal{H})=v(\mathcal{H}),\mathcal{H}$	
voloshin-hypergraph_F01209	169	■ 1.000	$\rho(\mathcal{H})$	$\rho(\mathcal{H})$	a transversal in $\mathcal{H}^*$ and vice versa, therefore
voloshin-hypergraph_F01210	169	■ 1.000	$\rho(\mathcal{H})=\bar{\alpha}(\mathcal{H})$	$\rho(\mathcal{H})=\bar{\alpha}(\mathcal{H})$	so $\rho(\mathcal{H})=2$. Vertices 1 and 3 form a maximum strongly independent se
voloshin-hypergraph_F01211	169	■ 0.522	$\rho(\mathcal{H})=2$	$\rho(\mathcal{H})=2$	Graph C_5 is an example of a hypergraph with $\tau=3>v$ and $\rho=3>\bar{\alpha}=\alpha=2$.

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voloshin-hypergraph_F01212	169	0.522	$\bar{\alpha}(\mathcal{H})=2$	$\bar{\alpha}(\mathcal{H}) = 2$	Graph C_5 is an example of a hypergraph with $\tau = 3 > 2 = v$ and $p = 3 > \bar{\alpha} = \alpha = 2$.
voloshin-hypergraph_F01213	169	0.991	$\tau=3>2=\nu$	$\tau = 3 > 2 = \nu$	Hypergraph minor: A hypergraph $\mathcal{H}'$ is a minor of a hypergraph $\mathcal{H}$ if it can be obtained from $\mathcal{H}$ by a sequence of weak or strong vertex or edge deletions. identification of
voloshin-hypergraph_F01214	169	0.991	$\rho=3>\bar{\alpha}=\alpha=2$	$\rho = 3 > \bar{\alpha} = \alpha = 2$	Hypergraph minor: A hypergraph $\mathcal{H}'$ is a minor of a hypergraph $\mathcal{H}$ if it can be obtained from $\mathcal{H}$ by a sequence of weak or strong vertex or edge deletions. identification of
voloshin-hypergraph_F01215	169	1.000	$X^{\prime}=\{1,2,3,4,5,7,8\}$	$X' = \{1, 2, 3, 4, 5, 7, 8\}$	in each hypergraph $\mathcal{H}$ induced by vertex subsets $X = \{x_1, x_2, x_3\}$.
voloshin-hypergraph_F01216	169	1.000	$\mathcal{D}^{\prime} =$	$\mathcal{D}' =$	in each hypergraph $\mathcal{H}$ induced by vertex subsets $X = \{x_1, x_2, x_3\}$.
voloshin-hypergraph_F01217	169	0.997	$v(\mathcal{H})$	$v(\mathcal{H})$	
voloshin-hypergraph_F01218	170	0.531	$\mathcal{v}(\mathcal{H})$	$\sqsubseteq(\mathcal{H})$	7. Find $\wp$
voloshin-hypergraph_F01219	170	1.000	$L(\mathcal{H})$	$L(\mathcal{H})$	$L(\mathcal{H}) = (\mathcal{D}, \mathcal{E})$, where $(D_i, D_j) \in \mathcal{E} \Leftrightarrow D_i \cap D_j \neq \emptyset$.
voloshin-hypergraph_F01220	170	1.000	$(\mathcal{H})_2$	$(\mathcal{H})_2$	Theorem 7.6.1 For any hypergraph $\mathcal{H}$,
voloshin-hypergraph_F01221	170	1.000	$L\left(\mathcal{H}^*\right)$	$L\left(\mathcal{H}^*\right)$	$L(\mathcal{H}) = (\mathcal{H}^*)_2$.
voloshin-hypergraph_F01222	171	0.931	$\alpha\left((\mathcal{H})_2\right) \cdot \square$	$\alpha\left((\mathcal{H})_2\right) \cdot \square$	
voloshin-hypergraph_F01223	171	0.684	$v\left(\mathcal{H}^*\right)=2$	$v\left(\mathcal{H}^*\right) = 2$	C_4 on vertices a, b, c, d by adding diagonal bd . Edges a and c in $\mathcal{H}$ form a matching with
voloshin-hypergraph_F01224	171	0.999	$\alpha\left(L\left(\mathcal{H}^*\right)\right)=2$	$\alpha\left(L\left(\mathcal{H}^*\right)\right) = 2$	with $\alpha(L$
voloshin-hypergraph_F01225	171	1.000	$\left(\mathcal{H}^*\right)_2$	$(\mathcal{H}^*)_2$	and $(\mathcal{H})_2$ are different graphs. A hypergraph $\mathcal{H}$ has the Helly property (is Helly , for short) if for every subfamily of its
voloshin-hypergraph_F01226	171	0.994	a, b, c, d	a, b, c, d	A hypergraph $\mathcal{H}$ has the Helly property (is Helly , for short) if for every subfamily of its edges the following implication holds: if every two edges of the subfamily have a nonempty
voloshin-hypergraph_F01227	171	0.994	$b d$	bd	A hypergraph $\mathcal{H}$ has the Helly property (is Helly , for short) if for every subfamily of its edges the following implication holds: if every two edges of the subfamily have a nonempty
voloshin-hypergraph_F01228	171	1.000	$\alpha(L(\mathcal{H}))=2$	$\alpha(L(\mathcal{H})) = 2$	A hypergraph $\mathcal{H}$ is called an intersecting family if all of its edges pairwise intersect. It
voloshin-hypergraph_F01229	171	0.574	$\boldsymbol{\omega}(L(\mathcal{H}))$	$\omega(L(\mathcal{H}))$	problem of finding a minimum transversal of a hypergraph $\mathcal{H}$, or, equivalently, a maximum
voloshin-hypergraph_F01230	172	1.000	$\tau(\mathcal{H})=\theta(L(\mathcal{H}))$	$\tau(\mathcal{H}) = \theta(L(\mathcal{H}))$	know, the last is equivalent to finding the chromatic number $\chi(L(\mathcal{H}))$ We can see that there is a relation between transversals of $\mathcal{H}$ and feasible partitions of the complementary graph
voloshin-hypergraph_F01231	172	1.000	$\chi(\overline{L(\mathcal{H})})$	$\chi(\overline{L(\mathcal{H})})$	finding $\tau(\mathcal{H})$.
voloshin-hypergraph_F01232	172	1.000	$\overline{L(\mathcal{H})}$	$\overline{L(\mathcal{H})}$	("triangle") which is not a star. Let us extend edge d and consider a hypergraph $\mathcal{H}$ in Figure 7.14. It becomes Helly with $L(\mathcal{H})$ and $\overline{L(\mathcal{H})}$ as in Figure 7.14. One can see that
voloshin-hypergraph_F01233	172	1.000	b, c	b, c	every edge (not necessarily maximal in size) and a corresponding vertex $x \in e$ and one minimum covering by cliques of $L(\mathcal{H})$ is: $\{a, b, d\}$ and $\{c\}$. It means that one minimal
voloshin-hypergraph_F01234	172	0.998	$\{a, b, d\}$	$\{a, b, d\}$	and vertex c with the second. There are several such optimal solutions. A hypergraph $\mathcal{H}$ is called conformal if all the maximal cliques of the graph $(\mathcal{H})_2$ are all
voloshin-hypergraph_F01235	172	0.998	$\{c\}$	$\{c\}$	and vertex c with the second. There are several such optimal solutions. A hypergraph $\mathcal{H}$ is called conformal if all the maximal cliques of the graph $(\mathcal{H})_2$ are all
voloshin-hypergraph_F01236	172	1.000	$\tau(\mathcal{H})=\theta(L(\mathcal{H}))=2$	$\tau(\mathcal{H}) = \theta(L(\mathcal{H})) = 2$	if and only if the edges of $\mathcal{H}$ are precisely the maximal cliques of 2-section $(\mathcal{H})_2$. In other
voloshin-hypergraph_F01237	172	0.998	$(\overline{L(\mathcal{H})})$	$(\overline{L(\mathcal{H})})$	words, any conformal hypergraph may be obtained from a simple graph by taking the set of
voloshin-hypergraph_F01238	172	0.998	$\chi(\overline{L(\mathcal{H})})=2$	$\chi(\overline{L(\mathcal{H})}) = 2$	words, any conformal hypergraph may be obtained from a simple graph by taking the set of

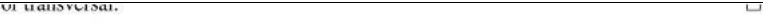
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01239	172	■ 1.000	$Y \subseteq D$	$Y \subseteq D$	
voloshin-hypergraph_F01240	172	■ 0.913	$G=(\mathcal{H})_2$	$G = (\mathcal{H})_2$	
voloshin-hypergraph_F01241	172	■ 1.000	$Z \subseteq X$	$Z \subseteq X$	
voloshin-hypergraph_F01242	172	■ 1.000	$Y \subseteq Z$	$Y \subseteq Z$	
voloshin-hypergraph_F01243	172	■ 1.000	$G_{\{Z\}}$	G_Z	
voloshin-hypergraph_F01244	172	■ 1.000	$Z=D$	$Z = D$	
voloshin-hypergraph_F01245	173	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	family $\mathcal{D}'$ of maximal edges of $\mathcal{H}$ and the family $\mathcal{C}$ of maximal cliques of G coincide, i.e.
voloshin-hypergraph_F01246	173	■ 1.000	$\mathcal{D}^{\prime}=\mathcal{C}$	$\mathcal{D}' = \mathcal{C}$	$\mathcal{D}$
voloshin-hypergraph_F01247	173	■ 1.000	$\mathcal{D}^{\prime} \subseteq \mathcal{C}$	$\mathcal{D}' \subseteq \mathcal{C}$	Case 1: Prove that $\mathcal{D}' \subseteq \mathcal{C}$. Let $D \in \mathcal{D}'$. By definition of G , all vertices of D are
voloshin-hypergraph_F01248	173	■ 1.000	$D \in \mathcal{D}^{\prime}$	$D \in \mathcal{D}'$	Case 1: Prove that $\mathcal{D}' \subseteq \mathcal{C}$. Let $D \in \mathcal{D}'$. By definition of G , all vertices of D are
voloshin-hypergraph_F01249	173	■ 0.615	$G_{\{D\}}$	G_D	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$. By the condition of Lemma, for C there exists an edge D' such that $C \subset D'$. So we have
voloshin-hypergraph_F01250	173	■ 0.615	$C \in \mathcal{C}$	$C \in \mathcal{C}$	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$. By the condition of Lemma, for C there exists an edge D' such that $C \subset D'$. So we have
voloshin-hypergraph_F01251	173	■ 0.615	$D \subseteq C$	$D \subseteq C$	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$. By the condition of Lemma, for C there exists an edge D' such that $C \subset D'$. So we have
voloshin-hypergraph_F01252	173	■ 1.000	$C \subseteq D^{\prime}$	$C \subseteq D'$	By the condition of Lemma, for C there exists an edge D' such that $C \subseteq D'$. So we have $D \subset C \subset D'$. But both D and D' are maximal what implies that $D = C = D'$ and $D \in \mathcal{C}$.
voloshin-hypergraph_F01253	173	■ 0.973	$D \subseteq C \subseteq D^{\prime}$	$D \subseteq C \subseteq D'$	Consequently, $\mathcal{D}' \subseteq \mathcal{C}$.
voloshin-hypergraph_F01254	173	■ 0.973	$D=C=D^{\prime}$	$D = C = D'$	Consequently, $\mathcal{D}' \subseteq \mathcal{C}$.
voloshin-hypergraph_F01255	173	■ 0.973	$D \in \mathcal{C}$	$D \in \mathcal{C}$	Consequently, $\mathcal{D}' \subseteq \mathcal{C}$.
voloshin-hypergraph_F01256	173	■ 1.000	$\mathcal{C} \subseteq \mathcal{D}^{\prime}$	$\mathcal{C} \subseteq \mathcal{D}'$	of $\mathcal{C}$ are pairwise adjacent. By the condition of Lemma, there exist an edge $D \in \mathcal{D}$ with $C \subset D \subset D'$. It follows that $D = D'$ and $C \subset D'$. So we have
voloshin-hypergraph_F01257	173	■ 1.000	$C \in \mathcal{C}$	$C \in \mathcal{C}$	of $\mathcal{C}$ are pairwise adjacent. By the condition of Lemma, there exist an edge $D \in \mathcal{D}$ with $C \subset D \subset D'$. It follows that $D = D'$ and $C \subset D'$. So we have
voloshin-hypergraph_F01258	173	■ 1.000	$C \subseteq D$	$C \subseteq D$	pairwise adjacent in G , there exists a maximal clique C' such that $D \subseteq C'$. Since both C and C' are maximal, and $C \subset D' \subset C'$, it follows that $C = D = C'$ and $C \in \mathcal{D}'$. Consequently,
voloshin-hypergraph_F01259	173	■ 1.000	$D \subseteq D^{\prime}$	$D \subseteq D'$	pairwise adjacent in G , there exists a maximal clique C' such that $D \subseteq C'$. Since both C and C' are maximal, and $C \subset D' \subset C'$, it follows that $C = D = C'$ and $C \in \mathcal{D}'$. Consequently,
voloshin-hypergraph_F01260	173	■ 1.000	$C^{\prime}$	C'	$\mathcal{C} \subseteq \mathcal{D}'$. □
voloshin-hypergraph_F01261	173	■ 1.000	$D^{\prime} \subseteq C^{\prime}$	$D' \subseteq C'$	$\mathcal{C} \subseteq \mathcal{D}'$. □
voloshin-hypergraph_F01262	173	■ 1.000	$C \subseteq D^{\prime} \subseteq C^{\prime}$	$C \subseteq D' \subseteq C'$	... $\mathcal{D}$ is a family of maximal edges of $\mathcal{H}$ and $\mathcal{C}$ is a family of maximal cliques of G .
voloshin-hypergraph_F01263	173	■ 1.000	$C=D=C^{\prime}$	$C = D = C'$	... $\mathcal{D}$ is a family of maximal edges of $\mathcal{H}$ and $\mathcal{C}$ is a family of maximal cliques of G .
voloshin-hypergraph_F01264	173	■ 1.000	$C \in \mathcal{D}^{\prime}$	$C \in \mathcal{D}'$	... $\mathcal{D}$ is a family of maximal edges of $\mathcal{H}$ and $\mathcal{C}$ is a family of maximal cliques of G .
voloshin-hypergraph_F01265	173	■ 1.000	$D_{\{1\}}, D_{\{2\}}, D_{\{3\}}$	D_1, D_2, D_3	

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voloshin-hypergraph_F01266	173	■ 1.000	$\left(D_{\{1\}} \cap D_{\{2\}}\right) \cup \left(D_{\{1\}} \cap D_{\{3\}}\right) \cup \left(D_{\{2\}} \cap D_{\{3\}}\right)$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3)$	7. Assume that the induction 7.4 holds and show that then the condition of Lemma 7.6.1
voloshin-hypergraph_F01267	173	■ 1.000	$ Y $	$ Y $	Since $ Y \geq 3$, choose vertices $x_1, x_2, x_3 \in Y$ and put $Y_i = Y - \{x_i\}, i = 1, 2, 3$. Notice that
voloshin-hypergraph_F01268	173	■ 1.000	$ Y =1$	$ Y = 1$	$Y = (Y_1 \cap Y_2) \cup (Y_1 \cap Y_3) \cup (Y_2 \cap Y_3)$
voloshin-hypergraph_F01269	173	■ 1.000	$ Y =2$	$ Y = 2$	$Y = (Y_1 \cap Y_2) \cup (Y_1 \cap Y_3) \cup (Y_2 \cap Y_3)$
voloshin-hypergraph_F01270	173	■ 1.000	$ Y <k, \ k \geq 3$	$ Y < k, k \geq 3$	By the induction hypothesis, there exist edges D_1, D_2, D_3 such that $Y_i \subseteq D_i, i = 1, 2, 3$.
voloshin-hypergraph_F01271	173	■ 1.000	$ Y \geq 3$	$ Y \geq 3$	$Y \subseteq (D_1 \cup D_2) \cup (D_1 \cup D_3) \cup (D_2 \cup D_3)$.
voloshin-hypergraph_F01272	173	■ 1.000	$x_{\{1\}}, \ x_{\{2\}}, \ x_{\{3\}} \in Y$	$x_1, x_2, x_3 \in Y$	$Y \subseteq (D_1 \cup D_2) \cup (D_1 \cup D_3) \cup (D_2 \cup D_3)$.
voloshin-hypergraph_F01273	173	■ 1.000	$Y_{\{i\}}=Y-\left\{x_{\{i\}}\right\}, \ i=1,2,3$	$Y_i = Y - \{x_i\}, i = 1, 2, 3$	$Y \subseteq (D_1 \cup D_2) \cup (D_1 \cup D_3) \cup (D_2 \cup D_3)$.
voloshin-hypergraph_F01274	173	■ 1.000	$Y_{\{i\}} \subseteq D_{\{i\}}, \ i=1,2,3$	$Y_i \subseteq D_i, i = 1, 2, 3$	Theorem 7.6.4 A hypergraph $\mathcal{H}$ is conformal if and only if its dual $\mathcal{H}^*$ is Helly
voloshin-hypergraph_F01275	174	■ 1.000	$x_{\{1\}}, \ x_{\{2\}}, \ \ldots, \ x_{\{k\}}$	$x_1, x_2, \dots, x_k$	Lemma 7.6.1, $\mathcal{H}$ is a conformal hypergraph. □
voloshin-hypergraph_F01276	174	■ 0.991	$x_{\{1\}}, \ x_{\{2\}}, \ \ldots, \ x_{\{k\}} \in D$	$x_1, x_2, \dots, x_k \in D$	The simplest example to show the relation between Helly property and conformality
voloshin-hypergraph_F01277	174	■ 0.991	$d \in X_{\{1\}}, \ d \in X_{\{2\}}, \ \ldots$	$d \in X_1, d \in X_2, \dots$	The simplest example to show the relation between Helly property and conformality
voloshin-hypergraph_F01278	174	■ 1.000	$d \in X_{\{k\}}$	$d \in X_k$	The simplest example to show the relation between Helly property and conformality is the triangle K_3 (see Figure 7.15). Apparently, K_3 itself is neither Helly nor a conformal
voloshin-hypergraph_F01279	174	■ 1.000	$d \in X_{\{1\}} \cap X_{\{2\}} \cap \cdots \cap X_{\{k\}}$	$d \in X_1 \cap X_2 \cap \dots \cap X_k$	The simplest example to show the relation between Helly property and conformality is the triangle K_3 (see Figure 7.15). Apparently, K_3 itself is neither Helly nor a conformal
voloshin-hypergraph_F01280	175	■ 0.996	$\mathcal{H}_{\{2\}} \cong \mathcal{H}_{\{1\}}^*$	$\mathcal{H}_2 \cong \mathcal{H}_1^*$	hypergraph $\mathcal{H}_2^*$. One can easily check that $\mathcal{H}_2 \cong \mathcal{H}_1^*$. At last, if we add to $\mathcal{H}_2$ an edge
voloshin-hypergraph_F01281	175	■ 0.999	$\mathcal{H}_{\{3\}}$	$\mathcal{H}_3$	containing all vertices, we obtain a hypergraph $\mathcal{H}_3$ which is both Helly and conformal.
voloshin-hypergraph_F01282	175	■ 0.995	$\mathcal{H} \cong \mathcal{H}^*$	$\mathcal{H} \cong \mathcal{H}^*$	A hypergraph is called self-dual if $\mathcal{H} \cong \mathcal{H}^*$; evidently, the incidence matrix of a self-
voloshin-hypergraph_F01283	175	■ 1.000	$\mathcal{H}_{\{3\}} \cong \mathcal{H}_{\{3\}}^*$	$\mathcal{H}_3 \cong \mathcal{H}_3^*$	
voloshin-hypergraph_F01284	175	■ 1.000	$n, \ p, \ q, \ r \geq 1$	$n, p, q, r \geq 1$	7. Prove that cube is and prism is not a Helly hypergraph.
voloshin-hypergraph_F01285	175	■ 1.000	$P_{\{n\}}, \ C_{\{n\}}, \ W_{\{n\}}, \ E_{\{n\}}, \ K_{\{n\}}, \ K_{\{p, \ q, \ r\}}^3$	$P_n, C_n, W_n, E_n, K_n, K_{p,q,r}^3$	7. Prove that cube is and prism is not a Helly hypergraph.
voloshin-hypergraph_F01286	175	■ 1.000	$n, \ r \geq 1$	$n, r \geq 1$	
voloshin-hypergraph_F01287	177	■ 1.000	$E_{\{3\}}$	E_3	
voloshin-hypergraph_F01288	178	■ 1.000	$L^{\prime}$	L'	by itself as a subtree, then the neighbors of this vertex in L form a clique, since as subtrees
voloshin-hypergraph_F01289	178	■ 1.000	$T-x$	$T - x$	introduce a chordless cycle. Hence L is chordal. □
voloshin-hypergraph_F01290	178	■ 1.000	$L^{\prime}=L$	$L' = L$	Proof. The usual proof is by induction; we present a short argument by Lehel (1972). Let
voloshin-hypergraph_F01291	178	■ 1.000	$\mathcal{F} \subseteq \mathcal{D}$	$\mathcal{F} \subseteq \mathcal{D}$	
voloshin-hypergraph_F01292	178	■ 1.000	$D \in \mathcal{F}$	$D \in \mathcal{F}$	

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voloshin-hypergraph_F01293	178	■ 1.000	$n(T)= X $	$n(T) = X $	
voloshin-hypergraph_F01294	178	■ 1.000	$n(T)-1$	$n(T) - 1$	
voloshin-hypergraph_F01295	178	■ 1.000	$D_{\{j\}}$	D_j	
voloshin-hypergraph_F01296	178	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	
voloshin-hypergraph_F01297	179	■ 1.000	$\mathcal{D}(x) \subseteq \mathcal{D}(y)$	$\mathcal{D}(x) \subseteq \mathcal{D}(y)$	
voloshin-hypergraph_F01298	179	■ 0.975	$\mathcal{H}=(Y, \mathcal{D})$	$\mathcal{H} = (Y, \mathcal{D})$	edges (operation inverse to weak deleting of the vertex) and constructing a host tree for $\mathcal{H}^*$.
voloshin-hypergraph_F01299	179	■ 1.000	$n=n(\mathcal{H})$	$n = n(\mathcal{H})$	Step 1: constructing hypergraph $\mathcal{H}$ and its 2-section $(\mathcal{H})_2$. Since $\mathcal{H}$ is a chordal
voloshin-hypergraph_F01300	179	■ 1.000	$n \leq 2$	$n \leq 2$	Step 1: constructing hypergraph $\mathcal{H}$ and its 2-section $(\mathcal{H})_2$. Since $\mathcal{H}$ is a chordal
voloshin-hypergraph_F01301	179	■ 1.000	$n>2$	$n > 2$	hypergraph, $(\mathcal{H})_2$ is a chordal graph. Therefore, $(\mathcal{H})_2$ has a simplicial vertex x . The closed
voloshin-hypergraph_F01302	179	■ 1.000	$\mathcal{H}_{\{0\}}$	$\mathcal{H}_0$	simplicial in $(\mathcal{H})_2$ and the edge $D_0 - \{x\}$ in $\mathcal{H}_0$ contains all neighbors of x in $\mathcal{H}$, we have
voloshin-hypergraph_F01303	179	■ 1.000	$\left(\mathcal{H}_{\{0\}}\right)_2$	$(\mathcal{H}_0)_2$	simplicial in $(\mathcal{H})_2$ and the edge $D_0 - \{x\}$ in $\mathcal{H}_0$ contains all neighbors of x in $\mathcal{H}$, we have
voloshin-hypergraph_F01304	179	■ 0.999	$D_{\{0\}}$	D_0	
voloshin-hypergraph_F01305	179	■ 1.000	$Y-\{x\}$	$Y - \{x\}$	
voloshin-hypergraph_F01306	179	■ 1.000	$\{D-\{x\}: D \in \mathcal{D}\}$	$\{D - \{x\} : D \in \mathcal{D}\}$	
voloshin-hypergraph_F01307	179	■ 1.000	$D_{\{0\}}-\{x\}$	$D_0 - \{x\}$	
voloshin-hypergraph_F01308	179	■ 1.000	$\left(\mathcal{H}_{\{0\}}\right)_2=(\mathcal{H})_2-x$	$(\mathcal{H}_0)_2 = (\mathcal{H})_2 - x$	
voloshin-hypergraph_F01309	180	■ 0.988	$\mathcal{H}_{\{0\}}\{ \}^*$	$\mathcal{H}_0^*$	subtree.
voloshin-hypergraph_F01310	180	■ 0.622	$\mathcal{D}^{\prime}=V\left(\mathcal{H}_{\{0\}}^*\right)$	$\mathcal{D}' = V(\mathcal{H}_0^*)$	
voloshin-hypergraph_F01311	180	■ 0.622	$x^{\prime} \in V\left(\mathcal{H}_{\{0\}}\right)=$	$x' \in V(\mathcal{H}_0) =$	
voloshin-hypergraph_F01312	180	■ 0.852	$\mathcal{D}\left(\mathcal{H}_{\{0\}}^*\right)$	$\mathcal{D}(\mathcal{H}_0^*)$	Step 3: induction step. It remains to show that $\mathcal{H}^*$ is also a hypertree. We do this in several steps. First, define a hypergraph $\mathcal{H}'$ by expanding the edge $D_0 - \{x\}$ to include x
voloshin-hypergraph_F01313	180	■ 0.985	$d_{\{0\}}$	d_0	than $D_0 - \{x\}$, obtained by deleting x from the edges of $\mathcal{D}(x)$ in $\mathcal{H}$. To obtain $\mathcal{H}'$ from $\mathcal{H}_0$, we must expand these edges to include x . For each vertex d of T that is a neighbor of d_0
voloshin-hypergraph_F01314	180	■ 1.000	$\mathcal{H}_{\{1\}}\{ \}^*$	$\mathcal{H}_1^*$	and corresponds to some $D \in \mathcal{D}'_x$, we expand
voloshin-hypergraph_F01315	180	■ 1.000	$\mathcal{D}_{\{x\}}^{\prime}$	$\mathcal{D}'_x$	d_0 . The pendant vertices of this star are vertices in $\mathcal{H}_2^*$ that correspond to edges in $\mathcal{H}_2'$ which
voloshin-hypergraph_F01316	180	■ 0.707	$D \in \mathcal{D}_{\chi}^{\prime}$	$D \in \mathcal{D}'_\chi$	d, d_0 and x , similarly, vertices x, y, z and c become edges x, y, z and c in dual $\mathcal{H}_0'$. Notice that X and C are the singletons. Host tree T is drawn by dashed lines.
voloshin-hypergraph_F01317	180	■ 1.000	$\mathcal{H}_2^*$	$\mathcal{H}_2^*$	
voloshin-hypergraph_F01318	180	■ 1.000	$D, D_{\{0\}}$	D, D_0	d or d_0 are the vertices of $\mathcal{H}_2'$ contained in D or D_0 , respectively, and we have $D \subset D_0$.
voloshin-hypergraph_F01319	180	■ 0.999	$d, d_{\{0\}}$	d, d_0	The vertex d in $\mathcal{H}_2^*$ occurs as a vertex of T not adjacent to d_0 . Adding the edge (d, d_0) to

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voloshin-hypergraph_F01320	180	■ 0.999	x, a, b	x, a, b	The vertex d in $\mathcal{H}_2^*$ occurs as a vertex of T not adjacent to d_0 . Adding the edge (d, d_0) to
voloshin-hypergraph_F01321	180	■ 0.999	X, A, B	X, A, B	The vertex d in $\mathcal{H}_2^*$ occurs as a vertex of T not adjacent to d_0 . Adding the edge (d, d_0) to
voloshin-hypergraph_F01322	180	■ 0.997	$D \in \mathcal{D}_{x'}^{\prime}$	$D \in \mathcal{D}'_x$	
voloshin-hypergraph_F01323	180	■ 1.000	$D \subset D_0$	$D \subset D_0$	
voloshin-hypergraph_F01324	180	■ 0.998	(d, d_0)	(d, d_0)	
voloshin-hypergraph_F01325	180	■ 0.735	(d, d')	(d, d')	
voloshin-hypergraph_F01326	180	■ 0.735	$d' \neq d_0$	$d' \neq d_0$	
voloshin-hypergraph_F01327	180	■ 1.000	T_3	T_3	—
voloshin-hypergraph_F01328	181	■ 1.000	$\mathcal{H}_3^*$	$\mathcal{H}_3^*$	that contains d and d' . In $\mathcal{H}_2$, the vertex x' belongs to D and D' . Since d is pendant to d_0 in
voloshin-hypergraph_F01329	181	■ 1.000	d'	d'	of the vertices of X' contains not only a and a' but also the pair from a' to a . This implies that the vertices of X' still form a subtree in T_3 .
voloshin-hypergraph_F01330	181	■ 1.000	$\mathcal{H}_3^*, \mathcal{H}_4^*, \dots$	$\mathcal{H}_3^*, \mathcal{H}_4^*, \dots$	edges in $\mathcal{H}$. By Lemma 6.1.2, $\mathcal{H}$ is rich, and hence these edges in $\mathcal{H}$ have a common vertex. Thus in $\mathcal{H}$ the elements of R lie in a common edge. This shows that the vertices of
voloshin-hypergraph_F01331	181	■ 1.000	$T_3, T_4, \dots$	$T_3, T_4, \dots$	edges in $\mathcal{H}$. By Lemma 6.1.2, $\mathcal{H}$ is rich, and hence these edges in $\mathcal{H}$ have a common vertex. Thus in $\mathcal{H}$ the elements of R lie in a common edge. This shows that the vertices of
voloshin-hypergraph_F01332	181	■ 0.998	$(\mathcal{H})_2 = L(\mathcal{H}^*)$	$(\mathcal{H})_2 = L(\mathcal{H}^*)$	
voloshin-hypergraph_F01333	181	■ 1.000	R	R	
voloshin-hypergraph_F01334	182	■ 0.995	$(\mathcal{H}^*)_2 = L(\mathcal{H})$	$(\mathcal{H}^*)_2 = L(\mathcal{H})$	edge set is the family of vertex sets of maximal cliques in graph G . Note that the 2-section of the clique hypergraph of a graph G is G . Further, by construction every clique hypergraph
voloshin-hypergraph_F01335	183	■ 0.831	A, B, C, D	A, B, C, D	hyperedges A, B, C, D , and E in $\mathcal{H}^*$ and each induce a subtree in T . One can easily see that
voloshin-hypergraph_F01336	183	■ 0.997	$G = L(\mathcal{H}^*)$	$G = L(\mathcal{H}^*)$	$G = L(\mathcal{H}^*)$. Drawings of $\mathcal{H}$ and $\mathcal{H}^*$ are simply two different drawings of a $(0,1)$ -matrix (namely, the incidence matrix of $\mathcal{H}$) and its transpose.
voloshin-hypergraph_F01337	183	■ 0.998	α	α	pertrees are called ‘‘arboreal hypergraphs’’ and chordal conformal hypergraphs are called
voloshin-hypergraph_F01338	183	■ 1.000	$L(\mathcal{H}) = (\mathcal{H}^*)_2$	$L(\mathcal{H}) = (\mathcal{H}^*)_2$	2. Given a hypertree, construct the line graph.
voloshin-hypergraph_F01339	183	■ 1.000	$G = L(\mathcal{H})$	$G = L(\mathcal{H})$	
voloshin-hypergraph_F01340	184	■ 1.000	$D = \{x\}$	$D = \{x\}$	an edge $D \in \mathcal{D}$ such that $D = \{x\}$ (i.e., D is a singleton). It means that x belongs to every
voloshin-hypergraph_F01341	184	■ 1.000	$y \in X, y \neq x$	$y \in X, y \neq x$	then called a twin vertex of x . In dual hypergraph $\mathcal{H}^*$, hyper-pendant vertex x becomes
voloshin-hypergraph_F01342	184	■ 1.000	$\mathcal{D}(x) = \emptyset$	$\mathcal{D}(x) = \emptyset$	hyper-pendant. Hyper-pendant vertex cannot be transversal, and transversal vertex cannot be hyper-pendant.
voloshin-hypergraph_F01343	184	■ 0.607	$\mathrm{T}(\mathcal{H}) = \mathrm{T}(\mathcal{H}_1)$	$\mathrm{T}(\mathcal{H}) = \mathrm{T}(\mathcal{H}_1)$	4. $\mathbf{v}(\mathcal{H}) = \mathbf{v}(\mathcal{H}_1)$
voloshin-hypergraph_F01344	184	■ 1.000	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$	5. if $\mathcal{H}$ is a hypertree
voloshin-hypergraph_F01345	184	■ 0.505	$v(\mathcal{H}) = v(\mathcal{H}_1)$	$v(\mathcal{H}) = v(\mathcal{H}_1)$	PROOF. 1. By definition, $\mathcal{D}(x) \subset \mathcal{D}(y) \subset \mathcal{D}$
voloshin-hypergraph_F01346	184	■ 1.000	$x \notin T$	$x \notin T$	dicts the condition of the case. Hence again, there exists a maximum independent set $S \subseteq X$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01347	184	0.730	$T_{\{1\}}=T-\{X\}$	$T_1 = T - \{X\}$	such that $x \in S$. Weak deletion of a vertex from a hypergraph cannot decrease the size of a minimum
voloshin-hypergraph_F01348	184	0.730	$ \left T_{\{1\}}\right = T -1=\tau(\mathcal{H})-1$	$ T_1 = T - 1 = \tau(\mathcal{H}) - 1$	such that $x \in S$. Weak deletion of a vertex from a hypergraph cannot decrease the size of a minimum
voloshin-hypergraph_F01349	184	1.000	$x \in X-T=S$	$x \in X - T = S$	Weak deletion of a vertex from a hypergraph cannot decrease the size of a minimum transversal, so $\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1)$. On the other hand, 1. i)
voloshin-hypergraph_F01350	184	1.000	$x \in T_{\{1\}}$	$x \in T_1$	Therefore, $\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$.
voloshin-hypergraph_F01351	184	1.000	$T_{\{2\}}=\left(T_{\{1\}}-\{x\}\right) \cup\{y\}$	$T_2 = (T_1 - \{x\}) \cup \{y\}$	Therefore, $\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$.
voloshin-hypergraph_F01352	184	0.844	$\tau(\mathcal{H}) \leq \tau\left(\mathcal{H}_{\{1\}}\right)$	$\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1)$	implies that every matching in $\mathcal{H}_1$ is a matching in $\mathcal{H}$, i.e., $v(\mathcal{H}) \geq v(\mathcal{H}_1)$.
voloshin-hypergraph_F01353	184	1.000	$\tau(\mathcal{H}) \geq \tau\left(\mathcal{H}_{\{1\}}\right)$	$\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1)$	
voloshin-hypergraph_F01354	184	0.996	$\tau(\mathcal{H})=\tau\left(\mathcal{H}_{\{1\}}\right)$	$\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$	
voloshin-hypergraph_F01355	184	0.966	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X , \alpha\left(\mathcal{H}_{\{1\}}\right)+\tau\left(\mathcal{H}_{\{1\}}\right)= X -1$	$\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X , \alpha(\mathcal{H}_1) + \tau(\mathcal{H}_1) = X - 1$	
voloshin-hypergraph_F01356	184	0.482	$\boldsymbol{\alpha}(\boldsymbol{\mathcal{H}})=\boldsymbol{\alpha}\left(\mathcal{H}_{\{1\}}\right)+1$	$\boldsymbol{\alpha}(\mathcal{H}) = \boldsymbol{\alpha}(\mathcal{H}_1) + 1$	
voloshin-hypergraph_F01357	184	0.511	$\mathcal{V}(\mathcal{H}) \leq \mathcal{V}\left(\mathcal{H}_{\{1\}}\right)$	$\mathcal{V}(\mathcal{H}) \leq \mathcal{V}(\mathcal{H}_1)$	
voloshin-hypergraph_F01358	184	0.827	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}\left(\mathcal{H}_{\{1\}}\right)$	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}(\mathcal{H}_1)$	
voloshin-hypergraph_F01359	185	0.998	$L(\mathcal{H})=$	$L(\mathcal{H}) =$	By Corollary 6.1.1, $\mathcal{H}$ is a hypertree if and only if it is rich and $L(\mathcal{H})$ is a chordal graph. Since $\mathcal{H}$ is hyper-hereditary, $\mathcal{H}_1$ is a Helly hypergraph too, and moreover $L(\mathcal{H}) =$
voloshin-hypergraph_F01360	185	1.000	$L\left(\mathcal{H}_{\{1\}}\right)$	$L(\mathcal{H}_1)$	$L(\mathcal{H}_1)$. Consequently, $\mathcal{H}_1$ is a hypertree
voloshin-hypergraph_F01361	185	0.927	$\tau(\mathcal{H})=\tau\left(\mathcal{H}_{\{1\}}\right)+1$	$\tau(\mathcal{H}) = \tau(\mathcal{H}_1) + 1$	$\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$
voloshin-hypergraph_F01362	185	1.000	$\alpha(\mathcal{H})=\alpha\left(\mathcal{H}_{\{1\}}\right)$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1)$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1)$
voloshin-hypergraph_F01363	185	0.804	$v(\mathcal{H})=v\left(\mathcal{H}_{\{1\}}\right)+1$	$v(\mathcal{H}) = v(\mathcal{H}_1) + 1$	5. if $\mathcal{H}$ is a hypertree,
voloshin-hypergraph_F01364	185	1.000	$\{x\} \in \mathcal{D}$	$\{x\} \in \mathcal{D}$	is a transversal of $\mathcal{H}$, what implies $v(\mathcal{H}) \geq v(\mathcal{H}_1) + 1$. On the other hand, if T is a minimal transversal of $\mathcal{H}$ then $T_0 = T - \{x\}$ is a transversal of $\mathcal{H}_1$ what implies $\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1) +$
voloshin-hypergraph_F01365	185	0.989	$T_{\{1\}} \subset X_{\{1\}}=X-\{x\}$	$T_1 \subset X_1 = X - \{x\}$	3. Since $\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X $, $\alpha(\mathcal{H}_1) + \tau(\mathcal{H}_1) = X - 1$, and $\tau(\mathcal{H}) = \tau(\mathcal{H}_1) + 1$, con-
voloshin-hypergraph_F01366	185	0.989	$T_{\{2\}}=T_{\{1\}} \cup\{x\} \subset X$	$T_2 = T_1 \cup \{x\} \subset X$	3. Since $\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X $, $\alpha(\mathcal{H}_1) + \tau(\mathcal{H}_1) = X - 1$, and $\tau(\mathcal{H}) = \tau(\mathcal{H}_1) + 1$, con-
voloshin-hypergraph_F01367	185	1.000	$\tau(\mathcal{H}) \leq \tau\left(\mathcal{H}_{\{1\}}\right)+1$	$\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1) + 1$	clude that $\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1)$.
voloshin-hypergraph_F01368	185	0.999	$T_{\{0\}}=T-\{x\}$	$T_0 = T - \{x\}$	4. Since x is a transversal vertex, for any maximum matching $\mathcal{F}$ of $\mathcal{H}_1$ the family
voloshin-hypergraph_F01369	185	0.999	$\tau(\mathcal{H}) \geq \tau\left(\mathcal{H}_{\{1\}}\right)+1$	$\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1) + 1$	4. Since x is a transversal vertex, for any maximum matching $\mathcal{F}$ of $\mathcal{H}_1$ the family
voloshin-hypergraph_F01370	185	0.999	$\mathcal{F} \cup\{x\}$	$\mathcal{F} \cup \{x\}$	connected hypergraph, each connected component is a hypertree,
voloshin-hypergraph_F01371	185	0.999	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}\left(\mathcal{H}_{\{1\}}\right)+1$	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}(\mathcal{H}_1) + 1$	connected hypergraph, each connected component is a hypertree,
voloshin-hypergraph_F01372	185	0.940	$\mathcal{V}(\mathcal{H})-1 \leq \mathcal{V}\left(\mathcal{H}_{\{1\}}\right)$	$\mathcal{V}(\mathcal{H}) - 1 \leq \mathcal{V}(\mathcal{H}_1)$	Proposition 8.2.1 If $\mathcal{H}=(X, \mathcal{D})$, $ X \geq 2$ is a hypertree, then it contains at least two ver-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01373	185	1.000	$\mathcal{H}=(X, \mathcal{D}), X \geq 2$	$\mathcal{H}=(X, \mathcal{D}), X \geq 2$	
voloshin-hypergraph_F01374	186	0.786	$\tau=\nu=0$	$\tau=\nu=0$	one vertex hypergraph; either it is a hyper-pendant (isolated) vertex, then $\tau=\nu=0$, or a
voloshin-hypergraph_F01375	186	0.753	$\tau=v=1$	$\tau=v=1$	one-vertex hypergraph; either it is a hyper-pendant (isolated) vertex, then $\tau=\nu=0$, or a transversal vertex then $\tau=v=1$. In all stens we have the equality so $\tau(\mathcal{H})=\nu(\mathcal{H})$. $\square$
voloshin-hypergraph_F01376	186	0.753	$\tau(\mathcal{H})=v(\mathcal{H})$	$\tau(\mathcal{H})=v(\mathcal{H})$	one-vertex hypergraph; either it is a hyper-pendant (isolated) vertex, then $\tau=\nu=0$, or a transversal vertex then $\tau=v=1$. In all stens we have the equality so $\tau(\mathcal{H})=\nu(\mathcal{H})$. $\square$
voloshin-hypergraph_F01377	186	0.974	$\chi=\omega$	$\chi=\omega$	any chordal graph is perfect. Hence for chordal graphs $\chi=\omega$ and $\theta=\alpha$. We know that
voloshin-hypergraph_F01378	186	0.974	$\theta=\alpha$	$\theta=\alpha$	any chordal graph is perfect. Hence for chordal graphs $\chi=\omega$ and $\theta=\alpha$. We know that
voloshin-hypergraph_F01379	186	0.988	$\theta(L(\mathcal{H}))=\alpha(L(\mathcal{H}))$	$\theta(L(\mathcal{H}))=\alpha(L(\mathcal{H}))$	if $\mathcal{H}$ is a hypertree, then $L(\mathcal{H})$ is chordal. Therefore $\theta(L(\mathcal{H}))=\alpha(L(\mathcal{H}))$. Because $\mathcal{H}$ is Helly, the cardinality of minimum transversal $\tau(\mathcal{H})=\theta(L(\mathcal{H}))$. By definition of $L(\mathcal{H})$
voloshin-hypergraph_F01380	186	0.946	$\nu(\mathcal{H})=\alpha(L(\mathcal{H}))$	$\nu(\mathcal{H})=\alpha(L(\mathcal{H}))$	$\tau(\mathcal{H})=\nu(\mathcal{H})$. Moreover, it holds for every partial subhypergraph $\mathcal{H}'\subseteq\mathcal{H}$ and for every
voloshin-hypergraph_F01381	186	0.650	$\tau(\mathcal{H})=\mathcal{V}(\mathcal{H})$	$\tau(\mathcal{H})=\mathcal{V}(\mathcal{H})$	induced subgraph of $L(\mathcal{H})$ as well.
voloshin-hypergraph_F01382	187	0.930	$\{3\}$	$\{3\}$	ordering of vertices is: 1, 3, 2, 4. At this point we start constructing the host tree beginning with T . Sequentially adding vertices in inverse order 4, 3, 2, 1 and connecting them to their
voloshin-hypergraph_F01383	187	0.822	$\{2\}$	$\{2\}$	twins. finally, we arrive to tree T which is the host tree for $\mathcal{H}$.
voloshin-hypergraph_F01384	187	0.822	$\mathcal{H}_4$	$\mathcal{H}_4$	twins. finally, we arrive to tree T which is the host tree for $\mathcal{H}$.
voloshin-hypergraph_F01385	187	1.000	$T_{\{4\}}$	T_4	
voloshin-hypergraph_F01386	188	1.000	$T=\emptysetset, \mathcal{F}=\emptysetset$	$T=\emptyset, \mathcal{F}=\emptyset$	$\mathcal{A}$. If the vertex is
voloshin-hypergraph_F01387	188	0.968	$T, \mathcal{F}$	$T, \mathcal{F}$	more general I
voloshin-hypergraph_F01388	189	1.000	$T=\{1,3,5\}$	$T=\{1,3,5\}$	minimum transversal $T=\{1,3,5\}$ and maximum matching $\mathcal{F}=\{\{1,0\},\{2,3\},\{4,5\}\}$, so $\tau(G)=\nu(G)$. Maximum matching $\mathcal{F}$ is shown by double lines in the second drawing of G .
voloshin-hypergraph_F01389	189	1.000	$\mathcal{F}=\{\{1,6\},\{2,3\},\{4,5\}\}$	$\mathcal{F}=\{\{1,6\},\{2,3\},\{4,5\}\}$	minimum transversal $T=\{1,3,5\}$ and maximum matching $\mathcal{F}=\{\{1,0\},\{2,3\},\{4,5\}\}$, so $\tau(G)=\nu(G)$. Maximum matching $\mathcal{F}$ is shown by double lines in the second drawing of G .
voloshin-hypergraph_F01390	189	0.769	$\tau(G)=\mathcal{V}(G)$	$\tau(G)=\mathcal{V}(G)$	$\tau(G)=\nu(G)$. Maximum matching $\mathcal{F}$ is shown by double lines in the second drawing of G . One can compare graph G with cycle C_5 where $\tau(C_5)=3>2=\nu(C_5)$.
voloshin-hypergraph_F01391	189	0.961	$\tau(C_5)=3>2=\nu(C_5)$	$\tau(C_5)=3>2=\nu(C_5)$	
voloshin-hypergraph_F01392	189	0.987	τ, α	τ, α	
voloshin-hypergraph_F01393	189	0.772	$\tau(\mathcal{H}), \alpha(\mathcal{H})$	$\tau(\mathcal{H}), \alpha(\mathcal{H})$	
voloshin-hypergraph_F01394	189	0.772	$\nu(\mathcal{H})$	$\nu(\mathcal{H})$	
voloshin-hypergraph_F01395	190	0.977	$\wedge(G)=$	$\wedge(G)=$	(see Section 2.1.) that for a simple connected graph G , the cyclomatic number $\wedge(G)=m-n+1$. The same formula generalizes to multigraphs if we add to G any number of
voloshin-hypergraph_F01396	190	1.000	$m-n+1$	$m-n+1$	$m-n+1$. The same formula generalizes to multigraphs if we add to G any number of loops or multiple edges: each such addition forms a unique cycle with respect to a chosen
voloshin-hypergraph_F01397	190	1.000	$[\mathcal{H}]_2$	$[\mathcal{H}]_2$	1. all loops and edges of size 2 of $\mathcal{H}$ are included in E ;
voloshin-hypergraph_F01398	190	1.000	≥ 3	≥ 3	Hence, $(\mathcal{H})_2=[\mathcal{H}]_2$ if and only if $\mathcal{H}$ itself is a simple graph. Further, let us call any
voloshin-hypergraph_F01399	190	1.000	$(\mathcal{H})_2=[\mathcal{H}]_2$	$(\mathcal{H})_2=[\mathcal{H}]_2$	a multi-forest, then T denotes a forest obtained from T by weak deletion of all loops and replacing each set of parallel edges by an edge connecting the same pair of vertices. Thus

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voloshin-hypergraph_F01400	190	1.000	$\bar{T}$	$\bar{T}$	multigraph $[\mathcal{H}]_2$. For brevity, throughout this section call every such a spanning multi-
voloshin-hypergraph_F01401	190	1.000	$\bar{T}=(T)_2$	$\bar{T}=(T)_2$	Proposition 8.3.1 For any forest T of a multigraph $[\mathcal{H}]_2$
voloshin-hypergraph_F01402	190	1.000	$ D -1$	$ D -1$	Definition 8.3.1 Generalized arithmetic number of a hypergraph $\mathcal{H}$ with respect to for
voloshin-hypergraph_F01403	190	0.994	$l(\mathcal{H},T)$	$l(\mathcal{H},T)$	
voloshin-hypergraph_F01404	191	1.000	$[\mathcal{H}]_2=\mathcal{H}$	$[\mathcal{H}]_2=\mathcal{H}$	$l(\mathcal{H},T)=\sum_{ D =2}(\lambda(\mathcal{H},T)-\lambda(\mathcal{H},T)-\lambda(\mathcal{H},T)-\lambda(\mathcal{H},T))$
voloshin-hypergraph_F01405	191	0.998	$w(T)= X -1=n-1$	$w(T)= X -1=n-1$	$w(T)= X -1=n-1$
voloshin-hypergraph_F01406	191	0.998	$\Lambda(\mathcal{H},T)$	$\Lambda(\mathcal{H},T)$	$D\in\mathcal{D}$
voloshin-hypergraph_F01407	191	1.000	$w(T)=3$	$w(T)=3$	complete graph generated by an edge D are split into two subsets: the
voloshin-hypergraph_F01408	192	0.999	E_D	E_D	is a forest. Let E_D be the set of edges of size 2 which are generated by the 2-section of D
voloshin-hypergraph_F01409	192	1.000	$c(T,D)$	$c(T,D)$	is a forest. Let E_D be the set of edges of size 2 which are generated by the 2-section of D and lie on T and $c(T,D)$ be the number of connected components in such a subgraph
voloshin-hypergraph_F01410	192	1.000	$ D \geq 2$	$ D \geq 2$	above, each vertex and for every edge D such that $ D \geq 2$, the above inequality holds.
voloshin-hypergraph_F01411	192	1.000	$c(T,D)=1$	$c(T,D)=1$	$\lambda(\mathcal{H},T)=\sum_{D\in\mathcal{D}}(D -1)-n(\mathcal{H},T)+\lambda(\mathcal{H},T)$
voloshin-hypergraph_F01412	192	0.960	$T,\Lambda(\mathcal{H},T)\geq 0$	$T,\Lambda(\mathcal{H},T)\geq 0$	
voloshin-hypergraph_F01413	192	1.000	$l(\mathcal{H},T)\geq 0$	$l(\mathcal{H},T)\geq 0$	
voloshin-hypergraph_F01414	192	0.913	$\Lambda(\mathcal{H},T)=0$	$\Lambda(\mathcal{H},T)=0$	
voloshin-hypergraph_F01415	193	1.000	$l(\mathcal{H},T)=0$	$l(\mathcal{H},T)=0$	Proof. $\Rightarrow$ Assume $\lambda(\mathcal{H},T)=0$. Formula (8.4) implies $c(T,D)=1$ for any edge $D\in\mathcal{D}$ and in addition $l(\mathcal{H},T)=0$. It means that 2-section of any edge $D\in\mathcal{D}$ induces a
voloshin-hypergraph_F01416	193	0.918	$w(T)=\sum_{D\in\mathcal{D}}(D -1)$	$w(T)=\sum_{D\in\mathcal{D}}(D -1)$	Theorem 8.3.3 (Voloshin, 1987) Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph, $x\in$
voloshin-hypergraph_F01417	193	0.918	$\wedge(\mathcal{H},T)=0$	$\wedge(\mathcal{H},T)=0$	Theorem 8.3.3 (Voloshin, 1987) Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph, $x\in$
voloshin-hypergraph_F01418	193	1.000	$\overline{T_1}$	$\overline{T_1}$	Theorem 8.3.4 (Voloshin, 1987) Let $\mathcal{H}=(X,\mathcal{D})$ be a hypergraph, $x\in$
voloshin-hypergraph_F01419	193	1.000	$w(T_1)=w(T)$	$w(T_1)=w(T)$	$w(T_1)=w(T)$
voloshin-hypergraph_F01420	193	1.000	$x\in X,T$	$x\in X,T$	$w(T_1)=w(T)$
voloshin-hypergraph_F01421	193	1.000	$[\mathcal{H}]_2,\mathcal{D}_T(x)$	$[\mathcal{H}]_2,\mathcal{D}_T(x)$	$\Lambda(\mathcal{H},T)=\Lambda(\mathcal{H},T_1)+ \mathcal{U}(X) - \mathcal{U}_T(X) -m_2. \tag{8.6}$
voloshin-hypergraph_F01422	193	1.000	m_2	m_2	$\Lambda(\mathcal{H},T)=\Lambda(\mathcal{H},T_1)+ \mathcal{U}(X) - \mathcal{U}_T(X) -m_2. \tag{8.6}$
voloshin-hypergraph_F01423	193	1.000	$\mathcal{H}-x$	$\mathcal{H}-x$	
voloshin-hypergraph_F01424	193	1.000	$ \mathcal{D}(x) =3, \mathcal{D}_T(x) =1$	$ \mathcal{D}(x) =3, \mathcal{D}_T(x) =1$	
voloshin-hypergraph_F01425	193	1.000	$m_2=1$	$m_2=1$	

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voloshin-hypergraph_F01426	194	1.000	$[\mathcal{H}]_2, x \in X$	$[\mathcal{H}]_2, x \in X$	
voloshin-hypergraph_F01427	194	1.000	$\left[\mathcal{H}_1\right]_2$	$[\mathcal{H}_1]_2$	
voloshin-hypergraph_F01428	194	1.000	$T_1^{\prime}$	T_1'	
voloshin-hypergraph_F01429	194	1.000	$w\left(T_1^{\prime}\right) \geq w\left(T_1\right)$	$w\left(T_1^{\prime}\right) \geq w\left(T_1\right)$	
voloshin-hypergraph_F01430	194	1.000	$w\left(T_1^{\prime}\right) \geq w\left(T_1\right)+1$	$w\left(T_1^{\prime}\right) \geq w\left(T_1\right)+1$	
voloshin-hypergraph_F01431	194	1.000	$\left \mathcal{D}_T(x)\right $	$\left \mathcal{D}_T(x)\right $	
voloshin-hypergraph_F01432	194	1.000	$T^{\prime}$	T'	
voloshin-hypergraph_F01433	194	1.000	$w\left(T_1^{\prime}\right)=w\left(T_1\right)$	$w\left(T_1^{\prime}\right)=w\left(T_1\right)$	
voloshin-hypergraph_F01434	195	1.000	$m_2=0$	$m_2=0$	
voloshin-hypergraph_F01435	195	1.000	$\left \mathcal{D}(x)\right =\left \mathcal{D}_T(x)\right $	$\left \mathcal{D}(x)\right =\left \mathcal{D}_T(x)\right $	
voloshin-hypergraph_F01436	195	0.919	$\Lambda\left(\mathcal{H}_1, T_1\right)=\Lambda\left(\mathcal{H}_1\right)$	$\Lambda\left(\mathcal{H}_1, T_1\right)=\Lambda\left(\mathcal{H}_1\right)$	
voloshin-hypergraph_F01437	195	0.919	$\Lambda\left(\mathcal{H}\right)=\Lambda\left(\mathcal{H}_1\right)$	$\Lambda\left(\mathcal{H}\right)=\Lambda\left(\mathcal{H}_1\right)$	
voloshin-hypergraph_F01438	195	0.876	$\wedge\left(\mathcal{H}\right)=0$	$\wedge\left(\mathcal{H}\right)=0$	
voloshin-hypergraph_F01439	195	1.000	$1 \Rightarrow 2$	$1 \Rightarrow 2$	
voloshin-hypergraph_F01440	195	0.923	$\Lambda\left(\mathcal{H}\right)$	$\Lambda\left(\mathcal{H}\right)$	
voloshin-hypergraph_F01441	195	0.993	$2 \Rightarrow 3$	$2 \Rightarrow 3$	
voloshin-hypergraph_F01442	195	1.000	$3 \Rightarrow 1$	$3 \Rightarrow 1$	
voloshin-hypergraph_F01443	195	0.914	$\wedge\left(\mathcal{H}\right)$	$\wedge\left(\mathcal{H}\right)$	
voloshin-hypergraph_F01444	196	1.000	$w\left(\mathcal{H}\right)$	$w\left(\mathcal{H}\right)$	
voloshin-hypergraph_F01445	196	0.666	$\Lambda\left(\mathcal{H}\right)=\sum_{D \in \mathcal{D}}\left(D -1\right)-w\left(\mathcal{H}\right)$	$\Lambda\left(\mathcal{H}\right)=\sum_{D \in \mathcal{D}}\left(D -1\right)-w\left(\mathcal{H}\right)$	
voloshin-hypergraph_F01446	197	0.999	$D_0, D_1, D_2, \ldots D_{t-1}$	$D_0, D_1, D_2, \ldots D_{t-1}$	
voloshin-hypergraph_F01447	198	0.408	$\tau\left(\mathcal{H}^{\prime}\right)=v\left(\mathcal{H}^{\prime}\right)$	$\tau\left(\mathcal{H}^{\prime}\right)=v\left(\mathcal{H}^{\prime}\right)$	
voloshin-hypergraph_F01448	198	0.408	$v\left(\mathcal{H}^{\prime}\right)=1$	$v\left(\mathcal{H}^{\prime}\right)=1$	
voloshin-hypergraph_F01449	198	0.999	$\tau\left(\mathcal{H}^{\prime}\right)=1$	$\tau\left(\mathcal{H}^{\prime}\right)=1$	
voloshin-hypergraph_F01450	198	1.000	>3	>3	
voloshin-hypergraph_F01451	199	0.999	$K_n, K_{m, n}, W_n$	$K_n, K_{m, n}, W_n$	

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voloshin-hypergraph_F01452	200	■ 1.000	$C_{\{n\}}, n \geq 3$	$C_n, n \geq 3$	5. A graph G is called an interval graph if there exists an interv
voloshin-hypergraph_F01453	201	■ 1.000	$\chi^{\prime}(\mathcal{H})$	$\chi'(\mathcal{H})$	color. As for graphs, we denote it by $\chi'(\mathcal{H})$. Clearly, $\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$. Now we have the
voloshin-hypergraph_F01454	201	■ 1.000	$\chi^{\prime}(\mathcal{H})=\chi(L(\mathcal{H}))$	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$	color. As for graphs, we denote it by $\chi'(\mathcal{H})$. Clearly, $\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$. Now we have the
voloshin-hypergraph_F01455	201	■ 1.000	$\chi^{\prime}(\mathcal{H})=\Delta(\mathcal{H})$	$\chi'(\mathcal{H}) = \Delta(\mathcal{H})$	edge-coloring property means that if we take dual hypergraph $\mathcal{H}^*$, then $\chi((\mathcal{H}^*)_2) =$
voloshin-hypergraph_F01456	201	■ 0.996	$\chi\left(\left(\left(\mathcal{H}^{\ast}\right)_2\right)_2\right)=$	$\chi((\mathcal{H}^*)_2) =$	$\omega((\mathcal{H}^*)_2)$.
voloshin-hypergraph_F01457	201	■ 1.000	$\omega\left(\left(\left(\mathcal{H}^{\ast}\right)_2\right)_2\right)$	$\omega((\mathcal{H}^*)_2)$	edges.
voloshin-hypergraph_F01458	201	■ 0.992	$\tau\left(\mathcal{H}^{\prime}\right)=\mathcal{V}\left(\mathcal{H}^{\prime}\right)=1$	$\tau(\mathcal{H}') = \mathcal{V}(\mathcal{H}') = 1$	
voloshin-hypergraph_F01459	201	■ 0.998	$\chi^{\prime}(\mathcal{H})=\Delta\left(\mathcal{H}^{\prime}\right)$	$\chi'(\mathcal{H}) = \Delta(\mathcal{H}')$	
voloshin-hypergraph_F01460	201	■ 0.998	$\chi\left(\left(\left(\mathcal{H}^{\ast}\right)_2\right)_2\right)=\omega\left(\left(\left(\mathcal{H}^{\ast}\right)_2\right)_2\right)$	$\chi((\mathcal{H}^*)_2) = \omega((\mathcal{H}^*)_2)$	
voloshin-hypergraph_F01461	201	■ 0.687	$\chi^{\prime}(\mathcal{H})=$	$\chi'(\mathcal{H}) =$	—
voloshin-hypergraph_F01462	202	■ 1.000	$\Delta(\mathcal{H})$	$\Delta(\mathcal{H})$	$\Delta(\mathcal{H})$. Since by the same reason the last equality holds for every $\mathcal{H}' \in \mathcal{H}$, the hypergraph
voloshin-hypergraph_F01463	202	■ 1.000	$K_{\{n\}}, K_{\{m, n\}}, C_{\{n\}}, W_{\{n\}}$	$K_n, K_{m,n}, C_n, W_n$	4. For each of the hypergraphs in Figure 9.4 find the chromatic index and respective
voloshin-hypergraph_F01464	203	■ 1.000	$d \in \mathcal{D}$	$d \in \mathcal{D}$	graph $B(\mathcal{H})$ with vertex set $X \cup \mathcal{D}$. The vertex $x \in X$ is adjacent to the vertex $d \in \mathcal{D}$ in
voloshin-hypergraph_F01465	204	■ 1.000	f_2	f_2	while f_2 is an unbounded face of size 4. If for every edge we put a vertex at the center and connect it with the original vertices of the very same edge, we obtain a plane embedding
voloshin-hypergraph_F01466	204	■ 0.725	$1, 2, 3$	$1, 2, 3$	Proposition 9.4.1 A hypergraph $\mathcal{H}$ is planar if and only if dual hypergraph $\mathcal{H}^*$ is planar.
voloshin-hypergraph_F01467	204	■ 1.000	$\mathcal{H}=(X,\mathcal{D}), X =n, \mathcal{D} =m$	$\mathcal{H} = (X, \mathcal{D}), X = n, \mathcal{D} = m$	$n + m$ vertices,
voloshin-hypergraph_F01468	204	■ 1.000	$n^{\prime}=$	$n' =$	
voloshin-hypergraph_F01469	204	■ 1.000	$n+m$	$n + m$	
voloshin-hypergraph_F01470	205	■ 1.000	$f^{\prime}=f$	$f' = f$	edges and $\mathcal{D} = f$ faces. Since $B(\mathcal{H})$ is a planar graph by Theorem 4.2.1, Euler's formula
voloshin-hypergraph_F01471	205	■ 1.000	$f_{\{1\}}, f_{\{2\}}, f_{\{3\}}, f_{\{4\}}, f_{\{5\}}$	f_1, f_2, f_3, f_4, f_5	
voloshin-hypergraph_F01472	205	■ 1.000	$f_{\{6\}}$	f_6	
voloshin-hypergraph_F01473	205	■ 1.000	$f_{\{1\}}, \ldots, f_{\{6\}}$	$f_1, \dots, f_6$	
voloshin-hypergraph_F01474	209	■ 1.000	≤ 1	≤ 1	the maximum number of different colors that may actually be used in any λ -coloring is at
voloshin-hypergraph_F01475	209	■ 1.000	$\chi(\mathcal{H})$	$\chi(\mathcal{H})$	obtain a usual classic graph coloring as studied in Chapter 5. Since in a proper coloring of
voloshin-hypergraph_F01476	209	■ 1.000	$\lambda > n$	$\lambda > n$	of size ≥ 2 . As the loops are ignored in graph coloring, the singletons are ignored in hyper-graph coloring. An important point, however, is that graph coloring, as we shall see, may
voloshin-hypergraph_F01477	209	■ 0.703	$S_{\{1\}}, S_{\{2\}}, \ldots, S_{\{k\}}$	$S_1, S_2, \dots, S_k$	
voloshin-hypergraph_F01478	209	■ 0.703	$S_{\{i\}}$	S_i	

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voloshin-hypergraph_F01479	210	■ 1.000	$\left S_{\{i\}}\right \leq \alpha(\mathcal{H})$	$ S_i \leq \alpha(\mathcal{H})$	cyclic number χ cannot be greater than $\tau + 1$.
voloshin-hypergraph_F01480	210	■ 0.999	$\tau+1$	$\tau + 1$	minimum partition of the vertex set into three color classes. Vertices of color 1 and 2 represent a minimum transversal, and vertices of color 3 represent a maximum independent set. The
voloshin-hypergraph_F01481	210	■ 0.824	$\tau+1=n-\alpha+1$	$\tau + 1 = n - \alpha + 1$	inequality $\alpha(\mathcal{H})\chi(\mathcal{H}) \geq n$ turns into $3 \cdot 2 \geq 5$, and $\chi(\mathcal{H}) \leq$
voloshin-hypergraph_F01482	210	■ 1.000	$\tau(\mathcal{H})=2$	$\tau(\mathcal{H}) = 2$	Strong colorings. A strong λ -coloring of $\mathcal{H}$ is a coloring of the vertices using at most
voloshin-hypergraph_F01483	210	■ 1.000	$\chi(\mathcal{H})=2$	$\chi(\mathcal{H}) = 2$	Strong colorings. A strong λ -coloring of $\mathcal{H}$ is a coloring of the vertices using at most
voloshin-hypergraph_F01484	210	■ 1.000	$\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$	$\alpha(\mathcal{H})\chi(\mathcal{H}) \geq n$	strong λ -coloring of $\mathcal{H}$. It follows that $\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$ because every strong coloring is also
voloshin-hypergraph_F01485	210	■ 1.000	$3 \cdot 2 \geq 5$	$3 \cdot 2 \geq 5$	strong λ -coloring of $\mathcal{H}$. It follows that $\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$ because every strong coloring is also
voloshin-hypergraph_F01486	210	■ 1.000	$\chi(\mathcal{H}) \leq \tau(\mathcal{H})+1$	$\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$	strong λ -coloring of $\mathcal{H}$. It follows that $\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$ because every strong coloring is also
voloshin-hypergraph_F01487	210	■ 1.000	$2 \leq 2+1$	$2 \leq 2 + 1$	strong λ -coloring of $\mathcal{H}$. It follows that $\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$ because every strong coloring is also
voloshin-hypergraph_F01488	210	■ 1.000	$\gamma(\mathcal{H})$	$\gamma(\mathcal{H})$	
voloshin-hypergraph_F01489	210	■ 0.991	$\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$	$\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$	
voloshin-hypergraph_F01490	211	■ 0.998	$S_{\{i\}}, \ i=1,2, \ldots, \lambda$	$S_i, i = 1, 2, \dots, \lambda$	Equitable colorings. An equitable λ -coloring of $\mathcal{H} = (\mathcal{X}, \mathcal{D})$ is a partition of $\mathcal{X}$ into λ stable sets $S_i, \ i = 1, 2, \dots, \lambda$, such that for every $D \in \mathcal{D}$ and for every i the following
voloshin-hypergraph_F01491	211	■ 1.000	$\lfloor r \rfloor$	$\lfloor r \rfloor$	where $\lfloor r \rfloor$ is the largest integer not greater than r , and $\lceil r \rceil$ is the smallest integer not smaller than r .
voloshin-hypergraph_F01492	211	■ 1.000	$\lceil r \rceil$	$\lceil r \rceil$	where $\lfloor r \rfloor$ is the largest integer not greater than r , and $\lceil r \rceil$ is the smallest integer not smaller than r .
voloshin-hypergraph_F01493	211	■ 1.000	$i=1,2, \ldots, \lambda$	$i = 1, 2, \dots, \lambda$	
voloshin-hypergraph_F01494	211	■ 1.000	$\lambda \geq \max_{D \in \mathcal{D}} D $	$\lambda \geq \max_{D \in \mathcal{D}} D $	Uniform colorings. For a hypergraph $\mathcal{H}$ of order n , a proper λ -coloring $(S_1, S_2, \dots, S_\lambda)$
voloshin-hypergraph_F01495	211	■ 1.000	$\left(S_{\{1\}}, S_{\{2\}}, \ldots, S_{\{\lambda\}}\right)$	$(S_1, S_2, \dots, S_\lambda)$	$\left\lfloor \frac{n}{\lambda} \right\rfloor \leq S_i \leq \left\lceil \frac{n}{\lambda} \right\rceil.$
voloshin-hypergraph_F01496	211	■ 1.000	$a_{\{j\}}, \ b_{\{j\}}$	a_j, b_j	such a way that for every $D_j \in \mathcal{D}$, and
voloshin-hypergraph_F01497	211	■ 1.000	$a_{\{j\}}=0$	$a_j = 0$	least two vertices of the same color. Hence D_j may be monochromatic. If it is monochromatic and we use at least two colors, then some color is missing and some color uses all
voloshin-hypergraph_F01498	211	■ 1.000	$b_{\{j\}}=\max \left\{1,\left D_{\{j\}}\right -1\right\}$	$b_j = \max \{1, D_j - 1\}$	least two vertices of the same color. Hence D_j may be monochromatic. If it is monochromatic and we use at least two colors, then some color is missing and some color uses all
voloshin-hypergraph_F01499	211	■ 1.000	$b_{\{j\}}=1$	$b_j = 1$	matic and we use at least two colors, then some color is missing and some color uses all the vertices of D_j . Therefore, in the language of 1-regular colorings, we must put $a_j = 0$.
voloshin-hypergraph_F01500	211	■ 1.000	$a_{\{j\}}=\left\lfloor\frac{\left D_{\{j\}}\right }{\lambda}\right\rfloor$	$a_j = \left\lfloor \frac{ D_j }{\lambda} \right\rfloor$	the vertices of D_j . Therefore, in the language of 1-regular colorings, we have $b_j = D_j $. The last is equivalent to having no constraint on the color
voloshin-hypergraph_F01501	211	■ 1.000	$b_{\{j\}}=\left\lceil\frac{\left D_{\{j\}}\right }{\lambda}\right\rceil$	$b_j = \left\lceil \frac{ D_j }{\lambda} \right\rceil$	the vertices of D_j . Therefore, in the language of 1-regular colorings, we have $b_j = D_j $. The last is equivalent to having no constraint on the color
voloshin-hypergraph_F01502	211	■ 0.999	$b_{\{j\}}=\left D_{\{j\}}\right $	$b_j = D_j $	
voloshin-hypergraph_F01503	211	■ 0.998	$\left S_{\{i\}} \cap D_{\{j\}}\right \geq 2$	$ S_i \cap D_j \geq 2$	
voloshin-hypergraph_F01504	212	■ 1.000	$T=$	$T =$	Proof. We use Algorithm 8.3.1 that computes the cyclomatic number of $\mathcal{H}$. Let $T =$

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voloshin-hypergraph_F01505	212	0.982	$\wedge(\mathcal{H})+2$	$\wedge(\mathcal{H}) + 2$	Next, we cite some significant results in
voloshin-hypergraph_F01506	212	0.881	h, k, l	h, k, l	of length $< l$.
voloshin-hypergraph_F01507	212	0.972	$\chi(\mathcal{H})=k$	$\chi(\mathcal{H}) = k$	A hypergraph $\mathcal{H}$ is edge-critical if it contains no isolated vertices, $\chi(\mathcal{H}) = k$, $k \geq 2$
voloshin-hypergraph_F01508	212	1.000	$<l$	$< l$	and we
voloshin-hypergraph_F01509	212	1.000	$\chi(\mathcal{H})=k, k \geq 3$	$\chi(\mathcal{H}) = k, k \geq 3$	Lemma 10.1.1 (Zykov, 1974) Any hypergraph $\mathcal{H}$ with $\chi(\mathcal{H}) \geq 3$ can be transformed into
voloshin-hypergraph_F01510	212	0.998	$\chi(\mathcal{H}')=k-1$	$\chi(\mathcal{H}') = k - 1$	an edge-critical hypergraph by weak deletion of edges and vertices of degree ≤ 1 .
voloshin-hypergraph_F01511	212	1.000	$\chi(\mathcal{H}) \geq 3$	$\chi(\mathcal{H}) \geq 3$	The intersections of edges in cycles appear to be also important.
voloshin-hypergraph_F01512	212	1.000	$\chi(\mathcal{H})=3$	$\chi(\mathcal{H}) = 3$	has three edges that share a common vertex, then $\chi(\mathcal{H}) = 2$.
voloshin-hypergraph_F01513	212	0.727	$\chi(\mathcal{H})>k(\chi(\mathcal{H}) \leq k)$	$\chi(\mathcal{H}) > k(\chi(\mathcal{H}) \leq k)$	of [2].
voloshin-hypergraph_F01514	212	0.999	$\alpha(\mathcal{H}), \tau(\mathcal{H})$	$\alpha(\mathcal{H}), \tau(\mathcal{H})$	
voloshin-hypergraph_F01515	213	1.000	$m(x, \mathcal{H})$	$m(x, \mathcal{H})$	
voloshin-hypergraph_F01516	213	1.000	$\mathcal{D}_1(x) \subseteq \mathcal{D}(x)$	$\mathcal{D}_1(x) \subseteq \mathcal{D}(x)$	
voloshin-hypergraph_F01517	214	1.000	$M(\mathcal{H})$	$M(\mathcal{H})$	3.3. Recall that ω denotes the maximum cardinality of a clique, and the Szekeres-Wilf
voloshin-hypergraph_F01518	214	0.585	$\boldsymbol{\omega}$	$\boldsymbol{\omega}$	number is at least $\omega - 1$, see Proposition 3.3.1. We have seen in Theorem 3.3.1 that when considered for graphs the value $M(G)$ is closely related to chordal graphs
voloshin-hypergraph_F01519	214	1.000	$c(x)$	$c(x)$	Algorithm 3.3.1 (Greedy hypergraph coloring)
voloshin-hypergraph_F01520	214	1.000	$c=(c(\text{c}\text{left}(x_1)\text{right}), \text{c}\text{left}(x_2)\text{right}), \text{ldots}, \text{c}\text{left}(x_n)\text{right})\text{right})$	$c = (c(x_1), c(x_2), \dots, c(x_n))$	INPUT: An arbitrary hypergraph $\mathcal{H} = (X, \mathcal{U})$, $X = \{1, 2, \dots, n\}$. OUTPUT: A proper coloring $c = (c(1), c(2), \dots, c(n))$ of $\mathcal{H}$
voloshin-hypergraph_F01521	214	1.000	$\mathcal{H}; c(x)=0$	$\mathcal{H}; c(x) = 0$	INPUT: An arbitrary hypergraph $\mathcal{H} = (X, \mathcal{U})$, $X = \{1, 2, \dots, n\}$. OUTPUT: A proper coloring $c = (c(1), c(2), \dots, c(n))$ of $\mathcal{H}$
voloshin-hypergraph_F01522	214	1.000	$\mathcal{H}=(X, \mathcal{D}), X=\{1,2, \text{ldots}, n\}$	$\mathcal{H} = (X, \mathcal{D}), X = \{1, 2, \dots, n\}$	and label it x_n .
voloshin-hypergraph_F01523	214	1.000	$c=(c(1), c(2), \text{ldots}, c(n))$	$c = (c(1), c(2), \dots, c(n))$	2. Put $i := i - 1$; if $i = 0$, then go to step 5.
voloshin-hypergraph_F01524	214	0.994	$C=(0,0, \text{ldots}, 0), i=n, \mathcal{H}_n=\mathcal{H}$	$C = (0, 0, \dots, 0), i = n, \mathcal{H}_n = \mathcal{H}$	3. Strongly delete the vertex x_{i+1} and form an induced subhypergraph $\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$.
voloshin-hypergraph_F01525	214	0.994	$\mathcal{H}_n$	$\mathcal{H}_n$	3. Strongly delete the vertex x_{i+1} and form an induced subhypergraph $\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$.
voloshin-hypergraph_F01526	214	0.998	$i:=i-1$	$i := i - 1$	3. Color x_1 with the first color, $c(x_1) = 1$, i
voloshin-hypergraph_F01527	214	1.000	$\mathcal{H}_i=\mathcal{H}_{i+1}-x_{i+1}$	$\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$	7. Color x_i in $\mathcal{H}$ with the smallest suitable color from $\{1, 2, \dots, n\}$; go to step 6.
voloshin-hypergraph_F01528	214	1.000	$\mathcal{H}_i$	$\mathcal{H}_i$	8. Output $c = (c(1), c(2), \dots, c(n))$. End.
voloshin-hypergraph_F01529	214	1.000	$c(\text{c}\text{left}(x_1)\text{right})=1, i=1$	$c(x_1) = 1, i = 1$	Remark. It is important to observe that the alg
voloshin-hypergraph_F01530	214	1.000	$i=n+1$	$i = n + 1$	case results in a new color and at any step a proper

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voloshin-hypergraph_F01531	214	■ 0.992	$\{1,2, \ldots, n\}$	$\{1,2,\ldots,n\}$	
voloshin-hypergraph_F01532	215	■ 1.000	$t \leq M(\mathcal{H})$	$t \leq M(\mathcal{H})$	$m(v, \mathcal{H}_t) = \min m(z, \mathcal{H}_t) = M(\mathcal{H}).$
voloshin-hypergraph_F01533	215	■ 1.000	$t \geq M(\mathcal{H})$	$t \geq M(\mathcal{H})$	$m(v, \mathcal{H}_t) = \min m(z, \mathcal{H}_t) = M(\mathcal{H}).$
voloshin-hypergraph_F01534	215	■ 0.973	$y \in Y$	$y \in Y$	
voloshin-hypergraph_F01535	215	■ 1.000	$\mathcal{H}_{\{k\}}$	$\mathcal{H}_k$	Consequently, $t = M(\mathcal{H})$.
voloshin-hypergraph_F01536	215	■ 1.000	$t=M(\mathcal{H})$	$t = M(\mathcal{H})$	
voloshin-hypergraph_F01537	215	■ 1.000	$1,2, \ldots, t$	$1,2,\ldots,t$	Notice that Theorem 10.2.1 is a direct generalization of the procedure for computing $M(G)$ described at the beginning of Section 3.3., and Corollary 10.2.1 is a direct general-
voloshin-hypergraph_F01538	215	■ 0.972	$D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{t\}}$	$D_1, D_2, \ldots, D_t$	ization of Theorem 5.6.1
voloshin-hypergraph_F01539	215	■ 1.000	$\mathcal{H}_{\{5\}}=\mathcal{H}$	$\mathcal{H}_5 = \mathcal{H}$	
voloshin-hypergraph_F01540	215	■ 1.000	$X=\{1,2,3,4,5\}$	$X = \{1,2,3,4,5\}$	
voloshin-hypergraph_F01541	216	■ 1.000	$\min \{2,2,2,1,1\}=1$	$\min\{2,2,2,1,1\} = 1$	At this point the algorithm starts coloring the hypergraph $\mathcal{H}$ by assigning color 1 to vertex 2.
voloshin-hypergraph_F01542	216	■ 0.774	$\mathcal{H}_{\{4\}}=\mathcal{H}_{\{5\}}-5$	$\mathcal{H}_4 = \mathcal{H}_5 - 5$	at Step 5. Next it reconstructs the original hypergraph by adding vertices in inverse ordering
voloshin-hypergraph_F01543	216	■ 1.000	$\mathcal{H}_{\{5\}}, \mathcal{H}_{\{4\}}, \mathcal{H}_{\{3\}}, \mathcal{H}_{\{2\}}$	$\mathcal{H}_5, \mathcal{H}_4, \mathcal{H}_3, \mathcal{H}_2$	
voloshin-hypergraph_F01544	216	■ 1.000	$\mathcal{H}_{\{1\}}=(\{2\}, \emptyset)$	$\mathcal{H}_1 = (\{2\}, \emptyset)$	
voloshin-hypergraph_F01545	217	■ 1.000	$M(\mathcal{H})=2$	$M(\mathcal{H}) = 2$	maximum among all minimal mono-degrees found by Algorithm is 2. Hence by Theorem 10.2.1, $M(\mathcal{H}) = 2$. At last, one can easily see that $\gamma(\mathcal{H})=2 < M(\mathcal{H})+1=3$.
voloshin-hypergraph_F01546	217	■ 1.000	$\chi(\mathcal{H})=2 \leq M(\mathcal{H})+1=3$	$\chi(\mathcal{H}) = 2 \leq M(\mathcal{H}) + 1 = 3$	maximum among all minimal mono-degrees found by Algorithm is 2. Hence by Theorem 10.2.1, $M(\mathcal{H}) = 2$. At last, one can easily see that $\gamma(\mathcal{H})=2 < M(\mathcal{H})+1=3$.
voloshin-hypergraph_F01547	218	■ 1.000	$X=\left\{x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right\}, n \geq 1$	$X = \{x_1, x_2, \ldots, x_n\}, n \geq 1$	Let $X = \{x_1, x_2, \ldots, x_n\}, n \geq 1$. For a finite set, and let $\mathcal{C} = \{C_1, C_2, \ldots, C_l\}$ and $\mathcal{D} = \{D_1, D_2, \ldots, D_m\}$ be two families of subsets of X such that the size of every member of
voloshin-hypergraph_F01548	218	■ 1.000	$\mathcal{C}=\left\{C_{\{1\}}, C_{\{2\}}, \ldots, C_{\{l\}}\right\}$	$\mathcal{C} = \{C_1, C_2, \ldots, C_l\}$	Let $X = \{x_1, x_2, \ldots, x_n\}, n \geq 1$. For a finite set, and let $\mathcal{C} = \{C_1, C_2, \ldots, C_l\}$ and $\mathcal{D} = \{D_1, D_2, \ldots, D_m\}$ be two families of subsets of X such that the size of every member of
voloshin-hypergraph_F01549	218	■ 1.000	$\mathcal{D} =$	$\mathcal{D} =$	Let $X = \{x_1, x_2, \ldots, x_n\}, n \geq 1$. For a finite set, and let $\mathcal{C} = \{C_1, C_2, \ldots, C_l\}$ and $\mathcal{D} = \{D_1, D_2, \ldots, D_m\}$ be two families of subsets of X such that the size of every member of
voloshin-hypergraph_F01550	218	■ 1.000	$\left\{D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{m\}}\right\}$	$\{D_1, D_2, \ldots, D_m\}$	Let $X = \{x_1, x_2, \ldots, x_n\}, n \geq 1$, be a finite set, and let $\mathcal{C} = \{C_1, C_2, \ldots, C_l\}$ and $\mathcal{D} = \{D_1, D_2, \ldots, D_m\}$ be two families of subsets of X such that the size of every member of
voloshin-hypergraph_F01551	218	■ 1.000	$\mathcal{C} \cup \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$ is at least 2. Each of $\mathcal{C}, \mathcal{D}$ may be empty, and any subset of vertices may be
voloshin-hypergraph_F01552	218	■ 1.000	$\mathcal{C}, \mathcal{D}$	$\mathcal{C}, \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$ is at least 2. Each of $\mathcal{C}, \mathcal{D}$ may be empty, and any subset of vertices may be
voloshin-hypergraph_F01553	218	■ 0.224	$\mathcal{C} \neq \emptyset$	$\mathcal{C} \neq \emptyset$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1,2,\ldots,l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01554	218	■ 0.224	$I=\{1,2, \ldots, l\}$	$I = \{1,2,\ldots,l\}$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1,2,\ldots,l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01555	218	■ 0.224	$\mathcal{D} \neq \emptyset$	$\mathcal{D} \neq \emptyset$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1,2,\ldots,l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01556	218	■ 1.000	$J=\{1,2, \ldots, m\}$	$J = \{1,2,\ldots,m\}$	denote $J = \{1,2,\ldots,$
voloshin-hypergraph_F01557	218	■ 0.993	$\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	Definition 10.3.1 A mixed hypergraph is a triple $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ where X is called the vertex set, also denoted by $V(\mathcal{H})$. $\mathcal{C}$ is the family of subsets called $\mathcal{C}$ -edges, also denoted

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voloshin-hypergraph_F01558	218	■ 1.000	$V(\mathcal{H}), \mathcal{C}$	$V(\mathcal{H}), \mathcal{C}$	by $\mathcal{C}(\mathcal{H})$, and $\mathcal{D}$ is the family of subsets called $\mathcal{D}$ -edges, also denoted by $\mathcal{D}(\mathcal{H})$.
voloshin-hypergraph_F01559	218	■ 1.000	$\mathcal{C}(\mathcal{H})$	$\mathcal{C}(\mathcal{H})$	
voloshin-hypergraph_F01560	218	■ 1.000	$c(x), x \in X$	$c(x), x \in X$	Definition 10.1.6 A proper λ -coloring of a mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is a mapping $c: X \rightarrow \{1, 2, \dots, \lambda\}$ such that the following two conditions hold:
voloshin-hypergraph_F01561	218	■ 0.833	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	2) every $D \in \mathcal{D}$ has at least two vertices of Different colors.
voloshin-hypergraph_F01562	218	■ 1.000	$c: X \rightarrow \{1, 2, \dots, \lambda\}$	$c: X \rightarrow \{1, 2, \dots, \lambda\}$	A coloring of the mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is a mapping $c: X \rightarrow \{1, 2, \dots, \lambda\}$ such that
voloshin-hypergraph_F01563	218	■ 0.952	$\mathcal{C} = \emptyset$	$\mathcal{C} = \emptyset$	coloring, then $\mathcal{H}$ is called edge-colorable .
voloshin-hypergraph_F01564	218	■ 1.000	$X = \{1, 2\}, \mathcal{C} = \{\{1, 2\}\}$	$X = \{1, 2\}, \mathcal{C} = \{\{1, 2\}\}$	
voloshin-hypergraph_F01565	218	■ 1.000	$\mathcal{D} = \{\{1, 2\}\}$	$\mathcal{D} = \{\{1, 2\}\}$	
voloshin-hypergraph_F01566	218	■ 0.848	$\mathcal{C} = \{1, 2\}$	$\mathcal{C} = \{1, 2\}$	
voloshin-hypergraph_F01567	218	■ 1.000	$\mathcal{D} = \{1, 2\}$	$\mathcal{D} = \{1, 2\}$	
voloshin-hypergraph_F01568	219	■ 0.999	c_1, c_2	c_1, c_2	Two proper colorings c_1, c_2 of a mixed hypergraph $\mathcal{H}$ are said to be different , if there exists at least one vertex $v \in X$ that changes the color, i.e. $c_1(v) \neq c_2(v)$. Hence any permutation of colors results in a new proper coloring since it represents a different mapping. Let
voloshin-hypergraph_F01569	219	■ 1.000	$c_1(x) \neq c_2(x)$	$c_1(x) \neq c_2(x)$	of colors results in a new proper coloring since it represents a different mapping. Let
voloshin-hypergraph_F01570	219	■ 1.000	$P(\mathcal{H}, \lambda)$	$P(\mathcal{H}, \lambda)$	graphs, we need to identify proper colorings which use all the colors; in this case we use i
voloshin-hypergraph_F01571	219	■ 1.000	$1 \leq i \leq n$	$1 \leq i \leq n$	The maximum number of colors may now be specified as follows.
voloshin-hypergraph_F01572	219	■ 1.000	$\bar{\chi}(\mathcal{H})$	$\bar{\chi}(\mathcal{H})$	Definition 10.1.7 A mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is called polychromatic if all the colors $c(y), y \in Y$, are pairwise different. In a proper coloring
voloshin-hypergraph_F01573	219	■ 1.000	$c(y), y \in Y$	$c(y), y \in Y$	Any strict i -coloring of $\mathcal{H}$ induces a partition of the vertex set X into i nonempty
voloshin-hypergraph_F01574	219	■ 0.991	$\dots, X_i$	$\dots, X_i$	Definition 10.1.8 The chromatic number
voloshin-hypergraph_F01575	219	■ 1.000	$\mathcal{C} = X_1 \cup X_2 \cup \dots \cup X_i$	$\mathcal{C} = X_1 \cup X_2 \cup \dots \cup X_i$	
voloshin-hypergraph_F01576	219	■ 1.000	$r_i(\mathcal{H}) = r_i, 1 \leq i \leq n$	$r_i(\mathcal{H}) = r_i, 1 \leq i \leq n$	is called the chromatic spectrum of the mixed hypergraph $\mathcal{H}$.
voloshin-hypergraph_F01577	220	■ 0.693	$\mathcal{X}$	$\mathcal{X}$	In fact, since we have lower and upper chromatic numbers for every colorable mixed hypergraph $\mathcal{H}$ there are no feasible partitions into less than γ color classes and there no
voloshin-hypergraph_F01578	220	■ 0.998	$\bar{\chi}$	$\bar{\chi}$	feasible partitions into more than $\bar{\chi}$ color classes. Therefore, chromatic spectrum generally
voloshin-hypergraph_F01579	220	■ 0.924	$r_i i!$	$r_i i!$	$\binom{\lambda}{i}$ ways to choose the subset of i colors. Consequently the number of proper λ -colorings
voloshin-hypergraph_F01580	220	■ 0.924	$\lambda \geq i$	$\lambda \geq i$	$\binom{\lambda}{i}$ ways to choose the subset of i colors. Consequently the number of proper λ -colorings
voloshin-hypergraph_F01581	220	■ 1.000	$\binom{\lambda}{i}$	$\binom{\lambda}{i}$	generated by all feasible partitions into i subsets is $\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$
voloshin-hypergraph_F01582	220	■ 0.993	$\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$	$\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$	$r_i \lambda^{(i)}$. To obtain the number of all proper λ -coloring it is sufficient then to add all these numbers for every $\gamma(\mathcal{H}) < i < \bar{\gamma}(\mathcal{H})$. Thus we arrive to the following conclusion:
voloshin-hypergraph_F01583	220	■ 0.998	$r_i \lambda^{(i)}$	$r_i \lambda^{(i)}$	numbers for every $\lambda(\mathcal{H}) \leq i \leq \bar{\lambda}(\mathcal{H})$, thus we arrive to the following conclusion:

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01584	220	■ 0.907	$\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	$\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	Proposition 10.3.1 The number of proper λ -colorings $P(\mathcal{H}, \lambda)$ of a colorable mixed hy
voloshin-hypergraph_F01585	220	■ 0.968	$r_{\{\bar{\chi}\}}$	$r_{\bar{\chi}}$	what proves the statement.
voloshin-hypergraph_F01586	220	■ 1.000	$\mathcal{H}=(X, \mathcal{C}, \emptyset)$	$\mathcal{H}=(X, \mathcal{C}, \emptyset)$	coloring of a subset of vertices.
voloshin-hypergraph_F01587	220	■ 0.996	$\mathcal{H}_{\mathcal{C}}=(X, \mathcal{C})$	$\mathcal{H}_{\mathcal{C}}=(X, \mathcal{C})$	
voloshin-hypergraph_F01588	220	■ 1.000	$\mathcal{H}=(X, \emptyset, \mathcal{D})$	$\mathcal{H}=(X, \emptyset, \mathcal{D})$	
voloshin-hypergraph_F01589	220	■ 1.000	$\mathcal{H}_{\mathcal{D}}=(X, \mathcal{D})$	$\mathcal{H}_{\mathcal{D}}=(X, \mathcal{D})$	
voloshin-hypergraph_F01590	220	■ 0.699	$\mathcal{H}_{\mathcal{C}}$	$\mathcal{H}_{\mathcal{C}}$	
voloshin-hypergraph_F01591	220	■ 0.699	$\mathcal{H}_{\mathcal{D}}$	$\mathcal{H}_{\mathcal{D}}$	
voloshin-hypergraph_F01592	221	■ 0.942	$\bar{\chi}(\mathcal{H}_{\mathcal{C}})$	$\bar{\chi}(\mathcal{H}_{\mathcal{C}})$	mixed hypergraph $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$, the following evident inequalities hold:
voloshin-hypergraph_F01593	221	■ 0.997	$\mathcal{H}_{\mathcal{C}}$	$\mathcal{H}_{\mathcal{C}}$	
voloshin-hypergraph_F01594	221	■ 0.742	$\chi(\mathcal{H}_{\mathcal{D}})=1, r_1(\mathcal{H}_{\mathcal{C}})=1$	$\chi(\mathcal{H}_{\mathcal{D}})=1, r_1(\mathcal{H}_{\mathcal{C}})=1$	$1 \leq \chi(\mathcal{H}_{\mathcal{D}}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_{\mathcal{C}}) \leq n.$ (10.3)
voloshin-hypergraph_F01595	221	■ 0.742	$\bar{\chi}(\mathcal{H}_{\mathcal{D}})=n(\mathcal{H}), r_n(\mathcal{H}_{\mathcal{D}})=1$	$\bar{\chi}(\mathcal{H}_{\mathcal{D}})=n(\mathcal{H}), r_n(\mathcal{H}_{\mathcal{D}})=1$	$1 \leq \chi(\mathcal{H}_{\mathcal{D}}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_{\mathcal{C}}) \leq n.$ (10.3)
voloshin-hypergraph_F01596	221	■ 0.585	$X=\{1,2,3,4,5\}, \quad \mathcal{C}=\{C_1, C_2\}=\{\{1,4,5\}, \{1,3,4\}\}$	$X=\{1,2,3,4,5\}, \quad \mathcal{C}=\{C_1, C_2\}=\{\{1,4,5\}, \{1,3,4\}\}$	vertices $\{1,3,4\}$ can be neither monochromatic nor polychromatic; since it has three elements, necessarily two colors must be used. A coloring which uses the minimum number
voloshin-hypergraph_F01597	221	■ 0.585	$\mathcal{D}=\{D_1, D_2\}=\{ \{1,4,5\}, \{1,3,4\} \}$	$\mathcal{D}=\{D_1, D_2\}=\{ \{1,4,5\}, \{1,3,4\} \}$	vertices $\{1,3,4\}$ can be neither monochromatic nor polychromatic; since it has three elements, necessarily two colors must be used. A coloring which uses the minimum number
voloshin-hypergraph_F01598	221	■ 1.000	$\bar{\chi}(\mathcal{H})=4$	$\bar{\chi}(\mathcal{H})=4$	is $c_2=(1,2,3,1,4)$. There are also other proper colorings. The $\mathcal{C}$ -hypergraph $\mathcal{H}_{\mathcal{C}}$ is
voloshin-hypergraph_F01599	221	■ 1.000	$C_2=D_1=\{1,3,4\}$	$C_2=D_1=\{1,3,4\}$	obtained from $\mathcal{H}$ by weak deletion of all $\mathcal{D}$ -edges; the $\mathcal{D}$ -hypergraph $\mathcal{H}_{\mathcal{D}}$ is obtained from
voloshin-hypergraph_F01600	221	■ 0.998	$c_1=(1,2,2,1,1)$	$c_1=(1,2,2,1,1)$	
voloshin-hypergraph_F01601	221	■ 0.911	$c_2=(1,2,3,1,4)$	$c_2=(1,2,3,1,4)$	It is easy to construct uncolorable mixed hypergraphs. Both $\mathcal{H}_1$ and $\mathcal{H}_2$, see Figure 10.7, are uncolorable mixed hypergraphs. $\mathcal{H}_1$ contains $\mathcal{D}$ -edges E , which require three different
voloshin-hypergraph_F01602	222	■ 0.808	$\bar{\chi}(\mathcal{H}_{\mathcal{C}}) < \chi(\mathcal{H}_{\mathcal{D}})$	$\bar{\chi}(\mathcal{H}_{\mathcal{C}}) < \chi(\mathcal{H}_{\mathcal{D}})$	then it is a bi-edge . For example, in Figure 10.6, the subset of vertices $\{1,3,4\}$ is a $\mathcal{C}$ -edge and a $\mathcal{D}$ -edge at the same time, i.e. it is a bi-edge. A mixed hypergraph $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is a
voloshin-hypergraph_F01603	222	■ 1.000	$\{1,3,4\}$	$\{1,3,4\}$	$\mathcal{H}'=(X, \mathcal{E})$ is the underlying hypergraph of $\mathcal{H}$. Using underlying hypergraphs, many
voloshin-hypergraph_F01604	222	■ 1.000	$\mathcal{C}=\mathcal{D}$	$\mathcal{C}=\mathcal{D}$	introduced for hypergraphs. A mixed hypergraph $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is called uniform if $\mathcal{C}=\mathcal{D}$ and r if it is a path in $\mathcal{H}_{\mathcal{C}}$ ($\mathcal{H}_{\mathcal{D}}$), respectively. $\mathcal{H}$ is an r-uniform mixed hypergraph if $\mathcal{H}'$ is an
voloshin-hypergraph_F01605	222	■ 0.806	$\mathcal{E}=\mathcal{C} \cup \mathcal{D}$	$\mathcal{E}=\mathcal{C} \cup \mathcal{D}$	
voloshin-hypergraph_F01606	222	■ 0.994	$\mathcal{H}'=(X, \mathcal{E})$	$\mathcal{H}'=(X, \mathcal{E})$	r -uniform hypergraph. $\mathcal{H}$ is connected if $\mathcal{H}'$ is connected; otherwise $\mathcal{H}$ is disconnected .
voloshin-hypergraph_F01607	222	■ 0.907	$\mathcal{H}_{\mathcal{C}}(\mathcal{H}_{\mathcal{D}})$	$\mathcal{H}_{\mathcal{C}}(\mathcal{H}_{\mathcal{D}})$	by strong deletions of the respective vertices.
voloshin-hypergraph_F01608	222	■ 1.000	$\mathcal{H}_Y=(Y, \mathcal{C}', \mathcal{D}')$	$\mathcal{H}_Y=(Y, \mathcal{C}', \mathcal{D}')$	independent ($\mathcal{D}$-stable , $\mathcal{D}$-independent) if it contains no $\mathcal{C}$ -edge (no $\mathcal{D}$ -edge) as a subset.
voloshin-hypergraph_F01609	222	■ 1.000	$\mathcal{C}'$	$\mathcal{C}'$	Exercises 10.3.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01610	222	■ 1.000	$R(\mathcal{H})$	$R(\mathcal{H})$	—
voloshin-hypergraph_F01611	223	■ 1.000	$\mathcal{H}=(X, \mathcal{C})$	$\mathcal{H}=(X, \mathcal{C})$	
voloshin-hypergraph_F01612	223	■ 1.000	$\mathcal{C}(x)$	$\mathcal{C}(x)$	
voloshin-hypergraph_F01613	223	■ 0.639	$\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$	$\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$	
voloshin-hypergraph_F01614	223	■ 0.639	$\mathcal{C}(x) \cap \mathcal{C}(y)$	$\mathcal{C}(x) \cap \mathcal{C}(y)$	
voloshin-hypergraph_F01615	224	■ 1.000	$o(x, \mathcal{H})=0$	$o(x, \mathcal{H})=0$	Find $o(x, \mathcal{H})=0$ implies that there is another vertex $y \in X$ which is contained in all C-edges containing x. The term “originality” is justified as follows. If the vertices of a
voloshin-hypergraph_F01616	224	■ 0.997	$ \mathcal{C}(x) -1$	$ \mathcal{C}(x) -1$	Definition 10.4.1 The value
voloshin-hypergraph_F01617	224	■ 1.000	$O(\mathcal{H})$	$O(\mathcal{H})$	if all the vertices of M have the same color as x and are reachable from x by C -pe
voloshin-hypergraph_F01618	224	■ 0.988	$M \subseteq X$	$M \subseteq X$	Now, in order to find a lower bound for the upper chromatic number, we propose a quasi-
voloshin-hypergraph_F01619	224	■ 1.000	$x \in M$	$x \in M$	greedy algorithm for an arbitrary $\mathcal{H}=(X, \mathcal{C})$. As for the lower chromatic number, the idea
voloshin-hypergraph_F01620	224	■ 1.000	$M \cap \mathcal{C}(x)$	$M \cap \mathcal{C}(x)$	greedy algorithm for an arbitrary $\mathcal{H}=(X, \mathcal{C})$. As for the lower chromatic number, the idea
voloshin-hypergraph_F01621	224	■ 0.783	M	M	is to find a good ordering of the vertices and greedily color the vertices step by step, making
voloshin-hypergraph_F01622	224	■ 1.000	$\mathcal{H}=(X, \mathcal{C}), X =n$	$\mathcal{H}=(X, \mathcal{C}), X =n$	using the local information. But this time, at each step we use a new color for the next
voloshin-hypergraph_F01623	224	■ 1.000	$i=n, \mathcal{H}_n=\mathcal{H}$	$i=n, \mathcal{H}_n=\mathcal{H}$	
voloshin-hypergraph_F01624	225	■ 0.706	$U=\{1\}$	$U=\{1\}$	5. Set the list of used colors $U=\{1\}$. color the vertex x_1 with the color 1: set $i=$
voloshin-hypergraph_F01625	225	■ 0.706	$i=$	$i=$	5. Set the list of used colors $U=\{1\}$. color the vertex x_1 with the color 1: set $i=$
voloshin-hypergraph_F01626	225	■ 0.999	$i=n$	$i=n$	6. Put $i:=i+1$; if $i=n$, then go to step 10; color the vertex x_i of $\mathcal{H}_i$ with color new;
voloshin-hypergraph_F01627	225	■ 1.000	$U:=U \cup \{$	$U:=U \cup \{$	put $U:=U \cup \{new\}$, $new:=new+1$.
voloshin-hypergraph_F01628	225	■ 1.000	$\}$	$\}$	put $U:=U \cup \{new\}$, $new:=new+1$.
voloshin-hypergraph_F01629	225	■ 1.000	$:=$	$:=$	put $U:=U \cup \{new\}$, $new:=new+1$.
voloshin-hypergraph_F01630	225	■ 0.758	$U:=U-\{$	$U:=U-\{$	9. Re-color all the vertices from the monochromatic component $MC(y)$ with tl
voloshin-hypergraph_F01631	225	■ 0.971	$M \cap \mathcal{C}(y)$	$MC(y)$	10. Re-number the colors of U in increasing order, and
voloshin-hypergraph_F01632	225	■ 1.000	U	U	Remark. The monochromatic component re-coloring u
voloshin-hypergraph_F01633	225	■ 1.000	$ X =n, \mathcal{C} =k$	$ X =n, \mathcal{C} =k$	leads to the k -chromatic chain re-coloring by Kempe, see 8.
voloshin-hypergraph_F01634	225	■ 1.000	$n \times k$	$n \times k$	steps 1-4 may be implemented, in the worst case, in $O(n \times k)$ steps. To check if the coloring is proper, it is necessary to look through every column, so
voloshin-hypergraph_F01635	225	■ 1.000	$O(n \times k)$	$O(nk)$	the complexity of step 7 is $O(nk)$. Simultaneously, one can label the wrongly colored C -
voloshin-hypergraph_F01636	225	■ 1.000	$O(n^2 \times k)$	$O(n^2k)$	edges. Finding the neighborhood of x_i and simultaneously identifying the vertex y takes

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voloshin-hypergraph_F01637	225	1.000	$O\left(n^3 k\right)$	$O\left(n^3 k\right)$	$O(nk)$ steps. Finding and re-coloring the monochromatic component
voloshin-hypergraph_F01638	225	1.000	$X=$	$X=$	Form the induced $\mathcal{C}$ -hypergraph $\mathcal{H}_3 = (X_3, \mathcal{C}_3)$ with $X_3 = \{3, 4, 5\}$, $\mathcal{C}_3 = \{C_3\}$. The
voloshin-hypergraph_F01639	225	0.852	$\{1, 2, 3, 4, 5\}$, $\mathcal{C}=\left\{C_{\{1\}}, C_{\{2\}}, C_{\{3\}}, C_{\{4\}}, C_{\{5\}}\right\}$, $C_{\{1\}}=\{1, 2, 3\}$, $C_{\{2\}}=\{2, 3, 4\}$, $C_{\{3\}}=\{3, 4, 5\}$	$\{1, 2, 3, 4, 5\}, \mathcal{C}=\left\{C_1, C_2, C_3, C_4, C_5\right\}, C_1=\{1, 2, 3\}, C_2=\{2, 3, 4\}, C_3=\{3, 4, 5\}$	first vertex with minimum originality is $x_3=3$.
voloshin-hypergraph_F01640	225	1.000	$C_{\{4\}}=\{4, 5, 1\}$	$C_4=\{4, 5, 1\}$	Form the induced $\mathcal{C}$ -hypergraph $\mathcal{H}_2 = (X_2, \mathcal{C}_2)$ with $X_2 = \{4, 5\}$, $\mathcal{C}_2 = \{\emptyset\}$. The first
voloshin-hypergraph_F01641	225	1.000	$C_{\{5\}}=\{5, 1, 2\}$	$C_5=\{5, 1, 2\}$	Form the induced $\mathcal{C}$ -hypergraph $\mathcal{H}_2 = (X_2, \mathcal{C}_2)$ with $X_2 = \{4, 5\}$, $\mathcal{C}_2 = \{\emptyset\}$. The first
voloshin-hypergraph_F01642	225	0.669	$x_{\{5\}}=1$	$x_5=1$	vertex with minimum originanty is $x_2=4$.
voloshin-hypergraph_F01643	225	0.967	$\mathcal{H}_{\{4\}}=\left(X_{\{4\}}, \mathcal{C}_{\{4\}}\right)$	$\mathcal{H}_4=\left(X_4, \mathcal{C}_4\right)$	
voloshin-hypergraph_F01644	225	0.967	$X_{\{4\}}=\{2, 3, 4, 5\}$, $\mathcal{C}_{\{4\}}=\left\{C_{\{2\}}, C_{\{3\}}\right\}$	$X_4=\{2, 3, 4, 5\}, \mathcal{C}_4=\left\{C_2, C_3\right\}$	
voloshin-hypergraph_F01645	225	1.000	$x_{\{4\}}=2$	$x_4=2$	
voloshin-hypergraph_F01646	225	1.000	$\mathcal{H}_{\{3\}}=\left(X_{\{3\}}, \mathcal{C}_{\{3\}}\right)$	$\mathcal{H}_3=\left(X_3, \mathcal{C}_3\right)$	
voloshin-hypergraph_F01647	225	1.000	$X_{\{3\}}=\{3, 4, 5\}$, $\mathcal{C}_{\{3\}}=\left\{C_{\{3\}}\right\}$	$X_3=\{3, 4, 5\}, \mathcal{C}_3=\left\{C_3\right\}$	
voloshin-hypergraph_F01648	225	1.000	$x_{\{3\}}=3$	$x_3=3$	
voloshin-hypergraph_F01649	225	1.000	$\mathcal{H}_{\{2\}}=\left(X_{\{2\}}, \mathcal{C}_{\{2\}}\right)$	$\mathcal{H}_2=\left(X_2, \mathcal{C}_2\right)$	
voloshin-hypergraph_F01650	225	1.000	$X_{\{2\}}=\{4, 5\}$, $\mathcal{C}_{\{2\}}=\{\emptyset\}$	$X_2=\{4, 5\}, \mathcal{C}_2=\{\emptyset\}$	
voloshin-hypergraph_F01651	225	1.000	$x_{\{2\}}=4$	$x_2=4$	
voloshin-hypergraph_F01652	226	0.984	$\mathcal{H}_{\{1\}}=\left(X_{\{1\}}, \mathcal{C}_{\{1\}}\right)$	$\mathcal{H}_1=\left(X_1, \mathcal{C}_1\right)$	means the vertex is not colored, $i=1, \dots, 5$. Set $c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$. Start coloring. Step 6: $c=$
voloshin-hypergraph_F01653	226	0.984	$X_{\{1\}}=\{5\}$, $\mathcal{C}_{\{1\}}=\{\emptyset\}$	$X_1=\{5\}, \mathcal{C}_1=\{\emptyset\}$	means the vertex is not colored, $i=1, \dots, 5$. Set $c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$. Start coloring. Step 6: $c=$
voloshin-hypergraph_F01654	226	1.000	$x_{\{1\}}=5$	$x_1=5$	$\left(0, 0, 0, 0, 1\right)$
voloshin-hypergraph_F01655	226	0.976	$c(i)$	$c(i)$	$\mathcal{C}$ -hypergraph $\mathcal{H}_2$.
voloshin-hypergraph_F01656	226	0.976	$i, c(i)=0$	$i, c(i)=0$	$\mathcal{C}$ -hypergraph $\mathcal{H}_2$.
voloshin-hypergraph_F01657	226	1.000	$i=1, \ldots, 5$	$i=1, \dots, 5$	Step 6: $c=(0, 0, 3, 2, 1)$. Step 7: $\mathcal{C}$ -edge
voloshin-hypergraph_F01658	226	1.000	$c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$	$c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$	Step 8: re-coloring: $c=(0, 0, 2, 2, 1)$. Set $c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$. Start coloring. Step 6: $c=$
voloshin-hypergraph_F01659	226	1.000	$c=$	$c=$	Step 8: re-coloring: $c=(0, 0, 2, 2, 1)$. Set $c=\left(c(1), c(2), c(3), c(4), c(5)\right)=\left(0, 0, 0, 0, 0\right)$. Start coloring. Step 6: $c=$
voloshin-hypergraph_F01660	226	0.921	$c=\left(0, 0, 0, 2, 1\right)$	$c=\left(0, 0, 0, 2, 1\right)$	Step 7: there are no monochromatic $\mathcal{C}$ -edges in the $\mathcal{C}$ -hypergraph $\mathcal{H}_3$. Step 6: $c=\left(0, 3, 2, 2, 1\right)$
voloshin-hypergraph_F01661	226	1.000	$c=\left(0, 0, 3, 2, 1\right)$	$c=\left(0, 0, 3, 2, 1\right)$	Step 6: $c=\left(4, 3, 2, 2, 1\right)$.
voloshin-hypergraph_F01662	226	0.991	$c=\left(0, 0, 2, 2, 1\right)$	$c=\left(0, 0, 2, 2, 1\right)$	Step 7: the $\mathcal{C}$ -edges C_1, C_4, C_5 are ea

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voloshin-hypergraph_F01663	226	■ 0.980	$c=(0,3,2,2,1)$	$c = (0, 3, 2, 2, 1)$	
voloshin-hypergraph_F01664	226	■ 0.964	$c=(4,3,2,2,1)$	$c = (4, 3, 2, 2, 1)$	
voloshin-hypergraph_F01665	226	■ 1.000	$C_{\{1\}}, C_{\{4\}}, C_{\{5\}}$	C_1, C_4, C_5	
voloshin-hypergraph_F01666	226	■ 1.000	$\mathcal{H}_{\{5\}}$	$\mathcal{H}_5$	
voloshin-hypergraph_F01667	226	■ 0.910	$c(5)=1: c=(1,3,2,2,1)$	$c(5) = 1 : c = (1, 3, 2, 2, 1)$	
voloshin-hypergraph_F01668	226	■ 1.000	$M C(3)=\{3,4\}: c=(1,3,1,1,1)$	$MC(3) = \{3, 4\} : c = (1, 3, 1, 1, 1)$	
voloshin-hypergraph_F01669	226	■ 0.948	$\mathcal{H}_{\{1\}}=\mathcal{H}$	$\mathcal{H}_1 = \mathcal{H}$	
voloshin-hypergraph_F01670	227	■ 1.000	$c=(1,2,1,1,1)$	$c = (1, 2, 1, 1, 1)$	
voloshin-hypergraph_F01671	227	■ 1.000	$C=(4,3,2,2,1)$	$C = (4, 3, 2, 2, 1)$	
voloshin-hypergraph_F01672	227	■ 0.982	$1,2,3,4,5 \ldots$	$1, 2, 3, 4, 5 \dots$	
voloshin-hypergraph_F01673	227	■ 0.999	$t \leq O(\mathcal{H})$	$t \leq O(\mathcal{H})$	
voloshin-hypergraph_F01674	227	■ 1.000	$t \leq O(\mathcal{H})-1$	$t \leq O(\mathcal{H}) - 1$	
voloshin-hypergraph_F01675	227	■ 1.000	$\geq t+1$	$\geq t + 1$	
voloshin-hypergraph_F01676	227	■ 1.000	$t=O(\mathcal{H})$	$t = O(\mathcal{H})$	
voloshin-hypergraph_F01677	227	■ 1.000	$O(\mathcal{H})+1$	$O(\mathcal{H}) + 1$	
voloshin-hypergraph_F01678	227	■ 1.000	$c(y)$	$c(y)$	
voloshin-hypergraph_F01679	227	■ 0.971	$b\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right) \mathcal{C}$	$b\left(x_i, \mathcal{H}_i\right) \mathcal{C}$	
voloshin-hypergraph_F01680	227	■ 0.990	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)=\left \mathcal{C}\left(x_{\{i\}}\right)\right -b\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right) \mathcal{C}$	$o\left(x_i, \mathcal{H}_i\right)=\left \mathcal{C}\left(x_i\right)\right -b\left(x_i, \mathcal{H}_i\right) \mathcal{C}$	
voloshin-hypergraph_F01681	227	■ 1.000	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)$	$o\left(x_i, \mathcal{H}_i\right)$	
voloshin-hypergraph_F01682	227	■ 1.000	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)+1$	$o\left(x_i, \mathcal{H}_i\right)+1$	
voloshin-hypergraph_F01683	227	■ 1.000	$i, 1 \leq i \leq n$	$i, 1 \leq i \leq n$	
voloshin-hypergraph_F01684	227	■ 1.000	$O(\mathcal{H})=0$	$O(\mathcal{H}) = 0$	
voloshin-hypergraph_F01685	227	■ 1.000	$ U =p$	$ U = p$	
voloshin-hypergraph_F01686	227	■ 1.000	$\bar{\chi}(\mathcal{H}) \geq p$	$\bar{\chi}(\mathcal{H}) \geq p$	
voloshin-hypergraph_F01687	228	■ 1.000	$n=2,3$	$n = 2, 3$	
voloshin-hypergraph_F01688	228	■ 1.000	$O\left(\mathcal{H}_{\{Y\}}\right)=0$	$O\left(\mathcal{H}_Y\right)=0$	

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voloshin-hypergraph_F01689	228	■ 1.000	$Y \subset X$	$Y \subset X$	
voloshin-hypergraph_F01690	228	■ 0.902	$\overline{\mathrm{X}}(\mathrm{H})$	$\overline{X}(\mathcal{H})$	
voloshin-hypergraph_F01691	229	■ 0.934	$\mathrm{C} \neq \emptyset$	$\mathcal{C} \neq \emptyset$	
voloshin-hypergraph_F01692	229	■ 0.938	$(X, \mathrm{C}, \mathrm{D})$	$(X, \mathcal{C}, \mathcal{D})$	
voloshin-hypergraph_F01693	229	■ 1.000	$C_{\mathrm{i}} \subseteq C_{\mathrm{j}}$	$C_i \subseteq C_j$	
voloshin-hypergraph_F01694	229	■ 1.000	$P(\mathrm{H}, \lambda) = P(\mathrm{H} - C_{\mathrm{j}}, \lambda), R(\mathrm{H}) = R(\mathrm{H} - C_{\mathrm{j}}), i, j \in I$	$P(\mathcal{H}, \lambda) = P(\mathcal{H} - C_j, \lambda), R(\mathcal{H}) = R(\mathcal{H} - C_j), i, j \in I$	
voloshin-hypergraph_F01695	229	■ 1.000	C_{j}	C_j	
voloshin-hypergraph_F01696	229	■ 1.000	$D_{\mathrm{i}} \subseteq D_{\mathrm{j}}$	$D_i \subseteq D_j$	
voloshin-hypergraph_F01697	229	■ 1.000	$P(\mathrm{H}, \lambda) = P(\mathrm{H} - D_{\mathrm{j}}, \lambda), R(\mathrm{H}) = R(\mathrm{H} - D_{\mathrm{j}}), i, j \in J$	$P(\mathcal{H}, \lambda) = P(\mathcal{H} - D_j, \lambda), R(\mathcal{H}) = R(\mathcal{H} - D_j), i, j \in J$	
voloshin-hypergraph_F01698	230	■ 1.000	$\mathrm{left}\{x_{\mathrm{k}}, x_{\mathrm{l}}\}\mathrm{right}\notin \mathrm{D}$	$\{x_k, x_l\} \notin \mathcal{D}$	
voloshin-hypergraph_F01699	230	■ 1.000	$\mathrm{left}\{x_{\mathrm{k}}, x_{\mathrm{l}}\}\mathrm{right}\notin \mathrm{C}$	$\{x_k, x_l\} \notin \mathcal{C}$	
voloshin-hypergraph_F01700	230	■ 0.931	x_{l}	x_l	
voloshin-hypergraph_F01701	230	■ 1.000	$C_{\mathrm{t}} = \mathrm{left}\{x_{\mathrm{k}}, x_{\mathrm{l}}\}\mathrm{right}$	$C_t = \{x_k, x_l\}$	
voloshin-hypergraph_F01702	230	■ 1.000	$t \in I$	$t \in I$	
voloshin-hypergraph_F01703	230	■ 1.000	$x_{\mathrm{k}}, x_{\mathrm{l}} \in X$	$x_k, x_l \in X$	
voloshin-hypergraph_F01704	230	■ 1.000	$C_{\mathrm{t}} \neq D_{\mathrm{s}}$	$C_t \neq D_s$	
voloshin-hypergraph_F01705	230	■ 1.000	$s \in J$	$s \in J$	
voloshin-hypergraph_F01706	230	■ 0.619	$\mathrm{H}_{\mathrm{1}} = \mathrm{left}(X_{\mathrm{1}}, \mathrm{C}^{\mathrm{1}}, \mathrm{D}^{\mathrm{1}}), X_{\mathrm{1}} = \mathrm{left}(X \backslash \mathrm{left}\{x_{\mathrm{k}}, x_{\mathrm{l}}\}\mathrm{right}) \cup \{y\}, \quad y$	$\mathcal{H}_1 = (X_1, \mathcal{C}^1, \mathcal{D}^1), X_1 = (X \setminus \{x_k, x_l\}) \cup \{y\}, \quad y$	
voloshin-hypergraph_F01707	230	■ 1.000	$x_{\mathrm{k}} \in D_{\mathrm{j}}$	$x_k \in D_j$	
voloshin-hypergraph_F01708	230	■ 1.000	$x_{\mathrm{l}} \in D_{\mathrm{j}}, j \in J$	$x_l \in D_j, j \in J$	
voloshin-hypergraph_F01709	230	■ 1.000	$D_{\mathrm{j}}^{\mathrm{1}} = \mathrm{left}(D_{\mathrm{j}} \backslash \mathrm{left}\{x_{\mathrm{k}}, x_{\mathrm{l}}\}\mathrm{right}) \cup \{y\}$	$D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$	
voloshin-hypergraph_F01710	230	■ 1.000	$D_{\mathrm{j}}^{\mathrm{1}} = D_{\mathrm{j}}$	$D_j^1 = D_j$	
voloshin-hypergraph_F01711	230	■ 1.000	$x_{\mathrm{k}} \in C_{\mathrm{i}}$	$x_k \in C_i$	
voloshin-hypergraph_F01712	230	■ 1.000	$x_{\mathrm{l}} \in C_{\mathrm{i}}, i \in I, i \neq t$	$x_l \in C_i, i \in I, i \neq t$	

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voloshin-hypergraph_F01713	230	■ 1.000	$C_{i}^1=\left(C_{i}\setminus\backslash\left\{x_k,x_l\right\},\ x_l\right)\cup\left\{y\right\}$	$C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$	Any C -edge of size 2 can be contracted unless it coincides with a $\mathcal{D}$ -edge of size 2.
voloshin-hypergraph_F01714	230	■ 1.000	$C_{i}^1=C_{i}$	$C_i^1 = C_i$	Any C -edge of size 2 can be contracted unless it coincides with a $\mathcal{D}$ -edge of size 2.
voloshin-hypergraph_F01715	230	■ 1.000	$n(\mathcal{H})=n\left(\mathcal{H}_1\right)+1$	$n(\mathcal{H}) = n(\mathcal{H}_1) + 1$	The algorithm that allows us to compute $P(\mathcal{H}, \lambda)$ and $R(\mathcal{H})$ for any mixed hypergraph
voloshin-hypergraph_F01716	230	■ 1.000	$R(\mathcal{H})=R\left(\mathcal{H}_1\right)$	$R(\mathcal{H}) = R(\mathcal{H}_1)$	The algorithm that allows us to compute $P(\mathcal{H}, \lambda)$ and $R(\mathcal{H})$ for any mixed hypergraph
voloshin-hypergraph_F01717	230	■ 1.000	$r_{i}(\mathcal{H})=r_{i}\left(\mathcal{H}_1\right)$	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$	$\mathcal{H} = (X, C, \mathcal{D})$ is a generalization of the connection-contraction algorithm for graphs (see [10]).
voloshin-hypergraph_F01718	230	■ 1.000	$i=1,2,\ldots,n-1$	$i = 1, 2, \dots, n - 1$	$\mathcal{H} = (X, C, \mathcal{D})$ is a generalization of the connection-contraction algorithm for graphs (see [10]).
voloshin-hypergraph_F01719	230	■ 1.000	$r_{n}(\mathcal{H})=0$	$r_n(\mathcal{H}) = 0$	$\mathcal{H} = (X, C, \mathcal{D})$ is a generalization of the connection-contraction algorithm for graphs (see [10]).
voloshin-hypergraph_F01720	230	■ 1.000	C_{t}	C_t	$\mathcal{H} = (X, C, \mathcal{D})$ is a generalization of the connection-contraction algorithm for graphs (see [10]).
voloshin-hypergraph_F01721	230	■ 0.982	$1,2,\ldots,n$	$1, 2, \dots, n$	
voloshin-hypergraph_F01722	231	■ 0.998	$L=Z=Y=\emptyset,\ R(\mathcal{H})=(\emptyset,\emptyset,\ldots,\emptyset),\ P(\mathcal{H},\lambda)=0,\ \chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=0$	$L = Z = Y = \emptyset, R(\mathcal{H}) = (0, 0, \dots, 0), P(\mathcal{H}, \lambda) = 0, \chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0$ and $P(\mathcal{H}, \lambda) = 0$ for any $\lambda \in \mathbb{R}$.
voloshin-hypergraph_F01723	231	■ 0.588	$i=1,2,\ldots,n$	$i = 1, 2, \dots, n$	0 Determining $\chi(\mathcal{H})$ and $\bar{\chi}(\mathcal{H})$ using $P(\mathcal{H})$
voloshin-hypergraph_F01724	231	■ 1.000	$\chi(\mathcal{H}),\ \bar{\chi}(\mathcal{H})$	$\chi(\mathcal{H}), \bar{\chi}(\mathcal{H})$	Proposition 10.5.1 For any mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$,
voloshin-hypergraph_F01725	231	■ 0.988	$X=\{1,2$	$X = \{1, 2$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$ and $P(\mathcal{H}, \lambda) = 0$ for any $\lambda \in \mathbb{R}$.
voloshin-hypergraph_F01726	231	■ 0.698	$3,4\},\ \mathcal{C}=\{C\}=\{\{1,2,3,4\}\},\ \mathcal{D}=\{D_1,D_2,D_3,D_4\}=\{\{1,2\},\{2,3\},\{3,4\},\{4,1\}\}$	$3, 4\}, C = \{C\} = \{\{1, 2, 3, 4\}\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$\chi = 2, \bar{\chi} = 3$, and the corresponding list L of three strict colorings in the form $(c(1), c(2), c(3), c(4))$ is the following (from the numbers are colors):
voloshin-hypergraph_F01727	231	■ 1.000	$\chi=2,\ \bar{\chi}=3$	$\chi = 2, \bar{\chi} = 3$	
voloshin-hypergraph_F01728	231	■ 0.999	$(c(1),\ c(2),\ c(3),\ c(4))$	$(c(1), c(2), c(3), c(4))$	
voloshin-hypergraph_F01729	233	■ 1.000	$X=\{1,2,3,4,5\},\ \quad \mathcal{C}=\{\{1,2,3\},\{1,3,4\}$	$X = \{1, 2, 3, 4, 5\},\quad \mathcal{C} = \{\{1, 2, 3\}, \{1, 3, 4\}$	$\{1, 4, 5\}, \{1, 5, 2\}\},\quad \mathcal{D} = \{\{3, 5\}\}$; we have that $\chi(\mathcal{H}) = 3$ and, after adding the $\mathcal{D}$ -edge
voloshin-hypergraph_F01730	233	■ 0.997	$\{1,4,5\},\{1,5,2\}\},\ \quad \mathcal{D}=\{\{3,5\}\}$	$\{1, 4, 5\}, \{1, 5, 2\}\},\quad \mathcal{D} = \{\{3, 5\}\}$	$\{2, 4\}$, we obtain the new mixed hypergraph $\mathcal{H}_1$ for which $\chi(\mathcal{H}_1) = 2$. It is easy to see
voloshin-hypergraph_F01731	233	■ 0.997	$\bar{\chi}(\mathcal{H})=3$	$\bar{\chi}(\mathcal{H}) = 3$	$\{2, 4\}$, we obtain the new mixed hypergraph $\mathcal{H}_1$ for which $\chi(\mathcal{H}_1) = 2$. It is easy to see
voloshin-hypergraph_F01732	233	■ 0.967	$\{2,4\}$	$\{2, 4\}$	that, in general, adding one C -edge to $\mathcal{H}$ can increase $\chi(\mathcal{H})$ and adding one $\mathcal{D}$ -edge can decrease $\bar{\chi}(\mathcal{H})$.
voloshin-hypergraph_F01733	233	■ 0.967	$\bar{\chi}\left(\mathcal{H}_1\right)=2$	$\bar{\chi}(\mathcal{H}_1) = 2$	that, in general, adding one C -edge to $\mathcal{H}$ can increase $\chi(\mathcal{H})$ and adding one $\mathcal{D}$ -edge can decrease $\bar{\chi}(\mathcal{H})$.
voloshin-hypergraph_F01734	233	■ 0.989	$C\in\mathcal{C}(\mathcal{D})$	$C \in \mathcal{C}(\mathcal{D})$	
voloshin-hypergraph_F01735	233	■ 1.000	$X=\{1,2,3,4\}$	$X = \{1, 2, 3, 4\}$	$R(\mathcal{H}) = R(\mathcal{H} - C_i) = (1, 7, 0, 0),\ i = 1, 2, 3, 4.$
voloshin-hypergraph_F01736	233	■ 1.000	$\mathcal{C}=\{\{1,2,3\}$	$\mathcal{C} = \{\{1, 2, 3\}$	$R(\mathcal{H}) = R(\mathcal{H} - C_i) = (1, 7, 0, 0),\ i = 1, 2, 3, 4.$
voloshin-hypergraph_F01737	233	■ 0.987	$\{1,3,4\},\{1,2,4\},\{2,3,4\}\}$	$\{1, 3, 4\}, \{1, 2, 4\}, \{2, 3, 4\}$	Moreover, if we add to $\mathcal{H}$ the family

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voloshin-hypergraph_F01765	237	■ 0.959	$\chi(\mathcal{H}_{\mathcal{D}}) \geq 2$	$\chi(\mathcal{H}_{\mathcal{D}}) \geq 2$	hypergraph $\mathcal{H}$ contains no $\mathcal{C}$ -edges, and therefore is not uncolorable. Hence, consider the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial $\mathcal{C}$ -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_F01766	237	■ 0.959	$\chi(\mathcal{H}_{\mathcal{D}}) \geq 3$	$\chi(\mathcal{H}_{\mathcal{D}}) \geq 3$	hypergraph $\mathcal{H}$ contains no $\mathcal{C}$ -edges, and therefore is not uncolorable. Hence, consider the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial $\mathcal{C}$ -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_F01767	237	■ 0.959	$\bar{\chi}(\mathcal{H}_C) \geq k + 3$	$\bar{\chi}(\mathcal{H}_C) \geq k + 3$	hypergraph $\mathcal{H}$ contains no $\mathcal{C}$ -edges, and therefore is not uncolorable. Hence, consider the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial $\mathcal{C}$ -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_F01768	237	■ 0.959	$n = k + 3$	$n = k + 3$	hypergraph $\mathcal{H}$ contains no $\mathcal{C}$ -edges, and therefore is not uncolorable. Hence, consider the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial $\mathcal{C}$ -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_F01769	237	■ 0.984	$\chi(\mathcal{H}_C) = n$	$\chi(\mathcal{H}_C) = n$	$(X, \mathcal{C})$ is a bistar , i.e., a $\mathcal{C}$ -hypergraph having two vertices, say x_1 and x_2 , that belong to
voloshin-hypergraph_F01770	237	■ 0.999	$\chi(\mathcal{H}_{\mathcal{D}}) = 2$	$\chi(\mathcal{H}_{\mathcal{D}}) = 2$	all $\mathcal{C}$ -edges. If this pair $\{x_1, x_2\}$ were not a $\mathcal{D}$ -edge in $\mathcal{H}_{\mathcal{D}}$, then we could color x_1, x_2
voloshin-hypergraph_F01771	237	■ 1.000	$n = k + 2$	$n = k + 2$	with the first color.
voloshin-hypergraph_F01772	237	■ 0.983	$\bar{\chi}(\mathcal{H}_C) = k + 2$	$\bar{\chi}(\mathcal{H}_C) = k + 2$	uncolorability of $\mathcal{H}$. Consequently, the pair $\{x_1, x_2\}$ is a $\mathcal{D}$ -edge in $\mathcal{H}$. Since $\mathcal{H}$ is an uncolorable hypergraph minimal under inclusion, no $\mathcal{C}$ -edge may coincide with $\{x_1, x_2\}$.
voloshin-hypergraph_F01773	237	■ 0.715	$\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$	$\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$	Consider an arbitrary proper 2-coloring of $\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$. It is, at the same time, a
voloshin-hypergraph_F01774	237	■ 0.715	$\mathcal{H}_C =$	$\mathcal{H}_C =$	Consider an arbitrary proper 2-coloring of $\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$. It is, at the same time, a
voloshin-hypergraph_F01775	237	■ 0.994	$(X, \mathcal{C})$	$(X, \mathcal{C})$	coloring of the initial mixed hypergraph $\mathcal{H}$, because each $\mathcal{C}$ -edge contains at least three
voloshin-hypergraph_F01776	238	■ 1.000	$k = 3$	$k = 3$	$k \leq k + 4$. Now, in order to prove the converse inequality $v(k) \leq k + 4$, we construct a series of
voloshin-hypergraph_F01777	238	■ 1.000	$v(k) =$	$v(k) =$	$k + 4$, $k = 0, 1, 2, \dots$. The construction will depend on the parity of k ; we first describe the particular cases $k = 0, 1$ that can be verified directly.
voloshin-hypergraph_F01778	238	■ 0.999	$v(k) \leq k + 4$	$v(k) \leq k + 4$	$k = 0$. Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}$
voloshin-hypergraph_F01779	238	■ 0.373	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_{\mathcal{D}}) = k$	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_{\mathcal{D}}) = k$	Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$.
voloshin-hypergraph_F01780	238	■ 0.373	$n =$	$n =$	Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$.
voloshin-hypergraph_F01781	238	■ 1.000	$k + 4$, $k = 0, 1, 2, \dots$	$k + 4$, $k = 0, 1, 2, \dots$	$k = 0$. Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}$
voloshin-hypergraph_F01782	238	■ 1.000	$k = 0, 1$	$k = 0, 1$	Consider $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4, 5\}$.
voloshin-hypergraph_F01783	238	■ 1.000	$k = 0$	$k = 0$	
voloshin-hypergraph_F01784	238	■ 1.000	$X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}$	$X = \{1, 2, 3, 4\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}, \mathcal{D} = \{\{1, 2\}$	$k = 2l$, $l \geq 1$. Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4, 5\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}\}$, $\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}\}$.
voloshin-hypergraph_F01785	238	■ 0.665	$\{2, 3\}, \{2, 4\}, \{3, 4\}$	$\{2, 3\}, \{2, 4\}, \{3, 4\}$	Construct the $\{1, 2, 3\}, \{2, 4\}, \{3, 4\}$.
voloshin-hypergraph_F01786	238	■ 1.000	$X = \{1, 2, 3, 4, 5\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}\}$	$X = \{1, 2, 3, 4, 5\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}\}$	to all $\mathcal{C}$ -edges, and therefore $\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$. Moreover, $\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$ is the odd
voloshin-hypergraph_F01787	238	■ 0.950	$\mathcal{D} = \{\{1, 2\}, \{3, 4\}, \{4, 5\}, \{3, 5\}\}$	$\mathcal{D} = \{\{1, 2\}, \{3, 4\}, \{4, 5\}, \{3, 5\}\}$	cycle $(2, 3, 4, \dots, k + 4, 2)$ with all
voloshin-hypergraph_F01788	238	■ 1.000	$k = 2l$, $l \geq 1$	$k = 2l$, $l \geq 1$	
voloshin-hypergraph_F01789	238	■ 0.999	$X = \{1, 2, 3, \dots, k + 4\}$, $\mathcal{C} =$	$X = \{1, 2, 3, \dots, k + 4\}, \mathcal{C} =$	
voloshin-hypergraph_F01790	238	■ 0.990	$\{1, 2, i\} : 3 \leq i \leq k + 4\}$	$\{1, 2, i\} : 3 \leq i \leq k + 4\}$	
voloshin-hypergraph_F01791	238	■ 0.990	$\mathcal{D} = \{\{i, i + 1\} : 1 \leq i \leq k + 3\} \cup \{k + 4, 2\}$	$\mathcal{D} = \{\{i, i + 1\} : 1 \leq i \leq k + 3\} \cup \{k + 4, 2\}$	

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voloshin-hypergraph_F01792	238	0.860	$\bar{\chi}(\mathcal{H}_C)=n-1=k+3$	$\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$	
voloshin-hypergraph_F01793	238	0.990	$(2,3,4, \ldots, k+4,2)$	$(2,3,4, \ldots, k+4,2)$	
voloshin-hypergraph_F01794	238	0.990	$\chi(\mathcal{H}_D)=3$	$\chi(\mathcal{H}_D) = 3$	
voloshin-hypergraph_F01795	239	0.999	$i, i=1,2, \ldots, n$	$i, i = 1, 2, \ldots, n$	Let us try to construct a proper coloring c of $\mathcal{H}$. As usual, $c(i)$ means the color of vertex i , $i = 1, 2, \ldots, n$. In any possible coloring of $\mathcal{H}$, vertices 1 and 2 have to be colored
voloshin-hypergraph_F01796	239	0.998	$c(1)=1, c(2)=2$	$c(1) = 1, c(2) = 2$	differently, say $c(1) = 1, c(2) = 2$. Since $\{2,3\}$ is a $\mathcal{D}$ -edge, we have $c(3) \neq c(2)$ and,
voloshin-hypergraph_F01797	239	0.998	$c(3) \neq c(2)$	$c(3) \neq c(2)$	differently, say $c(1) = 1, c(2) = 2$. Since $\{2,3\}$ is a $\mathcal{D}$ -edge, we have $c(3) \neq c(2)$ and,
voloshin-hypergraph_F01798	239	0.600	$c(3)=$	$c(3) =$	because of the C -edge $\{1,2,3\}$, the unique possibility for vertex 3 to get colored is $c(3) =$
voloshin-hypergraph_F01799	239	0.993	$c(1)=1$	$c(1) = 1$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1,2,4\}$, the unique
voloshin-hypergraph_F01800	239	0.993	$c(4) \neq c(3)$	$c(4) \neq c(3)$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1,2,4\}$, the unique
voloshin-hypergraph_F01801	239	0.993	$\{1,2,4\}$	$\{1,2,4\}$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1,2,4\}$, the unique
voloshin-hypergraph_F01802	239	1.000	$c(4)=c(2)=2$	$c(4) = c(2) = 2$	possibility for vertex 4 to be colored is $c(4) = c(2) = 2$. It is now clear that the colors have to alternate on the
voloshin-hypergraph_F01803	239	0.996	$(2,3,4, \ldots, k+4)$	$(2,3,4, \ldots, k+4)$	It is now clear that the colors have to alternate on the cycle $(2,3,4, \ldots, k+4)$. Since $c(k+3) = 1$ and $c(2) = 2$, we can color vertex $k+4$ neither with color 1 nor with color
voloshin-hypergraph_F01804	239	1.000	$c(k+3)=1$	$c(k+3) = 1$	2. However, any other color $c(k+4)$ is infeasible because of the C -edge $\{1,2,k+4\}$.
voloshin-hypergraph_F01805	239	1.000	$c(2)=2$	$c(2) = 2$	2. However, any other color $c(k+4)$ is infeasible because of the C -edge $\{1,2,k+4\}$.
voloshin-hypergraph_F01806	239	1.000	$k+4$	$k+4$	2. However, any other color $c(k+4)$ is infeasible because of the C -edge $\{1,2,k+4\}$.
voloshin-hypergraph_F01807	239	0.994	$c(k+4)$	$c(k+4)$	Consequently, $\mathcal{H}$ is uncolorable. One can easily check that it is minimal under inclusion.
voloshin-hypergraph_F01808	239	0.994	$\{1,2, k+4\}$	$\{1,2,k+4\}$	Consequently, $\mathcal{H}$ is uncolorable. One can easily check that it is minimal under inclusion.
voloshin-hypergraph_F01809	239	1.000	$k=2l+1, l \geq 1$	$k = 2l + 1, l \geq 1$	Construct $\mathcal{H} = \mathcal{H}_C \cup \mathcal{H}_D$.
voloshin-hypergraph_F01810	239	0.936	$\mathcal{D}=\{1,2\} \cup \{i, i+1\}: 3 \leq i \leq k+3\} \cup \{k+4,3\}$	$\mathcal{D} = \{1,2\} \cup \{i, i+1\} : 3 \leq i \leq k+3\} \cup \{k+4,3\}$	edges, so that $\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$. In the present case, $\mathcal{H}_D = (X, \mathcal{D})$ is a disconnected
voloshin-hypergraph_F01811	239	0.929	$(3,4, \ldots, k+4,3)$	$(3,4, \ldots, k+4,3)$	Let $c(1) = 1, c(2) = 2$. For $c(3)$ there are only two possibilities: $c(3) = 1$, or $c(3) = 2$. By symmetry we may assume $c(3) = 1$. Then, similarly to the argument above, we obtain
voloshin-hypergraph_F01812	239	1.000	$c(3)$	$c(3)$	Since vertex $k+4$ cannot be colored with any color (because of the C -edge $\{1,2,k+4\}$ and
voloshin-hypergraph_F01813	239	1.000	$c(3)=1$	$c(3) = 1$	Since vertex $k+4$ cannot be colored with any color (because of the C -edge $\{1,2,k+4\}$ and
voloshin-hypergraph_F01814	239	1.000	$c(3)=2$	$c(3) = 2$	Since vertex $k+4$ cannot be colored with any color (because of the C -edge $\{1,2,k+4\}$ and
voloshin-hypergraph_F01815	239	1.000	$c(4)=2, c(5)=1, c(6)=2$	$c(4) = 2, c(5) = 1, c(6) = 2$	is also easily seen. Hence, the theorem follows. $\square$
voloshin-hypergraph_F01816	239	1.000	$\{k+3, k+4\}$	$\{k+3, k+4\}$	$/X \setminus /X \setminus$
voloshin-hypergraph_F01817	239	1.000	$\{k+4,3\}$	$\{k+4,3\}$	$/X \setminus /X \setminus$
voloshin-hypergraph_F01818	239	1.000	$2 \leq l, m \leq n= X $	$2 \leq l, m \leq n = X $	Hence, $ C = \binom{n}{l}$ and $ \mathcal{D} = \binom{n}{m}$. Call $\mathcal{K}(n, l, m)$ the complete (l, m)-uniform

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																
voloshin-hypergraph_F01819	239	0.996	$ \mathcal{C} =\binom{n}{l}$	$ \mathcal{C} =\binom{n}{l}$																																																																	
voloshin-hypergraph_F01820	239	0.996	$ \mathcal{D} =\binom{n}{m}$	$ \mathcal{D} =\binom{n}{m}$																																																																	
voloshin-hypergraph_F01821	239	0.996	$\mathcal{K}(n, l, m)$	$\mathcal{K}(n, l, m)$																																																																	
voloshin-hypergraph_F01822	239	0.996	(l, m)	(l, m)																																																																	
voloshin-hypergraph_F01823	239	1.000	n, l, m	n, l, m																																																																	
voloshin-hypergraph_F01824	239	0.740	$\mathcal{K}(4, 4, 2)$	$\mathcal{K}(4, 4, 2)$																																																																	
voloshin-hypergraph_F01825	240	1.000	$n \leq (l-1)(m-1)$	$n \leq (l-1)(m-1)$	PROOF. $\Rightarrow$ Let $n \leq (l-1)(m-1)$. We color $m-1$ vertices with the first color, the next $m-1$ vertices with the second color etc. Since $n \leq (l-1)(m-1)$ this procedure requires																																																																
voloshin-hypergraph_F01826	240	1.000	$m-1$	$m-1$	PROOF. $\Rightarrow$ Let $n \leq (l-1)(m-1)$. We color $m-1$ vertices with the first color, the next $m-1$ vertices with the second color etc. Since $n \leq (l-1)(m-1)$ this procedure requires																																																																
voloshin-hypergraph_F01827	240	0.994	$l-1$	$l-1$	colorable.																																																																
voloshin-hypergraph_F01828	240	0.998	$n \geq (l-1)(m-1)+1$	$n \geq (l-1)(m-1)+1$	each m -tuple is a $\mathcal{D}$ -edge and each l -tuple is a $\mathcal{C}$ -edge, the number of vertices in any one																																																																
voloshin-hypergraph_F01829	240	0.998	$(l-1)(m-1)$	$(l-1)(m-1)$	smaller and smaller. This can be stated as the following corollary.																																																																
voloshin-hypergraph_F01830	240	1.000	$n+1$	$n+1$	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$																																																																
voloshin-hypergraph_F01831	240	1.000	$l=1$	$l=1$	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$																																																																
voloshin-hypergraph_F01832	240	1.000	$m=1$	$m=1$	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$																																																																
voloshin-hypergraph_F01833	240	1.000	n^2	n^2	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$																																																																
voloshin-hypergraph_F01834	240	1.000	$2 \leq l \leq n+1, 2 \leq m \leq n+1$	$2 \leq l \leq n+1, 2 \leq m \leq n+1$	uncolorable complete mixed hypergraphs of order $n+1$. Consequently, the total number N_n of complete uncolorable mixed hypergraphs on $n+1$ vertices equals																																																																
voloshin-hypergraph_F01835	240	1.000	$\mathcal{K}(n, l, m+1)$	$\mathcal{K}(n, l, m+1)$	of complete uncolorable mixed hypergraphs on $n+1$ vertices																																																																
voloshin-hypergraph_F01836	240	0.997	$l \geq 2$	$l \geq 2$	where the asymptotic equation is meant as n tends to infinity (here and next we use simple calculus formulas). Thus, the proportion of uncolorable complete mixed hypergraphs																																																																
voloshin-hypergraph_F01837	240	0.997	$\left\lfloor \frac{n}{l-1} \right\rfloor$	$\left\lfloor \frac{n}{l-1} \right\rfloor$	where the asymptotic equation is meant as n tends to infinity (here and next we use simple calculus formulas). Thus, the proportion of uncolorable complete mixed hypergraphs																																																																
voloshin-hypergraph_F01838	240	1.000	N_n	N_n	pie calculus formulas). Thus, the proportion of uncolorable complete mixed hypergraphs equals																																																																
voloshin-hypergraph_F01839	241	0.585	$l \setminus m$	$l \setminus m$	<table><tr><th>$l \setminus m$</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>2</td><td>—</td><td>—</td><td>—</td><td>—</td><td>—</td><td>—</td><td>—</td></tr><tr><td>3</td><td>—</td><td>—</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>4</td><td>—</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>5</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>6</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>7</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>8</td><td>—</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr></table>	$l \setminus m$	2	3	4	5	6	7	8	2	—	—	—	—	—	—	—	3	—	—	—	+	+	+	+	4	—	—	+	+	+	+	+	5	—	+	+	+	+	+	+	6	—	+	+	+	+	+	+	7	—	+	+	+	+	+	+	8	—	+	+	+	+	+	+
$l \setminus m$	2	3	4	5	6	7	8																																																														
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7	—	+	+	+	+	+	+																																																														
8	—	+	+	+	+	+	+																																																														
voloshin-hypergraph_F01840	241	1.000	$\mathcal{K}(8, l, m)$	$\mathcal{K}(8, l, m)$																																																																	
voloshin-hypergraph_F01841	241	0.926	l, m	l, m	way. When n tends to infinity, and if l, m are fixed, then the uncolorability zone sooner or																																																																

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voloshin-hypergraph_F01842	241	■ 0.999	$x, y \in D$	$x, y \in D$	
voloshin-hypergraph_F01843	242	■ 1.000	$\chi(\mathcal{H}) \leq 2$	$\chi(\mathcal{H}) \leq 2$	An example of uncolorable mixed hypertree is shown in Figure 10.18. Edges are drawn by dashed curves, and D -edges are drawn by solid curves; t
voloshin-hypergraph_F01844	243	■ 0.996	$\mathcal{UC}$	$\mathcal{UC}$	Equivalently, $\mathcal{H}$ is uc if it allows exactly one feasible partition of color classes. Let us agree that the expression “unique coloring” mean
voloshin-hypergraph_F01845	243	■ 1.000	$r=2$	$r = 2$	Proof. This follows directly from the definition of a proper coloring of a mixed
voloshin-hypergraph_F01846	244	■ 0.712	$x=$	$x =$	Definition 10.7.2 In a mixed hypertree $\mathcal{H} = (V, C, \mathcal{D})$, a sequence of vertices $x =$
voloshin-hypergraph_F01847	244	■ 0.977	$x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{k\}}=y, k \geq 1$	$x_0, x_1, \dots, x_k = y, k \geq 1$	Definition 10.7.2a In a mixed hypertree $\mathcal{H} = (V, C, \mathcal{D})$, a sequence of vertices $x =$
voloshin-hypergraph_F01848	244	■ 0.977	$x_{\{i\}} \neq x_{\{i+1\}}$	$x_i \neq x_{i+1}$	Definition 10.7.2a In a mixed hypertree $\mathcal{H} = (V, C, \mathcal{D})$, a sequence of vertices $x =$
voloshin-hypergraph_F01849	244	■ 0.977	$\left(x_{\{i\}}, x_{\{i+1\}}\right) \in \mathcal{D}$	$(x_i, x_{i+1}) \in \mathcal{D}$	Definition 10.7.2a In a mixed hypertree $\mathcal{H} = (V, C, \mathcal{D})$, a sequence of vertices $x =$
voloshin-hypergraph_F01850	244	■ 1.000	$x_{\{1\}}, \ldots, x_{\{k-1\}}$	$x_1, \dots, x_{k-1}$	
voloshin-hypergraph_F01851	244	■ 0.999	$x_{\{0\}}, x_{\{1\}}$	x_0, x_1	$\mathcal{H}_1$ and $\mathcal{H}_2$ are uc mixed hypergraphs with the unique strict coloring shown in the figure. In
voloshin-hypergraph_F01852	244	■ 1.000	$x_{\{2\}}, x_{\{3\}}$	x_2, x_3	every proper coloring of $\mathcal{H}_1$, the vertices x_0 and x_3 have different colors. In contrast, in any
voloshin-hypergraph_F01853	244	■ 1.000	$x_{\{0\}}, x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}$	x_0, x_1, x_2, x_3, x_4	every proper coloring of $\mathcal{H}_1$, the vertices x_0 and x_3 have different colors. In contrast, in any
voloshin-hypergraph_F01854	244	■ 1.000	$\left(x_{\{0\}}, x_{\{4\}}\right)$	(x_0, x_4)	vertex x_3 from having the color of vertex x_0 , and, $\mathcal{H}_2$ transports the color of vertex x_0 to vertex x_4 . One can easily see that for both invertors $\mathcal{I} = \bar{\mathcal{I}} = \mathcal{I}$, and moreover
voloshin-hypergraph_F01855	244	■ 0.979	$\chi=\bar{\chi}=2$	$\chi = \bar{\chi} = 2$	Embeddings into uc mixed hypergraphs. In $\mathcal{D}$ -hypergraphs, for each n we have the result. The following proposition is due to the second author of [10].
voloshin-hypergraph_F01856	245	■ 0.835	$c=X_{\{1\}} \cup \cdots \cup X_{\{t\}}$	$c = X_1 \cup \dots \cup X_t$	hypergraph and $c = x_1 \cup \dots \cup x_t$ be a strict coloring, then there exists a uc mixed hypergraph $\mathcal{H}' = (X', C', \mathcal{D}')$ with the following properties:
voloshin-hypergraph_F01857	245	■ 0.999	$\mathcal{H}' = \left(X', C', \mathcal{D}'\right)$	$\mathcal{H}' = (X', C', \mathcal{D}')$	Proof. Let $\mathcal{H} = (X, C, \mathcal{D})$ be a mixed hypertree. For each $c \in C$, let X_c be the set of vertices in X that are in c . Since $\mathcal{H}$ is a hypertree, the sets X_c are pairwise disjoint. We will construct $\mathcal{H}'$ by adding new vertices and edges to $\mathcal{H}$. For each $c \in C$, we will add a new vertex x_c to X and a new edge c' to C such that $x_c \in c'$ and $c' \cap c = \emptyset$. We will also add new edges to $\mathcal{D}'$ such that $\mathcal{H}'$ is a hypertree. The details of the construction are left to the reader.
voloshin-hypergraph_F01858	245	■ 0.999	$\chi\left(\mathcal{H}'\right)=\bar{\chi}\left(\mathcal{H}'\right)=t$	$\chi(\mathcal{H}') = \bar{\chi}(\mathcal{H}') = t$	Proof. To extend $\mathcal{H}$ into some $\mathcal{H}'$ in the required way, we first choose a dummy span-
voloshin-hypergraph_F01859	245	■ 1.000	$T_{\{i\}}$	T_i	disjoint. These invertors ensure that vertices from X_i will always have the same color.
voloshin-hypergraph_F01860	245	■ 0.737	$\left\{x=x_{\{1\}}, x_{\{2\}}, z\right\}$	$\{x = x_1, x_2, z\}$	Next, for each pair (i, j) such that X_i and X_j are two distinct classes of c , we choose
voloshin-hypergraph_F01861	245	■ 1.000	$X_{\{j\}}$	X_j	of $\mathcal{H}$, and should not be in X . Since colors on invertors alternate, the colors on an odd
voloshin-hypergraph_F01862	245	■ 0.591	$x_{\{i\}} \in X_{\{i\}}$	$x_i \in X_i$	invertor are all the same as the two colors at the endpoints. Hence, no new colors outside X
voloshin-hypergraph_F01863	245	■ 0.591	$x_{\{j\}} \in X_{\{j\}}$	$x_j \in X_j$	invertor are all the same as the two colors at the endpoints. Hence, no new colors outside X
voloshin-hypergraph_F01864	246	■ 1.000	$\mu=\left(z_{\{0\}}, z_{\{1\}}, \ldots, z_{\{k\}}=z_{\{0\}}\right), k \geq 6$	$\mu = (z_0, z_1, \dots, z_k = z_0), k \geq 6$	Let $\mu = (z_0, z_1, \dots, z_k = z_0), k \geq 6$ be a simple cyclic invertor in a mixed hypertree.
voloshin-hypergraph_F01865	246	■ 0.983	$z_{\{0\}} \neq z_{\{1\}} \neq z_{\{2\}} \neq z_{\{0\}}$	$z_0 \neq z_1 \neq z_2 \neq z_0$	Let $\mu = (z_0, z_1, \dots, z_k = z_0), k \geq 6$ be a simple cyclic invertor in a mixed hypertree. Without loss of generality, assume that $z_0 \neq z_1 \neq z_2 \neq z_0$. From the definition of a simple
voloshin-hypergraph_F01866	246	■ 1.000	$z_{\{0\}} \neq z_{\{2\}} \neq \ldots \neq z_{\{k-2\}}$	$z_0 \neq z_2 \neq \dots \neq z_{k-2}$	cyclic invertor, it follows that $z_0 \neq z_2 \neq \dots \neq z_{k-2}$ and $z_1 = z_3 = \dots = z_{k-1} = y$, where y is
voloshin-hypergraph_F01867	246	■ 1.000	$z_{\{1\}}=z_{\{3\}}=\ldots=z_{\{k-1\}}=y$	$z_1 = z_3 = \dots = z_{k-1} = y$	cyclic invertor, it follows that $z_0 \neq z_2 \neq \dots \neq z_{k-2}$ and $z_1 = z_3 = \dots = z_{k-1} = y$, where y is
voloshin-hypergraph_F01868	246	■ 0.907	$\left \mathcal{D}\right \leq n-2$	$ \mathcal{D} \leq n - 2$	Theorem 10.7.2 If $\mathcal{H} = (X, C, \mathcal{D})$ is a reduced mixed hypertree with $ \mathcal{D} \leq n - 2$, then

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voloshin-hypergraph_F01869	246	■ 1.000	$r_2(\mathcal{H}) \geq 2$	$r_2(\mathcal{H}) \geq 2$	
voloshin-hypergraph_F01870	246	■ 1.000	$T=(X, \mathcal{E})$	$T = (X, \mathcal{E})$	in T there exists an edge $e = \{x, y\} \notin \mathcal{D}$. Starting with vertices x, y , we can construct
voloshin-hypergraph_F01871	246	■ 1.000	$e=\{x, y\} \notin \mathcal{D}$	$e = \{x, y\} \notin \mathcal{D}$	two different colorings with colors 1 and 2 generating two different partitions. First, put
voloshin-hypergraph_F01872	246	■ 1.000	$c(x)=c(y)=1$	$c(x) = c(y) = 1$	2, 1, 2, ... Second, apply the same procedure starting with $c(x) = 1$ and $c(y) = 2$. □
voloshin-hypergraph_F01873	246	■ 0.941	$2, 1, 2, \ldots$	$2, 1, 2, \dots$	Theorem 10.7.3 (Nicolitsa, Voloshin, 2000) A reduced mixed hypertree $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is
voloshin-hypergraph_F01874	246	■ 0.941	$c(x)=1$	$c(x) = 1$	Theorem 10.7.3 (Nicolitsa, Voloshin, 2000) A reduced mixed hypertree $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is
voloshin-hypergraph_F01875	246	■ 0.941	$c(y)=2$	$c(y) = 2$	Theorem 10.7.3 (Nicolitsa, Voloshin, 2000) A reduced mixed hypertree $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is
voloshin-hypergraph_F01876	246	■ 0.982	$u, v \in X$	$u, v \in X$	a triple of pairwise different vertices x_j, x_{j+1}, x_{j+2} not belonging to $\mathcal{C}$. If there is no $\mathcal{D}$ -path
voloshin-hypergraph_F01877	246	■ 0.982	(u, v)	(u, v)	a triple of pairwise different vertices x_j, x_{j+1}, x_{j+2} not belonging to $\mathcal{C}$. If there is no $\mathcal{D}$ -path
voloshin-hypergraph_F01878	246	■ 1.000	$u=x_1, x_2, \ldots, x_p=v$	$u = x_1, x_2, \dots, x_p = v$	Assume now that in the sequence $u = x_1, x_2, \dots, x_p = v$, each pair of consecutive ver-
voloshin-hypergraph_F01879	246	■ 1.000	x_j, x_{j+1}, x_{j+2}	x_j, x_{j+1}, x_{j+2}	tices is a $\mathcal{D}$ -edge, and there is a triple of pairwise different vertices x_j, x_{j+1}, x_{j+2} such that
voloshin-hypergraph_F01880	246	■ 1.000	$\left\{x_j, x_{j+1}, x_{j+2}\right\} \notin \mathcal{C}$	$\{x_j, x_{j+1}, x_{j+2}\} \notin \mathcal{C}$	partition different from c , a contradiction.
voloshin-hypergraph_F01881	246	■ 1.000	x_{j+1}	x_{j+1}	partition different from c , a contradiction.
voloshin-hypergraph_F01882	246	■ 1.000	$c\left(x_j\right)=c\left(x_{j+2}\right)$	$c\left(x_j\right)=c\left(x_{j+2}\right)$	sequently, $\mathcal{H}$ is colored with at least three colors. But every mixed hypertree can also be colored with two colors, and we again have two different feasible partitions, a contradiction.
voloshin-hypergraph_F01883	246	■ 0.658	x_{j+2}	x_{j+2}	colored with two colors, and we again have two different feasible partitions, a contradiction. ⇐ Assume that any two vertices $x, y \in X$ are joined by an (x, y) -inverter. Suppose $\mathcal{H}$
voloshin-hypergraph_F01884	246	■ 1.000	$c\left(x_j\right) \neq c\left(x_{j+2}\right)$	$c\left(x_j\right) \neq c\left(x_{j+2}\right)$	inverter $x' = x_0, x_1, \dots, x_k = y'$. From the definition of an inverter, it follows that, if k is even, then in all possible colorings the vertices x' and y' have the same color. If k is odd, then in
voloshin-hypergraph_F01885	246	■ 1.000	$\left\{x_j, x_{j+1}\right\},\left\{x_{j+1}, x_{j+2}\right\} \in \mathcal{D}, c\left(x_j\right) \neq c\left(x_{j+1}\right) \neq c\left(x_{j+2}\right)$	$\left\{x_j, x_{j+1}\right\},\left\{x_{j+1}, x_{j+2}\right\} \in \mathcal{D}, c\left(x_j\right) \neq c\left(x_{j+1}\right) \neq c\left(x_{j+2}\right)$	inverter $x' = x_0, x_1, \dots, x_k = y'$. From the definition of an inverter, it follows that, if k is even, then in all possible colorings the vertices x' and y' have the same color. If k is odd, then in
voloshin-hypergraph_F01886	246	■ 1.000	$x^{\prime}, y^{\prime}$	x', y'	Corollary 10.7.1 If $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is a reduced uniquely colorable mixed hypertree with the host tree $T=(V, \mathcal{E})$, then $\mathcal{D}=\mathcal{E}$.
voloshin-hypergraph_F01887	246	■ 1.000	$c_1\left(x^{\prime}\right)=c_1\left(y^{\prime}\right)$	$c_1\left(x^{\prime}\right)=c_1\left(y^{\prime}\right)$	the host tree $T=(X, \mathcal{E})$, then $\mathcal{D}=\mathcal{E}$.
voloshin-hypergraph_F01888	246	■ 1.000	$c_2\left(x^{\prime}\right) \neq c_2\left(y^{\prime}\right)$	$c_2\left(x^{\prime}\right) \neq c_2\left(y^{\prime}\right)$	the host tree $T=(X, \mathcal{E})$, then $\mathcal{D}=\mathcal{E}$.
voloshin-hypergraph_F01889	246	■ 1.000	$\left(x^{\prime}, y^{\prime}\right)$	$\left(x', y'\right)$	the host tree $T=(X, \mathcal{E})$, then $\mathcal{D}=\mathcal{E}$.
voloshin-hypergraph_F01890	246	■ 0.994	$x^{\prime}=x_0, x_1, \ldots, x_k=y^{\prime}$	$x' = x_0, x_1, \dots, x_k = y'$	
voloshin-hypergraph_F01891	246	■ 1.000	$c\left(x^{\prime}\right)=c\left(y^{\prime}\right)$	$c\left(x^{\prime}\right)=c\left(y^{\prime}\right)$	
voloshin-hypergraph_F01892	246	■ 1.000	$c\left(x^{\prime}\right) \neq c\left(y^{\prime}\right)$	$c\left(x^{\prime}\right) \neq c\left(y^{\prime}\right)$	
voloshin-hypergraph_F01893	246	■ 1.000	$\mathcal{D}=\mathcal{E}$	$\mathcal{D} = \mathcal{E}$	—
voloshin-hypergraph_F01894	247	■ 1.000	$R(\mathcal{H})=$	$R(\mathcal{H}) =$	Recall that in a mixed hypertree $\mathcal{H}$, the $\mathcal{C}$ -edges $C \in \mathcal{C}$ is called <i>redundant</i> if $\mathcal{D} \cap C \neq \emptyset$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01921	248	■ 1.000	$\left X_i\right -1$	$ X_i - 1$	case, each L_i is a tree, $i = 1, 2, \dots, \chi$. Since every edge in L_i corresponds to some C -edge of
voloshin-hypergraph_F01922	248	■ 1.000	$i=1,2,\ldots,\chi$	$i = 1, 2, \dots, \chi$	$\mathcal{H}$, we obtain that the minimum number of C -edges is:
voloshin-hypergraph_F01923	248	■ 0.562	$ \mathcal{C} <n-\chi(\mathcal{H})$	$ \mathcal{C} < n - \chi(\mathcal{H})$	Corollary 10.1.1 <i>The minimum number of C-edges in any reduced unique, mixed hypertree $\mathcal{H} = (V, \mathcal{C}, \mathcal{D})$ is</i>
voloshin-hypergraph_F01924	248	■ 1.000	$ \mathcal{C} -n+2$	$ \mathcal{C} - n + 2$	elementary cycle in L_i generates some simple edge of $\mathcal{H}$ is redundant if it has size > 4 .
voloshin-hypergraph_F01925	248	■ 1.000	$T_i, \ i=1,2$	$T_i, i = 1, 2$	Remark: A redundant C -edge may become redundant after deleting some other C -
voloshin-hypergraph_F01926	248	■ 0.997	$ \mathcal{D} =n-1, \quad n-2 \leq \mathcal{C} \leq M$	$ \mathcal{D} = n - 1, \quad n - 2 \leq \mathcal{C} \leq M$	
voloshin-hypergraph_F01927	248	■ 0.996	$\bar{\chi}(\mathcal{H})=k$	$\bar{\chi}(\mathcal{H}) = k$	
voloshin-hypergraph_F01928	248	■ 0.996	$ \mathcal{D} $	$ \mathcal{D} $	
voloshin-hypergraph_F01929	248	■ 0.996	$ \mathcal{C} $	$ \mathcal{C} $	
voloshin-hypergraph_F01930	249	■ 1.000	$C=\{x, \ y, \ z\}$	$C = \{x, y, z\}$	Evidently, $\{x, y\}, \{y, z\} \in \mathcal{D}$. Consequently, $c(x) = c(z)$ in the unique partition $\mathcal{C}$ of $\mathcal{D}$. Strongly delete x from $\mathcal{H}$, thus obtaining $\mathcal{H} - x$. Clearly, $\mathcal{H} - x$ is a mixed hypertree.
voloshin-hypergraph_F01931	249	■ 1.000	$\{x, \ y\}, \{y, \ z\} \in \mathcal{D}$	$\{x, y\}, \{y, z\} \in \mathcal{D}$	$ C(\mathcal{H} - x) = C(\mathcal{H}) - 1$. If it is not uc, then it admits at least two strict colorings. In
voloshin-hypergraph_F01932	249	■ 1.000	$c(x)=c(z)$	$c(x) = c(z)$	$ C(\mathcal{H} - x) = C(\mathcal{H}) - 1$. If it is not uc, then it admits at least two strict colorings. In
voloshin-hypergraph_F01933	249	■ 1.000	$ \mathcal{C}(\mathcal{H}-x) = \mathcal{C}(\mathcal{H}) -1$	$ \mathcal{C}(\mathcal{H} - x) = \mathcal{C}(\mathcal{H}) - 1$	$c(x) = c(z)$. Therefore both colorings of $\mathcal{H} - x$ induce different strict colorings of the
voloshin-hypergraph_F01934	249	■ 1.000	$\left(x_n, \ x_{n-1}, \ \ldots, \ x_1\right)$	$(x_n, x_{n-1}, \dots, x_1)$	ordering, by applying the following:
voloshin-hypergraph_F01935	249	■ 0.637	$\mathcal{H}=(X, \ \mathcal{C}, \ \mathcal{D}), \ \sigma$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D}), \sigma$	Algorithm:
voloshin-hypergraph_F01936	249	■ 1.000	$L_1, \ L_2$	L_1, L_2	
voloshin-hypergraph_F01937	250	■ 1.000	$L_i, \ i=1,2$	$L_i, i = 1, 2$	6. $i := 1$.
voloshin-hypergraph_F01938	250	■ 1.000	$i:=1$	$i := 1$	8.
voloshin-hypergraph_F01939	250	■ 0.965	$i:=3-i$	$i := 3 - i$	unique (x, y) -invertor for any $x, y \in X$. Theorem 10.7.7 ensures, at any step of the algorithm,
voloshin-hypergraph_F01940	250	■ 0.565	$\sigma=$	$\sigma =$	the existence of a vertex, say x , such that every elimination of pendant x from
voloshin-hypergraph_F01941	250	■ 0.999	$T_1=L_1$	$T_1 = L_1$	the uc-ordering is:
voloshin-hypergraph_F01942	250	■ 1.000	$\left\{x_1, \ x_3, \ x_4\right\}, \left\{x_4, \ x_3, \ x_5\right\}, \left\{x_2, \ x_3, \ x_5\right\}$	$\{x_1, x_3, x_4\}, \{x_4, x_3, x_5\}, \{x_2, x_3, x_5\}$	
voloshin-hypergraph_F01943	250	■ 1.000	$\left\{x_2, \ x_5\right\}$	$\{x_2, x_5\}$	$\sigma = (x_5, x_3, x_4, x_2, x_1)$.
voloshin-hypergraph_F01944	250	■ 1.000	$x_1, \ x_2, \ x_4, \ x_3$	x_1, x_2, x_4, x_3	
voloshin-hypergraph_F01945	251	■ 1.000	$R(\mathcal{H})=(0,1,0, \ldots, \ 0)$	$R(\mathcal{H}) = (0, 1, 0, \dots, 0)$	(2) $P(\mathcal{H}, \kappa) = \kappa(\kappa - 1)$; (3) $\mathcal{H}$ is uniquely colored
voloshin-hypergraph_F01946	251	■ 0.994	$P(\mathcal{H}, \lambda)=\lambda(\lambda-1)$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1)$	(4) Every two vertices
voloshin-hypergraph_F01947	252	■ 0.995	$\mathcal{K}_n^r, \ r \geq 3$	$\mathcal{K}_n^r, r \geq 3$	proper colorings and its properties are not so nice as those of the clique $\mathcal{H} = (X, \emptyset, \binom{X}{2})$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01948	252	■ 0.998	$\mathcal{H}=\left(X, \emptyset, \binom{X}{r}\right)$	$\mathcal{H}=\left(X, \emptyset, \binom{X}{r}\right)$	In contrast, we find a natural notion of perfection of an arbitrary mixed hypergraph with respect to the lower chromatic number. Hence, when talking about graph perfection we mean perfection with respect to the lower chromatic number, while talking about hyper-
voloshin-hypergraph_F01949	252	■ 0.972	$\mathcal{H}=\left(X, \emptyset, \binom{X}{2}\right)$	$\mathcal{H}=\left(X, \emptyset, \binom{X}{2}\right)$	
voloshin-hypergraph_F01950	252	■ 0.986	$\alpha_c(\mathcal{H})$	$\alpha_c(\mathcal{H})$	maximum $\mathcal{D}$ -stable set which in tur
voloshin-hypergraph_F01951	252	■ 0.976	$\bar{\chi}(\mathcal{H})=\alpha_c(\mathcal{H})$	$\bar{\chi}(\mathcal{H})=\alpha_c(\mathcal{H})$	
voloshin-hypergraph_F01952	252	■ 1.000	$\chi(G) \geq \omega(G)$	$\chi(G) \geq \omega(G)$	
voloshin-hypergraph_F01953	253	■ 1.000	$\bar{\chi} \geq k+1$	$\bar{\chi} \geq k+1$	Proof. Let $\mathcal{H}=(X, C)$ be a 3-uniform C -hypergraph with $X=\{1, 2, \dots, 2k+5\}$, and
voloshin-hypergraph_F01954	253	■ 1.000	$X=\{1, 2, \ldots, 2k+5\}$	$X=\{1, 2, \dots, 2k+5\}$	In other words, $\mathcal{H}$ is a 3-uniform monostar having vertex 1 as the unique center, see Figure 10.23. Vertex 1 represents the minimum transversal, therefore a
voloshin-hypergraph_F01955	253	■ 1.000	$C_1=\{1, 2, 3\}, C_2=\{1, 4, 5\}, \ldots, C_{k+2}=\{1, 2k+4, 2k+5\}$	$C_1=\{1, 2, 3\}, C_2=\{1, 4, 5\}, \dots, C_{k+2}=\{1, 2k+4, 2k+5\}$	Figure 10.23. Vertex 1 represents the minimum transversal, therefore a form a stable set. Hence $\alpha_c(\mathcal{H})=2k+4$. Further, since no two edges hav
voloshin-hypergraph_F01956	253	■ 1.000	$\alpha_c(\mathcal{H})=2k+4$	$\alpha_c(\mathcal{H})=2k+4$	made numbers can be arbitrarily far from their respective bounds.
voloshin-hypergraph_F01957	253	■ 1.000	$k+2$	$k+2$	every induced subhypergraph $\mathcal{H}'$ (including $\mathcal{H}$ itself) the following equality holds:
voloshin-hypergraph_F01958	253	■ 1.000	$2k+5-(k+2)=$	$2k+5-(k+2)=$	every induced subhypergraph $\mathcal{H}'$ (including $\mathcal{H}$ itself) the following equality holds:
voloshin-hypergraph_F01959	253	■ 0.996	$k+3$	$k+3$	$\bar{\chi}(\mathcal{H}')=\alpha_c(\mathcal{H}').$ (10
voloshin-hypergraph_F01960	253	■ 0.996	$\bar{\chi}(\mathcal{H})=k+3$	$\bar{\chi}(\mathcal{H})=k+3$	$\bar{\chi}(\mathcal{H}')=\alpha_c(\mathcal{H}').$ (10
voloshin-hypergraph_F01961	253	■ 0.996	$\alpha_c(\mathcal{H})-\bar{\chi}(\mathcal{H})=2k+4-(k+3)=k+1>k$	$\alpha_c(\mathcal{H})-\bar{\chi}(\mathcal{H})=2k+4-(k+3)=k+1>k$	$\bar{\chi}(\mathcal{H}')=\alpha_c(\mathcal{H}').$ (10
voloshin-hypergraph_F01962	253	■ 0.957	$\mathcal{H}=\left(X, \binom{X}{r}, \mathcal{D}\right), r \geq 2$	$\mathcal{H}=\left(X, \binom{X}{r}, \mathcal{D}\right), r \geq 2$	
voloshin-hypergraph_F01963	253	■ 0.998	$\bar{\chi}(\mathcal{H}_Y)=r-1=\alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y)=r-1=\alpha_c(\mathcal{H}_Y)$	—
voloshin-hypergraph_F01964	253	■ 0.998	$ Y \geq r$	$ Y \geq r$	—
voloshin-hypergraph_F01965	253	■ 0.998	$\bar{\chi}(\mathcal{H}_Y)= Y =\alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y)= Y =\alpha_c(\mathcal{H}_Y)$	—
voloshin-hypergraph_F01966	253	■ 0.998	$ Y <r$	$ Y <r$	—
voloshin-hypergraph_F01967	254	■ 0.570	$\bar{\chi}(\mathcal{H})=n-1=\alpha_c(\mathcal{H})$	$\bar{\chi}(\mathcal{H})=n-1=\alpha_c(\mathcal{H})$	an different colors, so $\bar{\chi}(\mathcal{H})=n-1=\alpha_c(\mathcal{H})$. Consider any $T \subseteq X$. If $\mathcal{H}_T$ contains at least one C -edge, then $\bar{\chi}(\mathcal{H}_T)= T -1=\alpha_c(\mathcal{H}_T)$. Otherwise, $\bar{\chi}(\mathcal{H}_T)= T =\alpha_c(\mathcal{H}_T)$. $\square$
voloshin-hypergraph_F01968	254	■ 0.904	$\bar{\chi}(\mathcal{H}_Y)= Y -1=\alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y)= Y -1=\alpha_c(\mathcal{H}_Y)$	one C -edge, then $\bar{\chi}(\mathcal{H}_T)= T -1=\alpha_c(\mathcal{H}_T)$; Otherwise, $\bar{\chi}(\mathcal{H}_T)= T =\alpha_c(\mathcal{H}_T)$. $\square$ Minimal non-perfect mixed hypergraphs. Recall that we denote by $\mathcal{M}(d)$ the
voloshin-hypergraph_F01969	254	■ 0.904	$\bar{\chi}(\mathcal{H}_Y)= Y =\alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y)= Y =\alpha_c(\mathcal{H}_Y)$	one C -edge, then $\bar{\chi}(\mathcal{H}_T)= T -1=\alpha_c(\mathcal{H}_T)$; Otherwise, $\bar{\chi}(\mathcal{H}_T)= T =\alpha_c(\mathcal{H}_T)$. $\square$ Minimal non-perfect mixed hypergraphs. Recall that we denote by $\mathcal{M}(d)$ the
voloshin-hypergraph_F01970	254	■ 0.793	$\tau(\mathcal{H}_C)$	$\tau(\mathcal{H}_C)$	transversal number of $\mathcal{H}_C$, i.e., the cardinality of the smallest subset of vertices that contains
voloshin-hypergraph_F01971	254	■ 0.998	$\tau_2(\mathcal{H}_C)$	$\tau_2(\mathcal{H}_C)$	denote the double transversal number of $\mathcal{H}_C$ which is the cardinality of the smallest subset of vertices that contains at least two vertices from every C -edge of a mixed hypergraph
voloshin-hypergraph_F01972	254	■ 0.997	$\tau(\mathcal{H}_C)=1$	$\tau(\mathcal{H}_C)=1$	The edges of $\mathcal{C}$ monochrome and $\mathcal{C}$ edges have an empty intersection.
voloshin-hypergraph_F01973	254	■ 0.997	$\tau_2(\mathcal{H}_C) \geq 3$	$\tau_2(\mathcal{H}_C) \geq 3$	The edges of $\mathcal{C}$ monochrome and $\mathcal{C}$ edges have an empty intersection.

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voloshin-hypergraph_F01974	254	■ 0.977	$\tau_2(\mathcal{H}_C)=2$	$\tau_2(\mathcal{H}_C) = 2$	Theorem 10.8.3 A C -monostar $\mathcal{H} = (X, C, \mathcal{D})$ is not a perfect mixed hypergraph.
voloshin-hypergraph_F01975	254	■ 0.982	$\alpha_c(\mathcal{H})= X -1$	$\alpha_c(\mathcal{H}) = X - 1$	number of colors is not greater than $ X - 2$. This implies that $\bar{\chi}(\mathcal{H}) \leq X - 2$, so $\bar{\chi}(\mathcal{H}) \neq$
voloshin-hypergraph_F01976	254	■ 1.000	$ X -2$	$ X - 2$	not perfect subhypergraph that becomes perfect after strong deletion of a vertex represents
voloshin-hypergraph_F01977	254	■ 1.000	$\bar{\chi}(\mathcal{H}) \leq X -2$	$\bar{\chi}(\mathcal{H}) \leq X - 2$	not perfect subhypergraph that becomes perfect after strong deletion of a vertex represents
voloshin-hypergraph_F01978	254	■ 1.000	$\bar{\chi}(\mathcal{H}) \neq$	$\bar{\chi}(\mathcal{H}) \neq$	not perfect subhypergraph that becomes perfect after strong deletion of a vertex represents
voloshin-hypergraph_F01979	254	■ 0.899	$\alpha_c(\mathcal{H})$	$\alpha_c(\mathcal{H})$	]
voloshin-hypergraph_F01980	254	■ 0.997	$\mathcal{H}=(X, \mathcal{C}), X =n \geq 3, r \geq 2$	$\mathcal{H} = (X, \mathcal{C}), X = n \geq 3, r \geq 2$	pains of length $r-1$ on C_n . Thus, the usual cycle C_n (considered as a C -hypergraph) is C^2 for any $n \geq 3$. Note that the example given for Algorithm 10.4.1 (see Figure 10.9) is the
voloshin-hypergraph_F01981	254	■ 0.844	C_n^r	C_n^r	C_n for any $n \geq 3$. Note that the example given for Algorithm 10.4.1, (see Figure 10.9), is the cycloid C_5^3 considered as the mixed hypergraph $(X, C_5^3, \emptyset)$.
voloshin-hypergraph_F01982	254	■ 0.844	$X=\{0,1, \ldots, n-1\}$	$X = \{0, 1, \dots, n - 1\}$	C_n for any $n \geq 3$. Note that the example given for Algorithm 10.4.1, (see Figure 10.9), is the cycloid C_5^3 considered as the mixed hypergraph $(X, C_5^3, \emptyset)$.
voloshin-hypergraph_F01983	254	■ 0.844	$\mathcal{C}=\{i, i+1(\bmod n), \ldots, i+r-$	$\mathcal{C} = \{i, i + 1(\bmod n), \dots, i + r -$	C_n for any $n \geq 3$. Note that the example given for Algorithm 10.4.1, (see Figure 10.9), is the cycloid C_5^3 considered as the mixed hypergraph $(X, C_5^3, \emptyset)$.
voloshin-hypergraph_F01984	254	■ 1.000	$1(\bmod n)\}: i=0,1, \ldots, n-1\}$	$1(\bmod n)\}: i = 0, 1, \dots, n - 1\}$	
voloshin-hypergraph_F01985	254	■ 1.000	$C_n=(X, E)$	$C_n = (X, E)$	
voloshin-hypergraph_F01986	254	■ 1.000	$r-1$	$r - 1$	
voloshin-hypergraph_F01987	254	■ 1.000	C_n^2	C_n^2	
voloshin-hypergraph_F01988	254	■ 0.614	C_5^3	C_5^3	
voloshin-hypergraph_F01989	254	■ 0.614	$\left(X, C_5^3, \emptyset\right)$	$(X, C_5^3, \emptyset)$	
voloshin-hypergraph_F01990	254	■ 0.995	$C_n^r=(X, \mathcal{C}), 3 \leq r \leq n$	$C_n^r = (X, \mathcal{C}), 3 \leq r \leq n$	
voloshin-hypergraph_F01991	254	■ 0.995	$2r \geq n+2$	$2r \geq n + 2$	
voloshin-hypergraph_F01992	255	■ 1.000	$r=n$	$r = n$	PROOF. Case 1: $2r > n + 2$. Since for $r = n$ the theorem is evident, let $r < n - 1$. Hence,
voloshin-hypergraph_F01993	255	■ 1.000	$r \leq n-1$	$r \leq n - 1$	PROOF. Case 1: $2r > n + 2$. Since for $r = n$ the theorem is evident, let $r < n - 1$. Hence,
voloshin-hypergraph_F01994	255	■ 1.000	$\left C_i \cap C_j\right \geq 2, i, j \in I$	$ C_i \cap C_j \geq 2, i, j \in I$	$ C_i \cap C_j \geq 2, i, j \in I$. Let x_1 be a neighbor of x_2, x_3 be a neighbor of x_4 , and the pair $\{x_1, x_2\}$
voloshin-hypergraph_F01995	255	■ 1.000	$\left\{x_3, x_4\right\}$	$\{x_3, x_4\}$	be opposite to the pair $\{x_3, x_4\}$ on the host cycle of C_n^r . The inequality $2r \geq n + 2$ implies
voloshin-hypergraph_F01996	255	■ 1.000	$C_i \in \mathcal{C}$	$C_i \in \mathcal{C}$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_F01997	255	■ 1.000	$\left C_i \cap\left\{x_1, x_2\right\}\right \geq 2$	$ C_i \cap \{x_1, x_2\} \geq 2$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_F01998	255	■ 1.000	$\left C_i \cap\left\{x_3, x_4\right\}\right \geq 2$	$ C_i \cap \{x_3, x_4\} \geq 2$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_F01999	255	■ 1.000	$\bar{\chi}\left(C_n^r\right)<n-1$	$\bar{\chi}\left(C_n^r\right) < n - 1$	vertices belonging to all C -edges, so $\chi\left(C_n^r\right) < n - 1$. On the other hand, we can color x_1, x_2 with the first color, x_3, x_4 with the second color and all the remaining $n - 4$ vertices with all

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voloshin-hypergraph_F02000	255	■ 1.000	$x_{\{3\}}, x_{\{4\}}$	x_3, x_4	different colors. Therefore, $\bar{\chi}(C_n^r) = n - 2$. Since C_n^r is not a C -monostar, and vertices x_1
voloshin-hypergraph_F02001	255	■ 1.000	$n-4$	$n - 4$	different colors. Therefore, $\bar{\chi}(C_n^r) = n - 2$. Since C_n^r is not a C -monostar, and vertices x_1
voloshin-hypergraph_F02002	255	■ 1.000	$\bar{\chi}(C_n^r)=n-2$	$\bar{\chi}(C_n^r) = n - 2$	and x_3 form a minimum transversal, $\tau(C_n^r) = 2$. It implies $\alpha(C_n^r) = n - 2 = \bar{\chi}(C_n^r)$. At last,
voloshin-hypergraph_F02003	255	■ 1.000	$\tau(C_n^r)=2$	$\tau(C_n^r) = 2$	every induced subhypergraph of C_n^r is a C -bistar and, by Theorem 10.8.2, is perfect. Hence,
voloshin-hypergraph_F02004	255	■ 1.000	$\alpha(C_n^r)=n-2=\bar{\chi}(C_n^r)$	$\alpha(C_n^r) = n - 2 = \bar{\chi}(C_n^r)$	every induced subhypergraph of C_n^r is a C -bistar and, by Theorem 10.8.2, is perfect. Hence,
voloshin-hypergraph_F02005	255	■ 1.000	$2r=n+1$	$2r = n + 1$	$r \geq 5$. Since $\tau(C_n^r) = 2$, it follows that $\alpha(C_n^r) = n - 2$. We show that $\bar{\chi}(C_n^r) < n - 2$. For a
voloshin-hypergraph_F02006	255	■ 1.000	$r=3,4$	$r = 3, 4$	contradiction, assume that $\bar{\chi}(C_n^r) = n - 2$. Consider a strict coloring using $\bar{\chi}$ colors. There
voloshin-hypergraph_F02007	255	■ 1.000	$r \geq 5$	$r \geq 5$	are two possibilities: either one pair of vertices is colored with one color and another pair
voloshin-hypergraph_F02008	255	■ 1.000	$\alpha(C_n^r)=n-2$	$\alpha(C_n^r) = n - 2$	are two possibilities: either one pair of vertices is colored with one color and another pair
voloshin-hypergraph_F02009	255	■ 1.000	$\bar{\chi}(C_n^r)<n-2$	$\bar{\chi}(C_n^r) < n - 2$	are two possibilities: either one pair of vertices is colored with one color and another pair
voloshin-hypergraph_F02010	255	■ 0.988	$x_{\{1\}}, x_{\{2\}} \in X$	$x_1, x_2 \in X$	that $n_{12} + n_{34} \geq 1$. We have $n_{12} + n_{23} + n_{34} + n_{41} + 4 = n = 2r - 1$. Since any C -edge must
voloshin-hypergraph_F02011	255	■ 1.000	$n_{\{i\ j\}}$	n_{ij}	contain either $\{x_1, x_2\}$ or $\{x_3, x_4\}$, it follows that
voloshin-hypergraph_F02012	255	■ 1.000	$x_{\{j\}}, i, j=1,2,3,4$	$x_j, i, j = 1, 2, 3, 4$	$n_{12} + n_{23} + n_{34} + 2 < r$.
voloshin-hypergraph_F02013	255	■ 1.000	$n_{\{12\}}=0$	$n_{12} = 0$	$n_{12} + n_{23} + n_{34} + 2 < r$.
voloshin-hypergraph_F02014	255	■ 1.000	$n_{\{34\}}=0$	$n_{34} = 0$	$n_{12} + n_{23} + n_{34} + 2 < r$.
voloshin-hypergraph_F02015	255	■ 0.942	$ C \cap \{x_{\{1\}}, x_{\{2\}}\} \leq 1$	$ C \cap \{x_1, x_2\} \leq 1$	$n_{12} + n_{23} + n_{34} + 1 \leq n_{12} + n_{23} + n_{34} + 2 < r$.
voloshin-hypergraph_F02016	255	■ 0.942	$ C \cap \{x_{\{3\}}, x_{\{4\}}\} \leq 1$	$ C \cap \{x_3, x_4\} \leq 1$	$n_{12} + n_{23} + n_{34} + 1 \leq n_{12} + n_{23} + n_{34} + 2 < r$.
voloshin-hypergraph_F02017	255	■ 1.000	$n_{\{12\}}+n_{\{34\}} \geq 1$	$n_{12} + n_{34} \geq 1$	$n_{12} + n_{14} + n_{34} + 2 < r$.
voloshin-hypergraph_F02018	255	■ 1.000	$n_{\{12\}}+n_{\{23\}}+n_{\{34\}}+n_{\{41\}}+4=n-2r-1$	$n_{12} + n_{23} + n_{34} + n_{41} + 4 = n = 2r - 1$	$n_{12} + n_{14} + n_{34} + 2 < r$.
voloshin-hypergraph_F02019	255	■ 1.000	$2r-1+n_{\{12\}}+n_{\{34\}}<2r$	$2r - 1 + n_{12} + n_{34} < 2r$	clockwise in this order. Then x_1, x_2, x_3 is a minimum C -bitransversal. Let n_{ij} be the number
voloshin-hypergraph_F02020	255	■ 1.000	$r<r$	$r < r$	of vertices bet
voloshin-hypergraph_F02021	255	■ 1.000	C_{2r-1}^r	C_{2r-1}^r	$n_{12} + 1 + n_{23} < r$,
voloshin-hypergraph_F02022	255	■ 1.000	$r \geq 3$	$r \geq 3$	$n_{12} + 1 + n_{23} < r$.
voloshin-hypergraph_F02023	255	■ 1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	
voloshin-hypergraph_F02024	255	■ 1.000	$x_{\{j\}}, i, j=1,2,3$	$x_j, i, j = 1, 2, 3$	
voloshin-hypergraph_F02025	255	■ 1.000	$n_{\{12\}}+n_{\{23\}}+n_{\{31\}}+3=n-2r-1$	$n_{12} + n_{23} + n_{31} + 3 = n = 2r - 1$	
voloshin-hypergraph_F02026	256	■ 1.000	$r<5$	$r < 5$	Summing these inequalities implies $r < 5$, a contradiction. Consequently, C_{r-1}^r cannot be

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voloshin-hypergraph_F02027	256	■ 1.000	$2r \leq n$	$2r \leq n$	Case 3: $2r \leq n$. In this case, all C -edges containing a fixed vertex form an induced
voloshin-hypergraph_F02028	256	■ 0.995	$\mathcal{H}^r = (X, \mathcal{C})$	$\mathcal{H}^r = (X, \mathcal{C})$	
voloshin-hypergraph_F02029	256	■ 1.000	$2r + 2$	$2r + 2$	
voloshin-hypergraph_F02030	256	■ 1.000	$2r$	$2r$	in the same direction around the cycle, and so on, ..., edge C_{2r} is obtained when shifting C_{2r-1} by 1 around the cycle. In other words, edges $C_0, C_1, \dots, C_{2r}$ are all copies of C .
voloshin-hypergraph_F02031	256	■ 0.978	$\mathcal{C}_{2r}$	$\mathcal{C}_{2r}$	$2r$ vertices.
voloshin-hypergraph_F02032	256	■ 1.000	$\mathcal{C}_{2r-1}$	$\mathcal{C}_{2r-1}$	For $r = 3$ the example is shown in Figure 10.24. To avoid confusion, edges $C_2, \dots, C_6$ are not shown, and edges C_0 and C_1 are shown separately.
voloshin-hypergraph_F02033	256	■ 1.000	$\mathcal{C}_2, \mathcal{C}_3, \dots, \mathcal{C}_{2r}$	$\mathcal{C}_2, \mathcal{C}_3, \dots, \mathcal{C}_{2r}$	For $r = 3$ the example is shown in Figure 10.24. To avoid confusion, edges $C_2, \dots, C_6$ are not shown, and edges C_0 and C_1 are shown separately.
voloshin-hypergraph_F02034	256	■ 1.000	$\mathcal{C}_o$	$\mathcal{C}_o$	are not shown, and edges C_0 and C_1 are shown separately.
voloshin-hypergraph_F02035	256	■ 0.950	$\mathcal{C}_e$	$\mathcal{C}_e$	Theorem 10.8.5 (Král', 2003) The $\mathcal{C}$ -hypergraph $\mathcal{H}^r$ contains no cycloid on $2r - 1$ ver-
voloshin-hypergraph_F02036	256	■ 0.950	$\mathcal{H}^r$	$\mathcal{H}^r$	Theorem 10.8.5 (Král', 2003) The $\mathcal{C}$ -hypergraph $\mathcal{H}^r$ contains no cycloid on $2r - 1$ ver-
voloshin-hypergraph_F02037	256	■ 0.950	$r + 1$	$r + 1$	Theorem 10.8.5 (Král', 2003) The $\mathcal{C}$ -hypergraph $\mathcal{H}^r$ contains no cycloid on $2r - 1$ ver-
voloshin-hypergraph_F02038	256	■ 0.997	$r = 3$	$r = 3$	$\bar{\chi}(\mathcal{H}^r) = 2r - 4 < 2r - 3 = \alpha_{\mathcal{C}}(\mathcal{H}^r)$
voloshin-hypergraph_F02039	256	■ 0.997	$\mathcal{C}_2, \dots, \mathcal{C}_6$	$\mathcal{C}_2, \dots, \mathcal{C}_6$	$\bar{\chi}(\mathcal{H}^r) = 2r - 4 < 2r - 3 = \alpha_{\mathcal{C}}(\mathcal{H}^r)$
voloshin-hypergraph_F02040	256	■ 0.997	$2r - 1$	$2r - 1$	family of edges the following implication holds: if every two edges of the subfamily have
voloshin-hypergraph_F02041	257	■ 0.901	$\alpha(\mathcal{H}_{\mathcal{C}}) = n - 1$	$\alpha(\mathcal{H}_{\mathcal{C}}) = n - 1$	the same color, therefore no coloring with $n - 1$ colors exist. In all cases $\chi(\mathcal{H}) \neq n - 1$. $\square$
voloshin-hypergraph_F02042	257	■ 0.901	$\bar{\chi}(\mathcal{H}) =$	$\bar{\chi}(\mathcal{H}) =$	the same color, therefore no coloring with $n - 1$ colors exist. In all cases $\chi(\mathcal{H}) \neq n - 1$. $\square$
voloshin-hypergraph_F02043	257	■ 0.960	$\alpha_{\mathcal{C}}(\mathcal{H})$	$\alpha_{\mathcal{C}}(\mathcal{H})$	In fact, for every such subfamily $ \bigcap_{C \in \mathcal{C}_1} \geq 2$ b
voloshin-hypergraph_F02044	257	■ 0.998	$\mathcal{H}_{\mathcal{C}} = (X, \mathcal{C}, \emptyset)$	$\mathcal{H}_{\mathcal{C}} = (X, \mathcal{C}, \emptyset)$	$\mathcal{H}_{\mathcal{C}}$ is a bi-Helly hypergraph, and moreover, $C \cap C' \neq \emptyset$ implies $ C \cap C' \geq 2$ for any $C, C' \in \mathcal{C}$.
voloshin-hypergraph_F02045	257	■ 1.000	$\mathcal{C}_1 \subseteq \mathcal{C}$	$\mathcal{C}_1 \subseteq \mathcal{C}$	Consider the problem of finding a minimum transversal of $\mathcal{H}_{\mathcal{C}}$. For this, co
voloshin-hypergraph_F02046	257	■ 1.000	$C \cap C' \neq \emptyset$	$C \cap C' \neq \emptyset$	Consider the problem of finding a minimum transversal of $\mathcal{H}_{\mathcal{C}}$. For this, co
voloshin-hypergraph_F02047	257	■ 1.000	$C, C' \in \mathcal{C}_1$	$C, C' \in \mathcal{C}_1$	Consider the problem of finding a minimum transversal of $\mathcal{H}_{\mathcal{C}}$. For this, co
voloshin-hypergraph_F02048	257	■ 1.000	$ \bigcap_{C \in \mathcal{C}_1} C \geq 2$	$ \bigcap_{C \in \mathcal{C}_1} C \geq 2$	
voloshin-hypergraph_F02049	257	■ 1.000	$\mathcal{C}_1$	$\mathcal{C}_1$	
voloshin-hypergraph_F02050	257	■ 0.989	$ \bigcap_{C \in \mathcal{C}_1} C \geq 2$	$ \bigcap_{C \in \mathcal{C}_1} C \geq 2$	
voloshin-hypergraph_F02051	257	■ 0.989	$C, C' \in \mathcal{C}_1$	$C, C' \in \mathcal{C}_1$	
voloshin-hypergraph_F02052	257	■ 0.815	$L(\mathcal{H}_{\mathcal{C}}) = (\mathcal{C}, \mathcal{E})$	$L(\mathcal{H}_{\mathcal{C}}) = (\mathcal{C}, \mathcal{E})$	
voloshin-hypergraph_F02053	257	■ 0.815	$(C, C') \in \mathcal{E} \Leftrightarrow C \cap C' \neq \emptyset$	$(C, C') \in \mathcal{E} \Leftrightarrow C \cap C' \neq \emptyset$	

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voloshin-hypergraph_F02054	257	0.718	$K_{\{1\}}, K_{\{2\}}, \ldots, K_{\{t\}}$	$K_1, K_2, \dots, K_t$	
voloshin-hypergraph_F02055	257	0.930	$\tau\left(\mathcal{H}_{\mathcal{C}}\right)=t$	$\tau\left(\mathcal{H}_{\mathcal{C}}\right)=t$	—
voloshin-hypergraph_F02056	257	0.930	$\alpha_{\mathcal{C}}\left(\mathcal{H}\right)= X -t$	$\alpha_{\mathcal{C}}\left(\mathcal{H}\right)= X -t$	—
voloshin-hypergraph_F02057	258	1.000	$\left\{x_{\{i\}}, y_{\{i\}}\right\}$	$\left\{x_i, y_i\right\}$	$t+1, t+2, \dots, X -t$. Thus, we obtain a proper coloring of $\mathcal{H}$ with $ X -t=\alpha_{\mathcal{C}}(\mathcal{H})$
voloshin-hypergraph_F02058	258	1.000	$K_{\{i\}}, i=1,2, \ldots, t$	$K_i, i=1,2, \dots, t$	$t+1, t+2, \dots, X -t$. Thus, we obtain a proper coloring of $\mathcal{H}$ with $ X -t=\alpha_{\mathcal{C}}(\mathcal{H})$
voloshin-hypergraph_F02059	258	1.000	$\left\{x_{\{i\}}, y_{\{i\}}\right\} \notin \mathcal{D}$	$\left\{x_i, y_i\right\} \notin \mathcal{D}$	colors, and the theorem follows. $\square$
voloshin-hypergraph_F02060	258	1.000	$K_{\{i\}} \neq K_{\{j\}}$	$K_i \neq K_j$	colors, and the theorem follows. $\square$
voloshin-hypergraph_F02061	258	1.000	$\left\{x_{\{i\}}, y_{\{i\}}\right\} \cap \left\{x_{\{j\}}, y_{\{j\}}\right\}=\emptyset$	$\left\{x_i, y_i\right\} \cap \left\{x_j, y_j\right\}=\emptyset$	colors, and the theorem follows. $\square$
voloshin-hypergraph_F02062	258	0.961	$x_{\{1\}}, y_{\{1\}}$	x_1, y_1	Corollary 10.8.1 If $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is a perfect mixed hvnertree. $\nu\left(\mathcal{H}_{\mathcal{C}}\right)$ and $\tau\left(\mathcal{H}_{\mathcal{C}}\right)$ are the
voloshin-hypergraph_F02063	258	0.961	$x_{\{2\}}, y_{\{2\}}$	x_2, y_2	Corollary 10.8.1 If $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is a perfect mixed hvnertree. $\nu\left(\mathcal{H}_{\mathcal{C}}\right)$ and $\tau\left(\mathcal{H}_{\mathcal{C}}\right)$ are the
voloshin-hypergraph_F02064	258	0.961	$\ldots, x_{\{t\}}, y_{\{t\}}$	$\ldots, x_t, y_t$	Corollary 10.8.1 If $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ is a perfect mixed hvnertree. $\nu\left(\mathcal{H}_{\mathcal{C}}\right)$ and $\tau\left(\mathcal{H}_{\mathcal{C}}\right)$ are the
voloshin-hypergraph_F02065	258	0.536	$t+1, t+2, \ldots, X -t$	$t+1, t+2, \dots, X -t$	respectively, then
voloshin-hypergraph_F02066	258	0.536	$ X -t=\alpha_{\mathcal{C}}\left(\mathcal{H}\right)$	$ X -t=\alpha_{\mathcal{C}}\left(\mathcal{H}\right)$	respectively, then
voloshin-hypergraph_F02067	258	0.447	$\mathrm{v}\left(\mathcal{H}_{\mathcal{C}}\right)$	$\mathrm{v}\left(\mathcal{H}_{\mathcal{C}}\right)$	Proof. Since $\mathcal{H}_{\mathcal{C}}$ is also a hypertree, it fulfills the well known König property, see Theo-
voloshin-hypergraph_F02068	258	0.999	$X=\{1,2,3,4,5\}, \mathcal{C}=\left\{C_{\{1\}}, C_{\{2\}}, C_{\{3\}}\right\}, C_{\{1\}}=\{1,2,3\}, C_{\{2\}}=\{2,3,4,5\}, C_{\{3\}}=\{1,4,5\}$	$X=\{1,2,3,4,5\}, \mathcal{C}=\left\{C_1, C_2, C_3\right\}, C_1=\{1,2,3\}, C_2=\{2,3,4,5\}, C_3=\{1,4,5\}$	
voloshin-hypergraph_F02069	258	1.000	$C_{\{1\}} \cap C_{\{3\}}=\{1\}$	$C_1 \cap C_3=\{1\}$	
voloshin-hypergraph_F02070	258	0.934	$\bar{\chi}\left(\mathcal{H}\right)=\alpha_{\mathcal{C}}\left(\mathcal{H}\right)=3$	$\bar{\chi}\left(\mathcal{H}\right)=\alpha_{\mathcal{C}}\left(\mathcal{H}\right)=3$	
voloshin-hypergraph_F02071	259	1.000	$\tau_{\{2\}}\left(\mathcal{H}_{\{1\}}\right)$	$\tau_2\left(\mathcal{H}_1\right)$	
voloshin-hypergraph_F02072	259	1.000	$\tau_{\{2\}}\left(\mathcal{H}_{\{2\}}\right)$	$\tau_2\left(\mathcal{H}_2\right)$	
voloshin-hypergraph_F02073	260	1.000	$S\left(\mathcal{H}\right)$	$S\left(\mathcal{H}\right)$	Definition 10.9.1 A mixed hypergraph $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ has a gap at k if $\mathcal{S}(\mathcal{H})$ contains elements larger and smaller than k but omits k . The chromatic spectrum $\mathcal{D}(\mathcal{H})$ is called
voloshin-hypergraph_F02074	260	1.000	$r_{\{k\}}\left(\mathcal{H}\right)>0$	$r_k\left(\mathcal{H}\right)>0$	Definition 10.9.1 A mixed hypergraph $\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$ has a gap at k if $\mathcal{S}(\mathcal{H})$ contains elements larger and smaller than k but omits k . The chromatic spectrum $\mathcal{D}(\mathcal{H})$ is called
voloshin-hypergraph_F02075	260	1.000	$\{a, b\}$	$\{a, b\}$	$R(\mathcal{H})=(0.7.0.0)$
voloshin-hypergraph_F02076	260	1.000	$a, a^{\prime}$	$a, a^{\prime}$	
voloshin-hypergraph_F02077	260	1.000	$b, b^{\prime}$	$b, b^{\prime}$	
voloshin-hypergraph_F02078	260	0.997	$\left\{a, a^{\prime}, b, b^{\prime}\right\}$	$\left\{a, a^{\prime}, b, b^{\prime}\right\}$	and all seven feasible partitions are induced by the following 14 partitions:
voloshin-hypergraph_F02079	260	1.000	$a, a^{\prime}, b, b^{\prime}$	$a, a^{\prime}, b, b^{\prime}$	order $a, a^{\prime}, b, b^{\prime}$). 1122 1112 1121 1222 2122 1212 1221
voloshin-hypergraph_F02080	260	1.000	$\mathcal{H}=\left(X, K_4^3, K_4^4\right)$	$\mathcal{H}=\left(X, K_4^3, K_4^4\right)$	

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voloshin-hypergraph_F02081	260	■ 1.000	$X=\left\{a, a^{\prime}, b, b^{\prime}\right\}$	$X=\{a, a', b, b'\}$	
voloshin-hypergraph_F02082	261	■ 0.949	$a^{\prime}$	a'	10.28. To avoid any confusion, the C -edges are not shown. Denote the mixed hypergraph
voloshin-hypergraph_F02083	261	■ 0.949	$b^{\prime}$	b'	10.28. To avoid any confusion, the C -edges are not shown. Denote the mixed hypergraph
voloshin-hypergraph_F02084	261	■ 0.999	$\left\{a, a^{\prime}\right\}$	$\{a, a'\}$	remains to consider the proper colorings when each pair is monochromatic. Surprisingly,
voloshin-hypergraph_F02085	261	■ 0.998	$\left\{b, b^{\prime}\right\},\left\{c, c^{\prime}\right\}$	$\{b, b'\},\{c, c'\}$	there is only one coloring of such type (if we do not count the permutations of colors):
voloshin-hypergraph_F02086	261	■ 0.998	$\left\{d, d^{\prime}\right\}$	$\{d, d'\}$	there is only one coloring of such type (if we do not count the permutations of colors):
voloshin-hypergraph_F02087	261	■ 1.000	$\left\{a, a^{\prime}, c\right\}$	$\{a, a', c\}$	indeed, no two pairs of vertices may have the same color because each pair of pairs is a $\mathcal{D}$ -edge. see Figure 10.28. The fundamental conclusion at this point is that $\mathcal{H}$ has several
voloshin-hypergraph_F02088	261	■ 0.982	$r_{\{2\}} \geq 2$	$r_2 \geq 2$	
voloshin-hypergraph_F02089	261	■ 0.982	$r_{\{4\}}=1$	$r_4=1$	
voloshin-hypergraph_F02090	261	■ 0.982	$r_{\{3\}}=0$	$r_3=0$	
voloshin-hypergraph_F02091	262	■ 0.991	$\left\{a, c, c^{\prime}\right\},\left\{a, d, d^{\prime}\right\},\left\{b, c, c^{\prime}\right\},\left\{b, d, d^{\prime}\right\}$	$\{a, c, c'\},\{a, d, d'\},\{b, c, c'\},\{b, d, d'\}$	denoted by $\mathcal{H}_{2,4}$ where index means $\mathfrak{S}(\mathcal{H})=\{2,4\}$. Developing this idea further, the construction of minimal examples of mixed hyper-
voloshin-hypergraph_F02092	262	■ 1.000	$\mathcal{H}_{\{2,4\}}$	$\mathcal{H}_{2,4}$	that $2 \leq s \leq t-2$. In this subsection we construct a ... of $\mathcal{H}=\{2,4\}$ and prove that $\mathcal{H}$ has the property
voloshin-hypergraph_F02093	262	■ 1.000	$S(\mathcal{H})=\{2,4\}$	$S(\mathcal{H})=\{2,4\}$	that $2 \leq s \leq t-2$. In this subsection we construct a ... of $\mathcal{H}=\{2,4\}$ and prove that $\mathcal{H}$ has the property
voloshin-hypergraph_F02094	262	■ 1.000	s, t	s, t	$S(K_n)=\{n\}$. First, we describe the construction informally. Beginning with K_t , expand $t-$
voloshin-hypergraph_F02095	262	■ 1.000	$2 \leq s \leq t-2$	$2 \leq s \leq t-2$	2 of the vertices into pairs, leaving two special vertices unexpanded. The $\mathcal{D}$ -edge consisting
voloshin-hypergraph_F02096	262	■ 1.000	$\mathcal{H}_{s, t}$	$\mathcal{H}_{s, t}$	2 of the vertices into pairs, leaving two special vertices unexpanded. The $\mathcal{D}$ -edge consisting
voloshin-hypergraph_F02097	262	■ 0.998	$S\left(\mathcal{H}_{\{s, t\}}\right)=\{s, t\}$	$S\left(\mathcal{H}_{s, t}\right)=\{s, t\}$	of the two special vertices remains, and the other $\mathcal{D}$ -edges expand into $\mathcal{D}$ -edges of size 3 or 4 (special vertex plus pair or union of two pairs). Add as C -edges all triples consisting of
voloshin-hypergraph_F02098	262	■ 0.998	s	s	of the two special vertices remains, and the other $\mathcal{D}$ -edges expand into $\mathcal{D}$ -edges of size 3 or 4 (special vertex plus pair or union of two pairs). Add as C -edges all triples consisting of
voloshin-hypergraph_F02099	262	■ 1.000	$t-1$	$t-1$	... (special vertex plus pair or union of two pairs), leaving no C -edges unexpanded and
voloshin-hypergraph_F02100	262	■ 1.000	$2 t-s$	$2 t-s$	three vertices arising from two original vertices (special vertex plus pair, or three
voloshin-hypergraph_F02101	262	■ 1.000	$2 t-2$	$2 t-2$	from two pairs).
voloshin-hypergraph_F02102	262	■ 0.723	$\{2, t\}$	$\{2, t\}$	
voloshin-hypergraph_F02103	262	■ 0.723	$\left(X,\emptyset,\binom{X}{2}\right)$	$\left(X, \emptyset, \binom{X}{2}\right)$	
voloshin-hypergraph_F02104	262	■ 0.945	$S\left(K_{\{n\}}\right)=\{n\}$	$S\left(K_n\right)=\{n\}$	
voloshin-hypergraph_F02105	262	■ 0.945	$K_{\{t\}}$	K_t	
voloshin-hypergraph_F02106	262	■ 0.945	$t-$	$t-$	

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voloshin-hypergraph_F02107	263	■ 0.924	t=4	$t = 4$	This describes the construction completely, but we present it more formally to facilitate the proofs. The smallest instance is for $t = 4$: from K_4 , a 6-vertex mixed hypergraph with
voloshin-hypergraph_F02108	263	■ 0.999	[m]=\{1, \ldots, m\}	$[m] = \{1, \dots, m\}$	spectrum $\{2, 4\}$ is produced. Let $[m] = \{1, \dots, m\}$.
voloshin-hypergraph_F02109	263	■ 1.000	\mathcal{H}_{2, t}	$\mathcal{H}_{2,t}$	Construction 1. Define a hypergraph $\mathcal{H}_{2,t}$ with vertex set
voloshin-hypergraph_F02110	263	■ 1.000	x_{\{r\}} a_{\{i\}} b_{\{i\}}	$x_r a_i b_i$	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_F02111	263	■ 1.000	r \in \{1, 2\}	$r \in \{1, 2\}$	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_F02112	263	■ 1.000	i \in [t-2]	$i \in [t - 2]$	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_F02113	263	■ 1.000	a_{\{i\}} b_{\{i\}} a_{\{j\}} b_{\{j\}}	$a_i b_i a_j b_j$	of the sets of four triples contained in $\{a_i, b_i, a_j, b_j\}$. The $\mathcal{C}$ -edges in $\mathcal{H}_{2,t}$ are $T \cup W$. The
voloshin-hypergraph_F02114	263	■ 1.000	i, j \in [t-2]	$i, j \in [t - 2]$	of the sets of four triples contained in $\{a_i, b_i, a_j, b_j\}$. The $\mathcal{C}$ -edges in $\mathcal{H}_{2,t}$ are $T \cup W$. The
voloshin-hypergraph_F02115	263	■ 1.000	W	W	of the sets of four triples contained in $\{a_i, b_i, a_j, b_j\}$. The $\mathcal{C}$ -edges in $\mathcal{H}_{2,t}$ are $T \cup W$. The
voloshin-hypergraph_F02116	263	■ 1.000	\left\{a_{\{i\}}, b_{\{i\}}, a_{\{j\}}, b_{\{j\}}\right\}	$\{a_i, b_i, a_j, b_j\}$	$\mathcal{D}$ -edges are $T \cup U \cup \{x_1 x_2\}$.
voloshin-hypergraph_F02117	263	■ 1.000	T \cup W	$T \cup W$	$\mathcal{D}$ -edges are $T \cup U \cup \{x_1 x_2\}$.
voloshin-hypergraph_F02118	263	■ 1.000	T \cup U \cup \left\{x_{\{1\}} x_{\{2\}}\right\}	$T \cup U \cup \{x_1 x_2\}$	Lemma 10.9.1 $\chi(\mathcal{H}_{2,t}) =$
voloshin-hypergraph_F02119	263	■ 1.000	S\left(\mathcal{H}_{2, t}\right)=\{2, t\}	$S(\mathcal{H}_{2,t}) = \{2, t\}$	Proof. Let c be an arbitrary s -
voloshin-hypergraph_F02120	263	■ 1.000	c\left(a_{\{i\}}\right) \neq c\left(b_{\{i\}}\right)	$c(a_i) \neq c(b_i)$	a_i or b_i . Thus, in this case there are at most two colors. The existence of $\mathcal{D}$ -edges prevents
voloshin-hypergraph_F02121	263	■ 1.000	a_{\{i\}}	a_i	a proper 1-coloring, and setting all $c(a_i) = c(x_1) = 1$ and $c(b_i) = c(x_2) = 2$ completes a proper 2-coloring.
voloshin-hypergraph_F02122	263	■ 1.000	b_{\{i\}}	b_i	a proper 1-coloring, and setting all $c(a_i) = c(x_1) = 1$ and $c(b_i) = c(x_2) = 2$ completes a proper 2-coloring.
voloshin-hypergraph_F02123	263	■ 0.987	c\left(a_{\{i\}}\right)=c\left(x_{\{1\}}\right)=1	$c(a_i) = c(x_1) = 1$	force these colors to be distinct for all i , and the $\mathcal{D}$ -edges in T along with $\{x_1 x_2\}$ require
voloshin-hypergraph_F02124	263	■ 0.987	c\left(b_{\{i\}}\right)=c\left(x_{\{2\}}\right)=2	$c(b_i) = c(x_2) = 2$	force these colors to be distinct for all i , and the $\mathcal{D}$ -edges in T along with $\{x_1 x_2\}$ require
voloshin-hypergraph_F02125	263	■ 1.000	c\left(a_{\{i\}}\right)=c\left(b_{\{i\}}\right)	$c(a_i) = c(b_i)$	is proper. □
voloshin-hypergraph_F02126	263	■ 1.000	\left\{x_{\{1\}} x_{\{2\}}\right\}	$\{x_1 x_2\}$	In order to extend this construction to the lower chromatic number s , we use a simple
voloshin-hypergraph_F02127	263	■ 1.000	\left(X_{\{1\}}, \mathcal{H}_{\{1\}}, \mathcal{D}_{\{1\}}\right)	$(X_1, \mathcal{C}_1, \mathcal{D}_1)$	Lemma 10.9.2 If $\mathcal{H}_1$ and $\mathcal{H}_2$ are mixed hypergraphs, then the feasible set of the join of $\mathcal{H}_1$
voloshin-hypergraph_F02128	263	■ 0.739	\left(X_{\{2\}}, \mathcal{H}_{\{2\}}, \mathcal{D}_{\{2\}}\right)	$(X_2, \mathcal{C}_2, \mathcal{D}_2)$	and $\mathcal{H}_2$ is $\{i + j : i \in S(\mathcal{H}_1), j \in S(\mathcal{H}_2)\}$.
voloshin-hypergraph_F02129	263	■ 0.739	(X, C, \mathcal{D})	$(X, \mathcal{C}, \mathcal{D})$	and $\mathcal{H}_2$ is $\{i + j : i \in S(\mathcal{H}_1), j \in S(\mathcal{H}_2)\}$.
voloshin-hypergraph_F02130	263	■ 0.998	X=X_{\{1\}} \cup X_{\{2\}}, \mathcal{H}=\mathcal{H}_{\{1\}} \cup \mathcal{H}_{\{2\}}	$X = X_1 \cup X_2, \mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2$	
voloshin-hypergraph_F02131	263	■ 0.998	\mathcal{D}=\mathcal{D}_{\{1\}} \cup \mathcal{D}_{\{2\}} \cup R	$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup R$	
voloshin-hypergraph_F02132	263	■ 0.991	\left\{i+j: i \in S\left(\mathcal{H}_{\{1\}}\right), j \in S\left(\mathcal{H}_{\{2\}}\right)\right\}	$\{i + j : i \in S(\mathcal{H}_1), j \in S(\mathcal{H}_2)\}$	Call the join $\mathcal{H}'$ of an arbitrary mixed hypergraph $\mathcal{H}$ and $\mathcal{H}'$ the join of $\mathcal{H}$ and $\mathcal{H}'$.
voloshin-hypergraph_F02133	263	■ 1.000	z \notin X	$z \notin X$	$2t - s$, and this is sharp.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02134	263	■ 1.000	$\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$(r_1, r_2, \ldots, r_n)$	
voloshin-hypergraph_F02135	263	■ 1.000	$R\left(\mathcal{H}^{\prime}\right)=\left(0, r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$R\left(\mathcal{H}^{\prime}\right)=\left(0, r_1, r_2, \ldots, r_n\right)$	
voloshin-hypergraph_F02136	263	■ 1.000	$n \geq$	$n \geq$	
voloshin-hypergraph_F02137	264	■ 1.000	$n < 2 t-s$	$n < 2 t-s$	PROOF. Consider a proper coloring of $\mathcal{H}$ using k colors, where k is the smallest element of the feasible set larger than $t-1$. If $n < 2 t-s$, then using at least t colors requires having
voloshin-hypergraph_F02138	264	■ 0.996	$s+1$	$s+1$	at least $s+1$ color classes of size 1. Two such color classes can be combined to obtain a
voloshin-hypergraph_F02139	264	■ 0.999	$K_{\{s+1\}}$	K_{s+1}	in the feasible set, $\mathcal{H}$ contains K_{s+1} . Now $\mathcal{H}$ is not s -colorable; the contradiction yields
voloshin-hypergraph_F02140	264	■ 1.000	$n \geq 2 t-s$	$n \geq 2 t-s$	$n \geq$
voloshin-hypergraph_F02141	264	■ 1.000	$s=2$	$s=2$	For $s=2$, Lemma 10.9.1 shows that Construction 10.7.1 achieves the bound. For $s \geq 2$, define $\mathcal{H}_s$ to be the join of K_{s-2} and $\mathcal{H}_{s-2}$. By Lemma 10.9.2, the feasible set of $\mathcal{H}_s$ is
voloshin-hypergraph_F02142	264	■ 1.000	$s > 2$	$s > 2$	For $s=2$, Lemma 10.9.1 shows that Construction 10.7.1 achieves the bound. For $s \geq 2$, define $\mathcal{H}_s$ to be the join of K_{s-2} and $\mathcal{H}_{s-2}$. By Lemma 10.9.2, the feasible set of $\mathcal{H}_s$ is
voloshin-hypergraph_F02143	264	■ 0.996	$K_{\{s-2\}}$	K_{s-2}	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$. The number of vertices in $\mathcal{H}_{s,t}$ is $s-2+2(t-s+2)-2=$
voloshin-hypergraph_F02144	264	■ 0.996	$\mathcal{H}_{\{2, t-s+2\}}$	$\mathcal{H}_{2, t-s+2}$	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$. The number of vertices in $\mathcal{H}_{s,t}$ is $s-2+2(t-s+2)-2=$
voloshin-hypergraph_F02145	264	■ 1.000	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$	$2 t-s$. □
voloshin-hypergraph_F02146	264	■ 1.000	$s-2+2(t-s+2)-2=$	$s-2+2(t-s+2)-2=$	$2 t-s$. □
voloshin-hypergraph_F02147	264	■ 1.000	$t-1 > s \geq 2$	$t-1 > s \geq 2$	As we mentioned earlier, closer analysis allows one $\bar{C}$ -edge in the 6-vertex example to
voloshin-hypergraph_F02148	264	■ 1.000	$s \geq 2$	$s \geq 2$	As we mentioned earlier, closer analysis allows one $\bar{C}$ -edge in the 6-vertex example to
voloshin-hypergraph_F02149	264	■ 1.000	$t \geq 4$	$t \geq 4$	be dropped without changing the spectrum. Thus, 7 $\bar{C}$ -edges i
voloshin-hypergraph_F02150	264	■ 1.000	$t-s \geq 2$	$t-s \geq 2$	be dropped without changing the spectrum. Thus, 7 $\bar{C}$ -edges i
voloshin-hypergraph_F02151	264	■ 1.000	$n \geq t+(t-s) \geq 6$	$n \geq t+(t-s) \geq 6$	be dropped without changing the spectrum. Thus, 7 $\bar{C}$ -edges i
voloshin-hypergraph_F02152	264	■ 0.999	$7 \mathcal{H}\{C\}$	$7 \mathcal{C}$	$\{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4, 5\}$. Subsets $\{1, 2, 3\}, \{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}$ are the bi-
voloshin-hypergraph_F02153	264	■ 0.999	$6 \mathcal{H}\{D\}$	$6 \mathcal{D}$	$\{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4, 5\}$. Subsets $\{1, 2, 3\}, \{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}$ are the bi-
voloshin-hypergraph_F02154	264	■ 0.615	$X=\{1, 2, 3, 4, 5, 6\}, C=\{\{1, 2, 3\}$	$X=\{1, 2, 3, 4, 5, 6\}, C=\{\{1, 2, 3\}$	edges. In Figure 10.29, the bi-edges are drawn by doubled curves. One can easily check that the four strict 2-colorings in order of vertices 1, 2, 3, 4, 5, 6 are 112122, 112212, 212121,
voloshin-hypergraph_F02155	264	■ 0.783	$\{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4\}, \{3, 4, 5\}, \{2, 4, 5\}\}, \mathcal{H}\{D\}=\{\{1, 6\}, \{1, 2, 3\}, \{1, 4, 5\}$	$\{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4\}, \{3, 4, 5\}, \{2, 4, 5\}, \mathcal{D}=\{\{1, 6\}, \{1, 2, 3\}, \{1, 4, 5\}$	and the four strict 2-colorings in order of vertices 1, 2, 3, 4, 5, 6 are 112122, 112212, 212121, 212211, the unique strict 4-coloring is 122334, hence the chromatic spectrum
voloshin-hypergraph_F02156	264	■ 0.906	$X, \emptyset, \emptyset$	$X, \emptyset, \emptyset$	
voloshin-hypergraph_F02157	264	■ 1.000	$\{1, \ldots, n\}$	$\{1, \ldots, n\}$	
voloshin-hypergraph_F02158	265	■ 0.990	$X^{\prime}=\bigcup_{v \in X}\left\{v^{-}, v^{+}\right\}$	$X^{\prime}=\bigcup_{v \in X}\left\{v^{-}, v^{+}\right\}$	element 2 to the feasible set. This is what Construction 1 did to K_t , and the analysis here
voloshin-hypergraph_F02159	265	■ 0.957	$D^{\prime}=\bigcup_{v \in D}\left\{v^{-}, v^{+}\right\}$	$D^{\prime}=\bigcup_{v \in D}\left\{v^{-}, v^{+}\right\}$	generalizes Lemma 10.9.1.
voloshin-hypergraph_F02160	265	■ 0.957	$C^{\prime}=\left\{v^{-}: v \in C\right\}$	$C^{\prime}=\left\{v^{-}: v \in C\right\}$	generalizes Lemma 10.9.1.

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voloshin-hypergraph_ F02161	265	■ 0.999	$\left\{v^-, v^+, u^-\right\}$	$\{v^-, v^+, u^-\}$	Lemma 10.9.3 Let $\mathcal{H}$ be a mixed hypergraph with feasible set S . If $\chi(\mathcal{H}) \geq 2$, then the
voloshin-hypergraph_ F02162	265	■ 0.999	$\left\{v^-, v^+, u^+\right\}$	$\{v^-, v^+, u^+\}$	Lemma 10.9.3 Let $\mathcal{H}$ be a mixed hypergraph with feasible set S . If $\chi(\mathcal{H}) \geq 2$, then the
voloshin-hypergraph_ F02163	265	■ 1.000	$\chi(\mathcal{H}) \geq 2$	$\chi(\mathcal{H}) \geq 2$	setting $c(u^-) = c(v^-)$ and $c(u^+) = c(v^+)$ for all u . Since each member of $\mathcal{D}'$ consists of
voloshin-hypergraph_ F02164	265	■ 1.000	$S \cup \{2\}$	$S \cup \{2\}$	full pairs, the constraints on $\mathcal{D}$ -edges are satisfied. Also, each member of $\mathcal{C}'$ contains two
voloshin-hypergraph_ F02165	265	■ 0.986	$c(v^-) \neq c(v^+)$	$c(v^-) \neq c(v^+)$	It remains to consider proper colorings with $c(v^-) = c(v^+)$ for each vertex v of $\mathcal{H}$. Let
voloshin-hypergraph_ F02166	265	■ 1.000	v^-, v^+	v^-, v^+	$\tilde{c}$ be the coloring of $\mathcal{H}$ defined by $\tilde{c}(v) = c(v^-)$. For each member of $\mathcal{D}'$, the coloring
voloshin-hypergraph_ F02167	265	■ 1.000	$c(v^-)$	$c(v^-)$	$\tilde{c}$ be the coloring of $\mathcal{H}$ defined by $\tilde{c}(v) = c(v^-)$. For each member of $\mathcal{D}'$, the coloring
voloshin-hypergraph_ F02168	265	■ 1.000	$c(v^+)$	$c(v^+)$	constraint is satisfied by $\tilde{c}$ if and only if $\tilde{c}$ satisfies the constraint for the corresponding
voloshin-hypergraph_ F02169	265	■ 1.000	$c(u^-) = c(v^-)$	$c(u^-) = c(v^-)$	member of $\mathcal{D}$. The same statement holds for members of $\mathcal{C}'$ that arise from members of
voloshin-hypergraph_ F02170	265	■ 1.000	$c(u^+) = c(v^+)$	$c(u^+) = c(v^+)$	$\mathcal{C}$. By construction, the new triples in $\mathcal{C}'$ are automatically satisfied. Thus, c is a proper
voloshin-hypergraph_ F02171	265	■ 1.000	$c(v^-) = c(v^+)$	$c(v^-) = c(v^+)$	$\mathcal{C}$. By construction, the new triples in $\mathcal{C}'$ are automatically satisfied. Thus, c is a proper
voloshin-hypergraph_ F02172	265	■ 1.000	$\tilde{c}$	$\tilde{c}$	
voloshin-hypergraph_ F02173	265	■ 1.000	$\tilde{c}(v) = c(v^-)$	$\tilde{c}(v) = c(v^-)$	
voloshin-hypergraph_ F02174	266	■ 0.813	$R(\mathcal{H}')$	$R(\mathcal{H}')$	Using shiftings (joins with cliques) and doublings, one can produce all feasible sets.
voloshin-hypergraph_ F02175	266	■ 1.000	$n(\mathcal{H}^{\prime})=2 n(\mathcal{H})$	$n(\mathcal{H}') = 2n(\mathcal{H})$	
voloshin-hypergraph_ F02176	266	■ 1.000	$\mathcal{H}(T)$	$\mathcal{H}(T)$	For $ T > 1$ and $t > 2$, we let $\mathcal{H}(T)$ be the join of K_{t-2} with the mixed hypergraph
voloshin-hypergraph_ F02177	266	■ 1.000	$T=\{t\}$	$T = \{t\}$	$\mathcal{H}(T')$, where T' is obtained from T by subtracting $t - 2$ from each element. Lemma 10.9.2
voloshin-hypergraph_ F02178	266	■ 1.000	$\mathcal{H}(T)=K_t$	$\mathcal{H}(T) = K_t$	$\mathcal{H}(T')$, where T' is obtained from T by subtracting $t - 2$ from each element. Lemma 10.9.2
voloshin-hypergraph_ F02179	266	■ 1.000	$ T >1$	$ T > 1$	implies that this works. □
voloshin-hypergraph_ F02180	266	■ 1.000	$t=2$	$t = 2$	implies that this works. □
voloshin-hypergraph_ F02181	266	■ 1.000	$\mathcal{H}(T-\{2\})$	$\mathcal{H}(T - \{2\})$	Next we present a conceptual algorithm which, for an arbitrary increasii
voloshin-hypergraph_ F02182	266	■ 1.000	$t>2$	$t > 2$	integer numbers, constructs a mixed hypergraph having the respective feasible set.
voloshin-hypergraph_ F02183	266	■ 1.000	K_{t-2}	K_{t-2}	integer numbers, constructs a mixed hypergraph having the respective feasible set.
voloshin-hypergraph_ F02184	266	■ 0.999	$\mathcal{H}(\mathcal{T}^{\prime})$	$\mathcal{H}(T')$	Aloorithm 10.9.1 (<i>doublingo-shiftingo aloorithm</i>)
voloshin-hypergraph_ F02185	266	■ 0.999	$t-2$	$t - 2$	Aloorithm 10.9.1 (<i>doublingo-shiftingo aloorithm</i>)
voloshin-hypergraph_ F02186	266	■ 0.861	$S=\left\{n_1, n_2, \ldots, n_p\right\}$	$S = \{n_1, n_2, \dots, n_p\}$	2. If $i \notin S$, do elementary shifting.
voloshin-hypergraph_ F02187	266	■ 0.861	$n_1 \geq 2$	$n_1 \geq 2$	2. If $i \notin S$, do elementary shifting.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02188	266	■ 1.000	$i=n_{\{p\}}-1 ; \mathcal{H}=K_{\{3\}}$	$i = n_p - 1; \mathcal{H} = K_3$	
voloshin-hypergraph_F02189	266	■ 0.982	$i \not= 1$	$i \neq 1$	End.
voloshin-hypergraph_F02190	266	■ 0.999	$i \in S$	$i \in S$	
voloshin-hypergraph_F02191	266	■ 0.999	$i \not\in S$	$i \notin S$	
voloshin-hypergraph_F02192	266	■ 1.000	$i=i-1$	$i = i - 1$	
voloshin-hypergraph_F02193	267	■ 1.000	$\mathcal{H}=(X, \mathcal{E})$	$\mathcal{H} = (X, \mathcal{E})$	Hence $r_5 = 0$. If $r_4 \neq 0$, then in a strict 4-coloring of $\mathcal{H}$, there are at least two singleton
voloshin-hypergraph_F02194	267	■ 1.000	$X=\{1,2, \ldots, 6\}$	$X = \{1, 2, \dots, 6\}$	Hence $r_5 = 0$. If $r_4 \neq 0$, then in a strict 4-coloring of $\mathcal{H}$, there are at least two singleton
voloshin-hypergraph_F02195	267	■ 0.640	$\mathcal{E}$	$\mathcal{E}$	color classes which implies $r_3 \neq 0$, a contradiction. Hence, there are no gaps in $R(\mathcal{H})$.
voloshin-hypergraph_F02196	267	■ 0.640	$R(\mathcal{H})=\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{6\}}\right)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_6)$	color classes which implies $r_3 \neq 0$, a contradiction. Hence, there are no gaps in $R(\mathcal{H})$.
voloshin-hypergraph_F02197	267	■ 0.998	$r_{\{1\}}=0, r_{\{6\}}=0$	$r_1 = 0, r_6 = 0$	chromatic spectrum. Let $\mathcal{H} = (X, \mathcal{E})$ be a 3-uniform bhypergraph defined as follows (see Figure 10.30). Hyperedges are drawn as classic hyperedges: $\mathcal{E} = \{1, 2, 3, 4, 5, 6\}$.
voloshin-hypergraph_F02198	267	■ 0.998	$r_{\{5\}} \not= 0$	$r_5 \neq 0$	chromatic spectrum. Let $\mathcal{H} = (X, \mathcal{E})$ be a 3-uniform bhypergraph defined as follows (see Figure 10.30). Hyperedges are drawn as classic hyperedges: $\mathcal{E} = \{1, 2, 3, 4, 5, 6\}$.
voloshin-hypergraph_F02199	267	■ 1.000	$r_{\{4\}} \not= 0, r_{\{3\}} \not= 0, r_{\{2\}} \not= 0$	$r_4 \neq 0, r_3 \neq 0, r_2 \neq 0$	proper coloring, there are two possibilities for vertices 1 and 2: either $c(1) \neq c(2)$ or $c(1) =$
voloshin-hypergraph_F02200	267	■ 1.000	$r_{\{5\}}=0$	$r_5 = 0$	$c(2)$. Let $\{A, B, C, D, \dots\}$ be the set of colors. If $c(1) = A, c(2) = B$, then all the remaining
voloshin-hypergraph_F02201	267	■ 1.000	$r_{\{4\}} \not= 0$	$r_4 \neq 0$	$c(2)$. Let $\{A, B, C, D, \dots\}$ be the set of colors. If $c(1) = A, c(2) = B$, then all the remaining
voloshin-hypergraph_F02202	267	■ 1.000	$r_{\{3\}} \not= 0$	$r_3 \neq 0$	
voloshin-hypergraph_F02203	267	■ 1.000	$X=\{1,2, \ldots, 7\}, \mathcal{E}=\{\{1,2,3\}$	$X = \{1, 2, \dots, 7\}, \mathcal{E} = \{\{1, 2, 3\}$	
voloshin-hypergraph_F02204	267	■ 0.831	$\{1,2,5\}, \{1,2,6\}, \{1,2,7\}, \{3,4,2\}, \{3,4,5\}, \{3,4,6\} \setminus, \{3,4,7\}, \{5,6,7\} \setminus$	$\{1, 2, 5\}, \{1, 2, 6\}, \{1, 2, 7\}, \{3, 4, 2\}, \{3, 4, 5\}, \{3, 4, 6\}, \{3, 4, 7\}, \{5, 6, 7\}$	
voloshin-hypergraph_F02205	267	■ 1.000	$c(1) \not= c(2)$	$c(1) \neq c(2)$	
voloshin-hypergraph_F02206	267	■ 1.000	$c(1)=$	$c(1) =$	
voloshin-hypergraph_F02207	267	■ 1.000	$c(2)$	$c(2)$	—
voloshin-hypergraph_F02208	267	■ 1.000	$\{A, B, C, D, \ldots\}$	$\{A, B, C, D, \dots\}$	—
voloshin-hypergraph_F02209	267	■ 1.000	$c(1)=A, c(2)=B$	$c(1) = A, c(2) = B$	—
voloshin-hypergraph_F02210	268	■ 0.995	$\{A, B\}$	$\{A, B\}$	vertices are colored with the colors from $\{A, B\}$; i.e., all the colorings are 2-colorings. If
voloshin-hypergraph_F02211	268	■ 1.000	$c(1)=c(2)=A$	$c(1) = c(2) = A$	vertices are colored with the colors from $\{A, B\}$, i.e., all the colorings are 2-colorings. If $c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$; i.e., we
voloshin-hypergraph_F02212	268	■ 1.000	$c(3)=c(4)=B$	$c(3) = c(4) = B$	vertices are colored with the colors from $\{A, B\}$, i.e., all the colorings are 2-colorings. If $c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$; i.e., we
voloshin-hypergraph_F02213	268	■ 1.000	$c(5), c(6), c(7) \in \{C, D\}$	$c(5), c(6), c(7) \in \{C, D\}$	vertices are colored with the colors from $\{A, B\}$, i.e., all the colorings are 2-colorings. If $c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$; i.e., we
voloshin-hypergraph_F02214	268	■ 0.999	$\left.R(\mathcal{H})=(0,12,0,3,0,0,0), P(\mathcal{H}), \lambda=3 \lambda(\lambda-1) \left(\lambda^2-5 \lambda+10\right)\right)$	$R(\mathcal{H}) = (0, 12, 0, 3, 0, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1) (\lambda^2 - 5\lambda + 10))$	analysis shows that $R(\mathcal{H}) = (0, 12, 0, 3, 0, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)(\lambda^2 - 5\lambda + 10))$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02215	268	■ 0.999	$X=\{1,2, \ldots, 7\}$	$X = \{1, 2, \dots, 7\}$	$\mathcal{H} = (\mathcal{X}, \mathcal{E})$ be a 3-uniform hypergraph with $\mathcal{X} = \{1, 2, \dots, 7\}$ and $ \mathcal{E} \geq 6$. For a contradiction, suppose that $R(\mathcal{H}) = (r_1, r_2, \dots, r_7)$ has at least one gap. We immediately have
voloshin-hypergraph_F02216	268	■ 0.999	$ \mathcal{H} \leq 8$	$ \mathcal{E} \leq 8$	$\mathcal{H} = (\mathcal{X}, \mathcal{E})$ be a 3-uniform hypergraph with $\mathcal{X} = \{1, 2, \dots, 7\}$ and $ \mathcal{E} \geq 6$. For a contradiction, suppose that $R(\mathcal{H}) = (r_1, r_2, \dots, r_7)$ has at least one gap. We immediately have
voloshin-hypergraph_F02217	268	■ 1.000	$R(\mathcal{H})=\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{7\}}\right)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_7)$	$r_1 = r_7 = 0$. If $r_6 \neq 0$ (respectively $r_6 = 0, r_5 \neq 0$), then at least 3 color classes are singletons
voloshin-hypergraph_F02218	268	■ 0.995	$r_{\{1\}}=r_{\{7\}}=0$	$r_1 = r_7 = 0$	which can easily be combined to obtain a strict coloring with 3, 4 and 5 (respectively 3 and
voloshin-hypergraph_F02219	268	■ 0.995	$r_{\{6\}} \neq 0$	$r_6 \neq 0$	which can easily be combined to obtain a strict coloring with 3, 4 and 5 (respectively 3 and
voloshin-hypergraph_F02220	268	■ 0.995	$r_{\{6\}}=0, r_{\{5\}} \neq 0$	$r_6 = 0, r_5 \neq 0$	which can easily be combined to obtain a strict coloring with 3, 4 and 5 (respectively 3 and
voloshin-hypergraph_F02221	268	■ 0.960	$r_{\{7\}}=r_{\{6\}}=r_{\{5\}}=0, r_{\{4\}} \neq 0$	$r_7 = r_6 = r_5 = 0, r_4 \neq 0$	singleton color classes $\{1, 2\}$ and $\{3, 7\}$ is a contradiction. Hence, the only type of a coloring to examine is
voloshin-hypergraph_F02222	268	■ 0.531	$\{1,2,7\} \notin \mathcal{H}$	$\{1, 2, 7\} \notin \mathcal{E}$	a contradiction. So, we have that in any case
voloshin-hypergraph_F02223	268	■ 0.531	$\{3,4,7\} \notin \mathcal{H}, \{5,6,7\} \notin \mathcal{H}$	$\{3, 4, 7\} \notin \mathcal{E}, \{5, 6, 7\} \notin \mathcal{E}$	a contradiction. So, we have that in any case
voloshin-hypergraph_F02224	268	■ 0.998	(A, A, B, B, C, C, A)	(A, A, B, B, C, C, A)	$\{1, 2, 7\}, \{3, 4, 7\}, \{5, 6, 7\} \in \mathcal{E}$.
voloshin-hypergraph_F02225	268	■ 0.998	$(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$	$(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$	$\{1, 2, 7\}, \{3, 4, 7\}, \{5, 6, 7\} \in \mathcal{E}$.
voloshin-hypergraph_F02226	268	■ 1.000	$\{1,2,3\} \notin \mathcal{H}, \{1,2,4\} \notin \mathcal{H}, \{1,2,5\} \notin \mathcal{H}$	$\{1, 2, 3\} \notin \mathcal{E}, \{1, 2, 4\} \notin \mathcal{E}, \{1, 2, 5\} \notin \mathcal{E}$	ists: (B, A, C, C, B, B, A) (respectively (A, B, C, C, B, B, A)), a contradiction. Therefore $\{5, 6, 1\}, \{5, 6, 2\} \in \mathcal{E}$. If $\{1, 2, 6\} \in \mathcal{E}$, then all 8 bi-edges of $\mathcal{H}$ are exhausted and the 3-coloring
voloshin-hypergraph_F02227	268	■ 1.000	$\{3,4,1\}, \{3,4,2\} \notin \mathcal{H}$	$\{3, 4, 1\}, \{3, 4, 2\} \notin \mathcal{E}$	$\{3, 6, 2\} \in \mathcal{E}$. If $\{1, 2, 6\} \in \mathcal{E}$ then all 8 bi-edges of $\mathcal{H}$ are exhausted and the 3-coloring (A, A, B, B, C, C, C) completes the proof.
voloshin-hypergraph_F02228	268	■ 0.995	$\{3,4,1\} \in \mathcal{H}$	$\{3, 4, 1\} \in \mathcal{E}$	(A, A, B, B, C, C, C) completes the proof. Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02229	268	■ 0.995	$\{3,4,2\} \in \mathcal{H}$	$\{3, 4, 2\} \in \mathcal{E}$	(A, A, B, B, C, C, C) completes the proof. Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02230	268	■ 0.995	$\{3,4,5\}, \{3,4,6\} \notin \mathcal{H}$	$\{3, 4, 5\}, \{3, 4, 6\} \notin \mathcal{E}$	(A, A, B, B, C, C, C) completes the proof. Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02231	268	■ 0.745	$\{3,4,5\} \in \mathcal{H}$	$\{3, 4, 5\} \in \mathcal{E}$	ing 3-coloring exists: (A, B, A, A, C, C, B) (respectively (B, A, A, A, C, C, B)), a contradiction.
voloshin-hypergraph_F02232	268	■ 0.996	$\{3,4,6\} \in \mathcal{H}$	$\{3, 4, 6\} \in \mathcal{E}$	Therefore both $\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$. Hence all 8 bi-edges of $\mathcal{H}$ are exhausted, and the
voloshin-hypergraph_F02233	268	■ 0.996	$\{5,6,1\} \notin \mathcal{H}$	$\{5, 6, 1\} \notin \mathcal{E}$	Therefore both $\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$. Hence all 8 bi-edges of $\mathcal{H}$ are exhausted, and the
voloshin-hypergraph_F02234	268	■ 0.996	$\{5,6,2\} \notin \mathcal{H}$	$\{5, 6, 2\} \notin \mathcal{E}$	Therefore both $\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$. Hence all 8 bi-edges of $\mathcal{H}$ are exhausted, and the
voloshin-hypergraph_F02235	268	■ 0.999	$\{5,6,2\} \in \mathcal{H}$	$\{5, 6, 2\} \in \mathcal{E}$	We conclude the section with the following important results:
voloshin-hypergraph_F02236	268	■ 0.999	$\{1,2,6\} \in \mathcal{D}$	$\{1, 2, 6\} \in \mathcal{D}$	We conclude the section with the following important results:
voloshin-hypergraph_F02237	268	■ 1.000	$\{1,2,6\} \notin \mathcal{H}$	$\{1, 2, 6\} \notin \mathcal{E}$	THEOREM 10.24 (Khalil et al., 2009) If $\mathcal{H}$ is a mixed hypertree, then the chromatic spectrum is continuous.
voloshin-hypergraph_F02238	268	■ 1.000	$\{3,4,1\} \notin \mathcal{H}$	$\{3, 4, 1\} \notin \mathcal{E}$	THEOREM 10.24 (Khalil et al., 2009) If $\mathcal{H}$ is a mixed hypertree, then the chromatic spectrum is continuous.
voloshin-hypergraph_F02239	268	■ 1.000	$\{3,4,2\} \notin \mathcal{H}$	$\{3, 4, 2\} \notin \mathcal{E}$	THEOREM 10.24 (Khalil et al., 2009) If $\mathcal{H}$ is a mixed hypertree, then the chromatic spectrum is continuous.
voloshin-hypergraph_F02240	268	■ 0.999	$\{3,4,1\}, \{3,4,2\} \in \mathcal{H}$	$\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$	
voloshin-hypergraph_F02241	269	■ 0.523	$S\left(\mathcal{H}_{\mathcal{C}}\right)$	$S\left(\mathcal{H}_{\mathcal{C}}\right)$	Computer Projects 10.9. Write a program for the following algorithmic problems.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02242	269	■ 0.523	$S(\text{\texttt{\textbackslash left(\text{\texttt{\textbackslash mathcal\{H\}_\{\text{\texttt{\textbackslash mathcal\{D\}}\}\right)}$	$S(\mathcal{H}_{\mathcal{D}})$	Computer Projects 10.9. Write a program for the following algorithmic problems.
voloshin-hypergraph_F02243	269	■ 1.000	$S=\{2,3,5\}$	$S = \{2, 3, 5\}$	2. Implement Algorithm 10.2.1.
voloshin-hypergraph_F02244	269	■ 1.000	$S(\text{\texttt{\textbackslash mathcal\{H\}}})=5$	$S(\mathcal{H}) = S$	
voloshin-hypergraph_F02245	270	■ 1.000	$B(\text{\texttt{\textbackslash mathcal\{H\}}^{\text{\texttt{\textbackslash prime}}}\text{\texttt{\textbackslash right}})$	$B(\mathcal{H}')$	from Section 9.4., a hypergraph $\mathcal{H}'$ is planar if and only if bipartite representation $B(\mathcal{H}')$ is a planar graph.
voloshin-hypergraph_F02246	270	■ 0.991	B, C	B, C	contraction Algorithm 10.5.1. and bi-edges of size 2 lead to uncolorability: so, in general,
voloshin-hypergraph_F02247	270	■ 0.981	$\text{\texttt{\textbackslash mathcal\{H\}}}=(X, \text{\texttt{\textbackslash mathcal\{C\}}}, \text{\texttt{\textbackslash mathcal\{D\}}})=\text{\texttt{\textbackslash left(X, \text{\texttt{\textbackslash binom\{X\}\{3\}}, \text{\texttt{\textbackslash binom\{X\}\{2\}}\right\text{\texttt{\textbackslash right}})}$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{3}, \binom{X}{2}\right)$	The general structure of uncolorable planar mixed hypergraphs is unknown. However, if a planar mixed hypergraph is colorable, then naturally the problem of determining the
voloshin-hypergraph_F02248	270	■ 1.000	$X=\{1,2,3\}, \text{\texttt{\textbackslash mathcal\{C\}}}=\{C\}=\{\{1,2,3\}\}, \text{\texttt{\textbackslash mathcal\{D\}}}=\text{\texttt{\textbackslash left\{D_{1}, D_{2}, D_{3}\}\right\}}=$	$X = \{1, 2, 3\}, \mathcal{C} = \{C\} = \{\{1, 2, 3\}\}, \mathcal{D} = \{D_1, D_2, D_3\} =$	lower and upper chromatic number arises. Finding the lower chromatic number is difficult
voloshin-hypergraph_F02249	270	■ 0.991	$\{\{1,2\}, \{2,3\}, \{1,3\}\}$	$\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$	since if we allow $\mathcal{D}$ -edges of cardinality 2, it contains the four color problem as a special
voloshin-hypergraph_F02250	270	■ 0.991	$f_{\{1\}}, f_{\{2\}}, f_{\{3\}}, f_{\{4\}}$	f_1, f_2, f_3, f_4	since if we allow $\mathcal{D}$ -edges of cardinality 2, it contains the four color problem as a special
voloshin-hypergraph_F02251	270	■ 1.000	$G(\text{\texttt{\textbackslash mathcal\{H\}}})$	$G(\mathcal{H})$	
voloshin-hypergraph_F02252	272	■ 1.000	$S(n, k)$	$S(n, k)$	max coloring and there are exactly $j_{k-1}(G)$ maximal k -colorings of G .
voloshin-hypergraph_F02253	272	■ 1.000	$f_k(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})})$	$f_k(G^*)$	Proof. By the Jordan Curve Theorem 4.1.1, a given 2-factor consisting of $k-1$ disjoint
voloshin-hypergraph_F02254	272	■ 1.000	$f(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})})=\sum_{i \geq 1} f_{\{i\}}(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})})$	$f(G^*) = \sum_{i \geq 1} f_i(G^*)$	different color to each face must be the unique maximal coloring for this 2-factor,
voloshin-hypergraph_F02255	272	■ 1.000	$f_{\{k-1\}}(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})})$	$f_{k-1}(G^*)$	
voloshin-hypergraph_F02256	273	■ 0.995	$f(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})})$	$f(G^*)$	Corollary 10.10.2 Every bitriangulation G has exactly $f(G^*)$ strict 2-colorings. In gen-
voloshin-hypergraph_F02257	273	■ 1.000	$k=2$	$k = 2$	2-factor consisting of i cycles can be k -face-colored in exactly $S(i, k-1)$ ways. To see this,
voloshin-hypergraph_F02258	273	■ 1.000	$\lambda=2$	$\lambda = 2$	create a graph whose vertices are the faces in the dual of the 2-factor, and two vertices
voloshin-hypergraph_F02259	273	■ 0.994	$S(i, k-1)$	$S(i, k-1)$	are adjacent if and only if the corresponding faces are separated by a 2-factor. This graph
voloshin-hypergraph_F02260	273	■ 1.000	$r_{\{k\}}(T)$	$r_k(T)$	is connected and has i edges. By the Jordan curve theorem, it has exactly $i+1$ vertices.
voloshin-hypergraph_F02261	273	■ 1.000	$r_{\{k\}}(T)=S(e(T), k-1)$	$r_k(T) = S(e(T), k-1)$	usual recursion for the Stirling numbers, as shown by Theorem 5.4.5 and Corollary 5.4.5. For the second formula, recall that the chromatic polynomial for a tree on $i+1$ vertices is
voloshin-hypergraph_F02262	273	■ 1.000	$r_{\{1\}}(\text{\texttt{\textbackslash left(K_{\{1\}}\text{\texttt{\textbackslash right}})})=1$	$r_1(K_1) = 1$	For the second formula, recall that the chromatic polynomial for a tree on $i+1$ vertices is $\lambda(\lambda-1)^i$ (Theorem 5.5.1). $\square$
voloshin-hypergraph_F02263	273	■ 1.000	$r_{\{k\}}(\text{\texttt{\textbackslash left(K_{\{1\}}\text{\texttt{\textbackslash right}})})=0$	$r_k(K_1) = 0$	For the second formula, recall that the chromatic polynomial for a tree on $i+1$ vertices is $\lambda(\lambda-1)^i$ (Theorem 5.5.1). $\square$
voloshin-hypergraph_F02264	273	■ 1.000	$r_{\{k\}}(T)=(k-1) r_{\{k\}}(T-x)+r_{\{k-1\}}(T-x)$	$r_k(T) = (k-1)r_k(T-x) + r_{k-1}(T-x)$	
voloshin-hypergraph_F02265	273	■ 1.000	$\lambda(\lambda-1)^i$	$\lambda(\lambda-1)^i$	Proof. Since G^* is a
voloshin-hypergraph_F02266	273	■ 1.000	$\bar{\chi}(G)=1+\max \left\{k: f_{\{k\}}(\text{\texttt{\textbackslash left(G^{\text{\texttt{\textbackslash *}}}\text{\texttt{\textbackslash right}})} \geq 1\right\}$	$\bar{\chi}(G) = 1 + \max \{k : f_k(G^*) \geq 1\}$	the upper chromatic number must be maximal,
voloshin-hypergraph_F02267	273	■ 1.000	$\bar{\chi}(G)$	$\bar{\chi}(G)$	Proof. We may assume that the mixed hypergraph is a maximal bitriangulation. Since
voloshin-hypergraph_F02268	273	■ 1.000	$k=\bar{\chi}(G)$	$k = \bar{\chi}(G)$	Proof. We may assume that the mixed hypergraph is a maximal bitriangulation, since adding $\mathcal{C}$ - or $\mathcal{D}$ -edges only decreases the number of 2-colorings. Similarly, if G contains

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voloshin-hypergraph_F02269	273	■ 1.000	$f_{\{k-1\}}(\left(G^{*}\right)) \geq 1$	$f_{k-1}(G^{*}) \geq 1$	PROOF. We may assume that the mixed hypergraph is a maximal binypergraph, since adding $\mathcal{C}$ - or $\mathcal{D}$ -edges only decreases the number of 2-colorings. Similarly, if G contains
voloshin-hypergraph_F02270	273	■ 1.000	$S(k-1, i-1) \geq 1$	$S(k-1, i-1) \geq 1$	PROOF. We may assume that the mixed hypergraph is a maximal binypergraph, since adding $\mathcal{C}$ - or $\mathcal{D}$ -edges only decreases the number of 2-colorings. Similarly, if G contains
voloshin-hypergraph_F02271	273	■ 1.000	$2 \leq i \leq k$	$2 \leq i \leq k$	any faces of size larger than 3, then they can be divided into faces of size 3 by adding graph
voloshin-hypergraph_F02272	273	■ 1.000	$r_{\{i\}}(G) \geq 1$	$r_i(G) \geq 1$	any faces of size larger than 3, then they can be divided into faces of size 3 by adding graph
voloshin-hypergraph_F02273	274	■ 1.000	$\bar{\chi}(G)=2$	$\bar{\chi}(G) = 2$	PROOF. Suppose that G is uniquely colorable and free of edges of size 2. Again, we may assume that G is a bitriangulation. By Corollary 10.10.3, $\mathcal{H}(G) = \mathcal{D}$; then, by Corol-
voloshin-hypergraph_F02274	274	■ 1.000	$u \ v$	uv	the graph whose vertices are the Hamiltonian paths starting at u with edge uv . Two such
voloshin-hypergraph_F02275	274	■ 1.000	$a, \ b, \ c$	a, b, c	In the second 2-factor, faces 1 and 3 are inside the cycle and colored with color A; faces 2 and 4 are outside the cycle and colored with color B. At least, in the third 2-factor, faces 1 and 2 are inside the cycle and colored with color A; faces 3 and 4 are outside the cycle and
voloshin-hypergraph_F02276	274	■ 0.998	$K_4^{\ast}$	K_4^*	and 4 are outside the cycle and colored with color B. At last, in the third 2-factor, faces 1 and 2 are inside the cycle and colored with color A; faces 3 and 4 are outside the cycle and
voloshin-hypergraph_F02277	274	■ 0.998	$A \ B \ B \ A, \ A \ B \ A \ B$	$ABBA, ABAB$	
voloshin-hypergraph_F02278	274	■ 1.000	$A \ A \ B \ B$	$AABB$	
voloshin-hypergraph_F02279	274	■ 1.000	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=2$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$	
voloshin-hypergraph_F02280	274	■ 1.000	$R(\mathcal{H})=(0,3,0,0)$	$R(\mathcal{H}) = (0, 3, 0, 0)$	
voloshin-hypergraph_F02281	274	■ 1.000	$P(\mathcal{H}, \lambda)=3 \lambda(\lambda-1)$	$P(\mathcal{H}, \lambda) = 3\lambda(\lambda-1)$	—
voloshin-hypergraph_F02282	275	■ 0.978	$\mathcal{H}_{2,4}^{\prime}=(X, \mathcal{H}_{\mathcal{C}}, \mathcal{H}_{\mathcal{D}})$	$\mathcal{H}_{2,4}' = (X, \mathcal{C}, \mathcal{D})$	
voloshin-hypergraph_F02283	275	■ 0.991	$X=\{1,2, \ldots, 6\}, \mathcal{H}_{\mathcal{C}}=\left\{C_{\{1\}}, \ldots, C_{\{4\}}\right\}=\{\{1,2,3\},\{2,3,4\},\{2,4,5\},\{4,5,6\}\}$	$X = \{1, 2, \dots, 6\}, \mathcal{C} = \{C_1, \dots, C_4\} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{2, 4, 5\}, \{4, 5, 6\}\}$	
voloshin-hypergraph_F02284	275	■ 0.990	$\mathcal{H}_{\mathcal{D}}=\left\{D_{\{1\}}, \ldots, D_{\{6\}}\right\}=\{\{1,2\},\{1,5\},\{1,6\},\{2,4\},\{3,6\},\{4,6\}\}$	$\mathcal{D} = \{D_1, \dots, D_6\} = \{\{1, 2\}, \{1, 5\}, \{1, 6\}, \{2, 4\}, \{3, 6\}, \{4, 6\}\}$	
voloshin-hypergraph_F02285	275	■ 0.990	$\mathcal{H}_{2,4}^{\prime}$	$\mathcal{H}_{2,4}'$	
voloshin-hypergraph_F02286	276	■ 1.000	$S\left(\mathcal{H}_{2,4}^{\prime}\right)=\{2,4\}$	$S\left(\mathcal{H}_{2,4}'\right) = \{2, 4\}$	<i>feasible set</i> $S(\mathcal{H}') = \{2\}$
voloshin-hypergraph_F02287	276	■ 1.000	$c(2) \neq c(3)$	$c(2) \neq c(3)$	indicated. Let c be a strict coloring of $\mathcal{H}_{2,4}'$. If $c(2) \neq c(3)$, then the remaining colors are forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and
voloshin-hypergraph_F02288	276	■ 1.000	$\{1,3,4\} \cup\{2,5,6\}$	$\{1, 3, 4\} \cup \{2, 5, 6\}$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and $c(2) \neq c(4)$, which results in the strict 2-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see
voloshin-hypergraph_F02289	276	■ 1.000	$c(2)=c(3)$	$c(2) = c(3)$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and $c(2) \neq c(4)$, which results in the strict 2-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see
voloshin-hypergraph_F02290	276	■ 1.000	$c(4)=c(5)$	$c(4) = c(5)$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and $c(2) \neq c(4)$, which results in the strict 2-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see
voloshin-hypergraph_F02291	276	■ 1.000	$c(2) \neq c(4)$	$c(2) \neq c(4)$	$c(2) \neq c(4)$, which results in the strict 2-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see that these feasible nartitions are unique. $\square$
voloshin-hypergraph_F02292	276	■ 1.000	$\{1\} \cup\{2,3\} \cup\{4,5\} \cup\{6\}$	$\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$	$c(2) \neq c(4)$, which results in the strict 2-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see that these feasible nartitions are unique. $\square$
voloshin-hypergraph_F02293	276	■ 1.000	$\{s, s+1, \ldots, t\}$	$\{s, s+1, \dots, t\}$	
voloshin-hypergraph_F02294	276	■ 0.995	$1 \leq s \leq 4$	$1 \leq s \leq 4$	
voloshin-hypergraph_F02295	276	■ 0.995	$\{2,4,5, \ldots, t\}$	$\{2, 4, 5, \dots, t\}$	

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voloshin-hypergraph_F02296	276	■ 0.999	$S(G)=\{\chi(G)$	$S(G) = \{\chi(G)$	
voloshin-hypergraph_F02297	276	■ 1.000	$\chi(G)+1, \ldots, t\}$	$\chi(G) + 1, \dots, t\}$	—
voloshin-hypergraph_F02298	277	■ 1.000	$\{7, \ldots, t+2\}$	$\{7, \dots, t + 2\}$	an interval. When S has a gap at 3, consider the mixed hypergraph obtained from $\mathcal{H}_{2,4}$ by taking the vertices $\{7, \dots, t+2\}$ and placing them in the region containing vertices 2 and 3.
voloshin-hypergraph_F02299	277	■ 1.000	$\{\{2,3,7\}, \{2,3,8\}, \ldots, \{2,3, t+2\}\}$	$\{\{2,3,7\}, \{2,3,8\}, \dots, \{2,3, t + 2\}\}$	Then include the C -edges $\{\{2,3,7\}, \{2,3,8\}, \dots, \{2,3, t + 2\}\}$. If we have a 2-coloring on
voloshin-hypergraph_F02300	277	■ 1.000	$S \neq \emptyset$	$S \neq \emptyset$	graph $\mathcal{H}$, with feasible set $S \neq \emptyset$. If $t \in S$, then S trivially forms an interval. Let c be a strict t -coloring of $\mathcal{H}$, where t is the largest value in S . We will construct a planar mixed
voloshin-hypergraph_F02301	277	■ 1.000	$1 \in S$	$1 \in S$	graph $\mathcal{H}$, with feasible set $S \neq \emptyset$. If $t \in S$, then S trivially forms an interval. Let c be a strict t -coloring of $\mathcal{H}$, where t is the largest value in S . We will construct a planar mixed
voloshin-hypergraph_F02302	277	■ 0.999	$\{4,5, \ldots, t\} \subset S \left(\mathcal{H}' \right) \subset S$	$\{4,5, \dots, t\} \subset S(\mathcal{H}') \subset S$	the required form.
voloshin-hypergraph_F02303	277	■ 1.000	u, v	u, v	contains vertices u, v with $c(u) \neq c(v)$. We replace the region by a $\mathcal{D}$ -edge $\{u, v\}$. If the
voloshin-hypergraph_F02304	277	■ 1.000	$c(u)=c(v)$	$c(u) = c(v)$	region is a bi-edge, then it contains vertices u, v, w with $c(u) = c(v) \neq c(w)$. We replace
voloshin-hypergraph_F02305	277	■ 1.000	$\{u, v\}$	$\{u, v\}$	region is a bi-edge, then it contains vertices u, v, w with $c(u) = c(v) \neq c(w)$. We replace
voloshin-hypergraph_F02306	277	■ 1.000	$c(u) \neq c(v)$	$c(u) \neq c(v)$	the region by a C -edge $\{u, v\}$ and a $\mathcal{D}$ -edge $\{v, w\}$. The mixed hypergraph $\mathcal{H}'$ we obtain is planar and still has c as a strict t -coloring. Furthermore, in obtaining $\mathcal{H}'$ from $\mathcal{H}$ no
voloshin-hypergraph_F02307	277	■ 1.000	u, v, w	u, v, w	is planar and still has c as a strict t -coloring. Furthermore, in obtaining $\mathcal{H}'$ from $\mathcal{H}$ no coloring constraints are lost so that every coloring of $\mathcal{H}'$ is a coloring of $\mathcal{H}$.
voloshin-hypergraph_F02308	277	■ 1.000	$c(u)=c(v) \neq c(w)$	$c(u) = c(v) \neq c(w)$	is planar and still has c as a strict t -coloring. Furthermore, in obtaining $\mathcal{H}'$ from $\mathcal{H}$ no coloring constraints are lost so that every coloring of $\mathcal{H}'$ is a coloring of $\mathcal{H}$.
voloshin-hypergraph_F02309	277	■ 1.000	$\{v, w\}$	$\{v, w\}$	coloring constraints are lost so that every coloring of $\mathcal{H}'$ is a coloring of $\mathcal{H}$. We now have a planar mixed hypergraph with all edges (C - or $\mathcal{D}$ -) of size 2. Contract
voloshin-hypergraph_F02310	277	■ 1.000	$S(G)=$	$S(G) =$	<i>planar mixed hypergraphs may occur only at 3.</i>
voloshin-hypergraph_F02311	277	■ 0.990	$\{\chi(G), \chi(G)+1, \ldots, t\}$	$\{\chi(G), \chi(G) + 1, \dots, t\}$	Exercises 10.10.
voloshin-hypergraph_F02312	277	■ 0.990	$\{4,5, \ldots, t\} \subset S \left(\mathcal{H}' \right) \subset S$	$\{4,5, \dots, t\} \subset S(\mathcal{H}') \subset S$	Exercises 10.10.
voloshin-hypergraph_F02313	279	■ 1.000	$Z_{\{1\}}, Z_{\{2\}}$	Z_1, Z_2	Z_3 do not interfere.
voloshin-hypergraph_F02314	279	■ 1.000	$Z_{\{3\}}$	Z_3	Z_3 do not interfere.
voloshin-hypergraph_F02315	279	■ 1.000	$Z_{\{1\}}$	Z_1	In how many ways and in which ways can we assign frequencies to the zones? Construct
voloshin-hypergraph_F02316	279	■ 1.000	$Z_{\{2\}}$	Z_2	In how many ways and in which ways can we assign frequencies to the zones? Construct
voloshin-hypergraph_F02317	279	■ 0.793	$b, c\}$	$b, c\}$	modeled by the mixed hypergraph colorings without any loss. Construct a mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ in the following way, see Figure 11.1.
voloshin-hypergraph_F02318	279	■ 1.000	$\left\{ Z_{\{1\}}, Z_{\{2\}}, Z_{\{3\}}, a, b, c \right\}$	$\{Z_1, Z_2, Z_3, a, b, c\}$	or with a color of vertex c . The similar is true for all C -edges.
voloshin-hypergraph_F02319	279	■ 1.000	$\mathcal{C}=\left\{ \left\{ Z_{\{1\}}, a, c \right\}, \left\{ Z_{\{2\}}, a, b \right\}, \left\{ Z_{\{3\}}, b, c \right\} \right\}$	$\mathcal{C} = \{\{Z_1, a, c\}, \{Z_2, a, b\}, \{Z_3, b, c\}\}$	As a consequence, graph G admits at least one list coloring if and only if the mixed hypergraph $\mathcal{H}$ is colorable. Classify the games of colors (the values of frequencies and the number of colors) for the games of colors of graphs G and H .
voloshin-hypergraph_F02320	279	■ 1.000	$\left\{ Z_{\{1\}}, a, c \right\}$	$\{Z_1, a, c\}$	zones in the region), the lists of colors with assignments, and the total number of colors
voloshin-hypergraph_F02321	279	■ 0.998	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) =$	—
voloshin-hypergraph_F02322	280	■ 1.000	$R(\mathcal{H})=(0,0,3,0,0,0)$	$R(\mathcal{H}) = (0,0,3,0,0,0)$	THE RESOLVED PROBLEM

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voloshin-hypergraph_F02323	280	■ 1.000	$P(\mathcal{H}, \lambda) = 3\lambda(\lambda -$	$P(\mathcal{H}, \lambda) = 3\lambda(\lambda -$	
voloshin-hypergraph_F02324	280	■ 0.780	$1)(\lambda - 2)$	$1)(\lambda - 2)$	Consider the following example. Let us have $n = 4$ elementary jobs $X = \{x_1, x_2, x_3,$
voloshin-hypergraph_F02325	280	■ 0.780	$\left. Z_1\right\}, Z_2\right\}, Z_3\right\}$	(Z_1, Z_2, Z_3)	Consider the following example. Let us have $n = 4$ elementary jobs $X = \{x_1, x_2, x_3,$
voloshin-hypergraph_F02326	280	■ 1.000	$(a, b, c), (c, b, c), (c, a, c)$	$(a, b, c), (c, b, c), (c, a, c)$	$x_4\}$ which are to be
voloshin-hypergraph_F02327	280	■ 0.995	$n=4$	$n = 4$	$S_4 = \{y_2, y_4, y_5\}$. For any resource used by at least two jobs simultaneously, a penalty must be paid.
voloshin-hypergraph_F02328	280	■ 0.995	$X=\left. x_1\right\}, x_2\right\}, x_3\right\}$.	$X = \{x_1, x_2, x_3$	$S_4 = \{y_2, y_4, y_5\}$. For any resource used by at least two jobs simultaneously, a penalty must be paid.
voloshin-hypergraph_F02329	280	■ 0.987	$\left. x_4\right\}$	$x_4\}$	Management requirements are: jobs x_1, x_2, x_3 must be performed for at most two units
voloshin-hypergraph_F02330	280	■ 0.987	$m=5$	$m = 5$	Management requirements are: jobs x_1, x_2, x_3 must be performed for at most two units
voloshin-hypergraph_F02331	280	■ 0.987	$Y=\left. y_1\right\}, y_2\right\}, y_3\right\}$.	$Y = \{y_1, y_2, y_3$	Management requirements are: jobs x_1, x_2, x_3 must be performed for at most two units
voloshin-hypergraph_F02332	280	■ 1.000	$\left. y_4\right\}, y_5\right\}$	$y_4, y_5\}$	of time, and jobs x_1, x_2, x_4 must also be performed for a maximum two units of time without
voloshin-hypergraph_F02333	280	■ 1.000	$S\left. (x_i\right\)} \subseteq Y$	$S(x_i) \subseteq Y$	
voloshin-hypergraph_F02334	280	■ 1.000	$S_1=\left. y_1\right\}$	$S_1 = \{y_1\}$	
voloshin-hypergraph_F02335	280	■ 1.000	$S_2=\left. y_1\right\}, y_2\right\}, y_3\right\}$	$S_2 = \{y_1, y_2, y_3\}$	
voloshin-hypergraph_F02336	280	■ 1.000	$S_3=\left. y_3\right\}, y_4\right\}$	$S_3 = \{y_3, y_4\}$	
voloshin-hypergraph_F02337	280	■ 1.000	$S_4=\left. y_2\right\}, y_4\right\}, y_5\right\}$	$S_4 = \{y_2, y_4, y_5\}$	
voloshin-hypergraph_F02338	280	■ 1.000	$x_1\}, x_2\}, x_4\}$	x_1, x_2, x_4	—
voloshin-hypergraph_F02339	281	■ 1.000	$S_1\} \cap S_2\} \neq \emptyset$	$S_1 \cap S_2 \neq \emptyset$	$(\{y_1, y_2, y_4\}, \{y_1, y_2, y_5\}) = (\{y_1, y_2, y_3, y_4\}, \{y_1, y_2, y_3, y_4, y_5\})$. Thus, the mixed hypergraph $\mathcal{H}(X, \mathcal{C}, \mathcal{D})$ is obtained.
voloshin-hypergraph_F02340	281	■ 1.000	$D_1=$	$D_1 =$	$(\{y_1, y_2, y_4\}, \{y_1, y_2, y_5\}) = (\{y_1, y_2, y_3, y_4\}, \{y_1, y_2, y_3, y_4, y_5\})$. Thus, the mixed hypergraph $\mathcal{H}(X, \mathcal{C}, \mathcal{D})$ is obtained.
voloshin-hypergraph_F02341	281	■ 0.999	$S_2\} \cap S_3\} \neq \emptyset$	$S_2 \cap S_3 \neq \emptyset$	One can easily see that $\mathcal{H}$ is uncolorable. This means that no allocation of resources,
voloshin-hypergraph_F02342	281	■ 0.999	$D_2=\left. x_2\right\}, x_3\right\}$	$D_2 = \{x_2, x_3\}$	One can easily see that $\mathcal{H}$ is uncolorable. This means that no allocation of resources,
voloshin-hypergraph_F02343	281	■ 0.999	$S_2\} \cap S_4\} \neq \emptyset$	$S_2 \cap S_4 \neq \emptyset$	One can easily see that $\mathcal{H}$ is uncolorable. This means that no allocation of resources,
voloshin-hypergraph_F02344	281	■ 1.000	$D_3=\left. x_2\right\}, x_4\right\}$	$D_3 = \{x_2, x_4\}$	without any penalty is possible. However, if we violate, for example, the constraint ex-
voloshin-hypergraph_F02345	281	■ 1.000	$S_3\} \cap S_4\} \neq \emptyset$	$S_3 \cap S_4 \neq \emptyset$	without any penalty is possible. However, if we violate, for example, the constraint ex-
voloshin-hypergraph_F02346	281	■ 0.993	$D_4=\left. x_3\right\}, x_4\right\}$	$D_4 = \{x_3, x_4\}$	pressed by the $\mathcal{C}$ -edge C_1 , then we can color the vertices x_1, x_2, x_3, x_4 respectively with the colors 1, 2, 2, 1. Therefore, in this case, the minimum number of time units is 2, and the
voloshin-hypergraph_F02347	281	■ 0.993	$S_1\} \cap S_3\} = \emptyset$	$S_1 \cap S_3 = \emptyset$	pressed by the $\mathcal{C}$ -edge C_1 , then we can color the vertices x_1, x_2, x_3, x_4 respectively with the colors 1, 2, 2, 1. Therefore, in this case, the minimum number of time units is 2, and the
voloshin-hypergraph_F02348	281	■ 0.993	$S_1\} \cap S_4\} = \emptyset$	$S_1 \cap S_4 = \emptyset$	pressed by the $\mathcal{C}$ -edge C_1 , then we can color the vertices x_1, x_2, x_3, x_4 respectively with the colors 1, 2, 2, 1. Therefore, in this case, the minimum number of time units is 2, and the
voloshin-hypergraph_F02349	281	■ 1.000	$C_1=\left. x_1\right\}, x_2\right\}, x_3\right\}, C_2=$	$C_1 = \{x_1, x_2, x_3\}, C_2 =$	colors 1, 2, 2, 1. Therefore, in this case, the minimum number of time units is 2, and the maximum number of time units is also 3 for all the jobs to be performed. In any case, one

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voloshin-hypergraph_F02350	281	■ 1.000	$\left\{x_{\{1\}}, x_{\{2\}}, x_{\{4\}}\right\}$	$\{x_1, x_2, x_4\}$	penalty must be paid. If we are interested in the performance without any penalty, then one
voloshin-hypergraph_F02351	281	■ 1.000	$\mathcal{C}=\left\{C_{\{1\}}, C_{\{2\}}\right\}, \mathcal{D}=\left\{D_{\{1\}}, D_{\{2\}}, D_{\{3\}}, D_{\{4\}}\right\}$	$\mathcal{C}=\left\{C_1, C_2\right\}, \mathcal{D}=\left\{D_1, D_2, D_3, D_4\right\}$	penalty must be paid. If we are interested in the performance without any penalty, then one
voloshin-hypergraph_F02352	283	■ 0.998	$n=1,2,3, \ldots$	$n=1,2,3, \ldots$	2. For any $k>1$, prove that if P_k is true, then P_{k+1} is true.
voloshin-hypergraph_F02353	283	■ 0.998	$P_{\{1\}}, P_{\{2\}}, \ldots$	$P_1, P_2, \ldots$	2. For any $k>1$, prove that if P_k is true, then P_{k+1} is true.
voloshin-hypergraph_F02354	283	■ 1.000	$P_{\{1\}}$	P_1	equivalen
voloshin-hypergraph_F02355	283	■ 1.000	$P_{\{k\}}$	P_k	step. It can be abbreviated as $P_k \rightarrow P_{k+1}$ and called an implicatio
voloshin-hypergraph_F02356	283	■ 1.000	$P_{\{k+1\}}$	P_{k+1}	step. It can be abbreviated as $P_k \rightarrow P_{k+1}$ and called an implicatio
voloshin-hypergraph_F02357	283	■ 1.000	$n=1,2, \ldots$	$n=1,2, \ldots$	mathematical induction is a bright example how generalizations work in mathematics: instead of proving $P_1 \rightarrow P_2$, then $P_2 \rightarrow P_3$, and so on, for n to be fixed, one first proves the
voloshin-hypergraph_F02358	283	■ 1.000	$P_{\{k\}} \rightarrow P_{\{k+1\}}$	$P_k \rightarrow P_{k+1}$	Evidently, any positive integer greater than 1 may serve as the induction basis.
voloshin-hypergraph_F02359	283	■ 1.000	$P_{\{n\}}, n$	P_n, n	Summarizing these arguments, one can describe the following sequence of steps in
voloshin-hypergraph_F02360	283	■ 1.000	$P_{\{1\}} \rightarrow P_{\{2\}}$	$P_1 \rightarrow P_2$	1. Write down the statement P_n , i. e. the theorem to be proved.
voloshin-hypergraph_F02361	283	■ 1.000	$P_{\{2\}} \rightarrow P_{\{3\}}$	$P_2 \rightarrow P_3$	1. Write down the statement P_n , i. e. the theorem to be proved.
voloshin-hypergraph_F02362	283	■ 1.000	$P_{\{n_{\{0\}}\}}$	P_{n_0}	
voloshin-hypergraph_F02363	283	■ 1.000	$n_{\{0\}}=3$	$n_0=3$	—
voloshin-hypergraph_F02364	283	■ 1.000	$n_{\{0\}}=4$	$n_0=4$	—
voloshin-hypergraph_F02365	284	■ 1.000	$k+1$	$k+1$	P_k (for a graph on k vertices) holds. Finally, considering the inverse operation one prove that property P_{k+1} (for a graph on $k+1$ vertices) fulfills
voloshin-hypergraph_F02366	284	■ 1.000	$n \geq n_{\{0\}}$	$n \geq n_0$	
voloshin-hypergraph_F02367	285	■ 1.000	$P_{\{3\}}$	P_3	strong mathematical induction requires the following: prove that P_1 is true, and if P_2 is true, and if P_3 is true, and so on, and if P_k is true, then P_{k+1} is true. This can be abbreviated
voloshin-hypergraph_F02368	285	■ 1.000	$P_{\{1\}}, P_{\{2\}}, \ldots, P_{\{k\}} \rightarrow P_{\{k+1\}}$	$P_1, P_2, \ldots, P_k \rightarrow P_{k+1}$	as $P_1, P_2, \ldots, P_k \rightarrow P_{k+1}$. Implicitly it is clear that if we started at P_1 and arrived to P_k , then
voloshin-hypergraph_F02369	285	■ 1.000	$P_{\{2\}}, P_{\{3\}}, \ldots, P_{\{k-1\}}$	$P_2, P_3, \ldots, P_{k-1}$	all intermediate statements $P_2, P_3, \ldots, P_{k-1}$ must be true. In some complicated cases we
voloshin-hypergraph_F02370	285	■ 1.000	$f(n)$	$f(n)$	algorithm is run on a computer which is c_1 times faster, and the second algorithm is run on a computer which is c_2 times faster, then the first is better when $f(n)/c_1 < g(n)/c_2$ holds
voloshin-hypergraph_F02371	285	■ 1.000	$g(n)$	$g(n)$	a computer which is c_2 times faster, then the first is better when $f(n)/c_1 \geq g(n)/c_2$ holds for all $n > N_5$ where N_5 is some other natural number. This is equivalent to $f(n) < Cg(n)$
voloshin-hypergraph_F02372	285	■ 1.000	$f(n) \leq g(n)$	$f(n) \leq g(n)$	for some constant $C = c_1/c_2$. Evidently, it holds for any other constant $C' \geq C$ and any
voloshin-hypergraph_F02373	285	■ 1.000	$n \geq N_{\{1\}}$	$n \geq N_1$	for some constant $C = c_1/c_2$. Evidently, it holds for any other constant $C' \geq C$ and any
voloshin-hypergraph_F02374	285	■ 1.000	$f(n) / c_{\{1\}} \leq g(n) / c_{\{2\}}$	$f(n)/c_1 \leq g(n)/c_2$	inequality $f(n) \leq Cg(n)$ holds for some sufficiently large constant C and all $n \geq N$, then the
voloshin-hypergraph_F02375	285	■ 1.000	$n \geq N_{\{2\}}$	$n \geq N_2$	first algorithm is better. This reasoning explains the meaning of the following notation in comparison of the complexity of algorithms
voloshin-hypergraph_F02376	285	■ 1.000	$f(n) \leq C g(n)$	$f(n) \leq Cg(n)$	first algorithm is better. This reasoning explains the meaning of the following notation in comparison of the complexity of algorithms

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voloshin-hypergraph_F02377	285	■ 1.000	$C=c_{\{1\}} / c_{\{2\}}$	$C = c_1/c_2$	comparison of the complexity of algorithms. We say that $f(n)$ is $O(g(n))$ (read “$f(n)$ is big-oh of $g(n)$”). sometimes denoted by $f(n)$
voloshin-hypergraph_F02378	285	■ 1.000	$C^{\{\backslashprime\}} \geq C$	$C' \geq C$	comparison of the complexity of algorithms. We say that $f(n)$ is $O(g(n))$ (read “$f(n)$ is big-oh of $g(n)$”). sometimes denoted by $f(n)$
voloshin-hypergraph_F02379	285	■ 1.000	$N \geq N_{\{2\}}$	$N \geq N_2$	$= O(g(n))$) if there are constant C and a number N such that
voloshin-hypergraph_F02380	285	■ 1.000	$n \geq N$	$n \geq N$	$f(n) = O(g(n))$
voloshin-hypergraph_F02381	285	■ 1.000	$O(g(n)$	$O(g(n)$	of algorithms, and that time should not depend on the speed of a computer. The comparison
voloshin-hypergraph_F02382	285	■ 0.999	$=O(g(n))$)	$= O(g(n))$)	of algorithms is asymptotical , i.e., what occurs beginning with
voloshin-hypergraph_F02383	285	■ 1.000	$O(g(n))$	$O(g(n))$	
voloshin-hypergraph_F02384	285	■ 1.000	$g(n)=n$	$g(n) = n$	
voloshin-hypergraph_F02385	285	■ 1.000	$O(n)$	$O(n)$	—
voloshin-hypergraph_F02386	286	■ 1.000	$O\left(n^{\{k\}}\right)$	$O\left(n^k\right)$	polynomial-time and complexity is denoted by $O(n^k)$. If $g(n)$ is an exponential function
voloshin-hypergraph_F02387	286	■ 1.000	$O\left(a^{\{n\}}\right)$	$O\left(a^n\right)$	polynomial-time and complexity is denoted by $O(n^k)$. If $g(n)$ is an exponential function, then the algorithm is called exponential-time and complexity is denoted by $O(a^n)$ where
voloshin-hypergraph_F02388	286	■ 1.000	$a>1$	$a > 1$	$a > 1$. There is also a constant complexity denoted by $O(1)$, logarithmic complexity
voloshin-hypergraph_F02389	286	■ 1.000	$O(1)$	$O(1)$	$a > 1$. There is also a constant complexity denoted by $O(1)$, logarithmic complexity
voloshin-hypergraph_F02390	286	■ 1.000	$O(\backslashlog n)$	$O(\log n)$	denoted by $O(\log n)$ and even factorial complexity denoted by $O(n!)$.
voloshin-hypergraph_F02391	286	■ 1.000	$O(n!)$	$O(n!)$	denoted by $O(\log n)$ and even factorial complexity denoted by $O(n!)$.
voloshin-hypergraph_F02392	286	■ 0.798	$\left(n^{\{2\}}-n\right) / 2$	$\left(n^2-n\right) / 2$	every entry of the lower triangle it is 0 or 1. The number of such checks is $(n^2-n)/2$ which is the polynomial of n . Therefore the complexity of such procedure is $O(n^2)$.
voloshin-hypergraph_F02393	286	■ 1.000	$O\left(n^{\{2\}}\right)$	$O\left(n^2\right)$	However, many problems in graph theory are much more complex. The best known
voloshin-hypergraph_F02394	286	■ 1.000	$m n$	mn	edge list, then we may scan the lists and determine the degree of every vertex. In this case,
voloshin-hypergraph_F02395	286	■ 0.974	$O(m n)$	$O(mn)$	the total time is O (the sum of all edge cardinalities). As one can see, the complexity of many algorithms depends on the graph representation in computer memory.
voloshin-hypergraph_F02396	286	■ 1.000	O	O	In graphing a problem and time complexity of polynomial-time algorithms is called tractable , otherwise it is called intractable . Tractable problems form the so called class P
voloshin-hypergraph_F02397	286	■ 1.000	$n\left(G_{\{1\}}\right)=4, m\left(G_{\{1\}}\right)=3, n\left(G_{\{2\}}\right)=5, m\left(G_{\{2\}}\right)=7, n\left(G_{\{3\}}\right)=8, m\left(G_{\{3\}}\right)=12$	$n\left(G_1\right)=4, m\left(G_1\right)=3, n\left(G_2\right)=5, m\left(G_2\right)=7, n\left(G_3\right)=8, m\left(G_3\right)=12$	$\left(\begin{array}{cccc} & 0 & 1 & 0 & 0 \end{array}\right) \backslash$
voloshin-hypergraph_F02398	286	■ 1.000	$G_{\{1\}}:(1,1,1,3), G_{\{2\}}:(2,2,2,4,4), G_{\{3\}}:(2,2,3,3,3,3,4,4)$	$G_1:(1,1,1,3), G_2:(2,2,2,4,4), G_3:(2,2,3,3,3,3,4,4)$	$A(G, \backslash)=\left(\begin{array}{cccc} 1 & 0 & 1 & 1 \end{array}\right)$
voloshin-hypergraph_F02399	286	■ 1.000	$L\left(G_{\{1\}}\right)=\{\backslash\{2\}, \backslash\{1,3,4\}, \backslash\{2,4\}, \backslash\{2,3\}\backslash\}$	$L\left(G_1\right)=\{\{2\},\{1,3,4\},\{2,4\},\{2,3\}\}$	3.
voloshin-hypergraph_F02400	287	■ 0.998	$J\left(G_{\{1\}}\right)=\{\backslash\{1,2\}, \backslash\{2,3\}, \backslash\{3,4\}, \backslash\{2,4\}\backslash\}$	$J\left(G_1\right)=\{\{1,2\},\{2,3\},\{3,4\},\{2,4\}\}$	4. $J\left(G_1\right)=\{\{1,2\},\{2,3\},\{3,4\},\{2,4\}\}$
voloshin-hypergraph_F02401	287	■ 1.000	$G_{\{5\}}, G_{\{6\}}, G_{\{7\}}, G_{\{8\}}, G_{\{9\}}$	G_5, G_6, G_7, G_8, G_9	8. $G_1 \cong G_2, G_3 \cong G_4, G_7 \cong G_8 \cong G_9$. 9. When the names and order of vertices correspond
voloshin-hypergraph_F02402	287	■ 0.999	$G_{\{1\}} \cong G_{\{2\}}, G_{\{3\}} \cong G_{\{4\}}, G_{\{7\}} \cong G_{\{8\}} \cong G_{\{9\}}$	$G_1 \cong G_2, G_3 \cong G_4, G_7 \cong G_8 \cong G_9$	to the given isomorphism. 10. When the names and order of vertices correspond to any
voloshin-hypergraph_F02403	287	■ 1.000	$\eta(T)=2, \eta\left(C_{\{5\}}\right)=3, \eta\left(W_{\{5\}}\right)=4, \eta\left(K_{\{2,3\}}\right)=3\right.$	$\eta(T)=2, \eta\left(C_5\right)=3, \eta\left(W_5\right)=4, \eta\left(K_{2,3}\right)=3$	vector with all components equal to $n-1$.

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voloshin-hypergraph_F02484	287	■ 1.000	$K_{\{4,4\}}: C_{\{4\}}, C_{\{8\}}$	$K_{4,4} : C_4, C_8$	$\alpha(G_7) = \alpha(G_8) = \alpha(G_9) = 3$, 15. For graphs in Figure 1.21: $\tau(G_1) = \tau(G_2) = 2, \tau(G_3) =$
voloshin-hypergraph_F02485	287	■ 1.000	$C_{\{4\}}, C_{\{8\}}$	C_4, C_8	$\alpha(G_7) = \alpha(G_8) = \alpha(G_9) = 3$, 15. For graphs in Figure 1.21: $\tau(G_1) = \tau(G_2) = 2, \tau(G_3) =$
voloshin-hypergraph_F02486	287	■ 1.000	$C_{\{5\}}, C_{\{9\}}$	C_5, C_9	$\alpha(G_7) = \alpha(G_8) = \alpha(G_9) = 3$, 15. For graphs in Figure 1.21: $\tau(G_1) = \tau(G_2) = 2, \tau(G_3) =$
voloshin-hypergraph_F02487	287	■ 0.890	$\omega\left(G_{\{1\}}\right)=\omega\left(G_{\{2\}}\right)=\omega\left(G_{\{3\}}\right)=\omega\left(G_{\{4\}}\right)=3, \omega\left(G_{\{5\}}\right)=\omega\left(G_{\{6\}}\right)=\omega\left(G_{\{7\}}\right)=\omega\left(G_{\{8\}}\right)=\omega\left(G_{\{9\}}\right)=2$	$\omega\left(G_1\right)=\omega\left(G_2\right)=\omega\left(G_3\right)=\omega\left(G_4\right)=3, \omega\left(G_5\right)=\omega\left(G_6\right)=\omega\left(G_7\right)=\omega\left(G_8\right)=\omega\left(G_9\right)=2$	$\tau\left(G_4\right)=3, \tau\left(G_5\right)=\tau\left(G_6\right)=4, \tau\left(G_7\right)=\tau\left(G_8\right)=\tau\left(G_9\right)=2$, 16. For graphs in Figure 1.21: $\alpha\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=2, \alpha\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=4, \alpha\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=2$
voloshin-hypergraph_F02488	287	■ 1.000	$\alpha\left(G_{\{1\}}\right)=\alpha\left(G_{\{2\}}\right)=\alpha\left(G_{\{3\}}\right)=\alpha\left(G_{\{4\}}\right)=2, \alpha\left(G_{\{5\}}\right)=\alpha\left(G_{\{6\}}\right)=4$	$\alpha\left(G_1\right)=\alpha\left(G_2\right)=\alpha\left(G_3\right)=\alpha\left(G_4\right)=2, \alpha\left(G_5\right)=\alpha\left(G_6\right)=4$	$\tau\left(G_1\right)=\tau\left(G_2\right)=\tau\left(G_3\right)=\tau\left(G_4\right)=2, \tau\left(G_5\right)=\tau\left(G_6\right)=4, \tau\left(G_7\right)=\tau\left(G_8\right)=\tau\left(G_9\right)=2$, 17. If n is even: $\alpha\left(C_{\cdot}\right)=\tau\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=n / 2$. if n is odd: $\alpha\left(C_{\cdot}\right)=\nu\left(C_{\cdot}\right)=(n-1) / 2$.
voloshin-hypergraph_F02489	287	■ 0.803	$\boldsymbol{\alpha}\left(G_{\{7\}}\right)=\boldsymbol{\alpha}\left(G_{\{8\}}\right)=\boldsymbol{\alpha}\left(G_{\{9\}}\right)=3$	$\alpha\left(G_7\right)=\alpha\left(G_8\right)=\alpha\left(G_9\right)=3$	$\tau\left(C_n\right)=(n+1) / 2$, for all $n: \omega\left(C_n\right)=2$.
voloshin-hypergraph_F02410	287	■ 0.803	$\boldsymbol{\tau}\left(G_{\{1\}}\right)=\boldsymbol{\tau}\left(G_{\{2\}}\right)=2, \boldsymbol{\tau}\left(G_{\{3\}}\right)=$	$\tau\left(G_1\right)=\tau\left(G_2\right)=2, \tau\left(G_3\right)=$	$\tau\left(C_n\right)=(n+1) / 2$, for all $n: \omega\left(C_n\right)=2$.
voloshin-hypergraph_F02411	287	■ 0.997	$\tau\left(G_{\{4\}}\right)=3, \tau\left(G_{\{5\}}\right)=\tau\left(G_{\{6\}}\right)=4, \tau\left(G_{\{7\}}\right)=\tau\left(G_{\{8\}}\right)=\tau\left(G_{\{9\}}\right)=2$	$\tau\left(G_4\right)=3, \tau\left(G_5\right)=\tau\left(G_6\right)=4, \tau\left(G_7\right)=\tau\left(G_8\right)=\tau\left(G_9\right)=2$	
voloshin-hypergraph_F02412	287	■ 0.995	$\nu\left(G_{\{1\}}\right)=\nu\left(G_{\{2\}}\right)=\nu\left(G_{\{3\}}\right)=\nu\left(G_{\{4\}}\right)=2, \nu\left(G_{\{5\}}\right)=\nu\left(G_{\{6\}}\right)=4, \nu\left(G_{\{7\}}\right)=\nu\left(G_{\{8\}}\right)=\nu\left(G_{\{9\}}\right)=2$	$\nu\left(G_1\right)=\nu\left(G_2\right)=\nu\left(G_3\right)=\nu\left(G_4\right)=2, \nu\left(G_5\right)=\nu\left(G_6\right)=4, \nu\left(G_7\right)=\nu\left(G_8\right)=\nu\left(G_9\right)=2$	Section 1.7 $\nu\left(C_{\cdot}\right)=2, 4, 6, 2, 12, 2$
voloshin-hypergraph_F02413	287	■ 1.000	$\alpha\left(C_{\{n\}}\right)=\tau\left(C_{\{n\}}\right)=\nu\left(C_{\{n\}}\right)=n / 2$	$\alpha\left(C_n\right)=\tau\left(C_n\right)=\nu\left(C_n\right)=n / 2$	$\nu, \kappa\left(G_j\right)=4, \forall j, \kappa \leq 4, 1 \leq j \leq 6$.
voloshin-hypergraph_F02414	287	■ 1.000	$\alpha\left(C_{\{n\}}\right)=\nu\left(C_{\{n\}}\right)=(n-1) / 2$	$\alpha\left(C_n\right)=\nu\left(C_n\right)=(n-1) / 2$	$\nu, \kappa\left(G_j\right)=4, \forall j, \kappa \leq 4, 1 \leq j \leq 6$.
voloshin-hypergraph_F02415	287	■ 1.000	$\tau\left(C_{\{n\}}\right)=(n+1) / 2$	$\tau\left(C_n\right)=(n+1) / 2$	CHAPTER 2
voloshin-hypergraph_F02416	287	■ 1.000	$n: \omega\left(C_{\{n\}}\right)=2$	$n: \omega\left(C_n\right)=2$	CHAPTER 2
voloshin-hypergraph_F02417	287	■ 0.905	$\Lambda\left(E_{\{n\}}\right)=0, \Lambda\left(C_{\{n\}}\right)=1, \Lambda\left(K_{\{n\}}\right)=(n-1)(n-2) / 2, \Lambda\left(W_{\{n\}}\right)=n-1$	$\Lambda\left(E_n\right)=0, \Lambda\left(C_n\right)=1, \Lambda\left(K_n\right)=(n-1)(n-2) / 2, \Lambda\left(W_n\right)=n-1$	2. Minimum weight = 28. 3. Maximum weight = 49. 4. $n^{\alpha} \sim$.
voloshin-hypergraph_F02418	287	■ 0.782	$\operatorname{diam}(T)=9$	$\operatorname{diam}(T)=9$	
voloshin-hypergraph_F02419	287	■ 0.782	$=5$	$=5$	
voloshin-hypergraph_F02420	287	■ 0.604	$=28$	$=28$	
voloshin-hypergraph_F02421	287	■ 0.604	$=49$	$=49$	
voloshin-hypergraph_F02422	287	■ 0.745	$m=n$	$m=n$	
voloshin-hypergraph_F02423	287	■ 0.745	$\tau\left(K_{\{m, n\}}\right)=\nu\left(K_{\{m, n\}}\right)=\min \{m, n\}$	$\tau\left(K_{m, n}\right)=\nu\left(K_{m, n}\right)=\min \{m, n\}$	
voloshin-hypergraph_F02424	287	■ 0.745	$\tau=\nu=13$	$\tau=\nu=13$	
voloshin-hypergraph_F02425	288	■ 0.646	$\Theta\left(G_{\{1\}}\right)=\alpha\left(G_{\{1\}}\right)=3, \Theta\left(G_{\{2\}}\right)=$	$\Theta\left(G_1\right)=\alpha\left(G_1\right)=3, \Theta\left(G_2\right)=$	2. G_1 and G_3 are chordal; G_2 is not chordal. 3. G_2 . 7. $\theta\left(G_1\right)=\alpha\left(G_1\right)=3, \theta\left(G_2\right)=$

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voloshin-hypergraph_F02426	288	0.544	$\boldsymbol{\alpha}(G_2)=5, \boldsymbol{\theta}(G_3)=\boldsymbol{\alpha}(G_3)=3$	$\alpha(G_2)=5, \theta(G_3)=\alpha(G_3)=3$	$\alpha(G_2)=5, \theta(G_3)=\alpha(G_3)=3$. 8. For cube: 6. 9. From cube: 5.
voloshin-hypergraph_F02427	288	0.987	$M(G_1)=3, \omega(G_1)-1=2; M(G_2)=4, \omega(G_2)-1=2; M(G_3)=2, \omega(G_3)-1=2.2$	$M(G_1)=3, \omega(G_1)-1=2; M(G_2)=4, \omega(G_2)-1=2; M(G_3)=2, \omega(G_3)-1=2.2$	1. $M(G_1)=3, \omega(G_1)-1=2, M(G_2)=4, \omega(G_2)-1=2, M(G_3)=2, \omega(G_3)-1=2.2$. G_1, G_2 are not chordal; G_3 is chordal. 3. $M(C_n)=2, M(K_n)=n-1, M(W_n)=3$ for cube
voloshin-hypergraph_F02428	288	1.000	$M(C_n)=2, M(K_n)=n-1, M(W_n)=3$	$M(C_n)=2, M(K_n)=n-1, M(W_n)=3$	prism and Petersen graph: $M(G)=3$. 4. Take empty graph E_n with sufficient large n ; add
voloshin-hypergraph_F02429	288	1.000	$M(G)=3$	$M(G)=3$	a vertex and make it adjacent to all the vertices of E_n ; repeat the procedure. Observe that
voloshin-hypergraph_F02430	288	1.000	$\leq k-1$	$\leq k-1$	Section 3.4
voloshin-hypergraph_F02431	288	0.934	$\operatorname{diam}(G)=6$	$\operatorname{diam}(G)=6$	Section 4.1
voloshin-hypergraph_F02432	288	1.000	$n=3, 4, K_n$	$n=3, 4, K_n$	Section 4.1
voloshin-hypergraph_F02433	288	1.000	$n \geq 3, W_n$	$n \geq 3, W_n$	Section 4.1
voloshin-hypergraph_F02434	288	1.000	$n=4, 5$	$n=4, 5$	Section 4.1
voloshin-hypergraph_F02435	288	0.999	$G_1: 5; G_2: 7$	$G_1: 5; G_2: 7$	Section 4.3
voloshin-hypergraph_F02436	288	0.427	$K_{3,3}: 0, 1, 2, 3$	$K_{3,3}: 0, 1, 2, 3$	CHAPTER 5
voloshin-hypergraph_F02437	288	0.427	$K_5: 0, 1, 2, 3, 4$	$K_5: 0, 1, 2, 3, 4$	CHAPTER 5
voloshin-hypergraph_F02438	288	1.000	$K_n: n=3, 4; K_{m,n}: m=1, n \geq 3$	$K_n: n=3, 4; K_{m,n}: m=1, n \geq 3$	1. $\chi(E_n)=1, \chi(K_n)=n, \chi(K_{m,n})=2, \chi(I_n)=2, \chi(C_{2n})=2, \chi(C_{2n+1})=3, \chi(W_{2n})=4, \chi(W_{2n+1})=3$. 4. $\chi(C_3)=2, \chi(C_4)=3, \chi(C_5)=4, \chi(C_6)=5, \chi(C_7)=6, \chi(C_8)=7, \chi(C_9)=8$
voloshin-hypergraph_F02439	288	1.000	$m=2, n \geq 3; W_n: n \geq 4$	$m=2, n \geq 3; W_n: n \geq 4$	1. $\chi(E_n)=1, \chi(K_n)=n, \chi(K_{m,n})=2, \chi(I_n)=2, \chi(C_{2n})=2, \chi(C_{2n+1})=3, \chi(W_{2n})=4, \chi(W_{2n+1})=3$. 4. $\chi(C_3)=2, \chi(C_4)=3, \chi(C_5)=4, \chi(C_6)=5, \chi(C_7)=6, \chi(C_8)=7, \chi(C_9)=8$
voloshin-hypergraph_F02440	288	1.000	$\chi(E_n)=1, \chi(K_n)=n, \chi(K_{m,n})=2, \chi(T_n)=2, \chi(C_{2n})=2, \chi(C_{2n+1})=3, \chi(W_{2n})=4$	$\chi(E_n)=1, \chi(K_n)=n, \chi(K_{m,n})=2, \chi(T_n)=2, \chi(C_{2n})=2, \chi(C_{2n+1})=3, \chi(W_{2n})=4$	Section 5.4
voloshin-hypergraph_F02441	288	0.928	$\chi(W_{2n+1})=3$	$\chi(W_{2n+1})=3$	$S(7,7)=1; s(3,1)=2, s(3,2)=-3, s(3,3)=1$. 3.
voloshin-hypergraph_F02442	288	0.928	$\lambda \geq 136$	$\lambda \geq 136$	$S(7,7)=1; s(3,1)=2, s(3,2)=-3, s(3,3)=1$. 3.
voloshin-hypergraph_F02443	288	0.928	$\chi(G)=5$	$\chi(G)=5$	$S(7,7)=1; s(3,1)=2, s(3,2)=-3, s(3,3)=1$. 3.
voloshin-hypergraph_F02444	288	1.000	$P(E_4, \lambda)=\lambda^4; P(C_5, \lambda)=(\lambda-1)^5-(\lambda-1); P(W_4, \lambda)=\lambda^4; P(P_n, \lambda)=\lambda(\lambda-1)^{n-1}$	$P(E_4, \lambda)=\lambda^4; P(C_5, \lambda)=(\lambda-1)^5-(\lambda-1); P(W_4, \lambda)=\lambda^4; P(P_n, \lambda)=\lambda(\lambda-1)^{n-1}$	
voloshin-hypergraph_F02445	288	1.000	$S(7,1)=1, S(7,2)=63, S(7,3)=301, S(7,4)=350, S(7,5)=140, S(7,6)=21$	$S(7,1)=1, S(7,2)=63, S(7,3)=301, S(7,4)=350, S(7,5)=140, S(7,6)=21$	
voloshin-hypergraph_F02446	288	0.999	$S(7,7)=1; s(3,1)=2, s(3,2)=-3, s(3,3)=1$	$S(7,7)=1; s(3,1)=2, s(3,2)=-3, s(3,3)=1$	
voloshin-hypergraph_F02447	288	0.999	$P(G_1, \lambda)=\lambda(\lambda-1)^3(\lambda-2)^2(\lambda^2-$	$P(G_1, \lambda)=\lambda(\lambda-1)^3(\lambda-2)^2(\lambda^2-$	
voloshin-hypergraph_F02448	288	1.000	$3\lambda+3)^2; P(G_2, \lambda)=\lambda(\lambda-1)(\lambda-2)^4; P(G_3, \lambda)=\lambda(\lambda-1)(\lambda-2)^4(\lambda^2-3\lambda+3)$	$3\lambda+3)^2; P(G_2, \lambda)=\lambda(\lambda-1)(\lambda-2)^4; P(G_3, \lambda)=\lambda(\lambda-1)(\lambda-2)^4(\lambda^2-3\lambda+3)$	—

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voloshin-hypergraph_F02449	289	■ 1.000	$L(G)=\{\{2,4\}\}$	$L(G) = \{\{2, 4\}\}$	Section 3.3 3. Graph $G = (X, E)$ with $X = \{1, 2, 3, 4, 5\}$ and adjacency list $L(G) = \{\{2, 4\}\}$.
voloshin-hypergraph_F02450	289	■ 0.970	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$; the order of online coloring: 1,2,3,4,5.
voloshin-hypergraph_F02451	289	■ 0.970	$1, 2, 3, 4, 5$	$1, 2, 3, 4, 5$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$; the order of online coloring: 1,2,3,4,5.
voloshin-hypergraph_F02452	289	■ 1.000	$M=3$	$M = 3$	1. For Petersen graph: $M = 3$, therefore $\chi \leq 4$. Idea: switch the colors in the regions on one side of a newly added line.
voloshin-hypergraph_F02453	289	■ 1.000	$\chi \leq 4$	$\chi \leq 4$	1. For Petersen graph: $M = 3$, therefore $\chi \leq 4$. Idea: switch the colors in the regions on one side of a newly added line.
voloshin-hypergraph_F02454	289	■ 1.000	W_{2k-1}	W_{2k-1}	not, $k \geq 3$.
voloshin-hypergraph_F02455	289	■ 1.000	W_{2k}	W_{2k}	not, $k \geq 3$.
voloshin-hypergraph_F02456	289	■ 1.000	$\chi^{\prime}(K_n)=n$	$\chi^{\prime}(K_n) = n$	- . . -
voloshin-hypergraph_F02457	289	■ 1.000	$\chi^{\prime}(K_n)=n-1$	$\chi^{\prime}(K_n) = n - 1$	- . . -
voloshin-hypergraph_F02458	289	■ 1.000	$\chi^{\prime}(C_n)=3$	$\chi^{\prime}(C_n) = 3$	- . . -
voloshin-hypergraph_F02459	289	■ 0.997	$\chi^{\prime}(C_n)=2$	$\chi^{\prime}(C_n) = 2$	Section 5.9 2. $\chi^{\prime}(C_n) = 0$ if $\chi^{\prime}(C_n) = 1$, $\chi^{\prime}(W_n) = 1$. For Petersen graph: $\chi^{\prime} = 3$ for sub- $\chi^{\prime}$.
voloshin-hypergraph_F02460	289	■ 0.997	$\chi^{\prime}(W_n)=n-1$	$\chi^{\prime}(W_n) = n - 1$	Section 5.9 2. $\chi^{\prime}(C_n) = 0$ if $\chi^{\prime}(C_n) = 1$, $\chi^{\prime}(W_n) = 1$. For Petersen graph: $\chi^{\prime} = 3$ for sub- $\chi^{\prime}$.
voloshin-hypergraph_F02461	289	■ 0.997	$n \geq 4$. $\chi^{\prime}(K_{m,n})=\max\{m,n\}$	$n \geq 4. \chi^{\prime}(K_{m,n}) = \max\{m, n\}$	Section 5.9 2. $\chi^{\prime}(C_n) = 0$ if $\chi^{\prime}(C_n) = 1$, $\chi^{\prime}(W_n) = 1$. For Petersen graph: $\chi^{\prime} = 3$ for sub- $\chi^{\prime}$.
voloshin-hypergraph_F02462	289	■ 0.973	$\bar{\chi}^{\prime}(G)=8$	$\bar{\chi}^{\prime}(G) = 8$	CHAPTER 6
voloshin-hypergraph_F02463	289	■ 0.973	$\bar{\chi}^{\prime}(C_n)=1, \bar{\chi}^{\prime}(W_n)=n-1$	$\bar{\chi}^{\prime}(C_n) = 1, \bar{\chi}^{\prime}(W_n) = n - 1$	CHAPTER 6
voloshin-hypergraph_F02464	289	■ 0.973	$\bar{\chi}^{\prime}=7$	$\bar{\chi}^{\prime} = 7$	CHAPTER 6
voloshin-hypergraph_F02465	289	■ 0.973	$\bar{\chi}^{\prime}=6$	$\bar{\chi}^{\prime} = 6$	CHAPTER 6
voloshin-hypergraph_F02466	289	■ 1.000	$\bar{\chi}^{\prime}=5$	$\bar{\chi}^{\prime} = 5$	$\bar{\chi}^{\prime} = 5$
voloshin-hypergraph_F02467	289	■ 1.000	$K_n: n \geq 3$; $K_{m,n}: m=n$; $W_n: n \geq 4$	$K_n : n \geq 3; K_{m,n} : m = n; W_n : n \geq 4$	and connect it with incoming arcs from all sinks.
voloshin-hypergraph_F02468	289	■ 0.998	$n\left(\mathcal{H}_1\right)=6, m\left(\mathcal{H}_1\right)=5 ; n\left(\mathcal{H}_2\right)=6, m\left(\mathcal{H}_2\right)=11$	$n\left(\mathcal{H}_1\right) = 6, m\left(\mathcal{H}_1\right) = 5; n\left(\mathcal{H}_2\right) = 6, m\left(\mathcal{H}_2\right) = 11$	1. $\mathcal{H}^* = \{d_1, d_2, d_3, d_4\}$, $\mathcal{H}^*(d_1) = \{d_2, d_3, d_4\}$, $\mathcal{H}^*(d_2) = \{d_1, d_3, d_4\}$, $\mathcal{H}^*(d_3) = \{d_1, d_2, d_4\}$, $\mathcal{H}^*(d_4) = \{d_1, d_2, d_3\}$. $N(5) = \{2, 3, 4\}$, $N(6) = \emptyset$; in $\mathcal{H}^*$: $N(d_1) = \{d_2, d_3\}$, $N(d_2) = \{d_1, d_3, d_4\}$, $N(d_3) =$
voloshin-hypergraph_F02469	289	■ 1.000	$\Delta\left(\mathcal{H}_1\right)=3, \Delta\left(\mathcal{H}_2\right)=5$	$\Delta\left(\mathcal{H}_1\right) = 3, \Delta\left(\mathcal{H}_2\right) = 5$	$\{d_1, d_2, d_4, d_5\}, N(d_4) = \{d_2, d_3, d_5\}, N(d_5) = \{d_3, d_4\}$. 3. $r(\mathcal{H}) = r(\mathcal{H}^*) = 3$. 7. $L(G) =$
voloshin-hypergraph_F02470	289	■ 1.000	$r\left(\mathcal{H}_1\right)=3 ; r\left(\mathcal{H}_2\right)=4$	$r\left(\mathcal{H}_1\right) = 3; r\left(\mathcal{H}_2\right) = 4$	$\{d_1, d_2, d_4, d_5\}, N(d_4) = \{d_2, d_3, d_5\}, N(d_5) = \{d_3, d_4\}$. 3. $r(\mathcal{H}) = r(\mathcal{H}^*) = 3$. 7. $L(G) =$
voloshin-hypergraph_F02471	289	■ 1.000	$\mathcal{H}: N(1)=\{2,4\}, N(2)=\{1,3,4,5\}, N(3)=\{2,4,5\}, N(4)=\{1,2,3,5\}$	$\mathcal{H} : N(1) = \{2, 4\}, N(2) = \{1, 3, 4, 5\}, N(3) = \{2, 4, 5\}, N(4) = \{1, 2, 3, 5\}$	
voloshin-hypergraph_F02472	289	■ 0.773	$N(5)=\{2,3,4\}, N(6)=\emptyset$	$N(5) = \{2, 3, 4\}, N(6) = \emptyset$	
voloshin-hypergraph_F02473	289	■ 0.773	$\mathcal{H}^*: N\left(d_1\right)=\left\{d_2, d_3\right\}, N\left(d_2\right)=\left\{d_1, d_3, d_4\right\}, N\left(d_3\right)=$	$\mathcal{H}^* : N\left(d_1\right) = \left\{d_2, d_3\right\}, N\left(d_2\right) = \left\{d_1, d_3, d_4\right\}, N\left(d_3\right) =$	

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voloshin-hypergraph_F02474	289	■ 0.493	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$	
voloshin-hypergraph_F02475	289	■ 0.493	$r\left(\mathrm{mathcal{H}}\right)=r\left(\mathrm{mathcal{H}}^*\right)=3.7 \quad L(G)=$	$r(\mathcal{H})=r\left(\mathcal{H}^*\right)=3.7.L(G)=$	
voloshin-hypergraph_F02476	289	■ 0.398	$\left\{\left\{1,5\right\},\left\{1,2\right\},\left\{2,3\right\},\left\{2,4\right\},\left\{4,5\right\}\right\} ; L\left(G^*\right)=\left\{\left\{a, b\right\},\left\{b, c, d\right\},\left\{c\right\},\left\{d, e\right\},\left\{a, e\right\}\right\} . 9 . E_n^*$	$\left\{\left\{1,5\right\},\left\{1,2\right\},\left\{2,3\right\},\left\{2,4\right\},\left\{4,5\right\}\right\} ; L\left(G^*\right)=\left\{\left\{a, b\right\},\left\{b, c, d\right\},\left\{c\right\},\left\{d, e\right\},\left\{a, e\right\}\right\} . 9 . E_n^*$	
voloshin-hypergraph_F02477	290	■ 1.000	$L\left(K_4^3\right)=\left\{\left\{1,2,3\right\},\left\{1,3,4\right\},\left\{2,3,4\right\},\left\{1,2,4\right\}\right\}$	$L\left(K_4^3\right)=\left\{\left\{1,2,3\right\},\left\{1,3,4\right\},\left\{2,3,4\right\},\left\{1,2,4\right\}\right\}$	Section 7.5 4. $L\left(K_4^3\right)=\left\{\left\{1,2,3\right\},\left\{1,3,4\right\},\left\{2,3,4\right\},\left\{1,2,4\right\}\right\}$. 6. When order of the vertices is the
voloshin-hypergraph_F02478	290	■ 1.000	$1 \rightarrow 7 \rightarrow 11 \rightarrow 6$	$1 \rightarrow 7 \rightarrow 11 \rightarrow 6$	2. $1 \rightarrow 7 \rightarrow 11 \rightarrow 6$. 10. $\mathcal{H}^*$ is the Petersen graph.
voloshin-hypergraph_F02479	290	■ 0.990	$S=\left\{3,4,5,6,7,8,10,11,13,14\right\}$	$S=\left\{3,4,5,6,7,8,10,11,13,14\right\}$	ample, $T=\left\{1,2,9,12,15\right\}$. 8. $\tau(\mathcal{H})=5$. 9. For example, $\mathcal{D}'=\left\{\left\{6,11,15\right\},$
voloshin-hypergraph_F02480	290	■ 0.990	$\alpha\left(\mathrm{mathcal{H}}\right)=10$	$\alpha(\mathcal{H})=10$	ample, $T=\left\{1,2,9,12,15\right\}$. 8. $\tau(\mathcal{H})=5$. 9. For example, $\mathcal{D}'=\left\{\left\{6,11,15\right\},$
voloshin-hypergraph_F02481	290	■ 0.927	$T=\left\{1,2,9,12,15\right\}$	$T=\left\{1,2,9,12,15\right\}$	$\left\{9,13,14\right\},\left\{1,3,7\right\},\left\{2,8,4\right\}\right\}$. 10. $v(\mathcal{H})=4$. 11. $\rho(\mathcal{H})=6$.
voloshin-hypergraph_F02482	290	■ 0.927	$\tau\left(\mathrm{mathcal{H}}\right)=5$	$\tau(\mathcal{H})=5$	$\left\{9,13,14\right\},\left\{1,3,7\right\},\left\{2,8,4\right\}\right\}$. 10. $v(\mathcal{H})=4$. 11. $\rho(\mathcal{H})=6$.
voloshin-hypergraph_F02483	290	■ 0.927	$\mathrm{mathcal{D}}^{\prime}=\left\{\left\{6,11,15\right\}\right\}$	$\mathcal{D}'=\left\{\left\{6,11,15\right\}\right\}$	$\left\{9,13,14\right\},\left\{1,3,7\right\},\left\{2,8,4\right\}\right\}$. 10. $v(\mathcal{H})=4$. 11. $\rho(\mathcal{H})=6$.
voloshin-hypergraph_F02484	290	■ 0.279	$\left\{9,13,14\right\},\left\{1,3,7\right\},\left\{2,8,4\right\}\right\} \cdot \mathbf{10} . \mathbf{v}(\mathcal{H})=4$	$\left\{9,13,14\right\},\left\{1,3,7\right\},\left\{2,8,4\right\}\right\} . \mathbf{10} . v(\mathcal{H})=4$	Section 7.6
voloshin-hypergraph_F02485	290	■ 0.279	$\boldsymbol{\rho}(\mathcal{H})=6$	$\boldsymbol{\rho}(\mathcal{H})=6$	Section 7.6
voloshin-hypergraph_F02486	290	■ 0.971	$\tau\left(\mathrm{mathcal{H}}_1\right)=2, \alpha\left(\mathrm{mathcal{H}}_1\right)=4, \nu\left(\mathrm{mathcal{H}}_1\right)=2 ; \tau\left(\mathrm{mathcal{H}}_2\right)=3, \alpha\left(\mathrm{mathcal{H}}_2\right)=5, \nu\left(\mathrm{mathcal{H}}_2\right)=3$	$\tau\left(\mathcal{H}_1\right)=2, \alpha\left(\mathcal{H}_1\right)=4, \nu\left(\mathcal{H}_1\right)=2 ; \tau\left(\mathcal{H}_2\right)=3, \alpha\left(\mathcal{H}_2\right)=5, \nu\left(\mathcal{H}_2\right)=3$	Section 7.6
voloshin-hypergraph_F02487	290	■ 1.000	$\wedge\left(\mathrm{mathcal{H}}_1\right)=1, \wedge\left(\mathrm{mathcal{H}}_2\right)=0$	$\wedge\left(\mathcal{H}_1\right)=1, \wedge\left(\mathcal{H}_2\right)=0$	1. $K_n: n=1,2 ; K_m,$
voloshin-hypergraph_F02488	290	■ 1.000	$K_n: n=1,2 ; K_{m, n}: m, n \geq 1 ; W_n$	$K_n: n=1,2 ; K_{m, n}: m, n \geq 1 ; W_n$	1. Paths $P_n, n > 1$. 2. In any linear ordering of the vertices of $C_n, n > 3$, there is always
voloshin-hypergraph_F02489	290	■ 0.326	$\mathcal{D}=\left\{\left\{1,2,3\right\},\left\{1,2,4\right\},\left\{2,3,4\right\}\right\}$	$\mathcal{D}=\left\{\left\{1,2,3\right\},\left\{1,2,4\right\},\left\{2,3,4\right\}\right\}$	$\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\left\{1,2,3,4\right\}, \mathcal{D}=\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$.
voloshin-hypergraph_F02490	290	■ 0.326	$X=\left\{1,2,3\right\}$	$X=\left\{1,2,3\right\}$	$\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\left\{1,2,3,4\right\}, \mathcal{D}=\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$.
voloshin-hypergraph_F02491	290	■ 0.326	$\mathcal{D}=\left\{\left\{1,2,3\right\}\right\}$	$\mathcal{D}=\left\{\left\{1,2,3\right\}\right\}$	$\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\left\{1,2,3,4\right\}, \mathcal{D}=\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$.
voloshin-hypergraph_F02492	290	■ 1.000	$\left\{1,2\right\},\left\{2,3\right\},\left\{1,3\right\}\right\}$	$\left\{1,2\right\},\left\{2,3\right\},\left\{1,3\right\}\right\}$	5. $\mathcal{H}=(X, \mathcal{D})$ w
voloshin-hypergraph_F02493	290	■ 1.000	$P_n, n \geq 1$	$P_n, n \geq 1$	None. 4. $\chi'(\mathcal{H}_1)=3 ; \chi'(\mathcal{H}_2)=3$. 5. None.
voloshin-hypergraph_F02494	290	■ 1.000	$X=\left\{1,2,3,4\right\}, \mathrm{mathcal{D}}=$	$X=\left\{1,2,3,4\right\}, \mathcal{D}=$	
voloshin-hypergraph_F02495	290	■ 0.513	$\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$	$\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$	
voloshin-hypergraph_F02496	290	■ 0.513	$X=\left\{1,2,3,4\right\}, \mathrm{mathcal{D}}=\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$	$X=\left\{1,2,3,4\right\}, \mathcal{D}=\left\{\left\{1,4\right\},\left\{2,4\right\},\left\{3,4\right\}\right\}$	
voloshin-hypergraph_F02497	290	■ 0.644	$L\left(\mathrm{mathcal{H}}_2\right)=C_5$	$L\left(\mathcal{H}_2\right)=C_5$	

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voloshin-hypergraph_F02498	290	■ 1.000	$\chi^{\prime}(\mathcal{H}_1)=3$; $\chi^{\prime}(\mathcal{H}_2)=3$	$\chi'(\mathcal{H}_1)=3; \chi'(\mathcal{H}_2)=3$	
voloshin-hypergraph_F02499	291	■ 0.259	$\alpha(\mathcal{H})=6$; $\tau(\mathcal{H})=2$	$\alpha(\mathcal{H})=6; \tau(\mathcal{H})=2$	1. $\chi(\mathcal{H})=2$. 2. $\alpha(\mathcal{H})=0$, $\chi(\mathcal{H})=2$. 3. $\tau(\mathcal{H})=4$. 3. $\chi(\mathcal{H})=0$. 6. 190.
voloshin-hypergraph_F02500	291	■ 0.259	$\gamma(\mathcal{H})=4.5$	$\gamma(\mathcal{H})=4.5$	1. $\chi(\mathcal{H})=2$. 2. $\alpha(\mathcal{H})=0$, $\chi(\mathcal{H})=2$. 3. $\tau(\mathcal{H})=4$. 3. $\chi(\mathcal{H})=0$. 6. 190.
voloshin-hypergraph_F02501	291	■ 0.817	$M(\mathcal{H})=2$, $\chi(\mathcal{H}) \geq 3$	$M(\mathcal{H})=2, \chi(\mathcal{H}) \geq 3$	Section 10.3
voloshin-hypergraph_F02502	291	■ 0.713	$\chi(\mathcal{H})=2$, $\bar{\chi}(\mathcal{H})=3$	$\chi(\mathcal{H})=2, \bar{\chi}(\mathcal{H})=3$	Section 10.4
voloshin-hypergraph_F02503	291	■ 0.713	$\chi(\mathcal{H}_C)=1$, $\bar{\chi}(\mathcal{H}_C)=3$; $\chi(\mathcal{H}_D)=2$, $\bar{\chi}(\mathcal{H}_D)=5$	$\chi(\mathcal{H}_C)=1, \bar{\chi}(\mathcal{H}_C)=3; \chi(\mathcal{H}_D)=2, \bar{\chi}(\mathcal{H}_D)=5$	Section 10.4
voloshin-hypergraph_F02504	291	■ 0.693	$\bar{\chi}(\mathcal{H}) \geq 3$	$\bar{\chi}(\mathcal{H}) \geq 3$	Section 10.5 1. $P(\mathcal{H}, \lambda) = 3\lambda(3) + 6\lambda(2)$. $P(\mathcal{H}, \lambda) = (0, 6, 3, 0)$. $P(\mathcal{H}, \lambda) = \lambda(2)$. $R(\mathcal{H}) = (0, 1, 0, 0)$.
voloshin-hypergraph_F02505	291	■ 1.000	$P(\mathcal{H}_1, \lambda) = 3\lambda^{(3)} + 6\lambda^{(2)}$, $R(\mathcal{H}_1) = (0, 6, 3, 0)$; $P(\mathcal{H}_2, \lambda) = \lambda^{(2)}$, $R(\mathcal{H}_2) = (0, 1, 0, 0)$	$P(\mathcal{H}_1, \lambda) = 3\lambda^{(3)} + 6\lambda^{(2)}, R(\mathcal{H}_1) = (0, 6, 3, 0); P(\mathcal{H}_2, \lambda) = \lambda^{(2)}, R(\mathcal{H}_2) = (0, 1, 0, 0)$	1. $\chi(\mathcal{H}_D) = 3$, $\bar{\chi}(\mathcal{H}_C) = 3$.
voloshin-hypergraph_F02506	291	■ 0.577	$\chi(\mathcal{H}_D)=3$, $\bar{\chi}(\mathcal{H}_C)=3$	$\chi(\mathcal{H}_D)=3, \bar{\chi}(\mathcal{H}_C)=3$	76. 1003. 87. 1003.
voloshin-hypergraph_F02507	291	■ 0.278	$\alpha_C(\mathcal{H}_1)=3$, $\bar{\chi}(\mathcal{H}_1)=3$; $\alpha_C(\mathcal{H}_2)=3$, $\bar{\chi}(\mathcal{H}_2)=3$	$\alpha_C(\mathcal{H}_1)=3, \bar{\chi}(\mathcal{H}_1)=3; \alpha_C(\mathcal{H}_2)=3, \bar{\chi}(\mathcal{H}_2)=3$	$\alpha_C(\mathcal{H}_1)=3, \bar{\chi}(\mathcal{H}_1)=3; \alpha_C(\mathcal{H}_2)=3, \bar{\chi}(\mathcal{H}_2)=3$
voloshin-hypergraph_F02508	291	■ 1.000	$\tau_2(\mathcal{H}_1)=2$, $\tau_2(\mathcal{H}_2)=3$	$\tau_2(\mathcal{H}_1)=2, \tau_2(\mathcal{H}_2)=3$	Section 10.10 2. 11
voloshin-hypergraph_F02509	291	■ 0.962	$S(\mathcal{H}_C)=\{1, 2, 3, 4, 5, 6\}$, $S(\mathcal{H}_D)=\{2, 3, 4, 5, 6, 7, 8\}$	$S(\mathcal{H}_C)=\{1, 2, 3, 4, 5, 6\}, S(\mathcal{H}_D)=\{2, 3, 4, 5, 6, 7, 8\}$	12.4. Glossary of Additional Concepts
voloshin-hypergraph_F02510	291	■ 0.962	$S(\mathcal{H})=\{2, 3, 4, 5\}$	$S(\mathcal{H})=\{2, 3, 4, 5\}$	12.4. Glossary of Additional Concepts
voloshin-hypergraph_F02511	291	■ 1.000	$n \rightarrow \infty$	$n \rightarrow \infty$	—
voloshin-hypergraph_F02512	292	■ 1.000	$\overline{C_k}$	$\overline{C_k}$	4. Antikatale subgraph induced by $\overline{C}$
voloshin-hypergraph_F02513	292	■ 0.966	(n, k, λ)	(n, k, λ)	neighbor from S
voloshin-hypergraph_F02514	292	■ 1.000	$m(G) / n(G)$	$m(G)/n(G)$	• Eigenvalue of a graph: ϵ
voloshin-hypergraph_F02515	292	■ 1.000	$ X = \mathcal{D} =r^2-r+$	$ X = \mathcal{D} =r^2-r+$	
voloshin-hypergraph_F02516	293	■ 1.000	$f: V(G_1) \rightarrow V(G_2)$	$f: V(G_1) \rightarrow V(G_2)$	• Hypergraph removal lemma. For a given r -uniform hypergraph $\mathcal{H}$, in a uniform hyderergraph containing a bounded number of copies of $\mathcal{H}$, one
voloshin-hypergraph_F02517	293	■ 1.000	$n > m$	$n > m$	• Ramsey number $R(n, 2)$: the minimum n such that any 2-coloring of the edges of K_n
voloshin-hypergraph_F02518	293	■ 1.000	$R(p, q)$	$R(p, q)$	
voloshin-hypergraph_F02519	293	■ 1.000	K_p	K_p	
voloshin-hypergraph_F02520	293	■ 1.000	K_q	K_q	

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voloshin-hypergraph_F02521	293	■ 1.000	$R(3,3)=6$	$R(3,3)=6$	—
voloshin-hypergraph_F02522	294	■ 0.996	$S \leq S(v)$	$SQS(v)$	Steiner System $S(t,k,v)$: a k-uniform hypergraph $\mathcal{H}=(X,\mathcal{B})$ v edges called blocks such that any t distinct vertices appear toget
voloshin-hypergraph_F02523	294	■ 0.996	$S(3,4,v)$	$S(3,4,v)$	Steiner System $S(t,k,v)$: a k-uniform hypergraph $\mathcal{H}=(X,\mathcal{B})$ v edges called blocks such that any t distinct vertices appear toget
voloshin-hypergraph_F02524	294	■ 1.000	$S(t,k,v)$	$S(t,k,v)$	block
voloshin-hypergraph_F02525	294	■ 1.000	$\mathcal{H}=(X,\mathcal{B})$	$\mathcal{H}=(X,\mathcal{B})$	block
voloshin-hypergraph_F02526	294	■ 1.000	$ X =v$	$ X =v$	block
voloshin-hypergraph_F02527	294	■ 0.959	$\operatorname{Tr} \mathcal{H}$	$\operatorname{Tr} \mathcal{H}$	Turán graph: the complete multipartite graph with all parts of almost the same size
voloshin-hypergraph_F02528	294	■ 0.959	$\mathcal{H}=(X,\mathcal{E}), \operatorname{Tr} \mathcal{H}=(X,\mathcal{D})$	$\mathcal{H}=(X,\mathcal{E}), \operatorname{Tr} \mathcal{H}=(X,\mathcal{D})$	Turán graph: the complete multipartite graph with all parts of almost the same size
voloshin-hypergraph_F02529	294	■ 1.000	$T(n,p,r)$	$T(n,p,r)$	
voloshin-hypergraph_F02530	297	■ 0.998	$x,7$	$x,7$	$L(G)$, line graph, 112 $L(L(G))$, line graph, 154
voloshin-hypergraph_F02531	297	■ 0.998	$\Lambda(G)$	$\Lambda(G)$	$\alpha(\mathcal{H})$, stability (independ
voloshin-hypergraph_F02532	297	■ 0.333	$\alpha_c(\mathcal{H}), \mathcal{C}$	$\alpha_c(\mathcal{H}), \mathcal{C}$	
voloshin-hypergraph_F02533	297	■ 0.986	$\kappa(G)$	$\kappa(G)$	$i(G)$, multiplicity of a graph
voloshin-hypergraph_F02534	297	■ 0.933	$\tau_2(\mathcal{H}_C)$	$\tau_2(\mathcal{H}_C)$	τ -factor, 32
voloshin-hypergraph_F02535	297	■ 0.831	$T,174$	$T,174$	er of feasible
voloshin-hypergraph_F02536	297	■ 1.000	$m=m(\mathcal{H})$	$m=m(\mathcal{H})$	$\pi_i(\mathcal{H})$, number of feasible partitions, 203
voloshin-hypergraph_F02537	297	■ 1.000	$r_i(\mathcal{H})$	$r_i(\mathcal{H})$	$\mathcal{C}$ -perfection, perfection, 237
voloshin-hypergraph_F02538	297	■ 0.587	$\alpha_c(\mathcal{H}),236$	$\alpha_c(\mathcal{H}),236$	
voloshin-hypergraph_F02539	298	■ 0.994	$(\mathcal{H})_2,154$	$(\mathcal{H})_2,154$	$\mathcal{H}$ -section, $(\mathcal{H})/2$,
voloshin-hypergraph_F02540	298	■ 0.388	$\tau_2(\mathcal{H}_C),238$	$\tau_2(\mathcal{H}_C),238$	
voloshin-hypergraph_F02541	298	■ 0.793	$\chi'(\mathcal{H}),185$	$\chi'(\mathcal{H}),185$	chromatic number, $\chi(\mathcal{H}),193$
voloshin-hypergraph_F02542	298	■ 0.813	$\chi(G),80$	$\chi(G),80$	$P(\mathcal{H},\lambda),204$ chromatic polynomial, $P(G,\lambda),80$
voloshin-hypergraph_F02543	298	■ 0.896	$\chi(\mathcal{H}),193$	$\chi(\mathcal{H}),193$	chromatic polynomial, $\chi(G,\lambda),80$ chromatic spectrum, $P(\mathcal{H}),203$
voloshin-hypergraph_F02544	298	■ 0.999	$P(\mathcal{H},\lambda),204$	$P(\mathcal{H},\lambda),204$	hs, 115
voloshin-hypergraph_F02545	298	■ 0.658	$P(G,\lambda),80$	$P(G,\lambda),80$	lass 2 graphs, 115
voloshin-hypergraph_F02546	299	■ 0.803	$\operatorname{K}(G)$	$\operatorname{K}(G)$	contractible graph contraction, 214

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voloshin-hypergraph_F02550	301	■ 0.973	Δ , 6	Δ , 6	
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